

Unified Theory Based on Time-Wave Dynamics (The P-model)

Part IV: Reproducibility Audit, Public Artifact Registry, and Update Operations

Shunji Ogawa
ENTERSYSTEM Co., Ltd.

Document v2.0 2026/4/8 UTC

1 Purpose of This Document

Part IV is the **verification dossier** for converting the claims of the P-model into "verifiable artifacts" rather than "text". In the main text (Parts I-III), specific file and script names are minimized for readability. The verification entry points and results are summarized in this document.

This document must meet the following two conditions.

1. A third party can access the artifacts (JSON/CSV/PNG) on the public repository (the reference destination is unique).
2. The artifact can be reproduced using a reproduction procedure (script) (the entry point is unique).

2 Public Repository (Verification Scripts and Artifacts)

Public repository (GitHub):

- [EnterOgawa/p-model](#)
- [Releases](#) (fixed snapshots)
- [Basic citation snapshot: v0.1.2](#)
- [Strong field snapshot bridge: v0.1.3](#)

Main directories:

- [scripts/](#) (reproduction entry points)
- [output/public/](#) (public artifact)
- [output/public/cosmology/](#) (Part II: Cosmology/Astronomy)
- [output/public/quantum/](#) (Part III: Quantum, nuclear, physical properties, heat)
- [output/public/llr/](#)(LLR)
- [output/public/weak_field/](#) (weak field integration pack)
- [output/public/theory/](#) (Part I: Core mapping)
- Public artifact hash ledger (all items): `output/public/summary/public_outputs_manifest.json`
- Summary of published artifacts by topic: `output/public/summary/public_outputs_manifest_topics.csv`
- Public artifact continuity audit (previous snapshot difference): `output/public/summary/public_outputs_manifest_continuity.json`
- Public artifact continuity by-topic differences: `output/public/summary/public_outputs_manifest_continuity_topics.csv`
- Public citation set (for fixed release/tag): `output/public/summary/public_manifest_citation_set.json`
- Public citation set (delivery ledger): `output/public/summary/public_manifest_citation_set.csv`
- The citation unit for public snapshots is the public artifact (zip + manifest) (references to local-only areas are not used in the public text)
- β sensitivity full-rerun follow-up audit: `output/public/summary/beta_sensitivity_audit.json` / `output/public/summary/beta_sensitivity_audit.pdf`
- [strong_field_public_snapshot_v0.1.3.zip](#) (Fixed artifacts of strong field 4 systems)
- [strong_field_public_snapshot_v0.1.3_manifest.json](#) (ledger with hash)

Public-repository policy: (Note) EHT/GW/XRISM/Pulsar is not a tree/blob reference, but Fixed artifacts (zip + manifest) of v0.1.3 will be fixed references for citation.

- Only "verification scripts" and "public artifacts" are placed in the public repository.
- Long documents such as the main text of the paper and design memos will be published in separate media, and the main material will be referenced from the main text.

Artifact placement:

- Public products will be collected in output/public/ (subject to auditing and citation).
- Local-only items (intermediate products, heavy HTML/DOCX, etc.) are aggregated in the private area and are not referenced in the public text.

Caution (snapshot policy): The official policy is "base snapshot v0.1.2 + strong-field bridge v0.1.3". Existing links in the text will be maintained based on v0.1.2, and the four strong field systems (EHT/GW/XRISM/Pulsar) will refer to the fixed artifacts (zip + manifest) of v0.1.3.

3 How to Read the Verification Layers

Each verification block is read through the same contract:

- **What was fixed** (freeze: values, windows, sorting rules, etc.)
- **What was output** (difference between observation and theory, residuals, statistics, figures)
- **Under what conditions should it be rejected** (threshold, audit conditions, exception conditions)

Below, references to published artifacts and reproduction entrances (scripts) are shown as a minimum.

3.1 Verification Summary (Latest Snapshot)

Purpose: Provide a one-page snapshot of the latest judgment breakdown across the public ledgers of the Part II main table and the Part III-B quantum scoreboard. **Input:** validation_scoreboard.json (reference rows of Part II main table) and quantum_scoreboard.json (current rows tally in Part III-B quantum scoreboard). **Process:** Read the current counts from the Part II main table and the Part III-B quantum scoreboard, then display the combined-total row as the column-wise sum of the two ledgers. Only the aggregate item counts are fixed in the main text; row-level details remain delegated to the respective artifacts. **Indicator:** Report the counts in the five decision classes Pass / Watch / Reject / Inconclusive / Reference. In this snapshot, the Inconclusive count is 0.

Fixed status definitions used throughout Part IV:

- **Pass:** State where the main gate is satisfied and the rejection condition is not satisfied.
- **Watch:** The main gate is satisfied, but monitoring conditions (thin margin, waiting for additional data, system uncertainty) remain.
- **Reject:** A state in which the main gate does not hold or an explicit rejection condition is met (explicit rejection including theory negation).
- **Inconclusive:** Undetectable/undeterminable. Although the rejection conditions have not been met, it is not possible to confirm acceptance or rejection due to lack of necessary information.
- **Reference:** A line that is only a comparison reference and is not subject to the main judgment of acceptance or rejection.

JSON consistency rules: If inconclusive_n (or a synonymous key) is > 0 , treat it as **Inconclusive** and do not confuse it with **Reject**.

Results:

Aggregate population	Pass	Watch	Reject	Inconclusive	Reference	Source
Part II main table (classical/cosmology verification)	14	5	3	0	5	summary/validation_scoreboard.json
Part II + III combined total	20	17	6	0	5	summary/validation_scoreboard.json + summary/quantum_scoreboard.json

Continued on next page

Aggregate population	Pass	Watch	Reject	Inconclusive	Reference	Source
Part III-B quantum scoreboard (rows tally)	6	12	3	0	0	summary/ quantum_ scoreboard.json

Note: The Part II + III combined total is the column-wise sum of the Part II main table and the Part III-B quantum scoreboard (rows tally). The quantum side uses the current **rows** aggregation in quantum_scoreboard.json and therefore includes the operational rows Bell (planned) and Quantum (planned), so its population does not match the nodal map in 4.1.1. Within validation_scoreboard.json, table1_status_counts is treated as a separate aggregate ledger value.

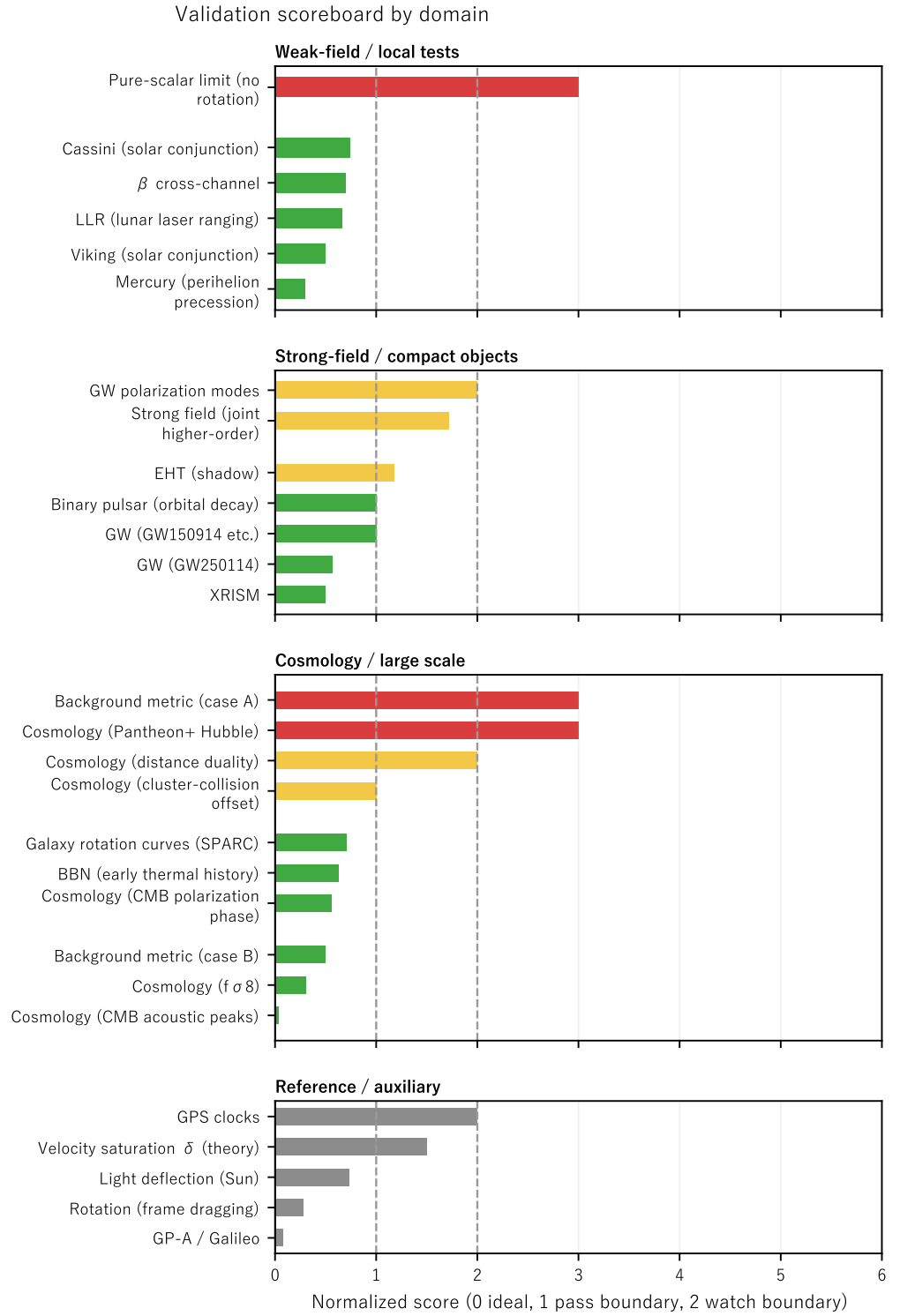


Figure 1: Comparison result for validation scoreboard.

Quantum validation scoreboard (green pass / yellow watch / red mismatch)

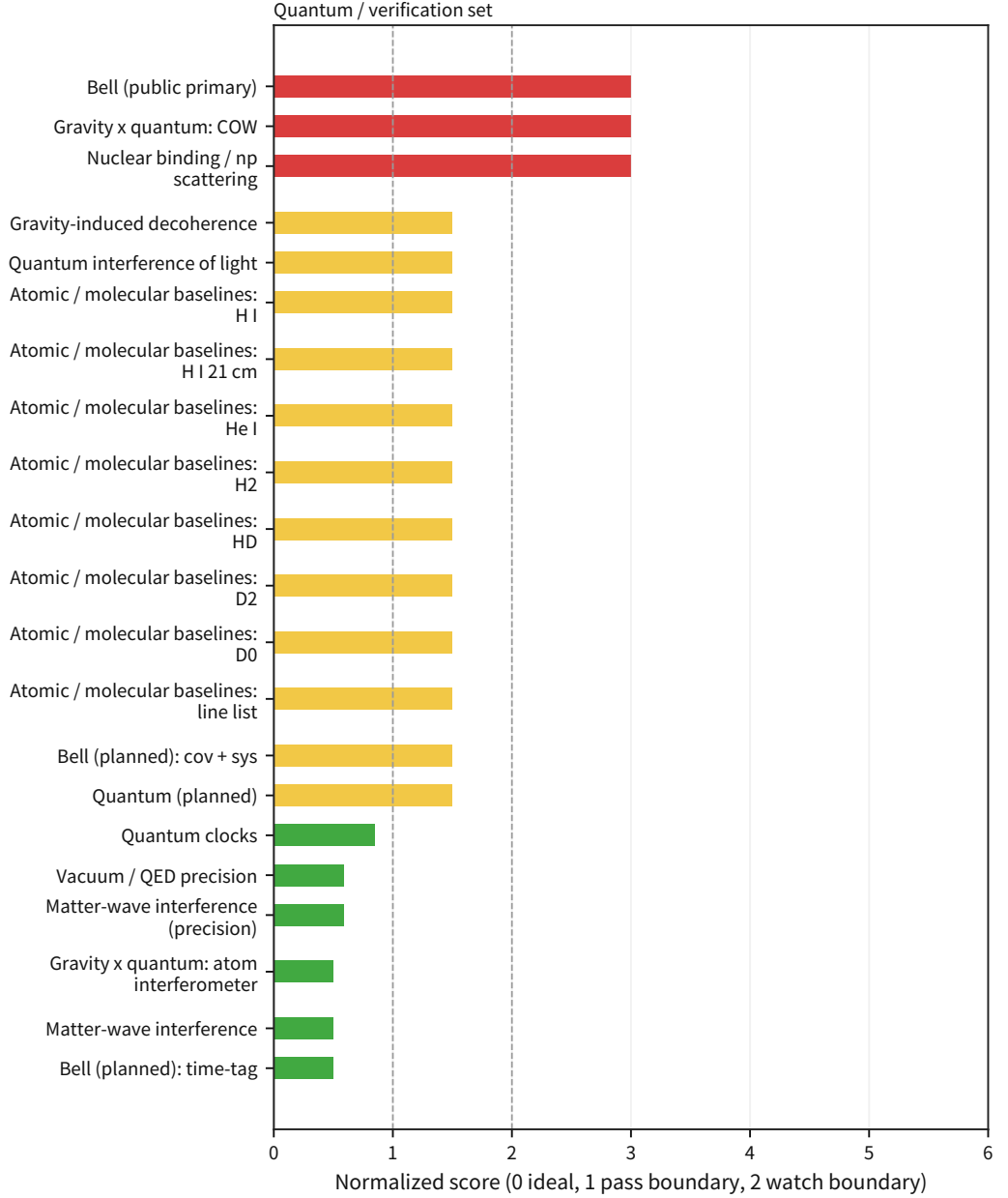


Figure 2: Comparison result for quantum scoreboard.

Note: The bar lengths in the figure are the displayed scores normalized to the decision band. The authoritative judgment is read from the main-gate and rejection-condition fields of each artifact JSON. In `summary/validation_scoreboard.json`, the raw index is stored in `rows[].scoreraw` and its normalization kind in `rows[].scorekind`; on the quantum side, `summary/quantum_scoreboard.json` stores the displayed score in `rows[].score` and the underlying metric in `rows[].metric`.

The fine-structure-constant selection theorem appears in Part IV only as diagnostic support for verification. The theorem statement itself belongs to Part III-A Section 2.6. $\alpha_{q*}(\beta_*) = \alpha_R(\beta_*)$ and its official reading, **a reading that leads to a single fine structure constant from a single freezing effect and the consistency of all physics**, remain authoritative there. Accordingly, any local table comparison or borrowed expression quoted in Part IV is not part of the main acceptance judgment and should be read only as a diagnostic aid.

3.2 Part III Verification Summary Table Row Label Alignment

The verification summary table here refers to the **Part III-B 4.1 scoreboard (row set of paper_table1_results.json)**. Fix the row labels (topic/observable) to the Part IV side to prevent misalignment between the main text reference and the summary row name.

No.	topic	observable	dataset/experiment	Part IV section
1	LLR (lunar laser ranging)	Residual RMS (median value, station $\times$ reflector)		4
2	LLR (lunar laser ranging: NGLR-1)	Residual RMS (median, station $\times$ NGLR-1)		4
3	Cassini (solar conjunction)	Doppler $y(t)$ (observation vs P-model)		6
4	Viking (solar conjunction)	Round Trip Shapiro Delay Peak		6
5	Mercury (perihelion precession)	Perihelion precession (arcseconds per century)		6
6	GPS (satellite clock)	Residual RMS (median, per satellite)		6
7	Light deflection (sun)	Published estimated value γ^* (PPN)		6
8	Gravitational redshift	Published deviation ϵ^* (weak field comparison)	GP-A (1976)	6
9	Gravitational redshift	Published deviation ϵ^* (weak field comparison)	Galileo 5/6 (2018)	6
10	Cosmology (distance duality)	$\eta^{(P)}(z=1) = D_L/((1+z)D_A)$		5
11	Cosmology (Tolman surface brightness)	Index n ($SB \propto (1+z)^{-n}$)	R band (2001)	8
12	Cosmology (Tolman surface brightness)	Index n ($SB \propto (1+z)^{-n}$)	I band (2001)	8
13	Cosmology (independent probe)	SN time dilation (p_t)		8
14	Cosmology (Independent Probe)	CMB Temperature Scaling (p_T)		8
15	JWST/MAST (spectral primary data)	$x_{1d \rightarrow z}$ (line identification; distance-indicator independent)		8
16	XRISM (BH/AGN)	Fe-K absorption line $\rightarrow v/c$ (SR Doppler)		9
17	XRISM (Galaxy cluster)	Fe-K emission line $\rightarrow z_{xray}$ (distance-indicator independent)/ σ_v (line width)		9
18	Cosmology (CMB polarization phase)	TT/EE/TE acoustic phase difference ($\Delta\varphi_{EE}, \Delta\varphi_{TE}$)		8
19	Cosmology (CMB acoustic peak)	TT acoustic peak (1~3 inverse identification, 4~6 holdout)		8
20	Cosmology (structure formation $f\sigma_8$)	Growth rate $f\sigma_8$ (effective friction of delayed branch Γ_{eff})		8

Continued on next page

No.	topic	observable	dataset/experiment	Part IV section
21	Cosmology (Pantheon+ Hubble diagram)	Pantheon+ $\mu(z)$ + CC $H(z)$ direct fit (H_{eff}^2 complete formula)		8
22	Cosmology (Galaxy cluster collision offset)	Lens center-gas center offset (Bullet system)		5
23	Frame drag	R_{drag} ratio	GP-B	7 / 9
24	Frame drag	R_{drag} ratio	LAGEOS	7 / 9
25	Rotation (frame drag)	Pure scalar limit (no rotation term) rejection gate		7 / 9
26	Gravitational waves (polarization mode)	Vector transverse waves vs scalar degeneracy (H1/L1/V1)		9
27	Background metric (caseB)	Vector expansion metric selection gate		9
28	Background metric (caseA)	Vector expansion metric selection gate		9
29	EHT (black hole shadow)	Ring diameter (M87*)		9
30	EHT (black hole shadow)	Ring diameter (Sgr A*)		9
31	Galaxy rotation curve (SPARC)	$V_{\text{obs}}(R)$ rotation curve fit (single- Υ)		5
32	Binary pulsar (orbital decay)	Matching $R = \text{Pdot_b(ops)}/\text{Pdot_b(P-model quad)}$ ($R=1$ matches)		9
33	Gravitational waves (GW150914)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
34	Gravitational waves (GW151226)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
35	Gravitational waves (GW170104)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
36	Gravitational waves (GW170817)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
37	Gravitational waves (GW190425)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
38	Gravitational waves (GW200115_042309)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
39	Gravitational waves (GW200129_065458)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
40	Gravitational waves (GW200224_222234)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
41	Gravitational waves (GW191216_213338)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9

Continued on next page

No.	topic	observable	dataset/experiment	Part IV section
42	Gravitational waves (GW200311_115853)	chirp phase ($f(t)$ extraction $\rightarrow$ quadrupole chirp law fit)		9
43	Gravitational waves (GW250114: Area theo- rem/QNM/IMR)	Significance of area theorem + ringdown QNM(220) + IMR matching		9
44	Strong field (high-order term simultaneous constraint)	Simultaneous constraint of EHT+GW+Pulsar+Fe- $K\alpha$ with a single λ_H		9
45	BBN (initial thermal history)	He-4 mass ratio Y_p (freezing temperature derivation)		11
46	Velocity saturation δ (theory)	$\gamma_{\max}$ (does not diverge at $v \rightarrow c$)		7
47	Quantum (Bell)	CHSH/CH indicator selection sensitivity audit		10
48	Quantum (Nuclear Mass)	Z-N Residual Map and Differential Predictions		11
49	Quantum (Nuclear Physics/BBN)	BBN multi-isotope gate (D/H, He-3/He-4, Li-7)		11
50	Quantum (Condensed Matter)	holdout audit (condensed)		12
51	Quantum (Thermodynamics)	Blackbody Baseline and Minimal Audit		13

Fixed number of rows: 51 (1:1 correspondence with Part III verification summary table rows)

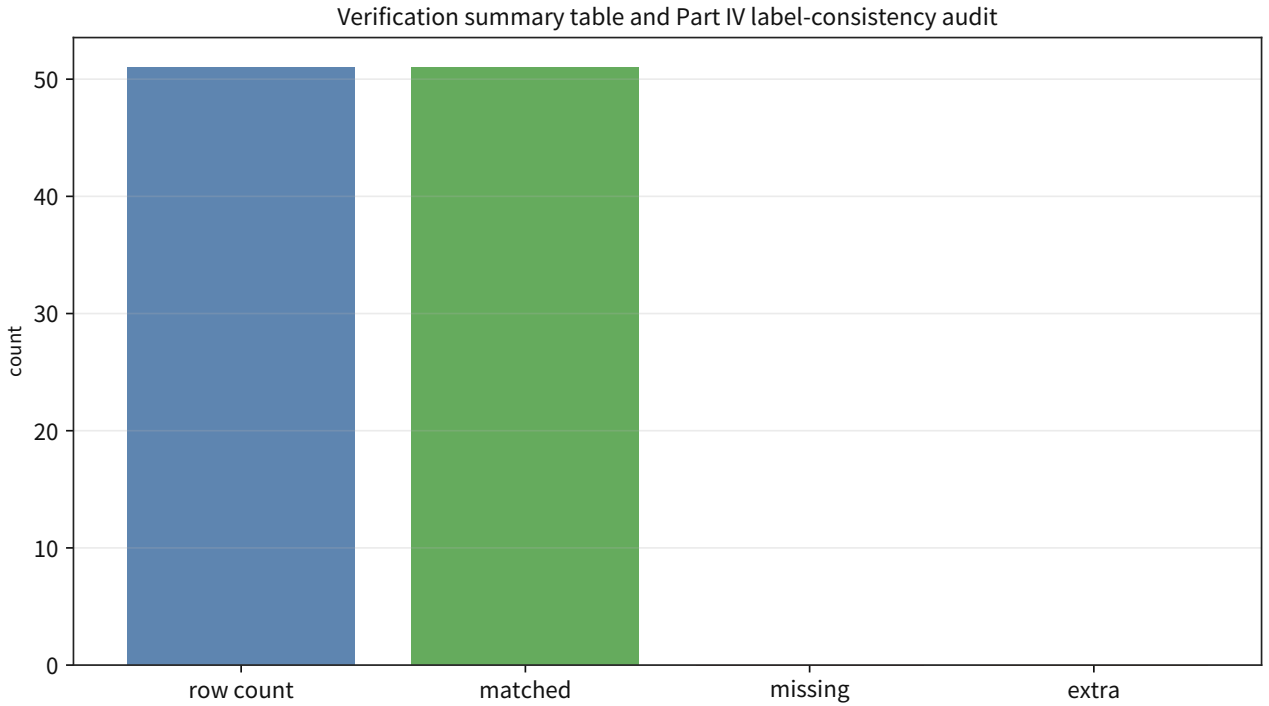


Figure 3: Part IV label-parity audit for the verification summary table.

Falsification conditions:

- If all rows of the verification summary table (topic/observable, with dataset/experiment if duplicated) do not exist in the fixed label table of Part IV in a 1:1 ratio, reject.
- If there is an extra row label (topic/observable/dataset) that does not exist in the verification summary table on the Part IV side, reject it.
- Reject if fixed number of rows (51) and `n_missing_rows=0` / `n_extra_rows=0` of audit JSON are not satisfied at the same time.

4 LLR (Lunar Laser Ranging): Time-Tag and Batch Audit

Purpose: Reanalyze the LLR primary data (EDC monthly NP2) on the same I/F, and fix in figures and statistics how the observation-prediction (TOF / Range) residuals change with procedure-dependent inputs such as time-tag interpretation and station-coordinate sources.

Public artifacts (GitHub):

- Directory: [output/public/llr/](#)
- Falsification pack (main gate: operational indicator $weightedRMS + normalizedresidual/chi^2 - like$, auxiliary gate: Shapiro term): [llr_falsification_pack.json](#)
- Operational indicator audit (fixed output so that the median RMS is not used as the main judgment): [output/public/llr/llr_operational_metrics_audit.json](#) / [output/public/llr/llr_operational_metrics_audit.csv](#) / [output/public/llr/llr_operational_metrics_audit.pdf](#)
- Shapiro's contribution (made worse by removal): Figure: [llr_shapiro_ablations_overall.pdf](#)
- Time tag selection (entrance audit by station): [llr_time_tag_best_by_station.json](#) / Figure: [llr_time_tag_selection_by_station.pdf](#)
- Outlier audit (time tag sensitivity/suspected target contamination): [llr_outliers_diagnosis_summary.json](#) / Figure: [llr_outliers_time_tag_sensitivity.pdf](#)
- Residual distribution (observation - model): Figure: [llr_residual_distribution.pdf](#)
- Root-cause decomposition by system (station/target/group and improvement amount rank): [output/public/llr/llr_systematics_root_cause.json](#) / [output/public/llr/llr_systematics_root_cause.csv](#) / [output/public/llr/locales/en/llr_systematics_root_cause.pdf](#)
- 4 ns Excess contribution audit (excess share of station/target/group): [output/public/llr/locales/en/llr_systematics_root_cause_over4ns.pdf](#)
- Station coordinate difference (pos+eop vs SLRLOG): Figure: [llr_station_coord_delta_pos_eop.pdf](#)

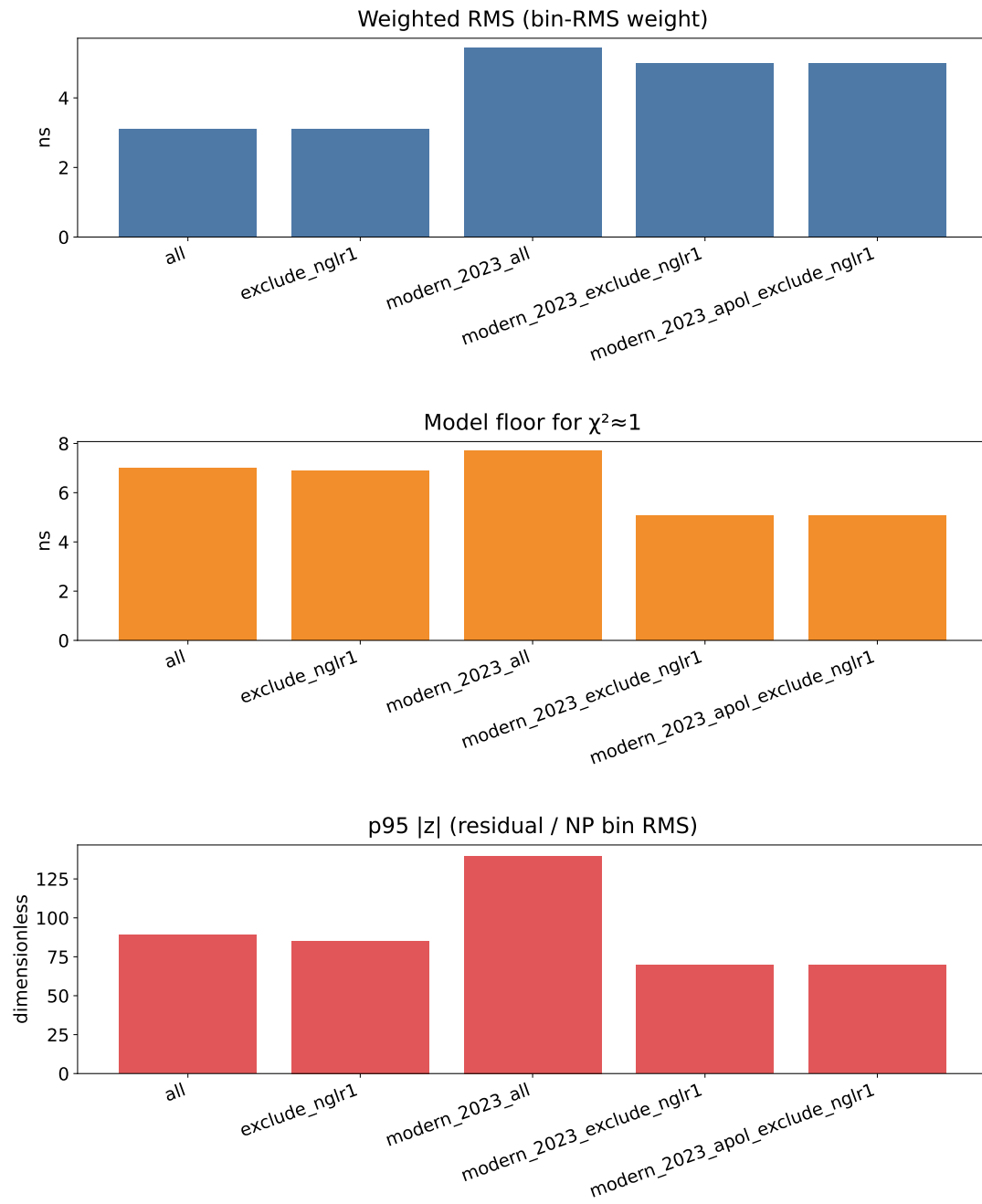


Figure 4: LLR operational-metrics audit

LLR residual root-cause decomposition

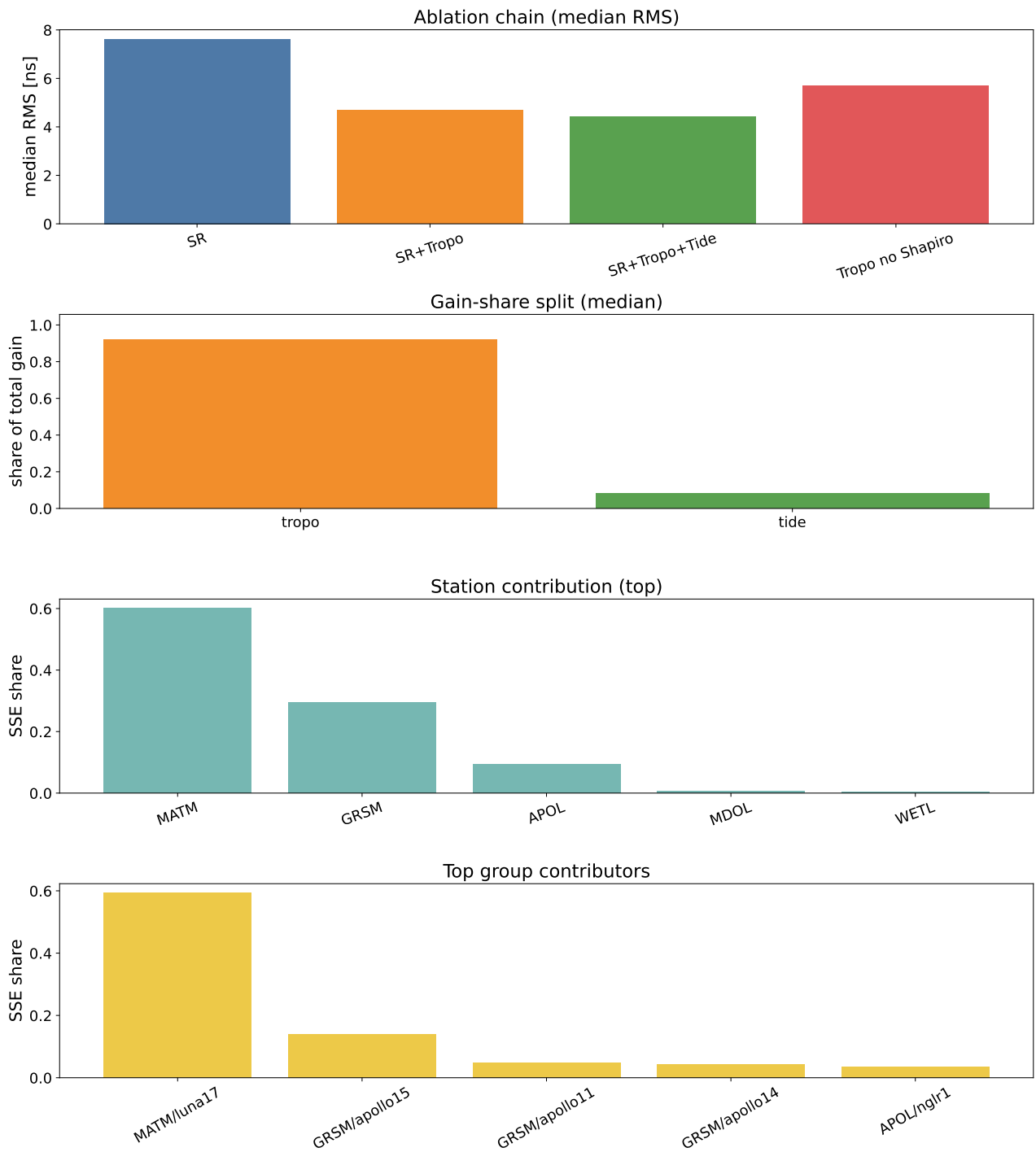


Figure 5: LLR root-cause decomposition audit

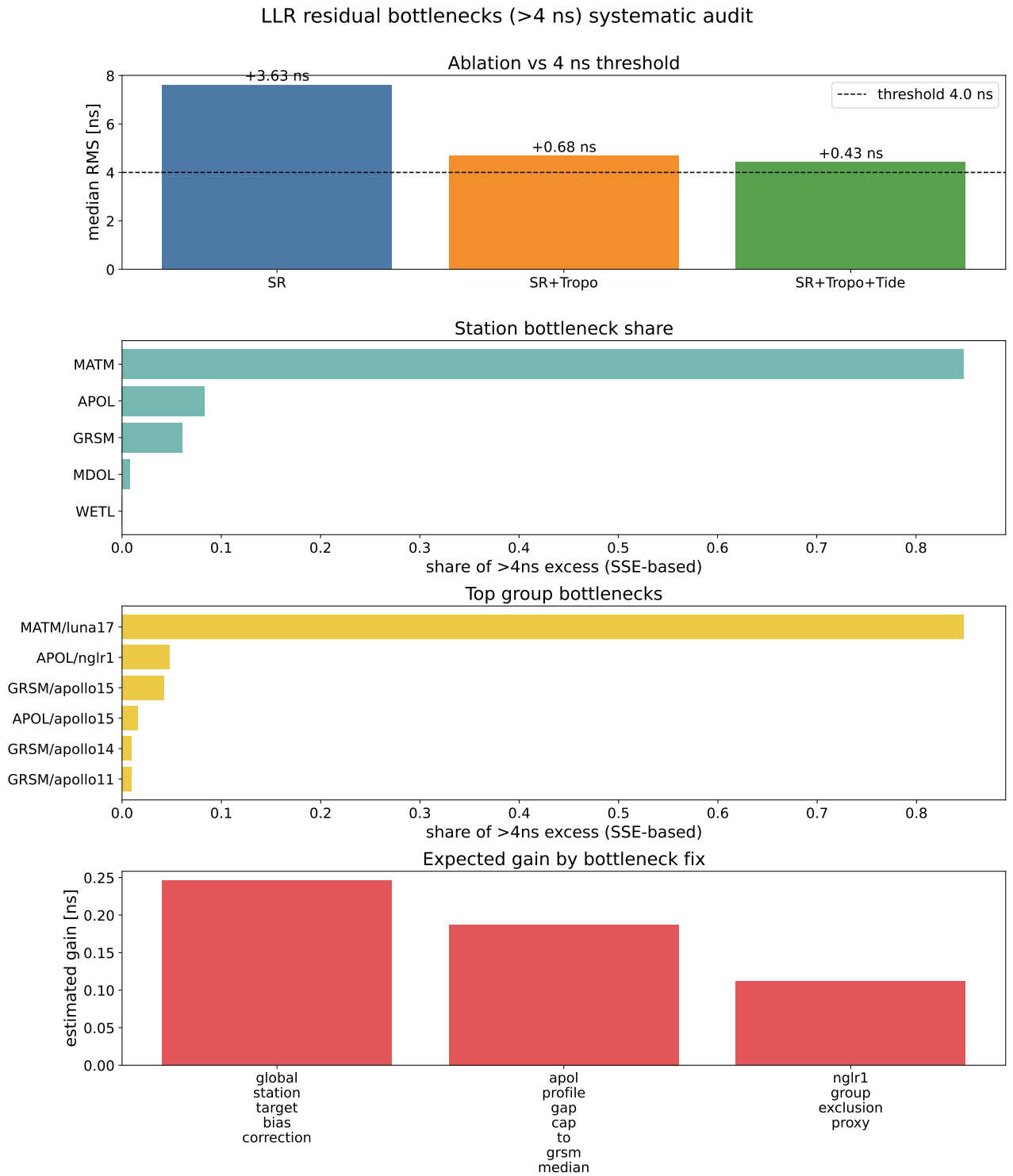


Figure 6: LLR excess-contribution audit over 4 ns

Reproduction entry points (scripts/GitHub):

- Batch acquisition (manifest generation) [scripts/llr/fetch_llr_edc_batch.py](#)
- Batch evaluation (generating figures and statistics) [scripts/llr/llr_batch_eval.py](#)
- Operational index audit (weighted RMS / normalized residual / χ^2) [scripts/llr/llr_operational_metrics_audit.py](#)

- Public pack generation (private $\rightarrow$ public promotion) [scripts/llr/llr_falsification_pack.py](#)

Caution: `--time-tag-mode auto` uses Horizons and requires a network. SSL verification may fail depending on the environment (in that case, use environment variables to avoid it). Also, use `output/llr/horizons_cache/` for offline re-execution.

5 Cosmology/Galaxies (Part II): DDR and SPARC (Public Artifacts)

Purpose: In Part II, we will first list the items (DDR and SPARC) that can be fixed on GitHub as public artifacts. Direct link to rejection condition pack and main figures.

Public artifacts (GitHub):

- Directory: [output/public/cosmology/](#)
- Distance duality (DDR): integration of publication constraints (visualization + indicators) [cosmology_distance_duality_constraints.pdf](#) / [cosmology_distance_duality_constraints_metrics.json](#)
- DDR: Re-evaluation using P-model indicator $\eta^{(P)}$ (figure + table) [cosmology_ddr_pmodel_eta_constraints.pdf](#) / [cosmology_ddr_pmodel_eta_constraints.json](#)
- DDR: Falsification-Condition Pack (Adoption/Rejection Gate) [cosmology_ddr_pmodel_falsification_pack.json](#)
- DDR: (1+z) Manual calculation trace (SNe Ia SALT2 / BAO distance estimation I/F audit)

Manual line-by-line trace for implicit expansion assumption in DDR pipelines

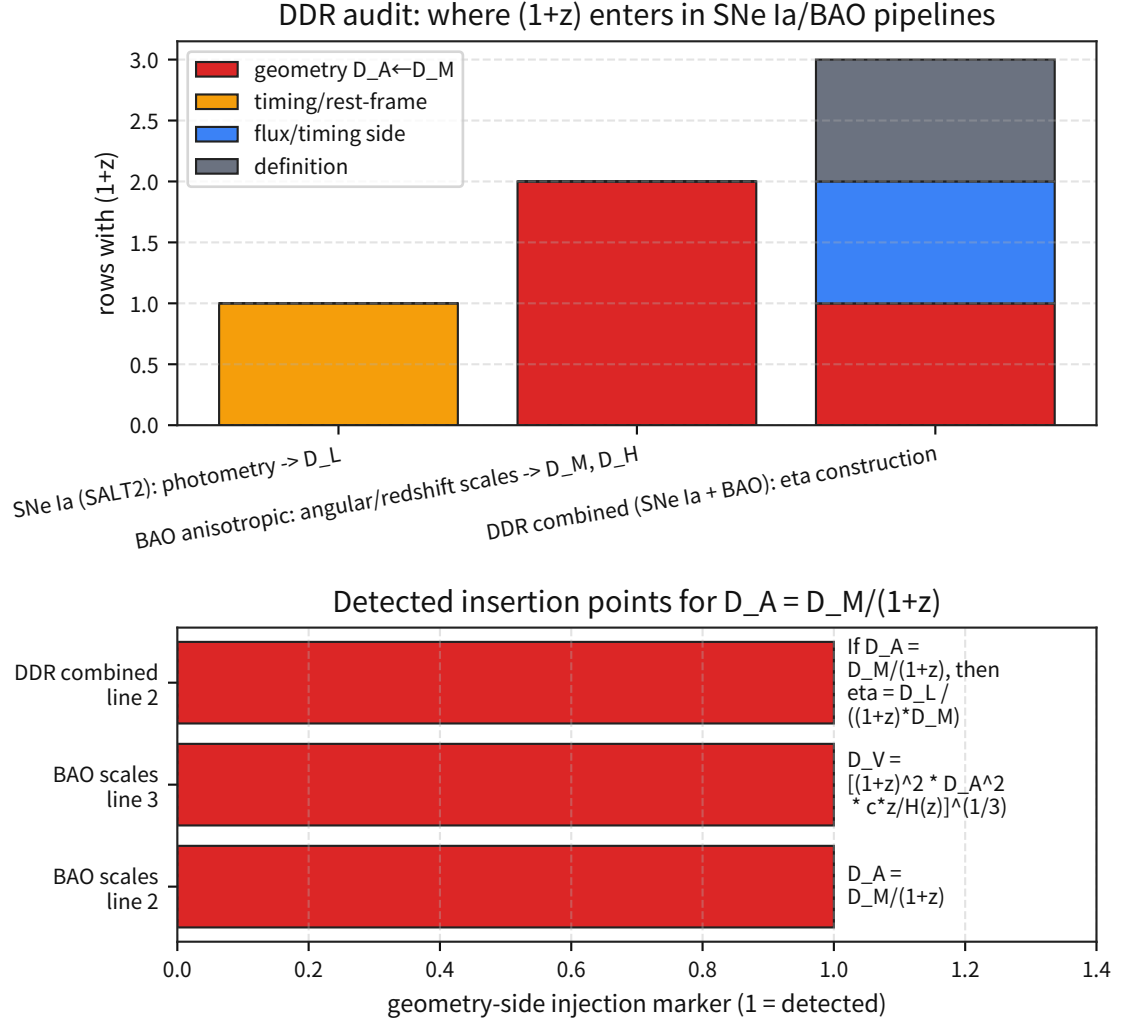


Figure 7: Comparison result for cosmology ddr 1pz manual trace audit.

output/public/cosmology/cosmology_ddr_1pz_manual_trace_audit.json / output/public/cosmology/cosmology_ddr_1pz_manual_trace_audit.csv

- Redshift (minimum fixed for P_{bg} mapping): [cosmology_redshift_pbg_metrics.json](#)
- DM elimination direct bridge route contract: output/public/quantum/mass_origin_direct_kappa_bridge_route_contract_metrics.json
- DM elimination direct bridge statement freeze: output/public/quantum/mass_origin_direct_kappa_bridge_statement_freeze_metrics.json
- DM elimination direct bridge retry / equality audit: output/public/quantum/mass_origin_dark_matter_postnewtonian_direct_bridge_retry_metrics.json / output/public/quantum/mass_origin_direct_kappa_sparc_equality_audit_metrics.json
- DM elimination radial-profile quantification / declaration gate: output/public/quantum/mass_

origin_postnewtonian_rotation_curve_profile_metrics.json / output/public/quantum/mass_
origin_dark_matter_elimination_declaration_gate_metrics.json

- SPARC (Galaxy Rotation): Falsification-Condition Pack (freeze / reject / indicator) [sparc_falsification_pack.json](#)
- SPARC: RAR reconstruction (fixed output of recalculation) [sparc_rar_reconstruction.csv](#)
- SPARC: freeze test (auditing procedural dependencies) [sparc_rar_freeze_test_metrics.json](#)
- SPARC: Rotation curve audit ($V_{obs}/V_{bar}/V_P$; single- Υ fit)

SPARC audit (single-Y fit)

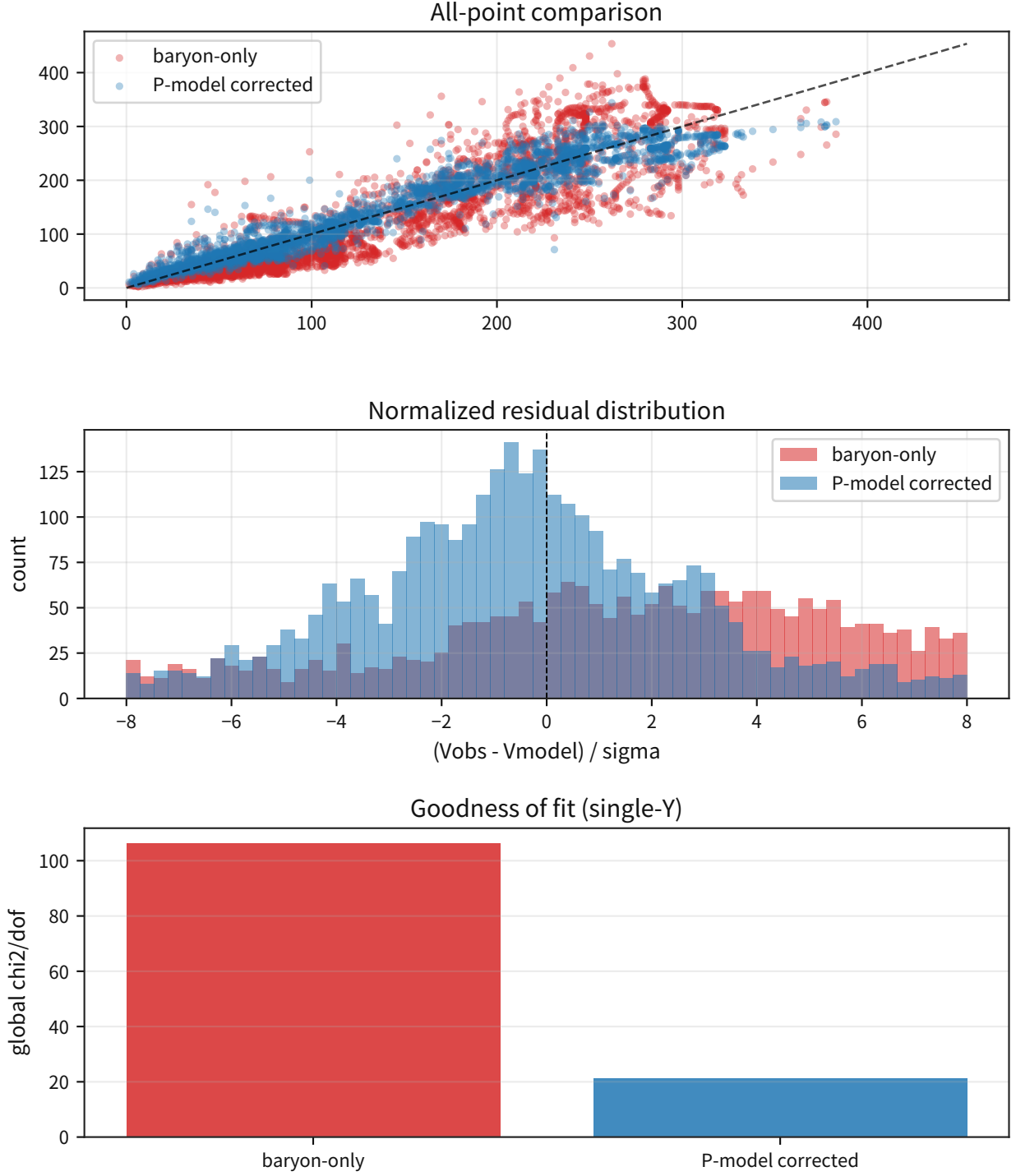


Figure 8: Comparison result for sparc rotation curve pmodel audit.

output/public/cosmology/sparc_rotation_curve_pmodel_audit_metrics.json / output/public/
cosmology/sparc_rotation_curve_pmodel_audit_galaxy_summary.csv

- Galaxy cluster collision offset (Bullet system; $\Delta x_{P\text{-gas}} / \Delta x_{P\text{-lens}}$ audit)

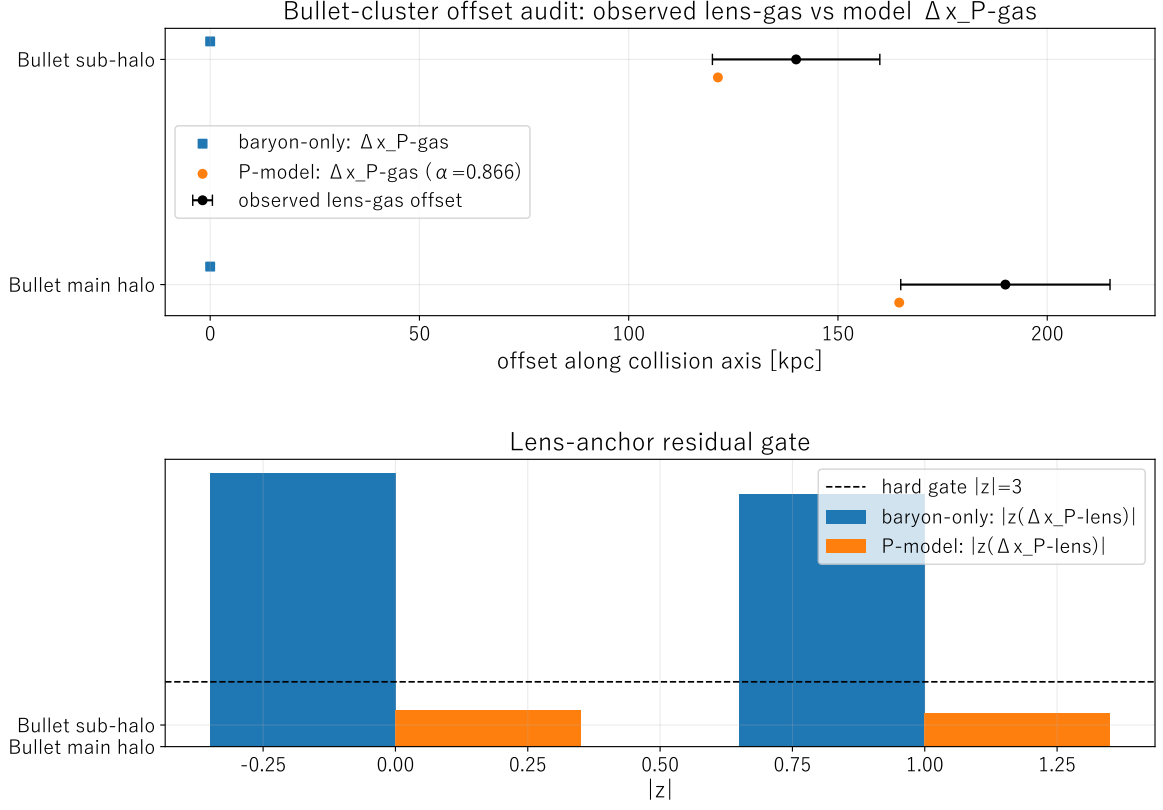


Figure 9: Comparison result for cosmology cluster collision p peak offset audit.

- output/public/cosmology/cosmology_cluster_collision_p_peak_offset_audit.json
- output/public/cosmology/cosmology_cluster_collision_p_peak_offset_audit.csv
- output/public/cosmology/collision_offset_observables_primary_template.csv
- output/public/cosmology/collision_offset_primary_registration_checklist.csv

(Current stage: overall_status=pass, main gate is pass, watch factor is resolved. primary_registration_readiness: decision_reason=ready primary_registration_checklist_summary: n_ready=2/2, n_missing_primary_refs=0, n_source_mode_pending=0 primary_map_update_watchpack: update_event_type=no_change, update_event_detected=false, event_counter=3, source_mode=primary_map_registered)

In this version, SPARC's $a_0 = \kappa_a c H_0^{(P)}$ is not treated as an operational fitting coefficient. The value $\kappa_a = 1/(2\pi)$ is derived from the late-time law of the background P-wave, $P_{\text{bg}}(t) \propto \exp[-H_0^{(P)}(t - t_0)]$, and matches the SPARC operational value to computer precision; the dark-matter-elimination branch is therefore fixed as a direct theoretical bridge. Here "same baryon I/F" means the same baryonic construction as in Part II Section 4.14, $V_{\text{bar}}^2(R) = V_{\text{gas}}^2 + \Upsilon V_{\text{disk}}^2 + (f_{\text{bulge}} \Upsilon) V_{\text{bulge}}^2$, with Υ as the only free parameter and f_{bulge} fixed. The operational test is then a cross-sample comparison under the same baryon I/F. If an independent galaxy sample meaningfully requires $\kappa_a \neq 1/(2\pi)$ while keeping $H_0^{(P)}$ fixed, reject the dark-matter-elimination branch.

Reproduction entry points (scripts/GitHub):

- DDR (publication constraint integration + $\eta^{(P)}$ mapping + pack generation): [scripts/cosmology/cosmology_distance_duality_constraints.py](#)
- DDR: (1+z) Manual calculation trace (SNe Ia/BAO line-by-row derivation audit): [scripts/cosmology/cosmology_ddr_1pz_manual_trace_audit.py](#)
- Redshift (minimum fixed for P_{bg} mapping): [scripts/cosmology/cosmology_redshift_pbg.py](#)
- DM elimination direct bridge (theoretical side freeze of $\kappa_a = 1/(2\pi)$): [scripts/quantum/mass_origin_dark_matter_direct_kappa_bridge_branch.py](#)
- SPARC (pack generation): [scripts/cosmology/sparc_falsification_pack.py](#)
- SPARC (freeze test audit): [scripts/cosmology/sparc_rar_freeze_test.py](#)
- SPARC (rotation curve audit; single- Υ fit): [scripts/cosmology/sparc_rotation_curve_pmodel_audit.py](#)
- Galaxy cluster collision offset audit: [scripts/cosmology/cosmology_cluster_collision_p_peak_offset_audit.py](#)

6 Weak-Field Integration (Cassini/Viking/Mercury/GPS/LLR): Public Pack

Purpose: When passing weak-field tests (Cassini/Viking/Mercury/GPS/LLR) with the same sign convention and frozen β (light propagation), fix the questions "Is it self-consistent even over long time spans and across multiple systems?" and "Under what conditions should it be rejected?" in a format that can be referenced together as a public artifact.

Public artifacts (GitHub):

- Integrated pack (falsification conditions, freeze, main indicators, list of deliverables): [output/public/weak_field/weak_field_falsification_pack.json](#)
- Long-term, multi-system integrated summary (diagram/JSON): [weak_field_longterm_consistency.pdf](#) / [weak_field_longterm_consistency.json](#)
- Test matrix (minimum fixed I/F: input/output/script): [weak_field_test_matrix.json](#)

Public artifacts by test (main directory):

- Cassini: [output/public/cassini/](#)
- Cassini primary data direct β audit (ODF raw, independent of PPN published values): [output/public/cassini/cassini_odf_beta_direct_fit_metrics.json](#) / [output/public/cassini/cassini_odf_beta_direct_fit_points.csv](#) / [output/public/cassini/cassini_odf_beta_direct_fit.pdf](#)
- Preprocessing dependency of Cassini direct fit (ODF/TDF horizontal comparison): [output/public/cassini/cassini_beta_direct_fit_cross_source_metrics.json](#) / [output/public/cassini/cassini_beta_direct_fit_cross_source_summary.csv](#) / [output/public/cassini/locales/en/cassini_beta_direct_fit_cross_source.pdf](#)
- Cassini ODF raw normalization I/F manifest (actual value ledger): [output/public/cassini/cassini_odf_raw_if_manifest.json](#) / [output/public/cassini/cassini_odf_raw_if_manifest.csv](#)
- Cassini ODF media correction field matching (TRK/SCE1 crosswalk): [output/public/cassini/cassini_odf_media_correction_field_map.json](#) / [output/public/cassini/cassini_odf_media_correction_field_map.csv](#)
- Cassini ODF External CSP acquisition availability ledger (IONCAL/TROPICAL/PLSMCAL): [output/public/cassini/cassini_odf_external_csp_availability.json](#) / [output/public/cassini/cassini_odf_external_csp_availability.csv](#)
- Cassini ODF/TDF convergence gate ledger (promotion judgment): [output/public/cassini/cassini_beta_direct_fit_cross_source_convergence.csv](#)
- Cassini media correction ancillary acquisition ledger (ION/TRO/CHPART): [output/public/cassini/cassini_csp_media_ancillary_availability.json](#) / [output/public/cassini/cassini_csp_media_ancillary_availability.csv](#)
- Media state binding audit (record-level coverage) to Cassini ODF lines: [output/public/cassini/cassini_odf_media_state_join_metrics.json](#) / [output/public/cassini/cassini_odf_media_state_join_summary.csv](#)

- Cassini sign/media closure audit (candidate refit + label observability): `output/public/cassini/cassini_odf_sign_media_closure_audit.json` / `output/public/cassini/cassini_odf_sign_media_closure_audit.csv`
- Cassini sign re-audit (record-level decomposition + window/quality re-evaluation): `output/public/cassini/cassini_odf_sign_recordlevel_reaudit.json` / `output/public/cassini/cassini_odf_sign_recordlevel_groups.csv` / `output/public/cassini/cassini_odf_sign_recordlevel_window_quality.csv`
- VLBI (AUA020; group delay primary data direct connection β audit): `output/public/vlbi/vlbi_17may01xa_beta_direct_fit_metrics.json` / `output/public/vlbi/locales/en/vlbi_17may01xa_beta_direct_fit.pdf`
- VLBI Session Identification Audit (17MAY01XA- > AUA020): `output/public/vlbi/vlbi_17may01xa_session_identity_audit.json` / `output/public/vlbi/vlbi_17may01xa_session_identity_audit.csv`
- VLBI nuisance sensitivity audit (*none/intercept/interceptlinear*): `output/public/vlbi/vlbi_17may01xa_beta_nuisance_sensitivity_metrics.json` / `output/public/vlbi/vlbi_17may01xa_beta_nuisance_sensitivity_summary.csv` / `output/public/vlbi/locales/en/vlbi_17may01xa_beta_nuisance_sensitivity.pdf`
- VLBI source-filter sensitivity audit (all vs 0229+131,0235+164): `output/public/vlbi/vlbi_17may01xa_beta_source_filter_sensitivity_metrics.json` / `output/public/vlbi/vlbi_17may01xa_beta_source_filter_sensitivity_summary.csv` / `output/public/vlbi/locales/en/vlbi_17may01xa_beta_source_filter_sensitivity.pdf`
- Viking:[output/public/viking/](#)
- Mercury:[output/public/mercury/](#)
- GPS:[output/public/gps/](#)
- Freezing parameter (β): [output/public/theory/](#)
- LLR (reference): [output/public/llr/](#)
- β derived terminal ledger (VLBI+LLR+MESSENGER cross-channel): `output/public/summary/beta_cross_channel_registry.json` / `output/public/summary/beta_cross_channel_registry.csv` / `output/public/summary/locales/en/beta_cross_channel_registry.pdf`

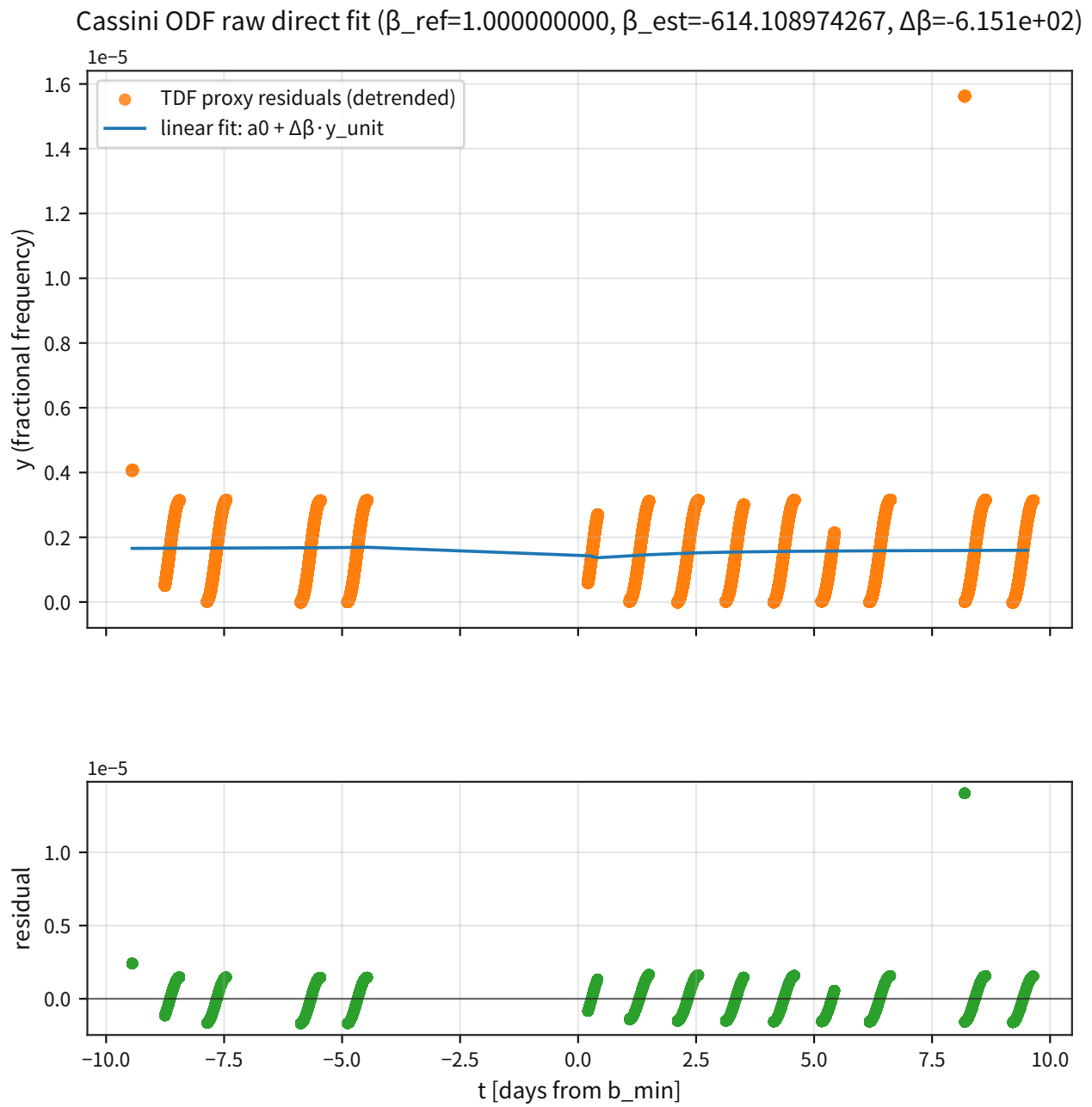


Figure 10: Cassini ODF raw direct β audit

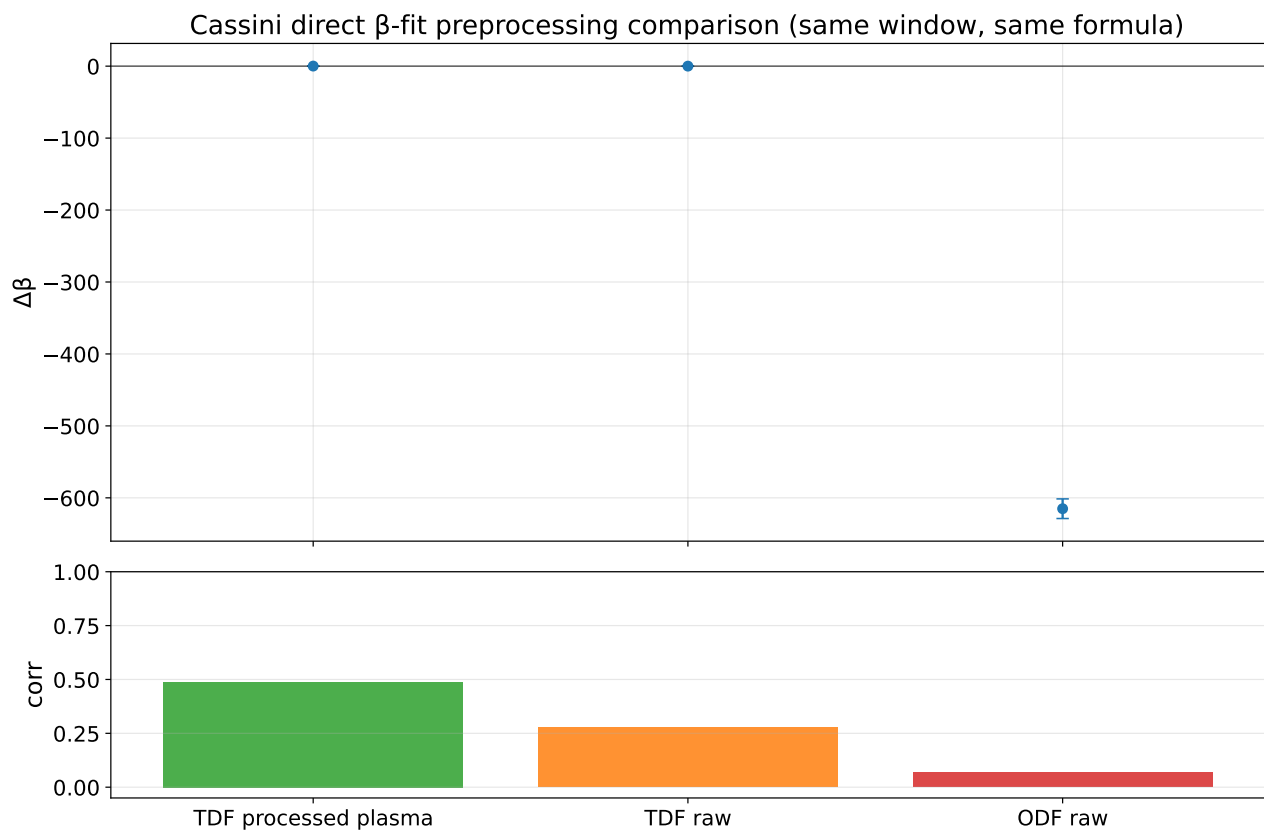


Figure 11: Cassini direct-fit cross-source audit

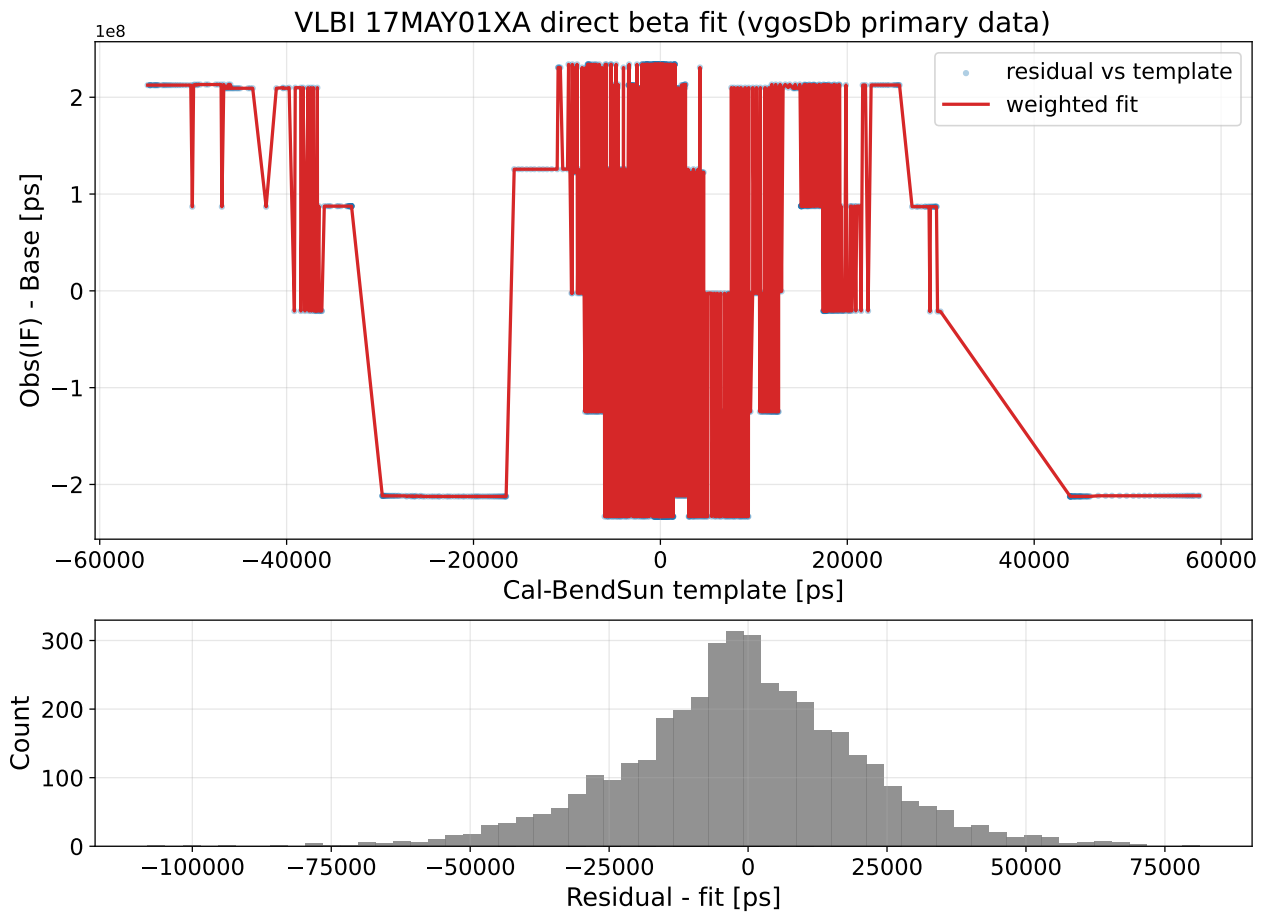


Figure 12: VLBI direct β fit audit

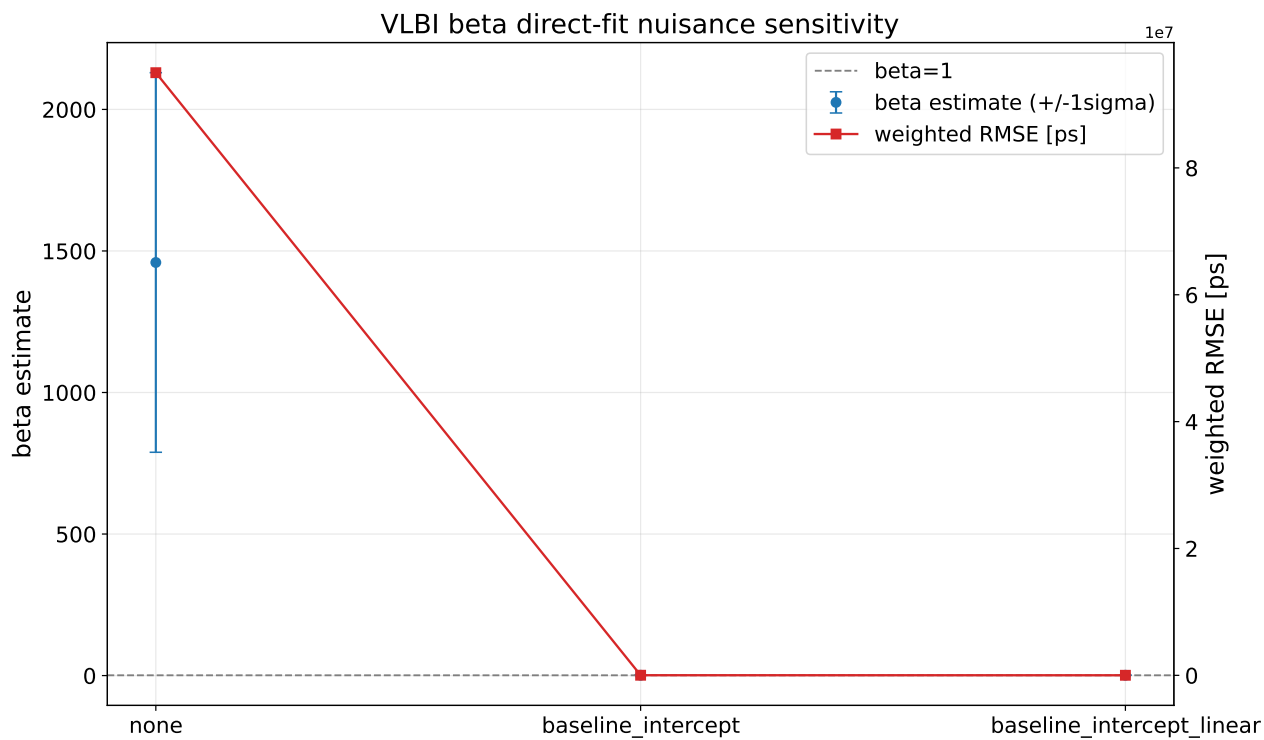


Figure 13: VLBI nuisance-sensitivity audit

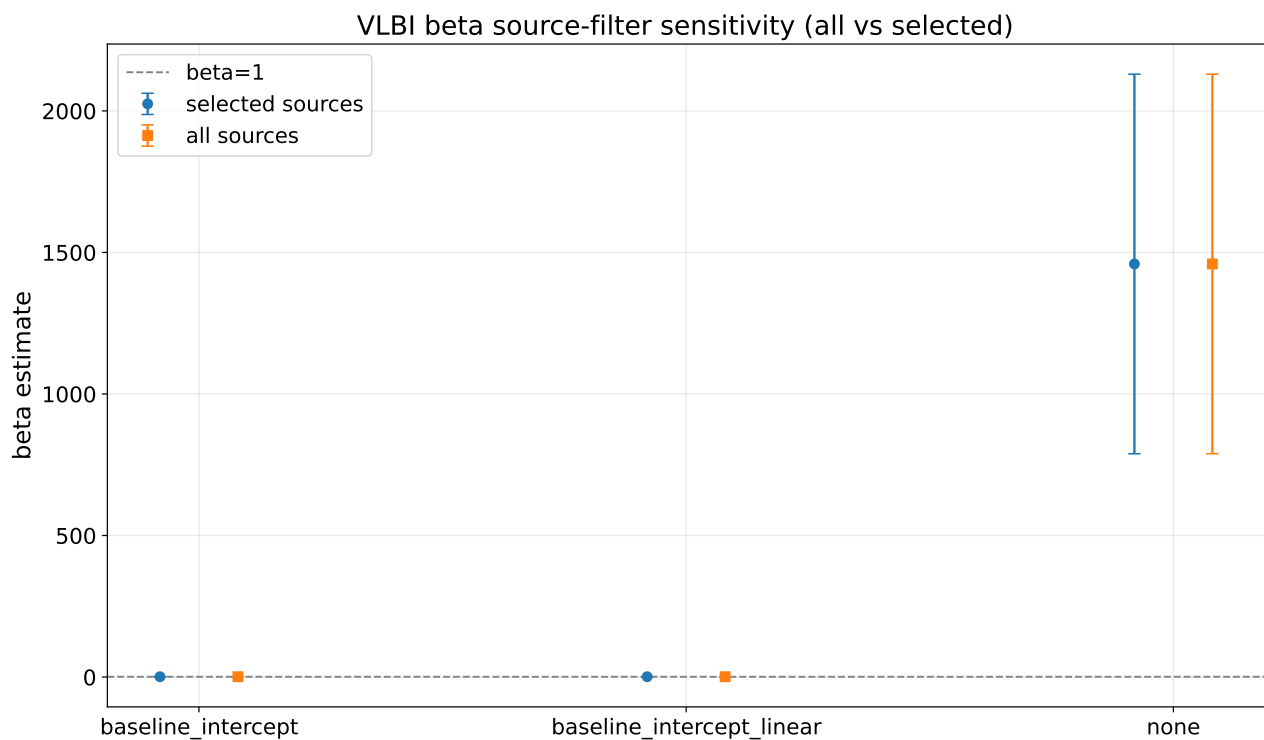


Figure 14: VLBI source-filter sensitivity audit

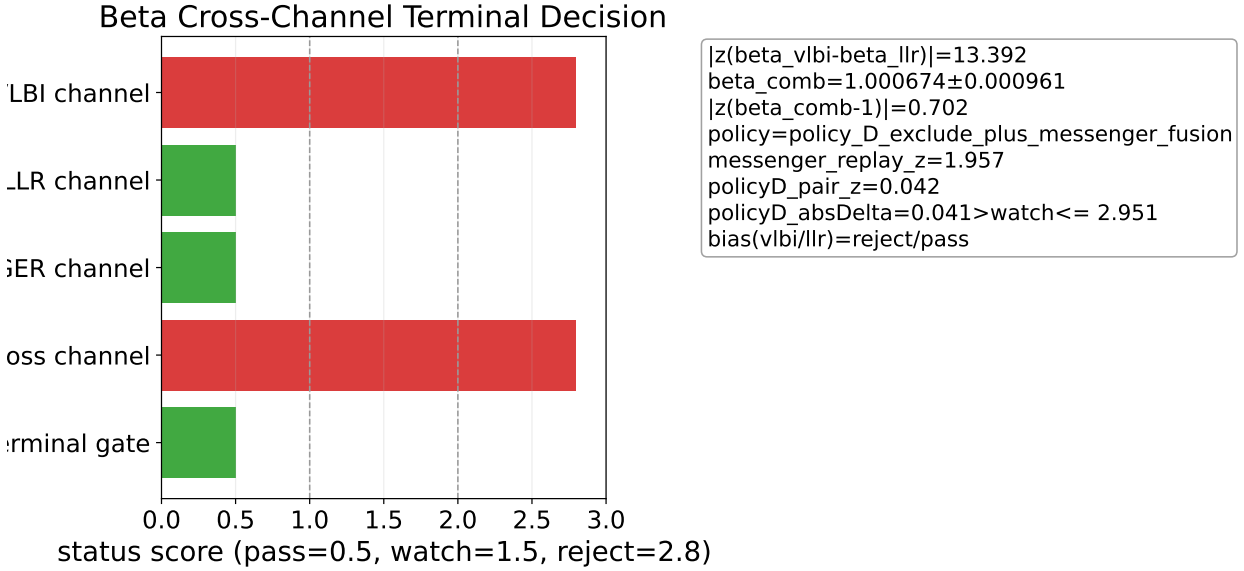


Figure 15: Cross-channel β terminal ledger

- VLBI subset refit (comparator eligibility): `output/public/vlbi/vlbi_beta_cross_consistency_subset_refit_metrics.json`
- LLR $\kappa \rightarrow \beta$ promotion gate: `output/public/llr/llr_kappa_llr_beta_promotion_gate_metrics.json`

Reproduction entry points (scripts/GitHub):

- Long-term/multi-system aggregation (recalculation is optional): [scripts/summary/weak_field_longterm_consistency.py](#)
- Rejection condition for integration (threshold gate): [scripts/summary/weak_field_falsification.py](#)
- Public pack generation (private $\rightarrow$ public promotion): [scripts/summary/weak_field_public_pack.py](#)

Caution: The weak-field pack above refers to the fixed snapshot (Release tag) v0.1.2 for quotation. If an update is required, update the snapshot with the next release. The direct fit of ODF raw is retained as an audit entrance directly connected to the primary data, and is not used for the main judgment of absolute β . The main judgment is made from TDF shape matching together with the scalar-limit gate. In horizontal comparisons (TDF plasma / TDF raw / ODF raw) using the same window and the same formula, the $\Delta\beta$ difference is large and preprocessing definition dependence is dominant, so it is not promoted to absolute β judgment until the ODF side I/F normalization is completed.

ODF raw normalization I/F:

Item	Fixed key	Judgment condition	Status
Band	band_id	Match observation series and template	Pass

Continued on next page

Item	Fixed key	Judgment condition	Status
Sign	doppler_sign_convention	Unified DSN sign convention	Watch
Media correction	media_correction_state	Fixed record-level observability with TRK/SCE1 matching + ancillary payload combination	Pass (external_csp_join_ready, coverage_any=1.0)
Time system	time_scale_id	UTC/TDB conversion rules into a single I/F	Pass

ODF/TDF convergence gate:

Gate	Threshold	Actual Measurement	Status
$ \Delta\beta_{\text{plasma}} - \Delta\beta_{\text{raw}} $	≤ 0.10	0.0770	Pass
$z_{\min}(\text{ODF/TDF})$	≤ 3	45.10	Watch
corr_{odf}	≥ 0.25	0.0693	Watch

sign/media closure audit:

Candidate/Audit	Results	Judgment
Best sign candidate (invert_sign + offset removal)	$z_{\min} = 33.01$, $\text{corr} = 0.0693$	Watch
media observability (14 ODF labels)	record-level flag not detected	Watch
TRK/SCE1 + ancillary combination	<i>sce1_{ancillary/ion,tro}</i> primary payload acquisition + CHPART window expansion + ODF combination (coverage_any=1.0)	Pass (hard-watch release)
record-level sign decomposition	groups=14, pass=2, pass_ratio=0.1429	Watch
Window/Quality	window=4.0 days, trim=0.01,	Watch (corr not reached)
Re-evaluation	$z_{\min} = 0.0141$, $\text{corr} = 5.07 \times 10^{-5}$	

The overall judgment is watch (Reason: if_manifest_not_closed, odf_delta_beta_not_converged, odf_shape_corr_too_low, sign_hypothesis_not_closed). Therefore, at present, ODF raw is treated as an audit entry point (Reference/Watch) and is not promoted to absolute β judgment.

VLBI (AUA020) Primary data direct audit:

Gate	Actual measurement	Condition
Session identification (17MAY01XA- > AUA020)	206/206 nc files match	Pass

Continued on next page

Gate	Actual measurement	Condition
selected(2 source)baseline_ intercept_linear	$\beta = 1.04023 \pm 0.01123$	
all sources baseline_intercept_linear	$\beta = 1.18669 \pm 0.06961$	
source-filter difference (best mode)	$\Delta\beta_{\text{all-selected}} = +0.14646,$ $ z = 2.08$	Watch

The current acceptance/rejection is Watch. While simultaneous estimation of nuisance can suppress the divergence of **none**, the source-filter dependence $|z| \approx 2$ will remain, so promotion to main judgment is conditional on reproducing the same I/F in an additional session.

VLBI all-sky consistency audit:

Gate	Actual measurement	Condition
all-sky 12 session(baseline_ intercept_linear)	$\chi_2/dof = 276.11$	Reject
all-sky 20 session (first half of May after XA addition)	$\chi_2/dof = 341.70$	Reject
all-sky 30 session (after adding XA at the end of April)	$\chi_2/dof = 310.36$	Reject
all-sky 35 session (2019/2020 May after XA addition)	$\chi_2/dof = 286.89$	Reject
High sensitivity subset ($\max Cal - BendSun \geq 10ns$, n=4)	$\chi_2/dof = 203.83$	Reject

output/public/vlbi/vlbi_allsky_beta_consistency_metrics.json is the main ledger, and the current All-sky audit judgment is fixed as Reject. In particular, since χ_2/dof does not decrease sufficiently even with a highly sensitive subset, simply increasing the number of sessions does not satisfy the consistency gate.

High sensitivity group factor analysis:

Session	Cal-BendSun Maximum absolute value (ns)	Controlling factor (z impact maximum absolute value)	Supplement
17MAY01XA	57.69	source 0235+164 (1.16)	<i>source/time/baseline</i> All less than Watch
18MAY01XA	38.92	source 0229+131 (9.81)	source Strongly dependent Watch/Reject contribution
20MAY06XA	11.45	time quartile Q3 (5.45)	Time zone dependence is dominant
20MAY04XA	10.11	source 0059+581 (3.14)	source and time contribute simultaneously

Referring to output/public/vlbi/vlbi_high_sensitivity_factor_decomposition_metrics.json, we confirmed that the main cause in the high-sensitivity group alternates between **source** and **time_quartile**

from session to session. Therefore, in the next stage, priority is given to stratified refit under the same source/time-band conditions, and it is determined whether all-sky Reject can be canceled.

High sensitivity group source $\times$ session $\Delta\beta$ matrix:

Gate	Actual measurement	Condition
High-sensitivity 4 session matrixization (min_source_points=20)	17MAY01XA, 18MAY01XA, 20MAY04XA, 20MAY06XA	Pass (configuration established)
Proxy index of consistency across matrices	$\chi_2/dof = 185.69$	Reject
key source 0229+131	$\chi_2/dof = 22.01$	Reject
key source 0235+164	$\chi_2/dof = 9.25$	Reject
Reference: 0955+476 / 1639-062	$\chi_2/dof = 0.51 / 1.39$	Pass

output/public/vlbi/vlbi_beta_source_session_matrix_metrics.json is the main ledger. We directly audited inter-year stability within the same source. Both 0229+131 and 0235+164 remain Reject; source structure alone is insufficient, and contamination by time-band systematics remains possible. Some sources still indicate Pass, so the next step is a stratified refit restricted to stable sources. The remaining task is to isolate whether the all-sky Reject is driven mainly by source dependence or by time dependence.

β derived termination audit (VLBI+LLR+MESSENGER cross-channel):

Gate	Actual measurement	Condition
VLBI comparator eligibility	eligible=false(reason=subset_consistency_reject)	Reject
LLR promotion gate	step1 - 5 = pass(promoted=true)	Pass
MESSENGER final gate	stage_j=pass, stage_e_replay=pass, bias_audit=pass	Pass
MESSENGER primary β (beta_lt)	$\beta_{lt} = 1.042153 \pm 0.983588$, $abs(z(\beta_{lt} - 1)) = 0.04286$	Pass
MESSENGER diagnostic β (beta_dyn)	$\beta_{dyn} = -8.543545 \pm 0.679605$, $abs(z(\beta_{dyn} - 1)) = 14.04$	Reject(diagnostic)
policy D pair gate(LLR vs MESSENGER)	$abs(z(\beta_{llr} - \beta_{lt}^{(MESSENGER)})) = 0.04217$, $watch/pass = pass/pass$	Pass
cross consistency(VLBI vs LLR)	$abs(z(beta_vlbi - beta_llr)) = 13.392$	Reject
beta minus one(active policy)	$\beta_{comb} = 1.000674 \pm 0.000961$, $abs(z(\beta_{comb} - 1)) = 0.702$	Pass
beta terminal(active policy)	policy_D_exclude_plus_messenger_fusion	Pass

Continued on next page

Gate	Actual measurement	Condition
policy B hold governance	policy_b_hold_status= watch (diagnostic value after release of hold)	Watch (diagnostic)
policy D promotion governance	policy_d_promotion_ status=pass, ready=true, blockers = []	Pass
policy D promotion reassessment	status=pass, order_match=true, blocker_delta=0, policy_alignment=true	Pass
policy recommended active	recommended_active_ policy_id=policy_D_ exclude_plus_messenger_ fusion	Pass
Stage I SRP proxy indicator registration	srp_proxy_registered=true (scenarios=all_proxies_on, srp_proxy_on)	Pass
policy D blocker resolution order	[] (resolved)	Pass
LLR gate execution order (bias→promotion)	[] (resolved)	Pass
LLR gate actions ordered	[] (resolved)	Pass
LLR gate minimal repro chain	llr_direct_fit->llr_ promotion_gate->beta_ cross_channel_registry-> beta_terminal_reject_ checklist	Pass
active policy update decision	active=policy_D_exclude_ plus_messenger_fusion, recommended=policy_D_ exclude_plus_messenger_ fusion, update_required=false	Pass
policy switch redcision(B→D)	decision=switch_to_policy_ D_now, switch_required_now=true, switch_allowed_now=true, switch_applied_now=true	Pass
policy switch branch wording	branch=switch_path_ applied	Pass
final watch statement(canonical)	statement_id=watch_ resolved_policy_alignment_ completed	Pass

Fixed output/public/summary/beta_cross_channel_registry.json as the terminal ledger, and the current judgment is beta_terminal=pass (active_policy=policy_D_exclude_plus_messenger_fusion). Policy matrix confirmed $A = reject, B = pass, C = pass, D = pass, E = pass$. MESSENGER

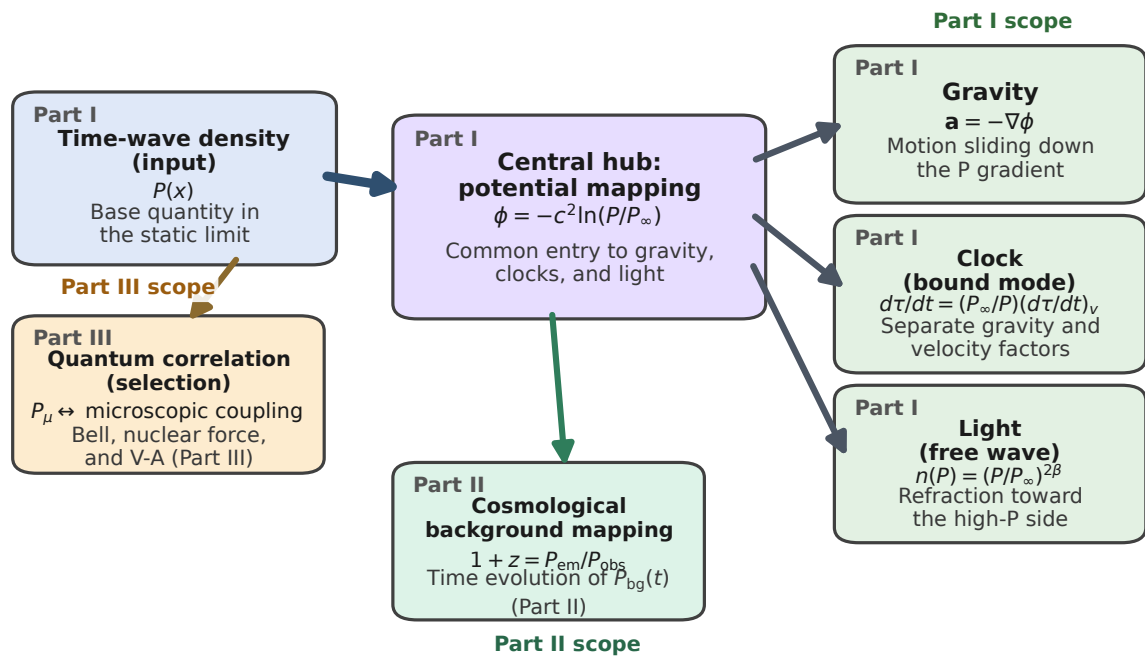
separated and fixed `beta_lt` as the main estimate and `beta_dyn` as the diagnostic value for the lack of a non-gravitational acceleration model. LLR blockers `llr_bias_gate`, `llr_promotion_gate`, `policy_d_bias_gate` have been gradually resolved, and policy D promotion has been applied. While retaining the execution order (`llr_bias_gate->llr_promotion_gate->policy_d_bias_gate`) in the audit ledger, the current status is fixed as `resolved(blockers=[])`. The shortest reproduction chain is `python-Bscripts/llr/llr_kappa_llr_direct_fit.py -> python-Bscripts/llr/llr_kappa_llr_beta_promotion_gate.py -> python-Bscripts/summary/beta_cross_channel_registry.py -> python-Bscripts/summary/beta_terminal_reject_checklist.py` and the first action is `close_llr_bias_audit_first`. Note: The script name `reject` comes from the naming during initial development, and currently it is used as a general-purpose checker that recalculates all gates for termination determination (the termination determination itself is fixed to `Pass`). Here, `beta_messenger` refers to `beta_lt` (light-time main estimate). `beta_dyn` is a diagnostic value for insufficient non-gravitational acceleration model and is not used for pair gate. Therefore, the current active policy is fixed at `policy_D_exclude_plus_messenger_fusion`.

7 Part I (Core Mapping): Definition Figures and Frozen Parameters

Purpose: In Part I, we will clarify the basis diagram of the fixed mapping and freezing parameters along with the reproduction entrance.

Public artifacts (GitHub):

- Directory: [output/public/theory/](#)
- Freezing parameters (β etc.): [frozen_parameters.json](#)
- Overview of core mapping: [pmodel_core_mapping_overview.pdf](#)
- Conceptual comparison (P-model vs. reference frame): [pmodel_core_concept_comparison.pdf](#)



Part I fixes the mapping and β_{frozen} ; Parts II/III run falsification audits with those frozen values.

Figure 16: Core mapping overview for Part I

Conceptual comparison table: P-model vs. reference frame:

Perspective	Reference frame (GR)	P-model
Essence of gravity	Curvature of spacetime	Response to P gradient
Description of motion	Along geodesic	Sliding down slope P
Propagation of light	Light follows space-time	Light is refracted toward the high P side
Time dilation	Space-time metric	P ratio (P_∞/P)

Continued on next page

Perspective	Reference frame (GR)	P-model
Redshift	Spatial expansion	Time change of background P

Note: This comparison summarizes the point that the P-model describes gravity not as a spatiotemporal geometry, but as a response to spatial changes in the temporal wave density P . In the weak field, they reproduce the same observables, but their conceptual starting points are independent.

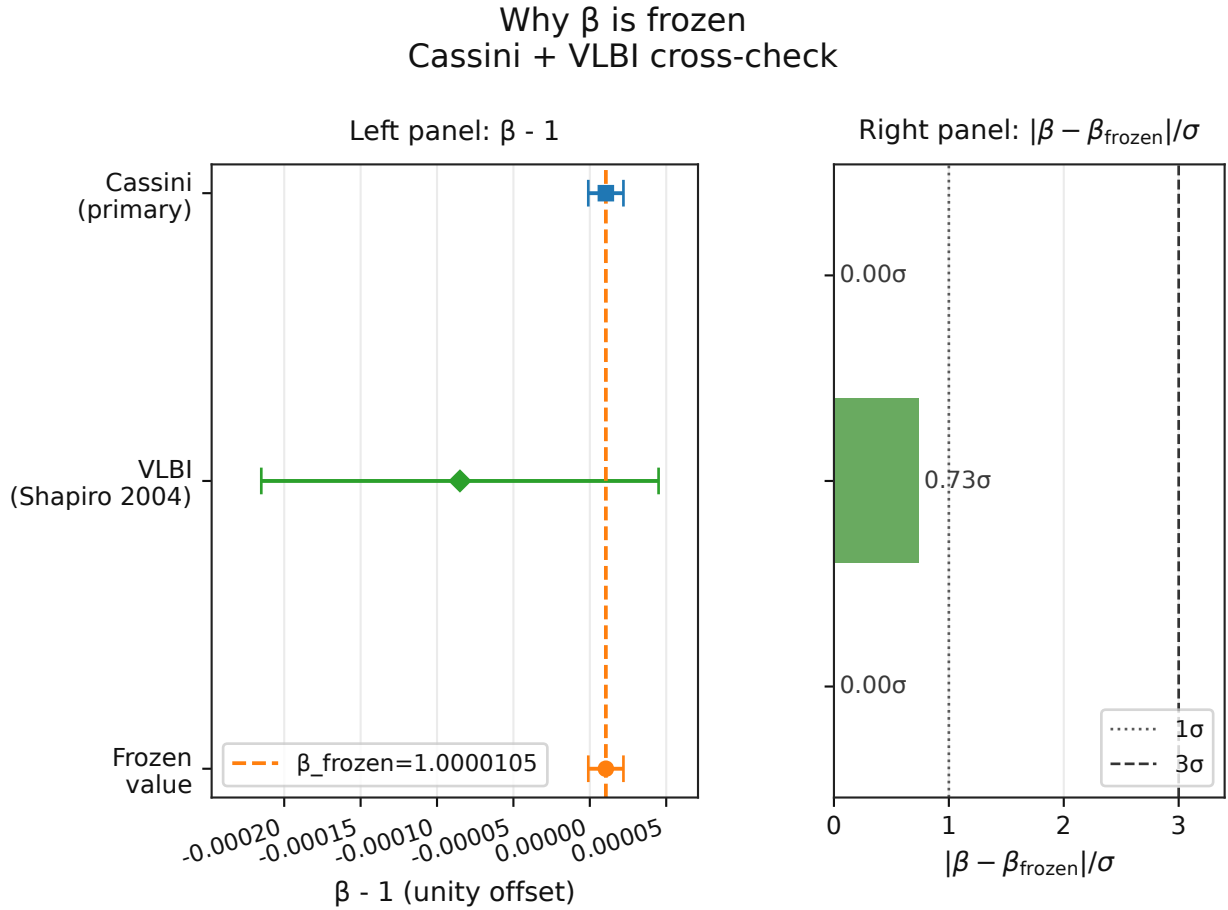


Figure 17: Freeze rationale for the Part I core mapping

Reproduction entry points (scripts/GitHub):

- [scripts/theory/pmodel_core_mapping_overview.py](#)
- [scripts/theory/pmodel_core_concept_comparison.py](#)
- [scripts/theory/freeze_parameters.py](#)

8 Cosmology (BAO/SN/CMB/AP): Public Artifacts

Purpose: Cosmology verification other than DDR/SPARC (BAO first-order statistics, SN time dilation, CMB temperature scaling, AP, etc.) Make outputs and inlets explicit as public artifacts.

Public artifacts (GitHub):

- Directory: [output/public/cosmology/](#)
- BAO primary statistics (ξ_ℓ /peakfit): [cosmology_bao_xi_multipole_peakfit.pdf](#)
- BAO cross-check (catalog vs publication): [cosmology_bao_catalog_vs_published_crosscheck.pdf](#)
- DESI DR1 BAO promotion check: [cosmology_desi_dr1_bao_promotion_check_public.pdf](#)
- BAO cross-integration (BOSS Stage A + DESI Stage B): [output/public/cosmology/locales/en/cosmology_bao_xi_cross_integration.pdf](#) / [output/public/cosmology/locales/en/cosmology_bao_xi_cross_integration_metrics.json](#) / [output/public/cosmology/locales/en/cosmology_bao_xi_cross_integration_summary.csv](#) Supplement: (stage_a.max_abs_z=1.063, stage_b.promoted=false, overall_status=watch, part4_scoreboard_update.recommend_update=false)
- SN time dilation: [cosmology_sn_time_dilation_constraints.pdf](#)
- Pantheon+ $\mu(z)$ / Cosmic Chronometers $H(z)$ Direct comparison (H_{eff}^2 complete expression): [output/public/cosmology/locales/en/cosmology_pantheonplus_cc_direct_fit.pdf](#) / [output/public/cosmology/locales/en/cosmology_pantheonplus_cc_direct_fit_metrics.json](#) / [output/public/cosmology/locales/en/cosmology_pantheonplus_cc_direct_fit_summary.csv](#) Note: ($\Delta AIC = AIC_{\text{baseline}} - AIC_P = -4.00$, winner=baseline_lcdm, current status is Watch/Reject)
- CMB temperature scaling: [cosmology_cmb_temperature_scaling_constraints.pdf](#)
- CMB acoustic peak (1st to 3rd reproduction + 4th to 6th holdout audit): [output/public/cosmology/locales/en/cosmology_cmb_acoustic_peak_reconstruction.pdf](#) / [output/public/cosmology/locales/en/cosmology_cmb_acoustic_peak_reconstruction_metrics.json](#) / [output/public/cosmology/locales/en/cosmology_cmb_acoustic_peak_reconstruction_falsification_pack.json](#) Note: (core gate: pass, extended diagnostic: pass)
- CMB TT full C_ℓ fit ($\ell \leq 2500$; operational audit): [output/public/cosmology/locales/en/cosmology_cmb_tt_full_cl_fit.pdf](#) / [output/public/cosmology/locales/en/cosmology_cmb_tt_full_cl_fit_metrics.json](#) / [output/public/cosmology/locales/en/cosmology_cmb_tt_full_cl_fit_summary.csv](#) Supplement: ($\Delta AIC = AIC_{\text{baseline}} - AIC_P = -7.018$, winner=baseline_lcdm, judgment=Reject)
- CMB TT/TE/EE simultaneous full C_ℓ fit ($\ell \leq 2500$; operational audit): [output/public/cosmology/cosmology_cmb_ttee_full_cl_fit.pdf](#) / [output/public/cosmology/cosmology_cmb_ttee_full_cl_fit_metrics.json](#) / [output/public/cosmology/cosmology_cmb_ttee_full_cl_fit_summary.csv](#) Supplement: ($\Delta AIC = AIC_{\text{baseline}} - AIC_P = -13.575$, winner=baseline_lcdm, shape_improvement=0.000, channel_scaling_improvement=2.425, judgment=Reject)

- CMB polarization phase (TE/EE; main gate fixed): output/public/cosmology/locales/en/cosmology_cmb_polarization_phase_audit.pdf / output/public/cosmology/locales/en/cosmology_cmb_polarization_phase_audit_metrics.json / output/public/cosmology/locales/en/cosmology_cmb_polarization_phase_audit_falsification_pack.json Note: (phase_hard_gate = ($|\Delta\phi_{EE}-0.5| \leq 0.12$) and ($|\Delta\phi_{TE}-0.25| \leq 0.12$) and (TEsignalalternation) is required and fixed. If either is not true, reject. (Use $\maxabs(\Delta\ell_{EE}) \leq 50$ / $\maxabs(\Delta\ell_{TE}) \leq 55$ for watch/pass judgment only when the main gate is established)
 - Galaxy cluster collision offset (Bullet system): output/public/cosmology/locales/en/cosmology_cluster_collision_p_peak_offset_audit.pdf / output/public/cosmology/cosmology_cluster_collision_p_peak_offset_audit.json Note: (Bullet is fixed. Non-Bullet primary registration is managed by watch)
 - Galaxy cluster collision τ/ξ Derived chain audit (assignment B): output/public/cosmology/locales/en/cosmology_tau_derivation_chain_audit.pdf / output/public/cosmology/locales/en/cosmology_tau_derivation_chain_audit.json / output/public/cosmology/locales/en/cosmology_tau_derivation_chain_audit.csv Note: (tau_origin_mode=derived_from_pmu_kernel, xi_mode=derived, ad_hoc_parameter_count=0 fixed expression tracking)
 - Galaxy cluster collision L_{path} Scaling law prediction (task 2): output/public/cosmology/locales/en/cosmology_lpath_scaling_law_prediction.pdf / output/public/cosmology/locales/en/cosmology_lpath_scaling_law_prediction.json / output/public/cosmology/locales/en/cosmology_lpath_scaling_law_prediction.csv Supplement: (Fixed universal ratio for $L_{\text{int}} = g_P P_\mu J^\mu$ origin, $\Pi_0 = \tau_{\text{int}}/\tau_{\text{free}} - 1$, $\rho/v/T$ ratio)
 - Galaxy cluster collision transfer template (Bullet reevaluation + non-Bullet application/reuse I/F): output/public/cosmology/locales/en/cosmology_cluster_collision_lpath_transfer_template.pdf
 - output/public/cosmology/locales/en/cosmology_cluster_collision_lpath_transfer_template.json
 - output/public/cosmology/locales/en/cosmology_cluster_collision_lpath_transfer_template.csv
 - output/public/cosmology/cluster_collision_nonbullet_primary_registration_checklist.csv
 - output/public/cosmology/cluster_collision_nonbullet_primary_registration_template.csv
 - output/public/cosmology/cluster_collision_nonbullet_primary_registration_apply_report.csv
 - output/public/cosmology/cluster_collision_lpath_transfer_input_template.csv
- Supplement: Bullet reevaluation is fixed as pass, non-Bullet is fixed as watch for unregistered observation, and hash-update monitoring uses primary_dataset_update_watchpack with blocked_priority_class=low_blocked_missing_primary_data.
- AP(Alcock–Paczynski): [cosmology_alcock_paczynski_constraints.pdf](#)
 - Structure formation $f\sigma_8$ (effective friction of delayed branch Γ_{eff} audit): output/public/cosmology/locales/en/cosmology_fsigma8_growth_mapping.pdf / output/public/cosmology/locales/en/cosmology_fsigma8_growth_mapping_metrics.json / output/public/cosmology/locales/en/cosmology_fsigma8_growth_mapping_falsification_pack.json

- JWST/NIRSpec primary spectrum update monitoring (watchpack awaiting release): [output/public/cosmology/jwst_spectra_release_waitlist.csv](#) / [output/public/cosmology/jwst_spectra_release_waitlist.json](#) / [output/public/cosmology/jwst_spectra_integration_metrics.json](#)
- Distance-indicator reach limit: [cosmology_distance_indicator_reach_limit.pdf](#)
- Distance-indicator error budget: [cosmology_distance_indicator_error_budget.pdf](#)
- Reconnection parameter space: [cosmology_reconnection_parameter_space.pdf](#)

BAO ξ_l cross integration (BOSS Stage A + DESI Stage B)

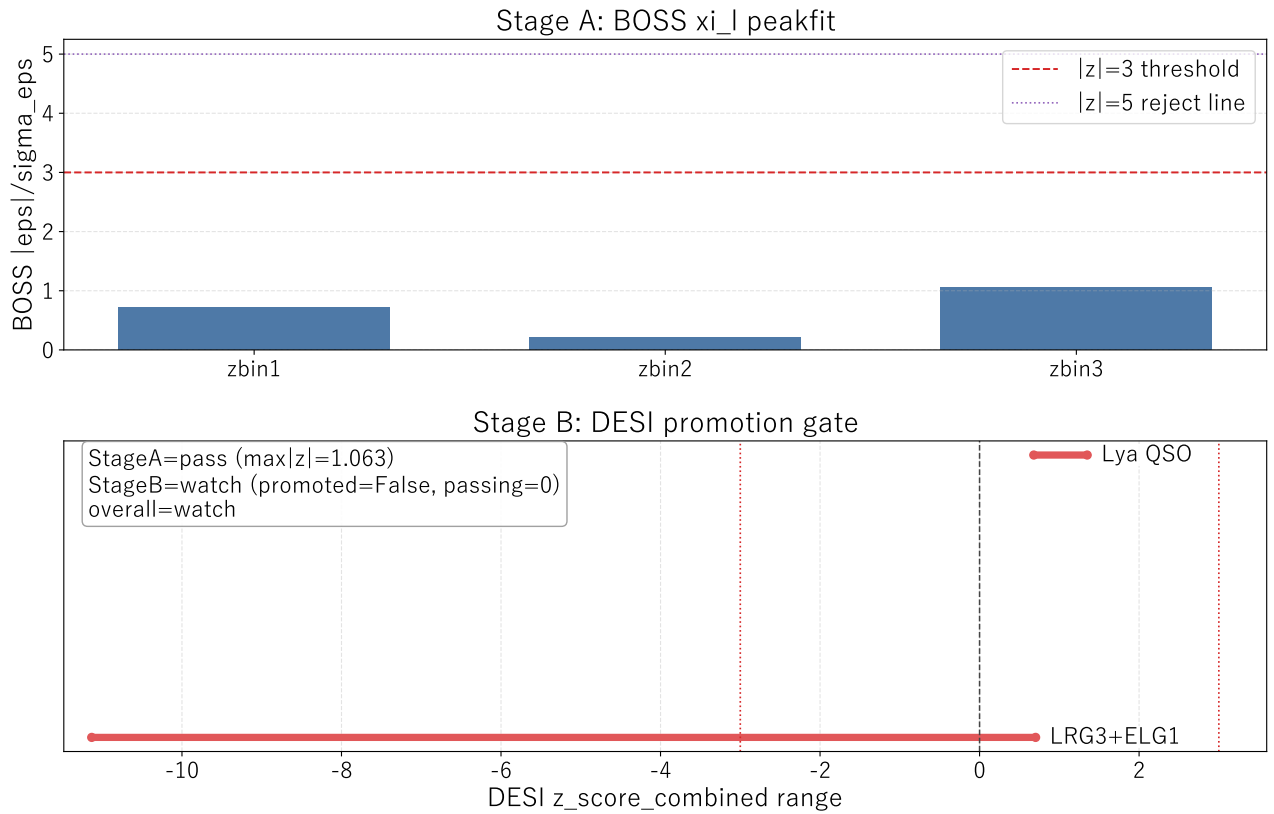
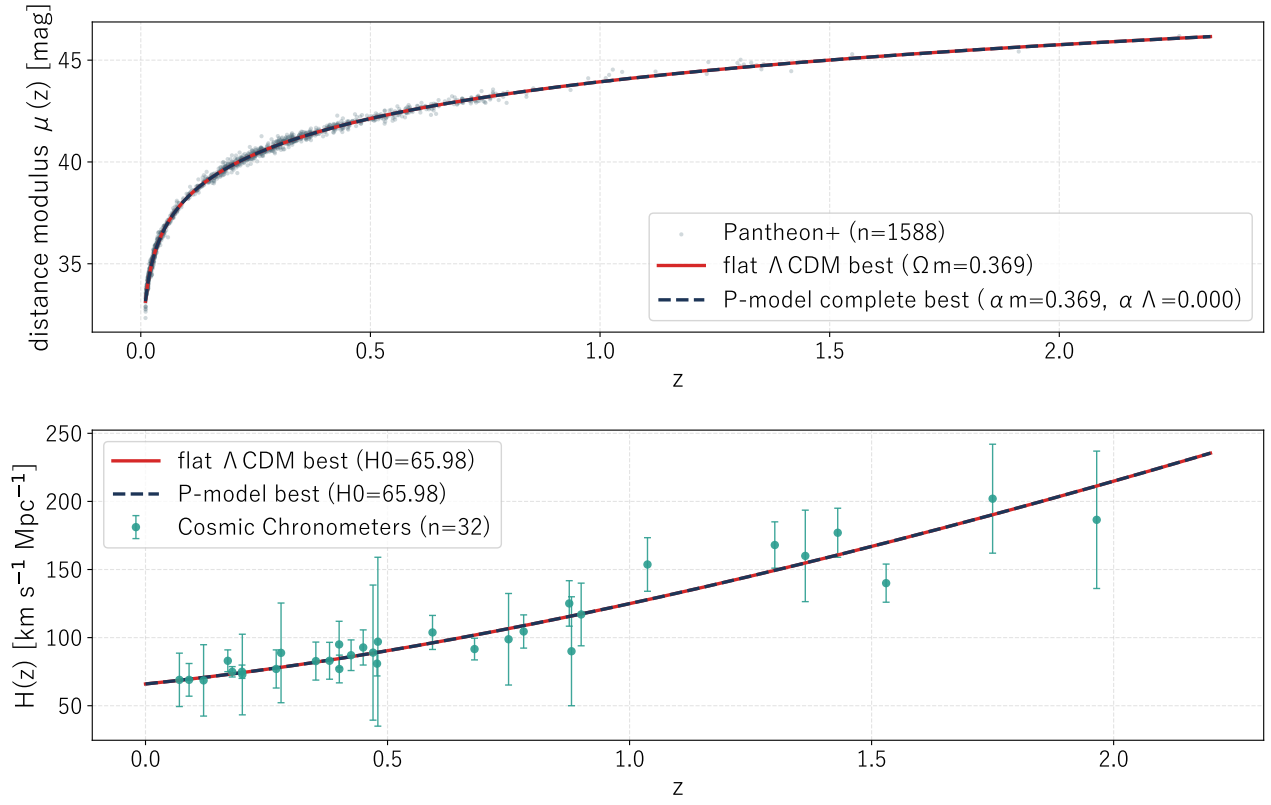


Figure 18: BAO cross-integration audit

Pantheon+ $\mu(z)$ and Cosmic Chronometers $H(z)$ direct comparison (same I/F)



χ^2_{total} : Λ CDM=725.25, P-model=725.25; $\Delta \text{AIC} = \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P}} = -4.00$ (positive favors P-model)

Figure 19: Pantheon+ and cosmic-chronometer direct fit

P-model CMB acoustic-peak audit (ℓ 1- ℓ 3 fit + ℓ 4- ℓ 6 holdout)

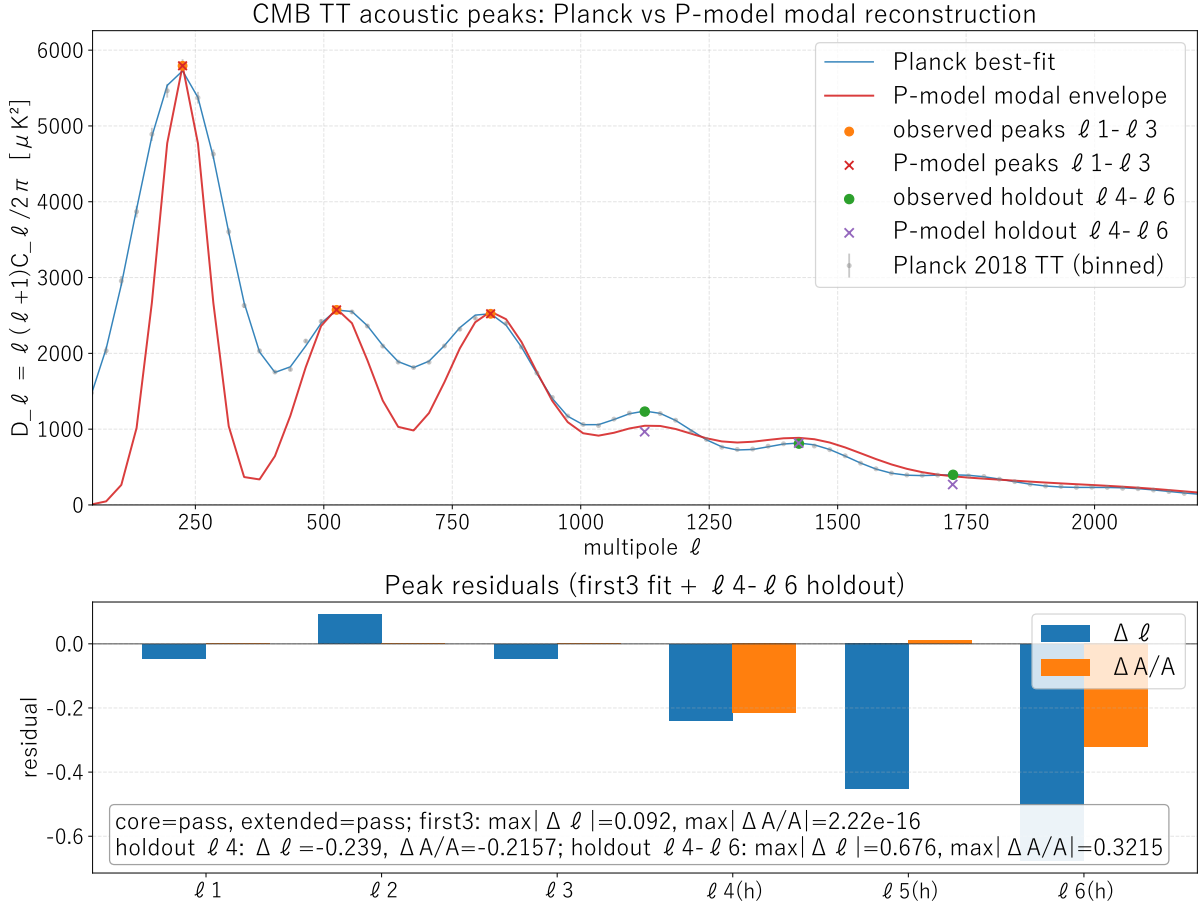


Figure 20: CMB acoustic-peak reconstruction audit

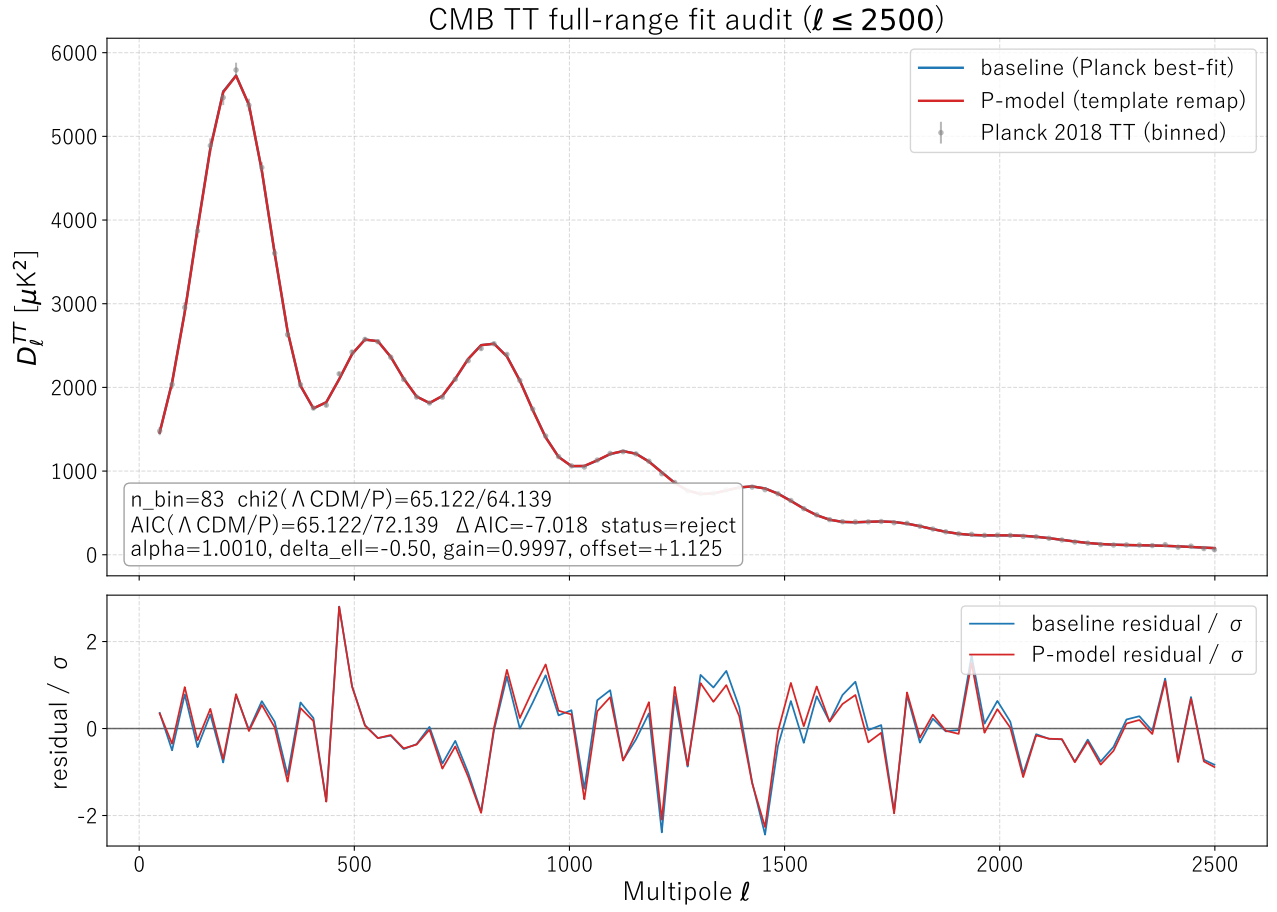
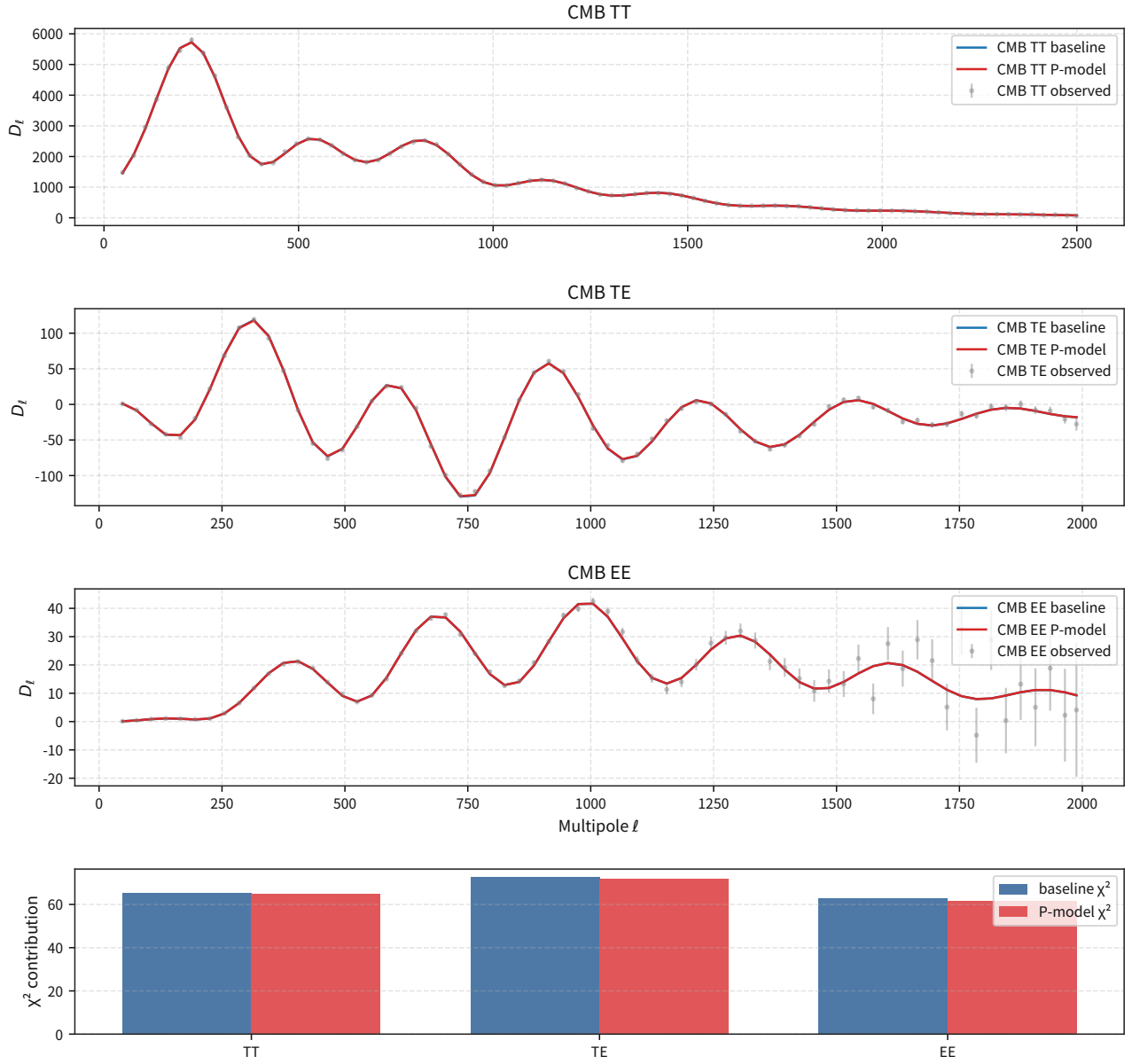


Figure 21: CMB TT full C_{ℓ} fit audit

CMB TT/TE/EE simultaneous full-range fit audit



n_bin_total=215 / $\chi^2(\Lambda\text{CDM}/P)=200.765/198.340$ / $AIC(\Lambda\text{CDM}/P)=200.765/214.340$
 $\Delta AIC=-13.575$ status=reject / shape_gain=0.000 / channel_scaling_gain=2.425
 $AIC_{\text{penalty}}=16.000$ / $\alpha=1.0000$, $\delta_{\ell}=+0.00$

Figure 22: CMB TT/TE/EE full C_ℓ fit audit

CMB polarization transfer audit (Stokes extension + Thomson source)

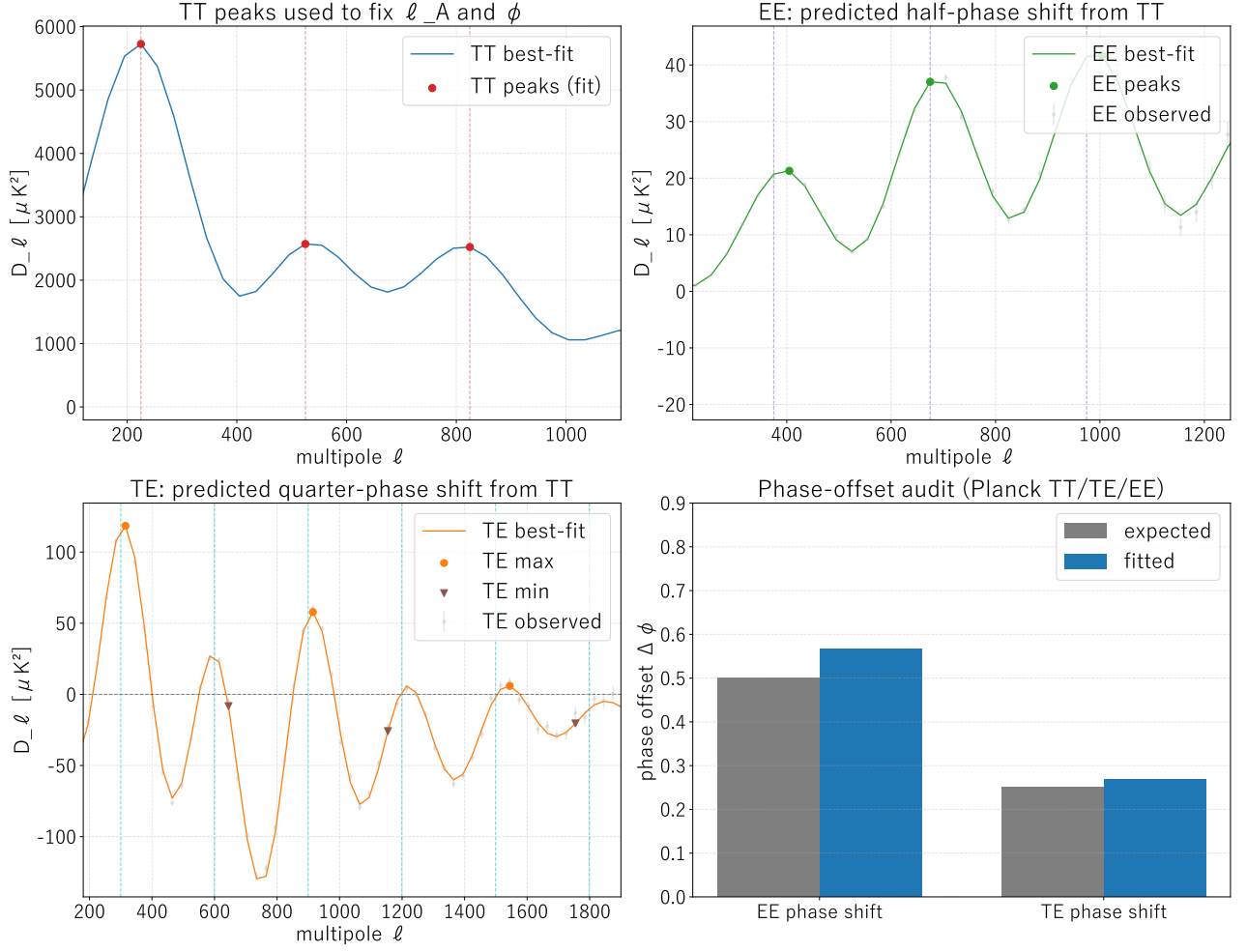


Figure 23: CMB polarization-phase audit

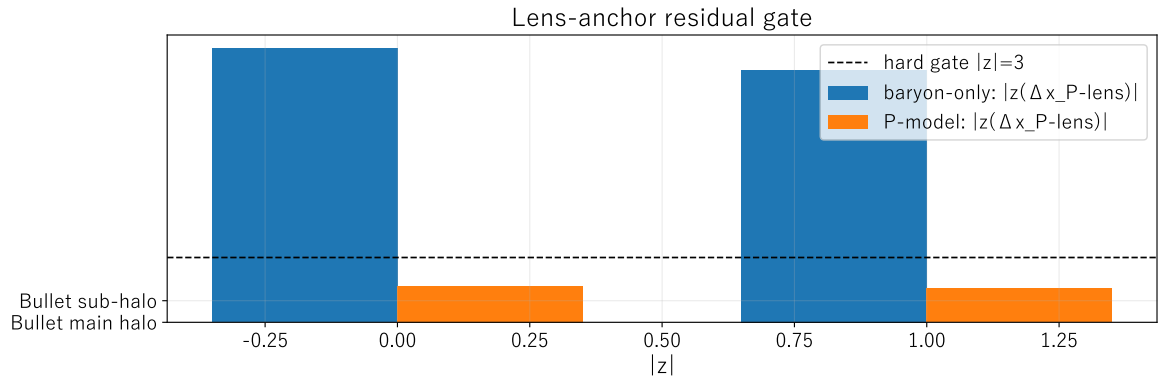
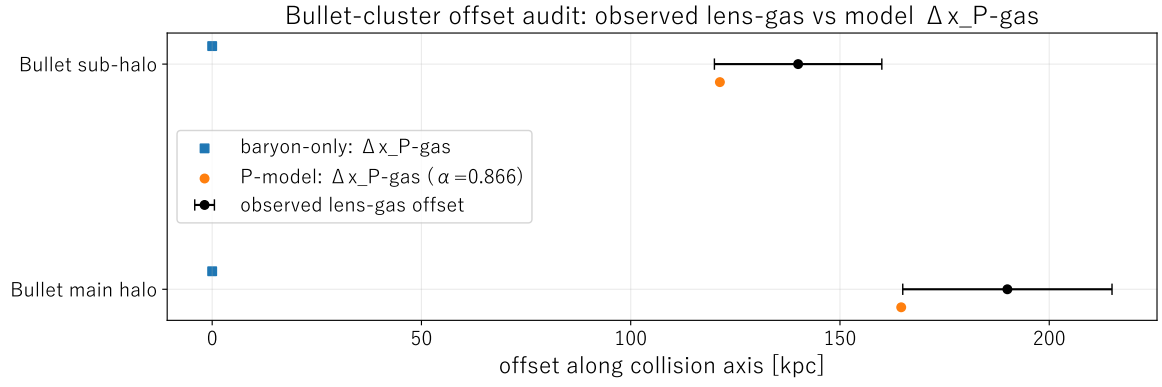
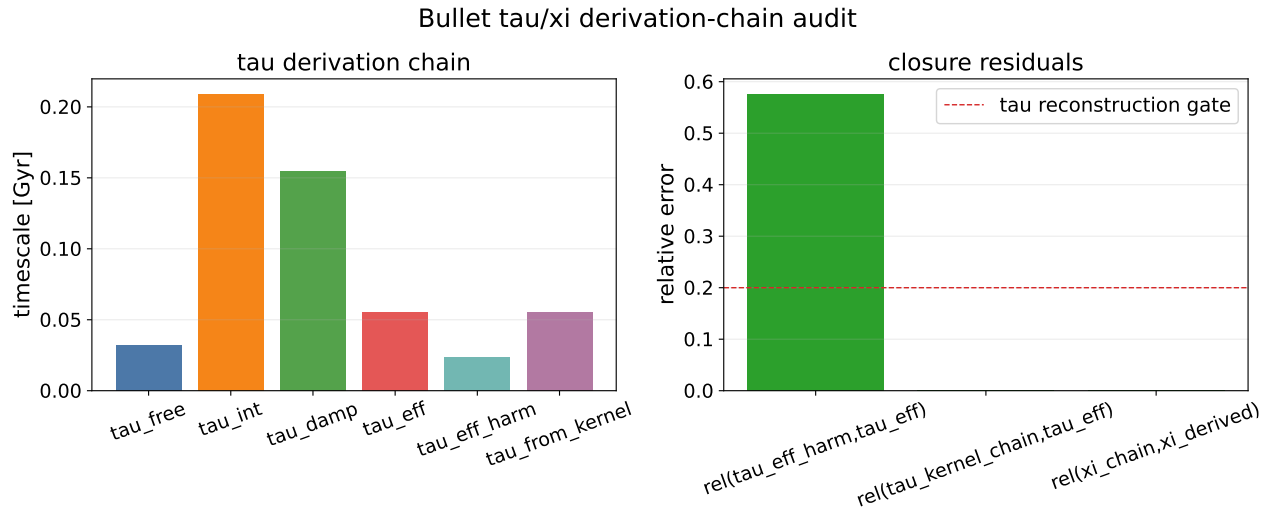


Figure 24: Galaxy-cluster collision offset audit



status=watch, decision=tau_derivation_chain_fixed_but_reconstruction_margin_watch, hard_fail_n=0
tau_mode=derived_from_pmu_kernel, kernel_mode=double_exp_derived, xi_mode=derived
tau_reconstruction_rel_error=0.123944 (gate<=0.200)
epsilon_slow=0.206926 (gate<=0.250)
rel_tau_kernel=0.000000, rel_xi_chain=0.000000, rel_fit_xi=0.000000

Figure 25: Galaxy-cluster collision τ/ξ derivation-chain audit

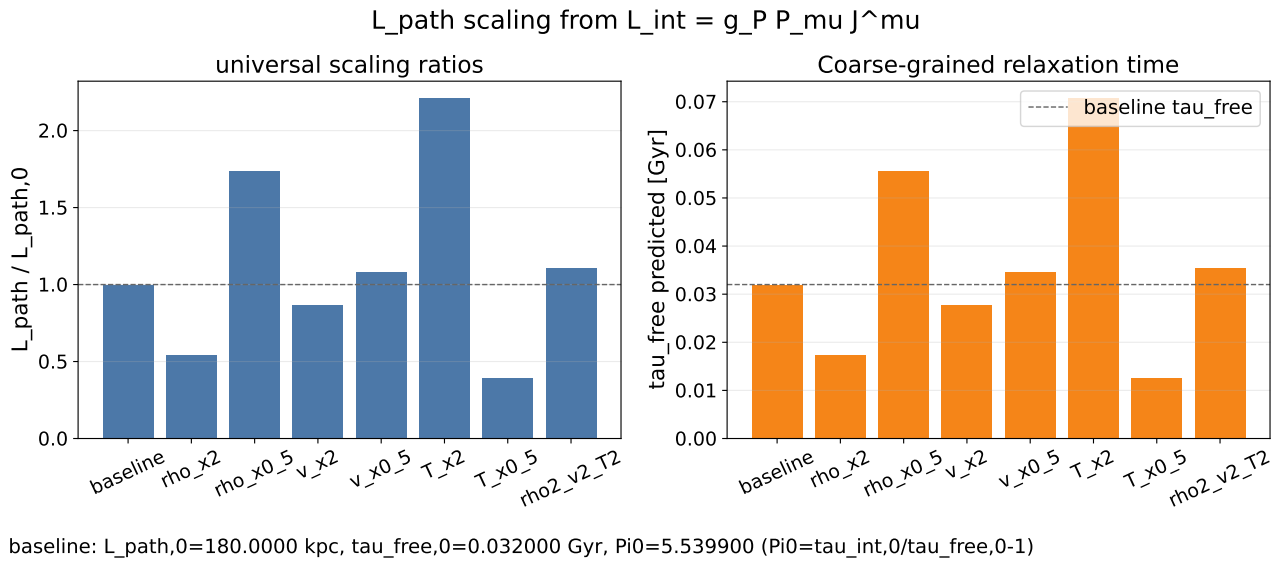


Figure 26: Galaxy-cluster collision L_{path} scaling-law prediction

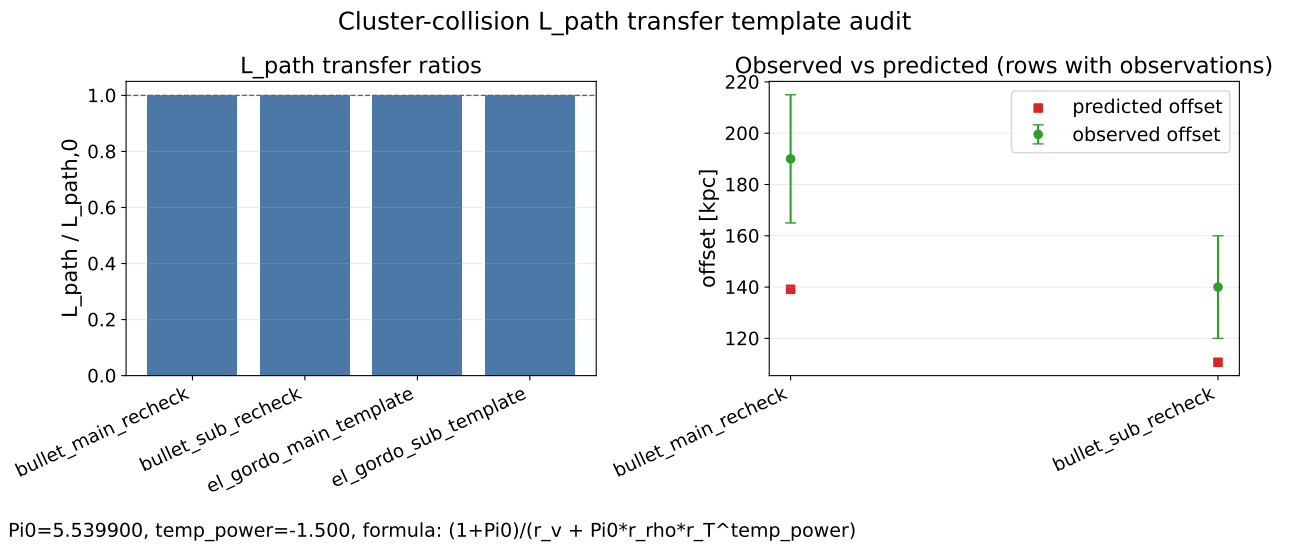


Figure 27: Galaxy-cluster collision transfer-template audit

Step 8.7.18.2: $f\sigma_8$ growth mapping with delayed P response

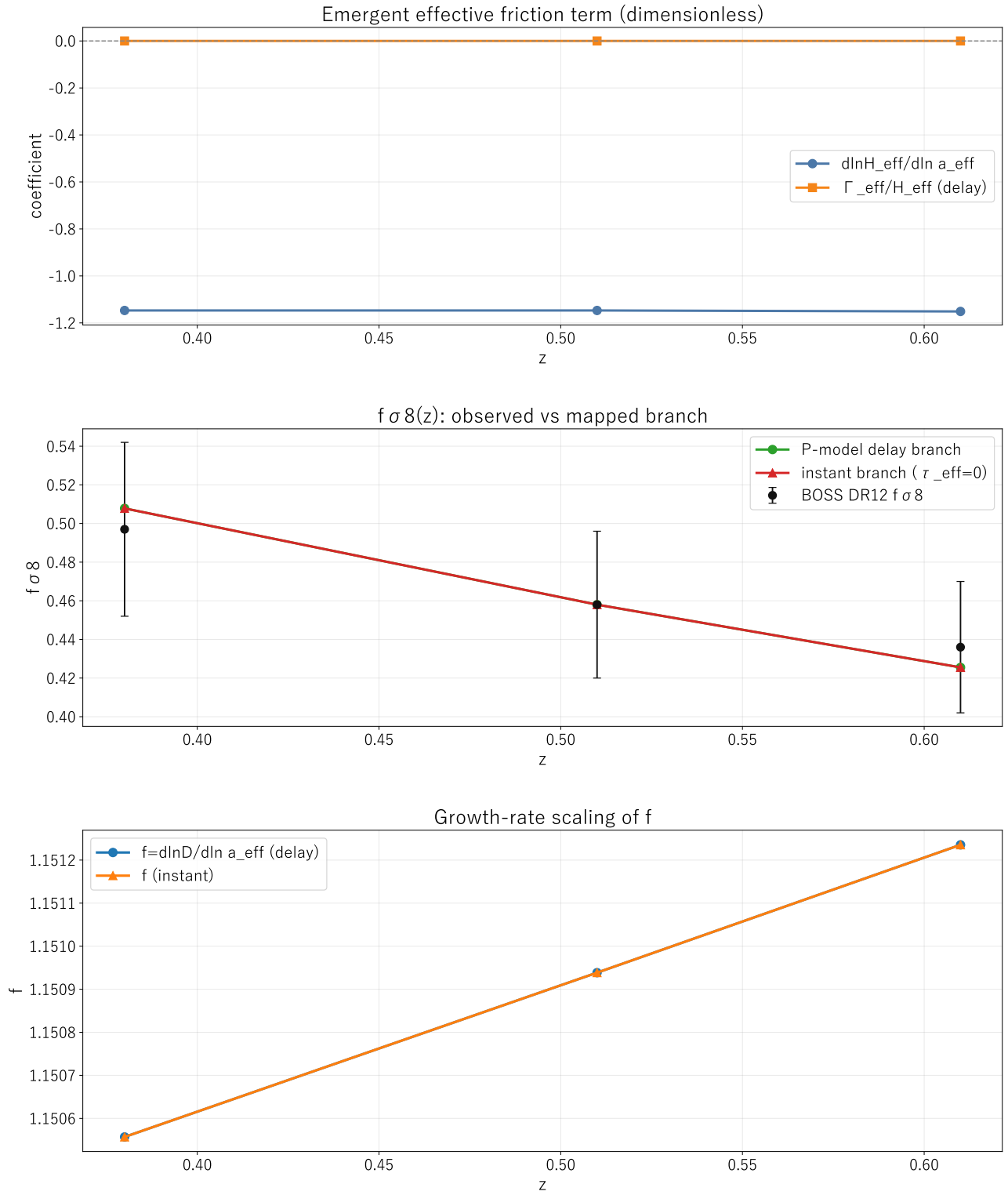


Figure 28: Structure-formation $f\sigma_8$ mapping audit

Reproduction entry points (scripts/GitHub):

- [scripts/cosmology/cosmology_bao_xi_multipole_peakfit.py](#)
- [scripts/cosmology/cosmology_bao_catalog_vs_published_crosscheck.py](#)

- [scripts/cosmology/cosmology_desi_dr1_bao_promotion_check.py](#)
- [scripts/cosmology/cosmology_bao_xi_cross_integration.py](#)
- [scripts/cosmology/cosmology_sn_time_dilation_constraints.py](#)
- [scripts/cosmology/cosmology_pantheonplus_cc_direct_fit.py](#)
- [scripts/cosmology/cosmology_cmb_temperature_scaling_constraints.py](#)
- [scripts/cosmology/cosmology_cmb_acoustic_peak_reconstruction.py](#)
- [scripts/cosmology/cosmology_cmb_polarization_phase_audit.py](#)
- [scripts/cosmology/cosmology_cluster_collision_p_peak_offset_audit.py](#)
- [scripts/cosmology/cosmology_tau_derivation_chain_audit.py](#)
- [scripts/cosmology/cosmology_lpath_scaling_law_prediction.py](#)
- [scripts/cosmology/cosmology_cluster_collision_lpath_transfer_template.py](#)
- [scripts/cosmology/cosmology_alcock_paczynski_constraints.py](#)
- [scripts/cosmology/cosmology_fsigma8_growth_mapping.py](#)
- [scripts/cosmology/fetch_mast_jwst_spectra.py](#)
- [scripts/cosmology/jwst_spectra_integration.py](#)
- [scripts/cosmology/jwst_spectra_release_waitlist.py](#)
- [scripts/cosmology/cosmology_distance_indicator_reach_limit.py](#)
- [scripts/cosmology/cosmology_distance_indicator_error_budget.py](#)
- [scripts/cosmology/cosmology_reconnection_parameter_space.py](#)

Correspondence from expression to audit value (Bullet τ/ξ):

Body expression (Part I/II)	Audit JSON key
$\tau_{\text{free}} = L_{\text{path}}/c_w$	tau_origin_block.derived_components_gyr.tau_free
$\tau_{\text{int}} = \text{sqrt}(<(\chi v)^2> / <(d(\chi v)/dt)^2>)$	tau_origin_block.derived_components_gyr.tau_int
$\tau_{\text{damp}} = \text{sqrt}(<v^2> / <(dv/dt)^2>)$	tau_origin_block.derived_components_gyr.tau_damp
$\tau_{\text{eff}} = \tau_{\text{free}} + \eta\tau_{\text{kernel}}$	tau_origin_block.derived_components_gyr.tau_eff
$\xi = (\tau_{\text{int}} + \tau_{\text{damp}})/\tau_{\text{eff}}$	tau_origin_block.derived_xi / fit.xi_mode=derived
$\Pi_0 = \tau_{\text{int}}/\tau_{\text{free}} - 1$	reference_state.pi0_median
$L_{\text{path}} = c_w/(v/L_{\text{corr}} + A_{\text{col}}rhoT^{-3/2})$	prediction_equations.lpath_balance

Continued on next page

Body expression (Part I/II)	Audit JSON key
$L_{\text{path}}/L_{\text{path}}, 0 =$	<code>prediction_equations.universal_ratio</code>
$(1 + \Pi_0)/(r_v + \Pi_0 r_{\text{rho}} r_T^{-3/2})$	
$\Delta_{\text{x_pred}}(\text{case}) =$	<code>equations.offset_transfer</code>
$\Delta_{\text{x_ref}}(\text{anchor}) * (L_{\text{path}}/L_{\text{path}}, 0)$	
$(1/c_w^2)\partial_{tt}u_L - \partial_{xx}u_L = S_L$	<code>chain.equations.wave_projected</code>
$\tau_{\text{free}}\partial_{tt}u_L + \partial_t u_L = D\partial_{xx}u_L + c_w^2 S_L$	<code>chain.equations.telegraph_with_</code> <code>memory_friction</code>
$\epsilon_{\text{slow}} = \tau_{\text{free}}/\min(\tau_{\text{int}}, \tau_{\text{damp}})$	<code>chain.stored.epsilon_slow</code>
External limbs prohibited	<code>tau_origin_block.ad_hoc_parameter_</code> <code>count=0 / tau_origin_block.tau_</code> <code>manual_injection=false</code>

9 Strong-Field / High-Energy (EHT/GW/XRISM/Pulsar): Public Artifacts

Purpose: Organize differential predictions in the strong gravity/high energy region (EHT/GW/XRISM/pulsars) as public artifacts.

Public artifacts (GitHub):

Item	Contents
Strong field snapshot bridging	Fixed artifacts (zip + manifest) of v0.1.3 are fixed as citation standards

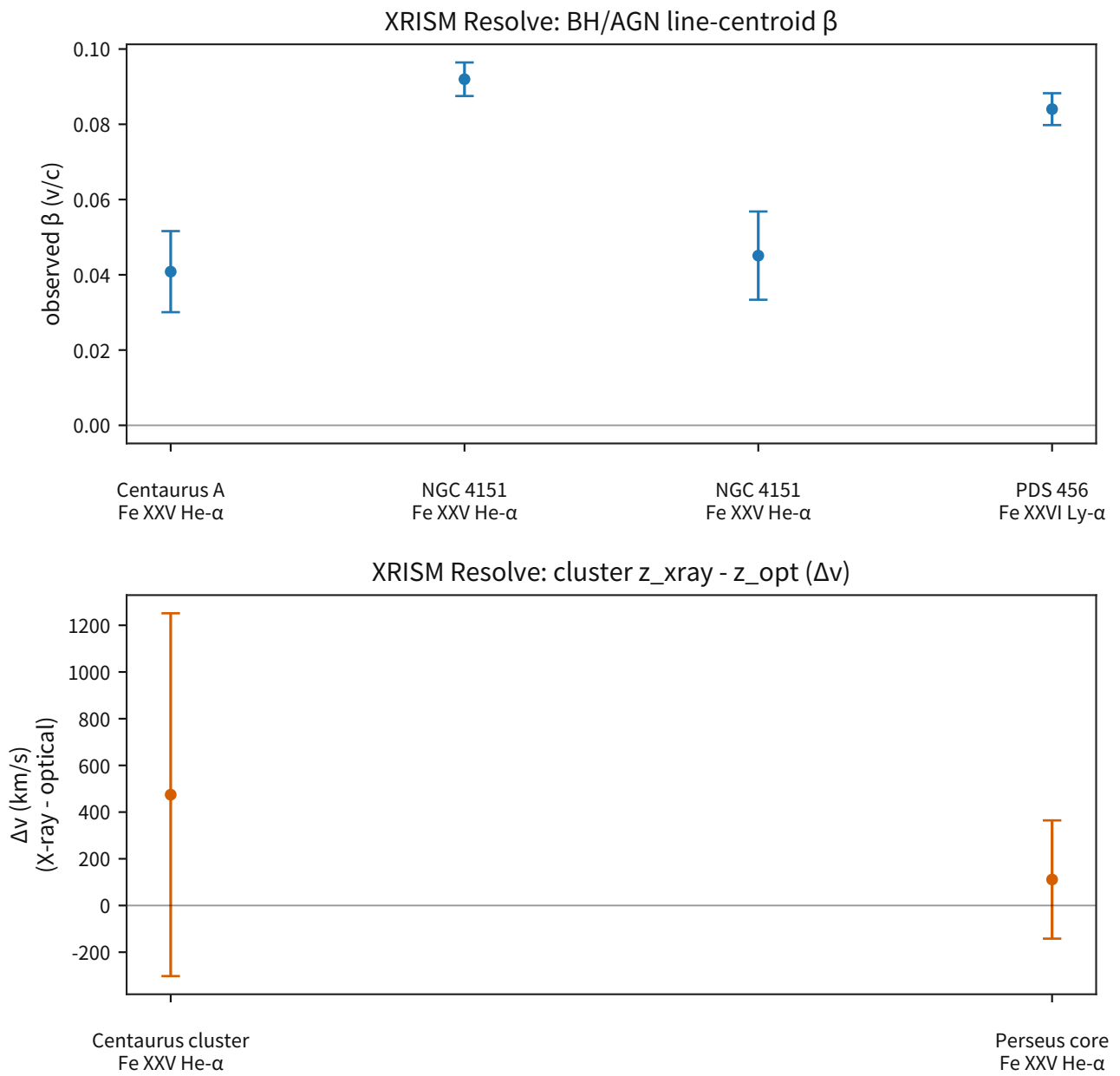


Figure 29: XRISM Resolve summary.

Falsification conditions:

- If `strong_field_public_snapshot_v0.1.3_manifest.json` and `sha256` of the actual file do not match, reject it.
- If any of `eht` / `gw` / `xrism` / `pulsar` is missing in the zip, it will be rejected.
- If the `blob/tree main` reference is reintroduced in Part IV, the fixed snapshot operation will be abandoned.
- Fixed references for the 4 strong fields will be released as release assets of v0.1.3. [strong_field_public_snapshot_v0.1.3.zip](#) / [strong_field_public_snapshot_v0.1.3_manifest.json](#)
- EHT (Shadow/Ring): `eht_shadow_compare` / `eht_m87_persistent_shadow_ring_diameter` / `eht_kappa_fit` and others
- EHT (first principles κ radiative transport envelope): `eht_kappa_first_principles_transfer`
(Fix $\kappa_{\text{fp}}(\text{center}) = 1$, σ_{fp} as the sum of squares of emit-gradient + absorption + scattering, $z = (\kappa_{\text{fit}} - \kappa_{\text{fp}})/\sigma_{\text{fp}}$ gate determines acceptance or rejection)
- Fixed artifact: `output/public/eht/eht_kappa_first_principles_transfer.json` / `output/public/eht/eht_kappa_first_principles_transfer.csv` /

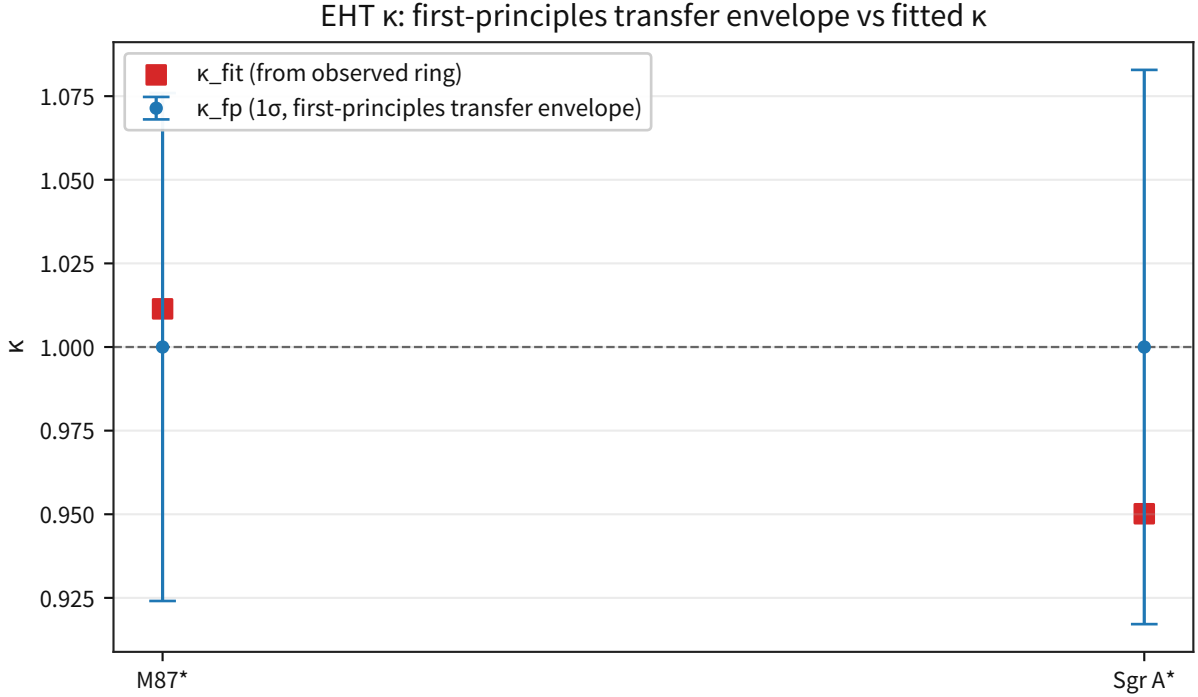


Figure 30: Comparison result for EHT kappa first principles transfer.

- GW (waveform/ringdown/area theorem): `gw150914_chirp_phase` / `gw250114_ringdown_qnm_fit` / `gw250114_area_theorem_test` and others

- GW (inter-detector amplitude ratio audit): gw150914_h1_l1_amplitude_ratio (optimal lag -5.13 ms, $|corr|_{max} = 0.749$, $median(|H1|/|L1|) = 1.469$)
- GW (H1/L1 multi-event isomorphic audit): gw_h1_l1_multi_event_amplitude_audit (n_usable_events=2, median_abs_corr_usable=0.701, overall_status=pass)
- GW (3 detector polarization corr_use_min stage audit): gw_polarization_corr_gate_stage_audit (corr_use_min=0.05 is fixed for operation, ≥ 0.08 is inconclusive)
- GW (local research of 3-detector polarization watch boundaries): refixed $pass/watch = 0$ in 24 trials of corr_use_min=0.05..0.07, response_floor=0.05/0.08, min_ring=6/8, $relax = 2/4$.
Minimum scalar_overlap_proxy=0.500, scalar_overlap_proxy=0.0 in n_usable_events ≥ 2 condition is only n_usable_events=1.
- Fixed artifact: output/public/gw/locales/en/gw_polarization_h1_l1_v1_network_tuning_audit_refine_step87198.json / output/public/gw/locales/en/gw_polarization_h1_l1_v1_network_tuning_audit_refine_step87198.csv /

GW polarization network tuning audit

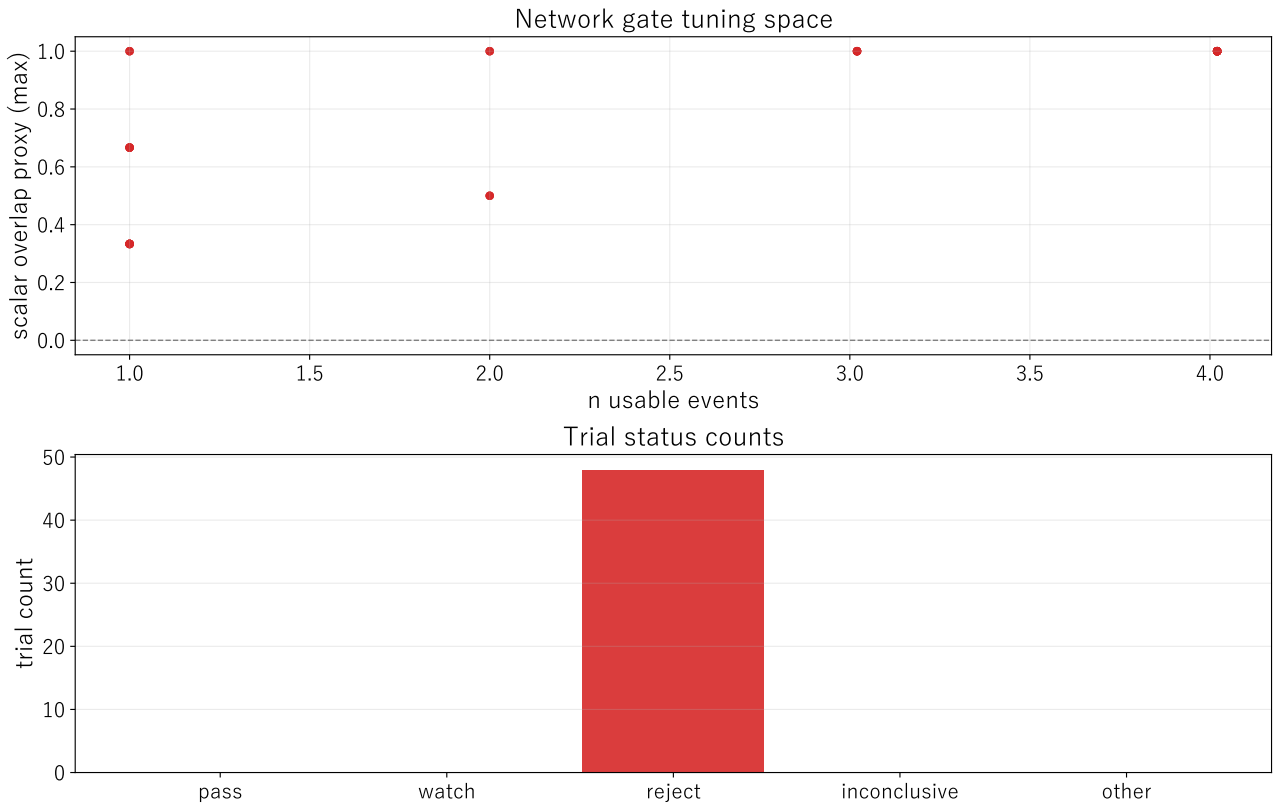


Figure 31: Comparison result for GW polarization h1 l1 v1 network tuning audit refine step87198.

- GW (3-detector polarization coverage extended audit): scan 7 subsets of 2-4 events with GW200129_065458 as the anchor. The target conditions (n_usable_events ≥ 2 and scalar_overlap_proxy < 0.5) are still difficult to reach; the current result is gate_hits=0/7, best_u2=0.500 (GW200129_065458, GW200311_115853), overall_status=watch.

- Fixed artifact: `output/public/gw/locales/en/gw_polarization_network_coverage_expansion_audit.json` / `output/public/gw/locales/en/gw_polarization_network_coverage_expansion_audit.csv` /

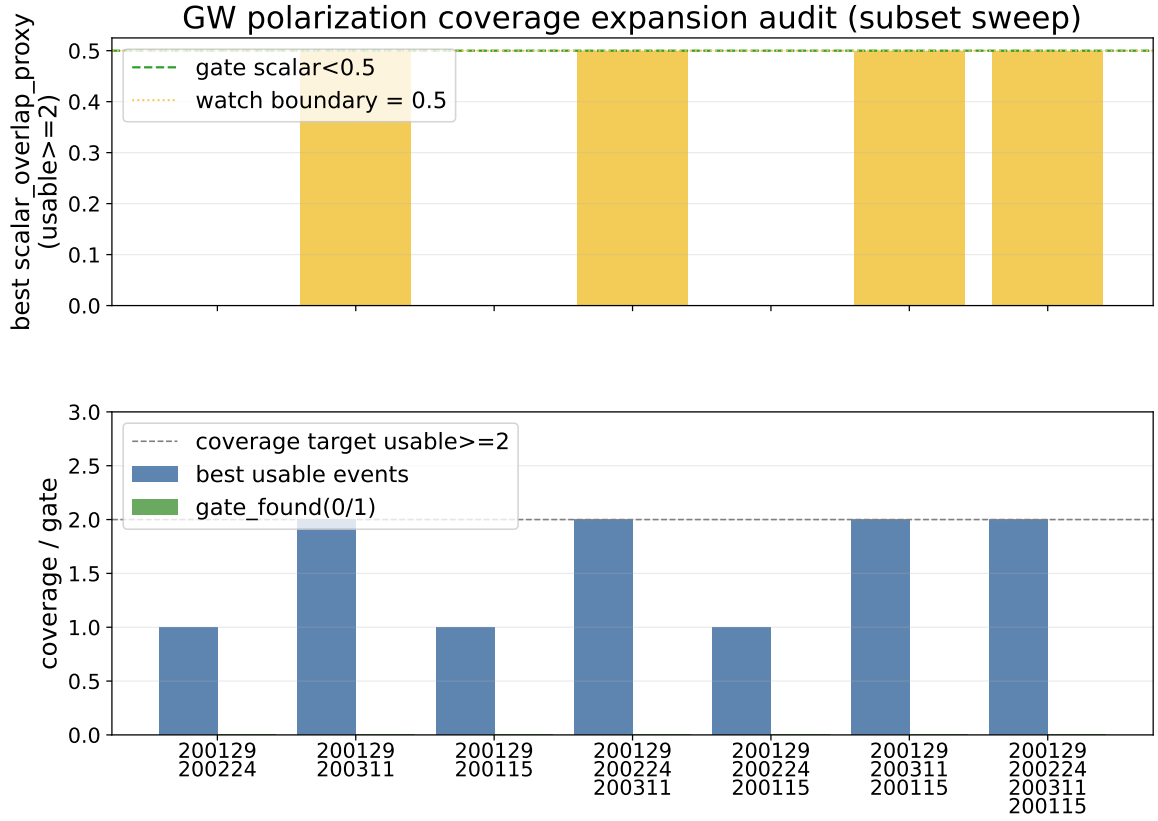


Figure 32: Comparison result for GW polarization network coverage expansion audit.

- GW (3 detector public event input-hash monitoring): record input-hash updates as watch points for rerun candidates. When needed, rerun the coverage-extension audit on the same I/F and test whether `gate_hits` or `best_u2` improves. Currently `update_event_type=no_change`, `rerun_triggered=false`, `event_counter=1`, `overall_status=watch`.
- Fixed artifact: `output/public/gw/locales/en/gw_polarization_network_event_update_watch.json` / `output/public/gw/locales/en/gw_polarization_network_event_update_watch.csv` /

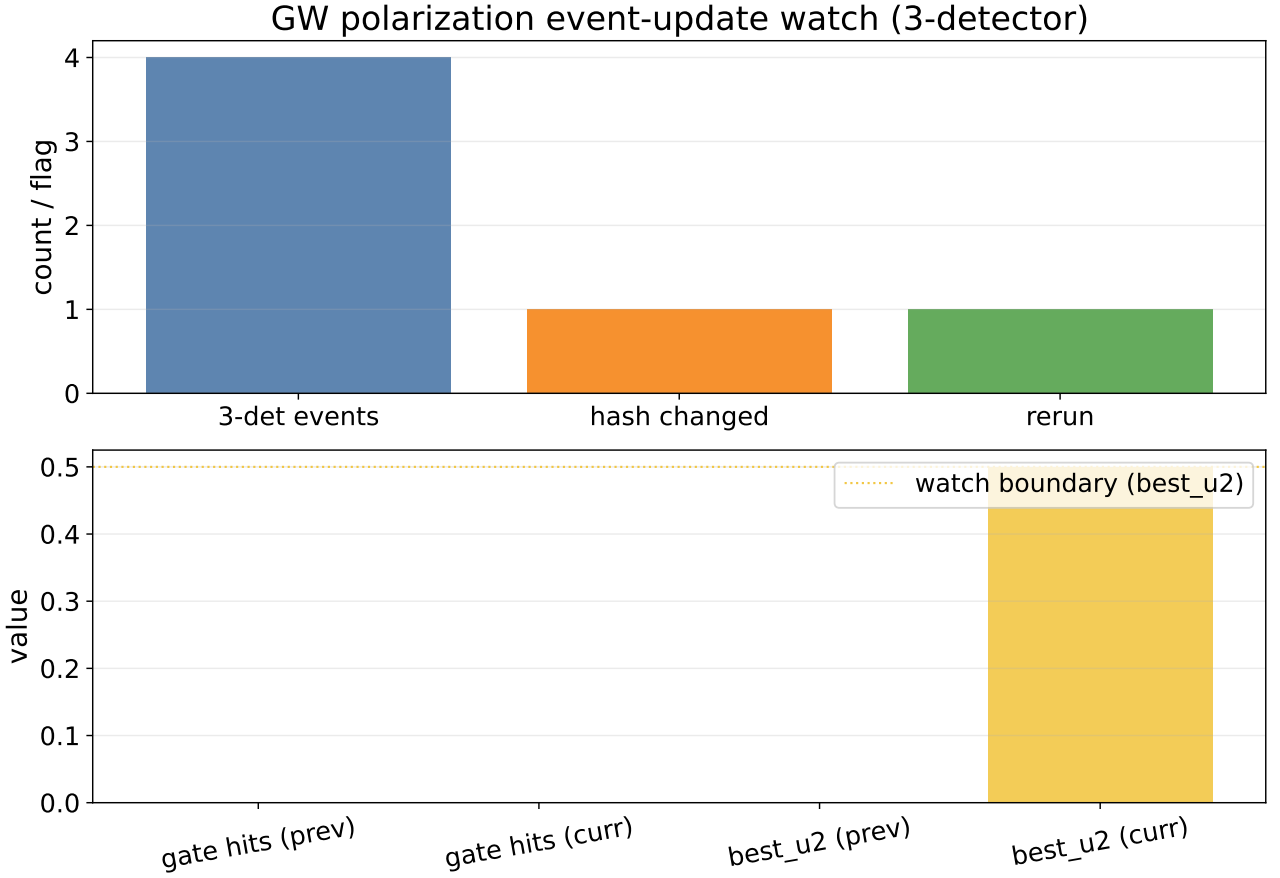


Figure 33: Comparison result for GW polarization network event update watch.

- Strong field higher-order term simultaneous constraint (EHT+GW+Pulsar+Fe-K α): pmodel_strong_field_higher_order_audit

Supplement: ($\lambda_{\text{joint}} = -4.97 \times 10^{-5} \pm 6.63 \times 10^{-5}$, $\Delta\text{AIC} = -1.44$, GW isomorphism index 0.749/0.701, area-theorem significance 4.447σ , number of Pulsar observation points $n = 4$, Fe-K α number of observation points $n = 5$, cross-domain direct-covariance ΔAIC difference -0.0031, and, on the κ side, an additional precision requirement corresponding to an approximately 80.8% reduction (information-content gain $\times 27.1$). The current verdict is Watch. The machine-readable detailed key is the Part IV ledger output/public/theory/locales/en/pmodel_strong_field_higher_order_audit.json.)

- Fixed artifact: output/public/theory/locales/en/pmodel_strong_field_higher_order_audit.json / output/public/theory/locales/en/pmodel_strong_field_higher_order_audit.csv /

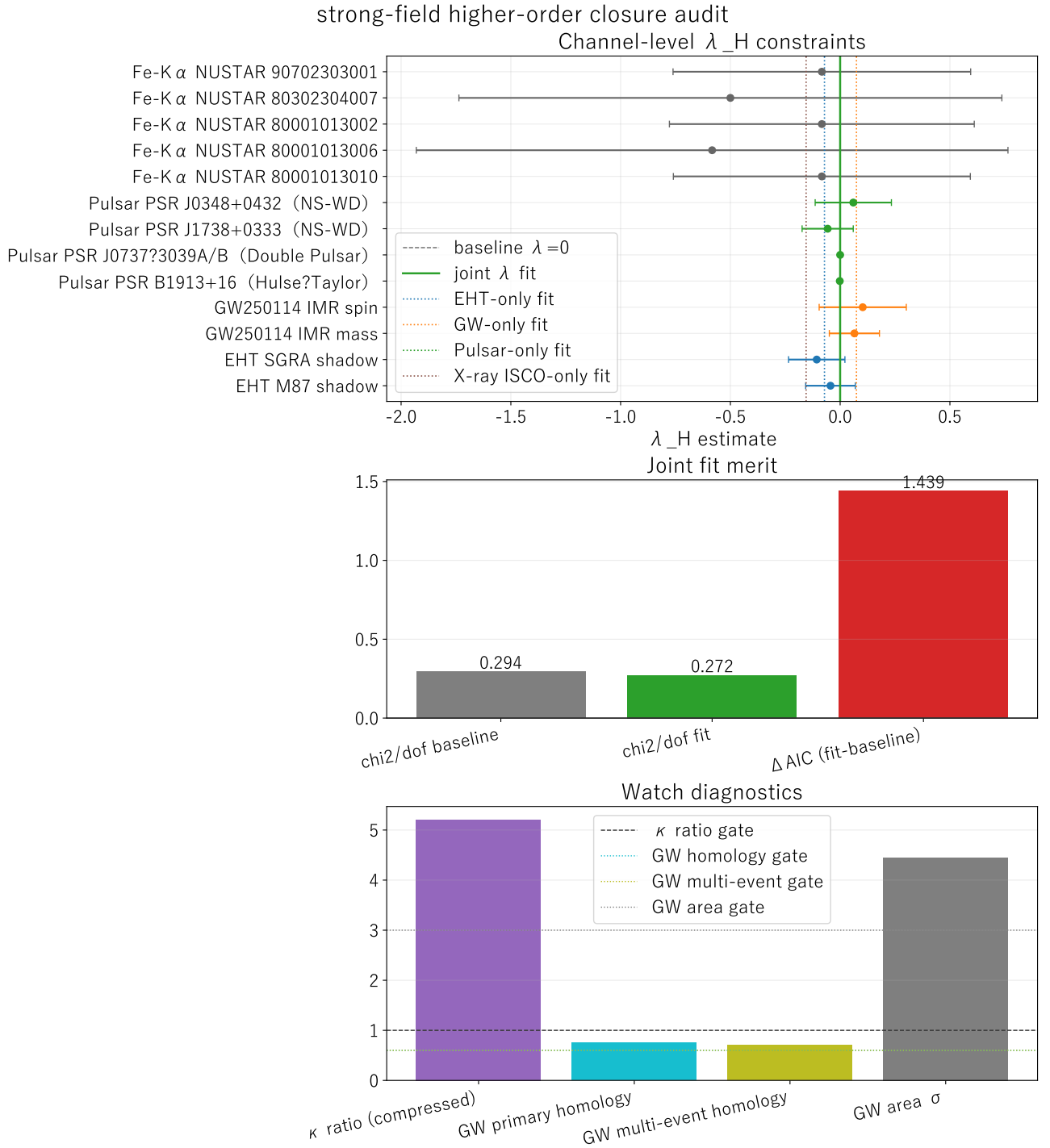


Figure 34: Comparison result for pmodel strong field higher order audit.

- Direct audit of rotating strong-field vacuum solution (steady axisymmetric; $P_t(r, \theta), P_\phi(r, \theta)$): `pmodel_rotating_bh_photon_ring_direct_audit`

Supplement: (The PDE and separation solution of $\partial_\mu F^{\mu\nu} = 0$ expanded with $J^\mu = 0, m_P \rightarrow 0$ are fixed. $\chi_2/dof = 0.371$, $\max |z_{\text{ring_diameter}}| = 0.516$, `axisymmetric_boundary_closure_pass=true`)

- Fixed artifact: `output/public/theory/locales/en/pmodel_rotating_bh_photon_ring_direct_audit.json` / `output/public/theory/locales/en/pmodel_rotating_bh_photon_ring_direct_audit.csv` /

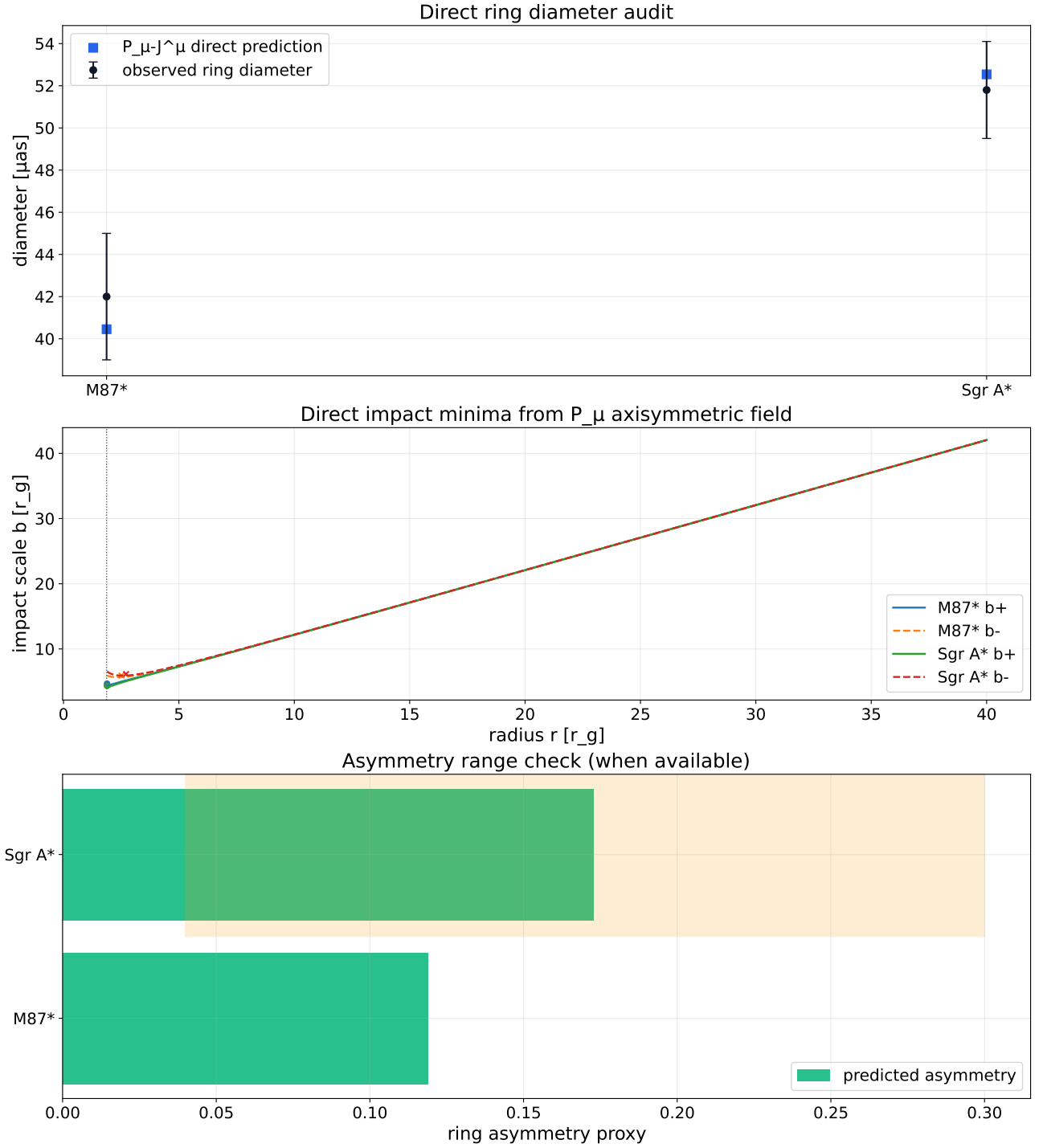


Figure 35: Comparison result for pmodel rotating bh photon ring direct audit.

- Nonlinear PDE derivation chain audit under effective metric: `pmodel_effective_metric_nonlinear_pde_derivation_audit`

Supplement: (Audit of the same notation for Part I 2.7.4 / the Part II joint strong-field higher-order-constraint section / Part IV 9, `nonlinear_pde_closure_pass=true`, `boundary_closure_pass=true`, Fixed traceability of $\max f_{\text{rel, std}}^{(\text{caseB})} = \max f_{\text{rel, std}}^{(\text{direct})} = 1.2527 \times 10^{-16}$)

- Fixed artifact: `output/public/theory/locales/en/pmodel_effective_metric_nonlinear_pde_derivation_audit.json` / `output/public/theory/locales/en/pmodel_effective_metric_nonlinear_pde_derivation_audit.csv` /

Step 8.7.32.10: effective-metric nonlinear PDE derivation audit

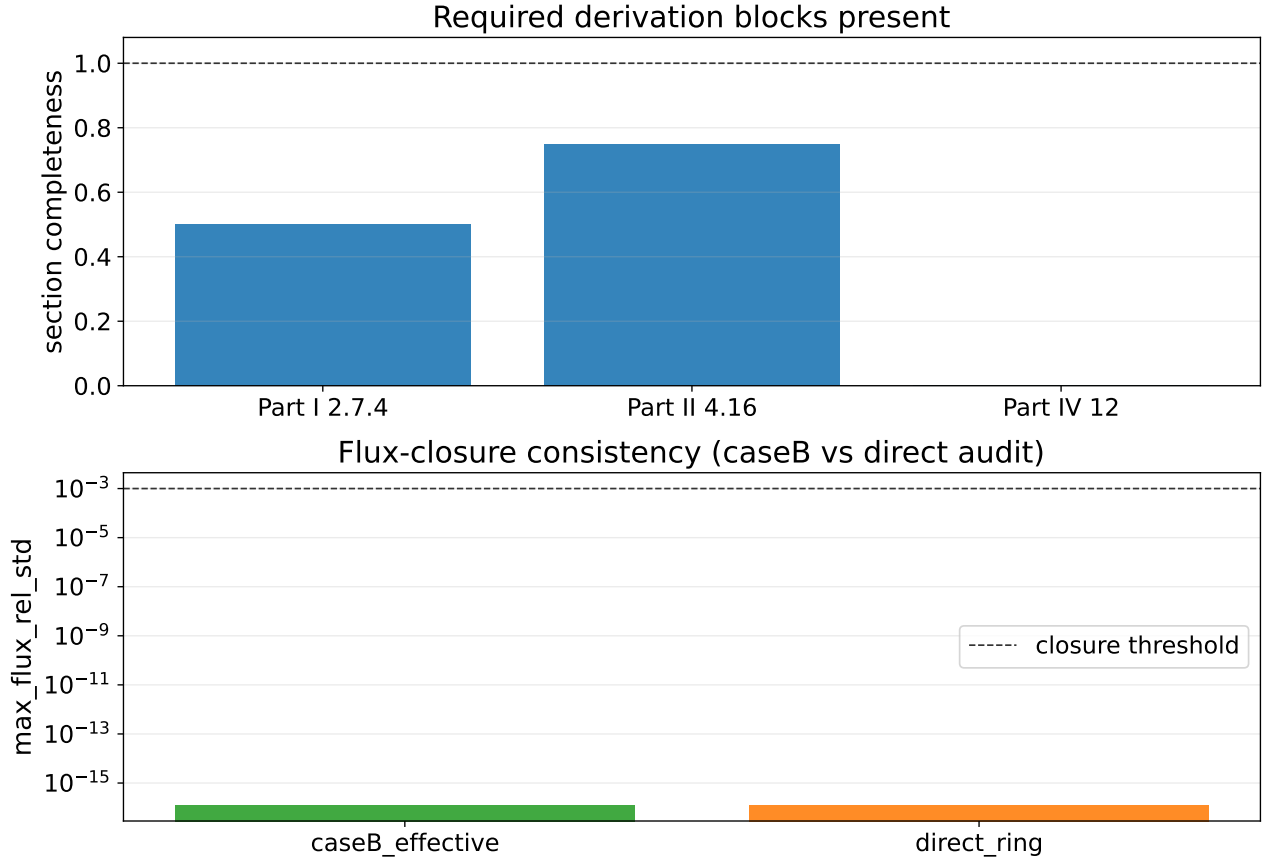


Figure 36: Comparison result for pmodel effective metric nonlinear pde derivation audit.

- $\mathcal{N}_0^{(2)}[u, P_t, P_\phi]$ Explicit source term and δP_t Perturbation solution audit: pmodel_effective_metric_n0_source_solution_audit

Supplement: P_t here refers to the time component of P_μ , and the distant reference background field is separated using the P_∞ / P_{bg} notation. Note: fix $N_0^{(2)} = N_{metric}^{(2)} + N_\phi^{(2)}$, construct δP_t from the spherical-average source $\bar{N}_0(r)$ by an integral solution, and freeze the key audit values as follows: core difference 4.6278%, secondary source-term contribution approximately +0.1806%, composite difference 4.8084%, $\max |\delta P_t| = 2.03 \times 10^{-3}$, overall status Pass.

- Fixed artifact: output/public/theory/locales/en/pmodel_effective_metric_n0_source_solution_audit.json / output/public/theory/locales/en/pmodel_effective_metric_n0_source_solution_audit.csv /

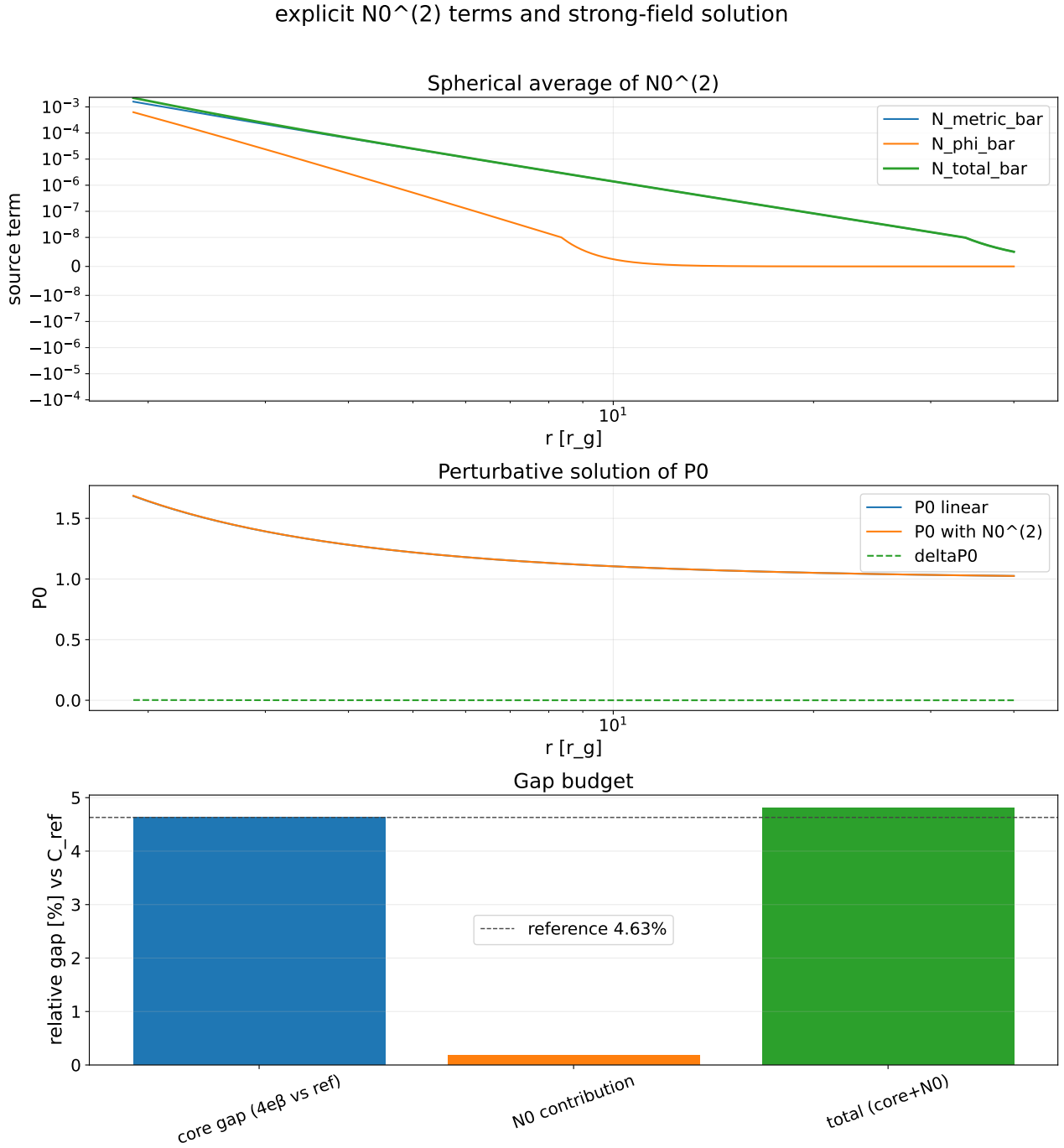


Figure 37: Comparison result for pmodel effective metric n_0 source solution audit.

- XRISM (Fe- $K\alpha$ /ISCO proxy indicator, etc.): `xrism_resolve_summary / fek_relativistic_broadening_isco_constraints`
- Pulsar (binary collapse): `binary_pulsar_orbital_decay`

Reproduction entry points (scripts/GitHub):

- EHT: [scripts/ehl/ehl_shadow_compare.py](#)
- EHT (error budget): [scripts/ehl/ehl_kappa_error_budget.py](#)

- EHT (first principles κ radiative transport envelope): [scripts/eh/eh_kappa_first_principles_transfer.py](#)
- GW:[scripts/gw/gw150914_chirp_phase.py](#)
- GW (H1/L1 amplitude ratio): [scripts/gw/gw150914_h1_l1_amplitude_ratio.py](#)
- GW (H1/L1 multi-event isomorphism): [scripts/gw/gw_h1_l1_multi_event_amplitude_audit.py](#)
- GW (area theorem): [scripts/gw/gw_area_theorem.py](#)
- GW (3 detector polarization corr gate stage audit): [scripts/gw/gw_polarization_corr_gate_stage_audit.py](#)
- GW (3 detector public event input hash monitoring): [scripts/gw/gw_polarization_network_event_update_watch.py](#)
- Strong field higher-order term simultaneous constraint: [scripts/theory/pmodel_strong_field_higher_order_audit.py](#)
- Rotating strong-field vacuum solution (steady axisymmetric PDE + direct ring observables): [scripts/theory/pmodel_rotating_bh_photon_ring_direct_audit.py](#)
- Effective metric nonlinear PDE derived chain audit: [scripts/theory/pmodel_effective_metric_nonlinear_pde_derivation_audit.py](#)
- $\mathcal{N}_0^{(2)}$ Explicit source term + δP_t Perturbation solution audit: [scripts/theory/pmodel_effective_metric_n0_source_solution_audit.py](#)
- XRISM:[scripts/xrism/xrism_integration.py](#)
- XRISM (ISCO proxy indicator): [scripts/xrism/fek_relativistic_broadening_isco_constraints.py](#)
- Pulsar: [scripts/pulsar/binary_pulsar_orbital_decay.py](#)

10 Quantum (Bell): Cross-Dataset Integration and Systematics Budget

Purpose: Audit the possibility that an "apparent violation ($S > 2$)" of the Bell inequality can be manufactured by arbitrary selection or window choices, by decomposing it into a systematic error budget and checking cross-dataset consistency.

Public artifacts (GitHub):

- Directory: [output/public/quantum/bell/](#)
- Bell's final summary (falsification conditions, freeze, key indicators): [falsification_pack.json](#) / Figure (Summary): [falsification_pack.pdf](#)
- Systematic error decomposition of 15 items (operation budget): [systematics_decomposition_15items.json](#) / Figure (Budget + Correlation): [systematics_decomposition_15items.pdf](#)
- Cross-dataset covariance (including window/offset sweep): [cross_dataset_covariance.json](#) / Figure (covariance): [cross_dataset_covariance.pdf](#)
- Quantum connection KPI (Bell/interference/common judgment of physical properties/thermal holdout): [quantum_connection_shared_kpi.json](#) / Figure (common KPI): [quantum_connection_shared_kpi.pdf](#)
- Bell $\leftrightarrow$ interference bridge table (separation of sorting origin and physics origin): [quantum_connection_bridge_table.json](#) / Figure (bridge table): [quantum_connection_bridge_table.pdf](#)
- Born 2.6.2 A/B crossing gate (identity determination of the COW/atom/HOM proxy indicator): [quantum_connection_born_ab_gate.json](#) / Figure (A/B judgment): [quantum_connection_born_ab_gate.pdf](#)
- Observation proxy-indicator constraint for Born rule route A (machine judgment of A continuation/A rejection $\rightarrow$ B transition): [output/public/quantum/locales/en/born_route_a_proxy_constraints_pack.json](#) / Figure (Comparison result for born route A proxy constraints pack.): [output/public/quantum/locales/en/born_route_a_proxy_constraints_pack.pdf](#)
- Principle of action $\rightarrow$ EL derivation audit (covariance/invariance check under local phase transformation): [output/public/quantum/locales/en/action_principle_el_derivation_audit.json](#) / Figure (Comparison result for action principle EL derivation audit.): [output/public/quantum/locales/en/action_principle_el_derivation_audit.pdf](#)
- Non-relativistic limit (2.6 $\rightarrow$ 2.5) audit (approximate order gate of $\epsilon_v^2/\epsilon_\phi/\epsilon_{env}$): [output/public/quantum/locales/en/nonrelativistic_reduction_schrodinger_mapping_audit.json](#) / Figure (Comparison result for nonrelativistic reduction schrodinger mapping audit.): [output/public/quantum/locales/en/nonrelativistic_reduction_schrodinger_mapping_audit.pdf](#)
- Derivation-derived parameter falsification condition pack (Bell/interference/mapping to physical properties/heat): [output/public/quantum/locales/en/derivation_parameter_falsification_pack.json](#) / Figure (Comparison result for derivation parameter falsification pack.): [output/public/quantum/locales/en/derivation_parameter_falsification_pack.pdf](#)

- Derivation $\rightarrow$ observation chain lock audit (consistency judgment of EL/nonrel/Born A/B/bridge/shared/derivation pack): output/public/quantum/locales/en/derivation_observable_chain_lock_audit.json / Figure (Comparison result for derivation observable chain lock audit.): output/public/quantum/locales/en/derivation_observable_chain_lock_audit.pdf
- Derivation $\rightarrow$ observation chain lock audit (Part IV judgment after watch policy): output/public/quantum/derivation_observable_chain_lock_audit_watch_policy.json
- Reference skeleton of action + Closure audit of derived extension of Part III-A ($L_{\text{total}} - > EL - > observables$; Noether/integration gate of non-relativistic mapping): output/public/quantum/locales/en/lagrangian_noether_observable_closure_audit.json / Figure (Comparison result for lagrangian noether observable closure audit.): output/public/quantum/locales/en/lagrangian_noether_observable_closure_audit.pdf
- Drift audit of derived extension of action reference framework + Part III-A (fixing recalculation requirements upon gate deviation): output/public/quantum/locales/en/lagrangian_noether_observable_closure_drift_audit.json / Figure (Comparison result for lagrangian noether observable closure drift audit.): output/public/quantum/locales/en/lagrangian_noether_observable_closure_drift_audit.pdf
- Working principle closure audit of rotational coupling L_{rot} (independent prior + holdout prediction gate): output/public/quantum/lagrangian_noether_rotational_closure_audit.json / Figure (Comparison result for lagrangian noether rotational closure audit.): output/public/quantum/lagrangian_noether_rotational_closure_audit.pdf
- Cross-cutting update of quantum covariance (Bell/interference/physical properties/thermal block covariance): output/public/quantum/locales/en/quantum_connection_block_covariance.json / Reference figure (block covariance): output/public/quantum/locales/en/quantum_connection_block_covariance.pdf

Item	Contents
Minimum synchronization of quantum connections	Raw judgment of common KPI is due to Bell pairing and is reject, bridge table is watch, Born A/B is A_{continue} , and the main mode of block covariance is interference (principal factor loading ≈ 0.9999). In Part IV, watch-policy reads Bell pairing-only reject as a monitoring judgment that does not hinder progress
Condensed-matter / thermal holdout	KPI targets are included_n=106 / reject_n=0 / inconclusive_n=0 (excluded_n=1). $Si\alpha(T)$ (Debye+Einstein basis) is excluded from KPI aggregation as it is only for diagnosis

Supplement: the same block-covariance figure is fixed as the synchronized reference.

Falsification conditions:

- If channels.condensed_thermal_holdout.kpi.reject_n of quantum_connection_shared_kpi.json is > 0 or inconclusive_n is > 0 , reject the pass of physical property/thermal holdout.
- If decision.route_a_gate of quantum_connection_born_ab_gate.json is A_reject, or if dominant_loading <0.9 in quantum_connection_block_covariance.json, discard the current quantum connection order (including A-route continuation).
- If decision.shared_gate_policy of quantum_part4_closeout_sync_pack.json is not watch_if_bell_pairing_only, or decision.part4_quantum_sync_status is not pass, reject the arrangement that reads Bell pairing-only reject as a monitoring judgment that does not hinder progress.

Verifiable artifacts / figures:

- Dynamic collapse simulation of measurement problem (nonlinear measurement equation; 2 states + 3ch pointer + environment dissipation): output/public/quantum/quantum_measurement_dynamic_collapse_simulation_metrics.json / Figure (time series + convergence distribution): output/public/quantum/quantum_measurement_dynamic_collapse_simulation.pdf
- Quantum closeout boundary audit (Born/measurement/entanglement boundary fixing): output/public/quantum/quantum_v11_plus_scope_closeout_audit_metrics.json / output/public/quantum/quantum_v11_plus_scope_closeout_cases.csv
- Part IV synchronous pack (fixed replacement of legacy gate and closeout boundary): output/public/quantum/quantum_part4_closeout_sync_pack.json / output/public/quantum/quantum_part4_closeout_sync_rows.csv
- Support aspects of the selection theorem for fine structure constants (sign alternation, internal rank 1 matching, piecewise extension of harmonic gratings, paper summary diagram of support for fine structure constants at fixed q): output/public/quantum/v2_trial2_theorem_support_summary_metrics.json / Figure (supporting surface of the fine structure constant selection theorem): output/public/quantum/v2_trial2_theorem_support_summary.pdf
- Summary of theoretical criteria with coupled-localization conditions for the W/Z mass reference points with the two-component coupled Q-ball (family (17,1,1), positive coupled-localization, numerical solution and spectral scale, paper summary diagram of charge-window expansion): output/public/quantum/v2_trial3_weak_checkpoint_summary_metrics.json / Figure (theoretical standard based on the coupled-localization condition at the W/Z mass checkpoints): output/public/quantum/v2_trial3_weak_checkpoint_summary.pdf
- Note: The v11_plus stem in the artifact filename / script name is for compatibility, and the text should be read against the closeout-boundary audit listed above.
- Gravity-induced decoherence (Part III 4.5; Self-Gravity Scale Audit): output/public/quantum/gravity_induced_decoherence_metrics.json / Figure ($\chi \sim (E_G/\hbar)T$): output/public/quantum/gravity_induced_decoherence.pdf
- Pikovski structural comparison: output/public/quantum/gravity_induced_decoherence_pikovski_comparison_metrics.json / output/public/quantum/gravity_induced_decoherence_pikovski_comparison_rows.csv

- Atom interferometer differential surrogate index: `output/public/quantum/atom_interferometer_pikovski_phase_difference_metrics.json` / `output/public/quantum/atom_interferometer_pikovski_phase_difference_rows.csv`
- Optical lattice clock differential observable: `output/public/quantum/optical_clock_differential_frequency_metrics.json` / `output/public/quantum/optical_clock_differential_frequency_rows.csv`
- Aggregate differential table artifacts: `output/public/quantum/gravity_quantum_differential_prediction_table_metrics.json` / `output/public/quantum/gravity_quantum_differential_prediction_table_rows.csv`

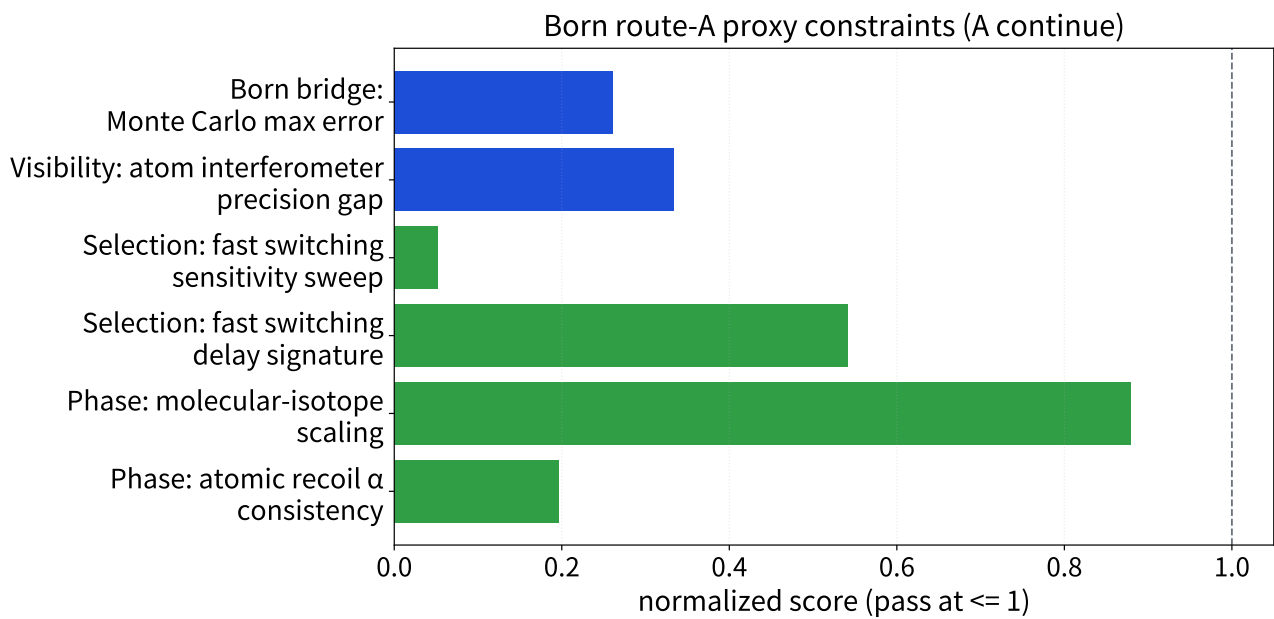


Figure 38: Born route-A proxy-constraint pack

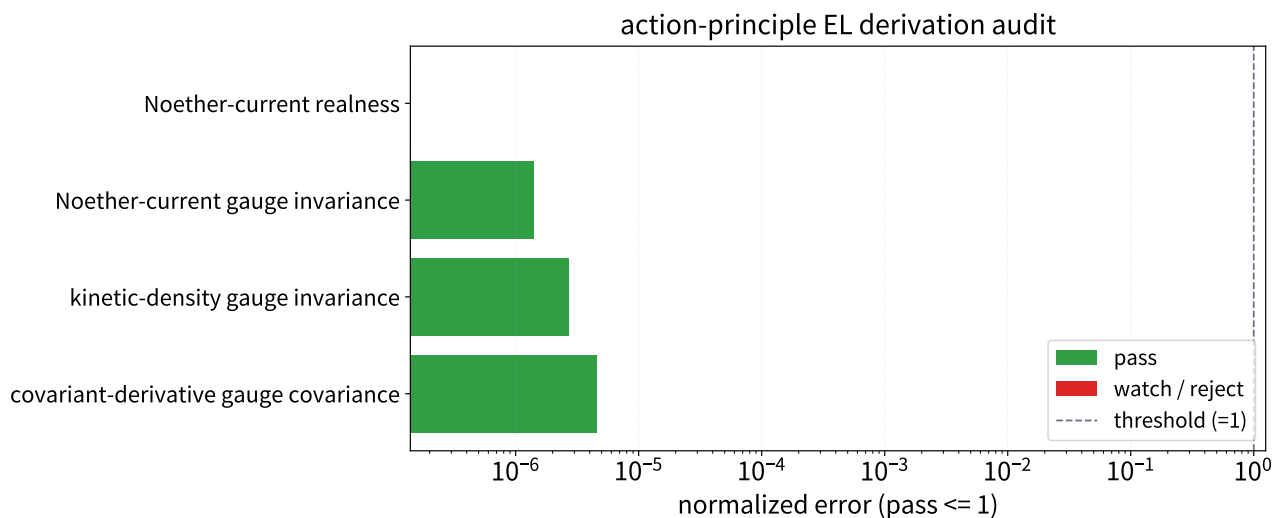


Figure 39: Action-principle to EL derivation audit

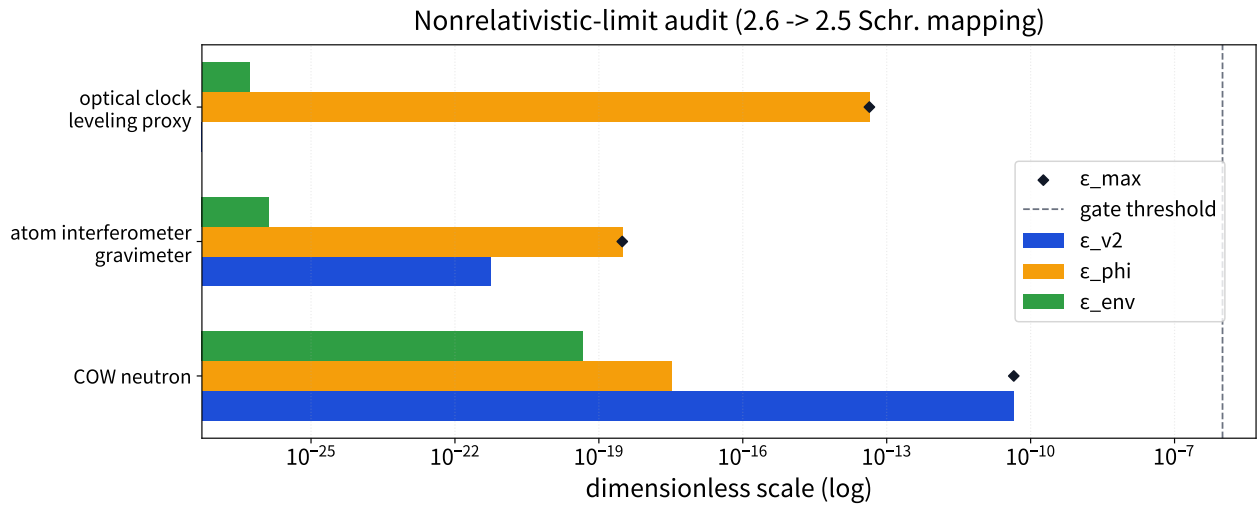


Figure 40: Non-relativistic reduction to the Schrödinger mapping audit

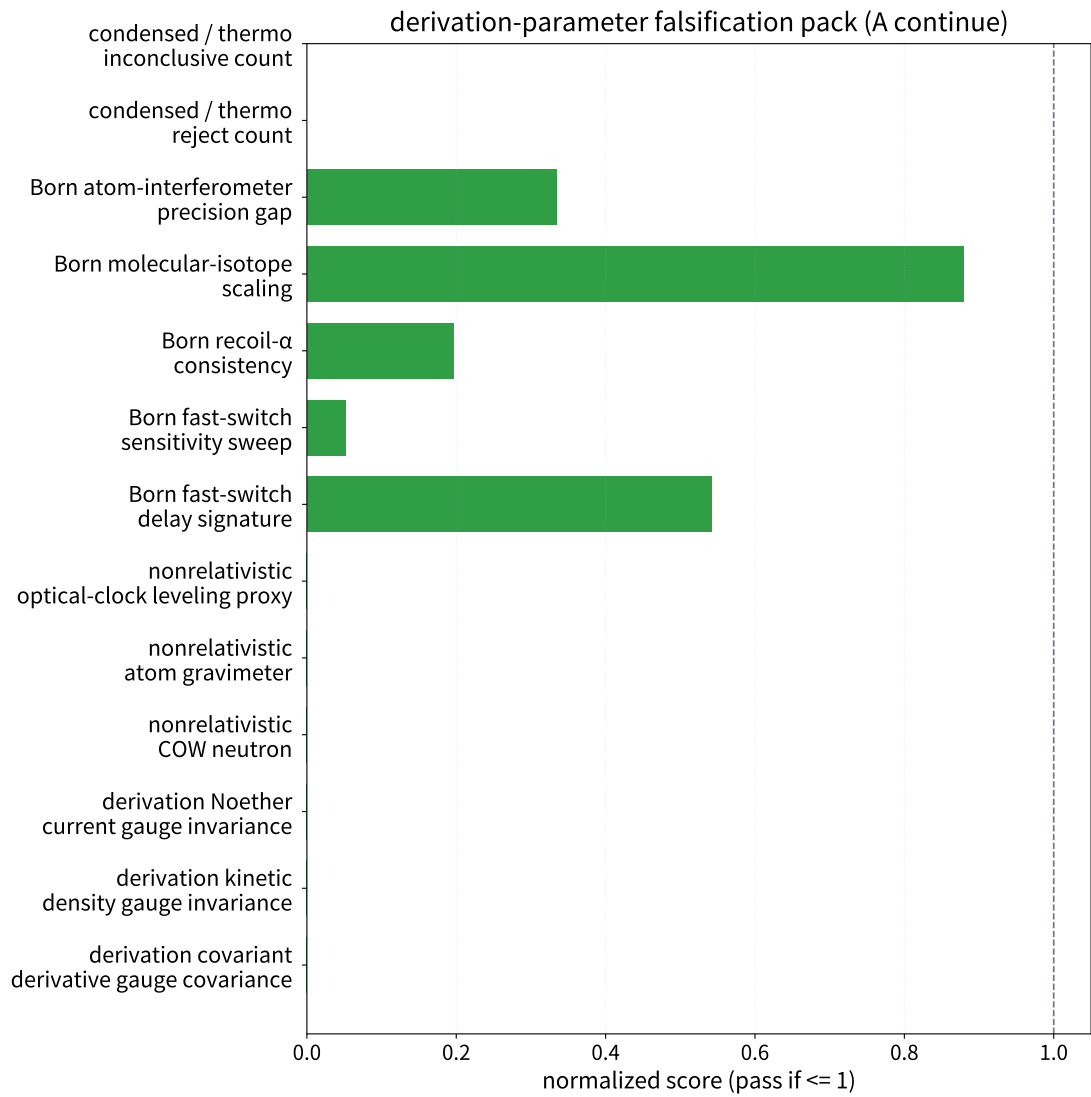


Figure 41: Derivation-derived parameter falsification-condition pack

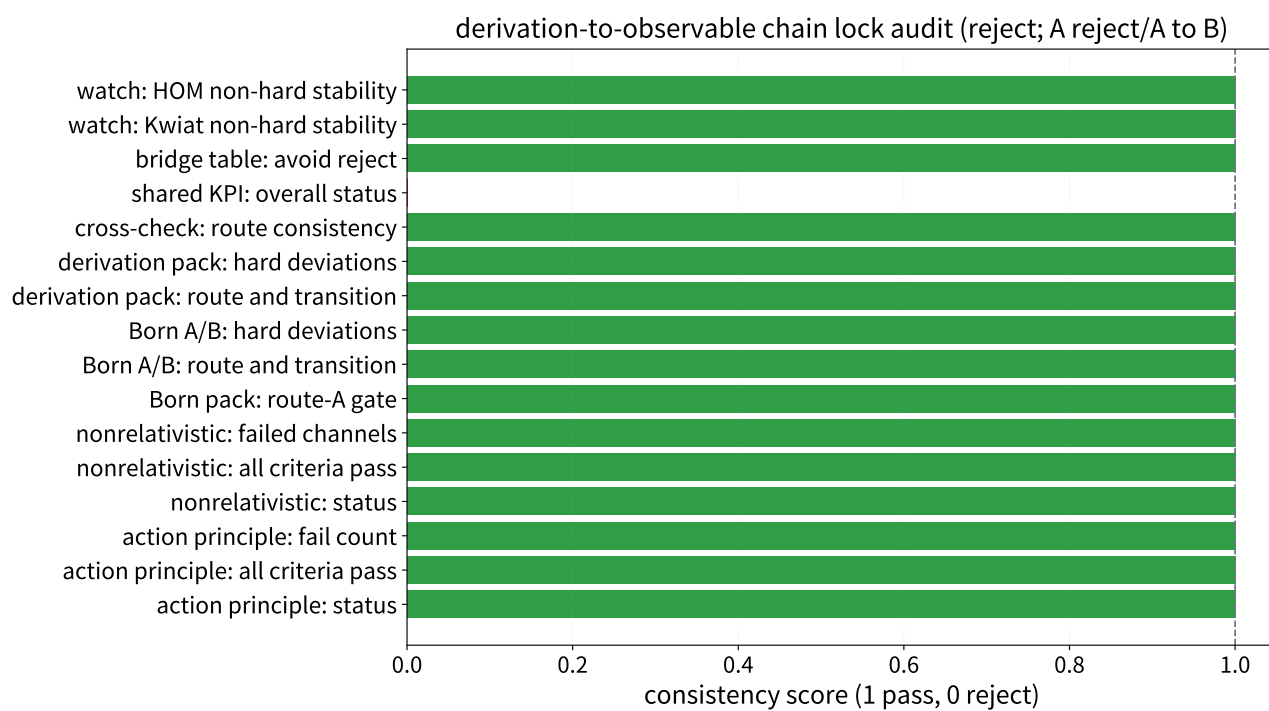


Figure 42: Derivation-to-observation chain lock audit

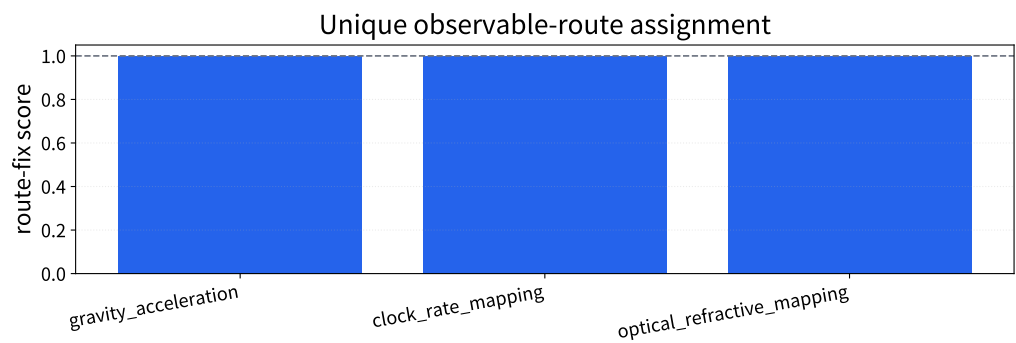
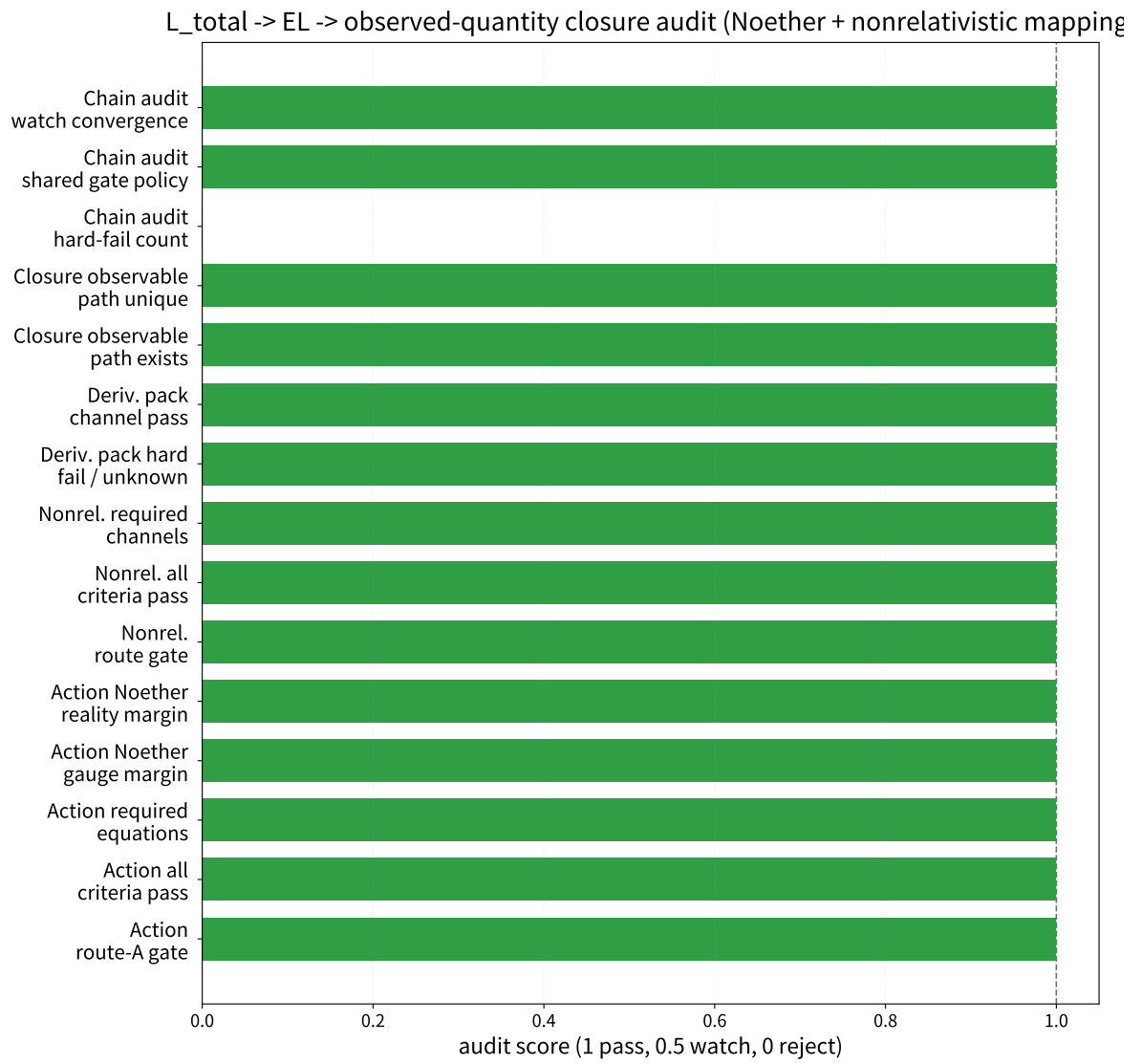


Figure 43: Lagrangian-Noether observable-closure audit

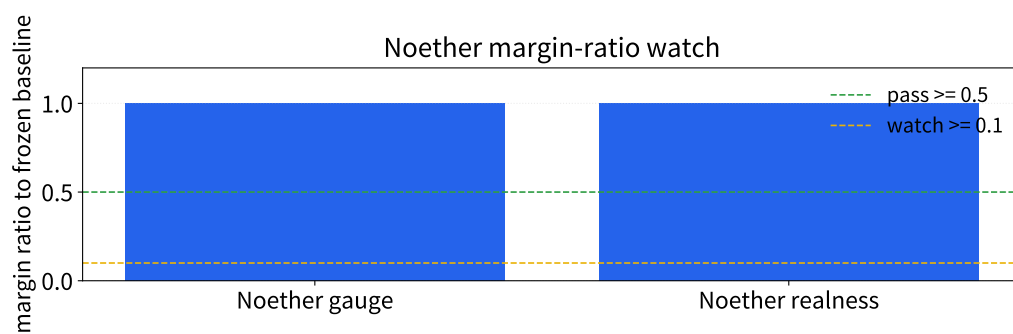
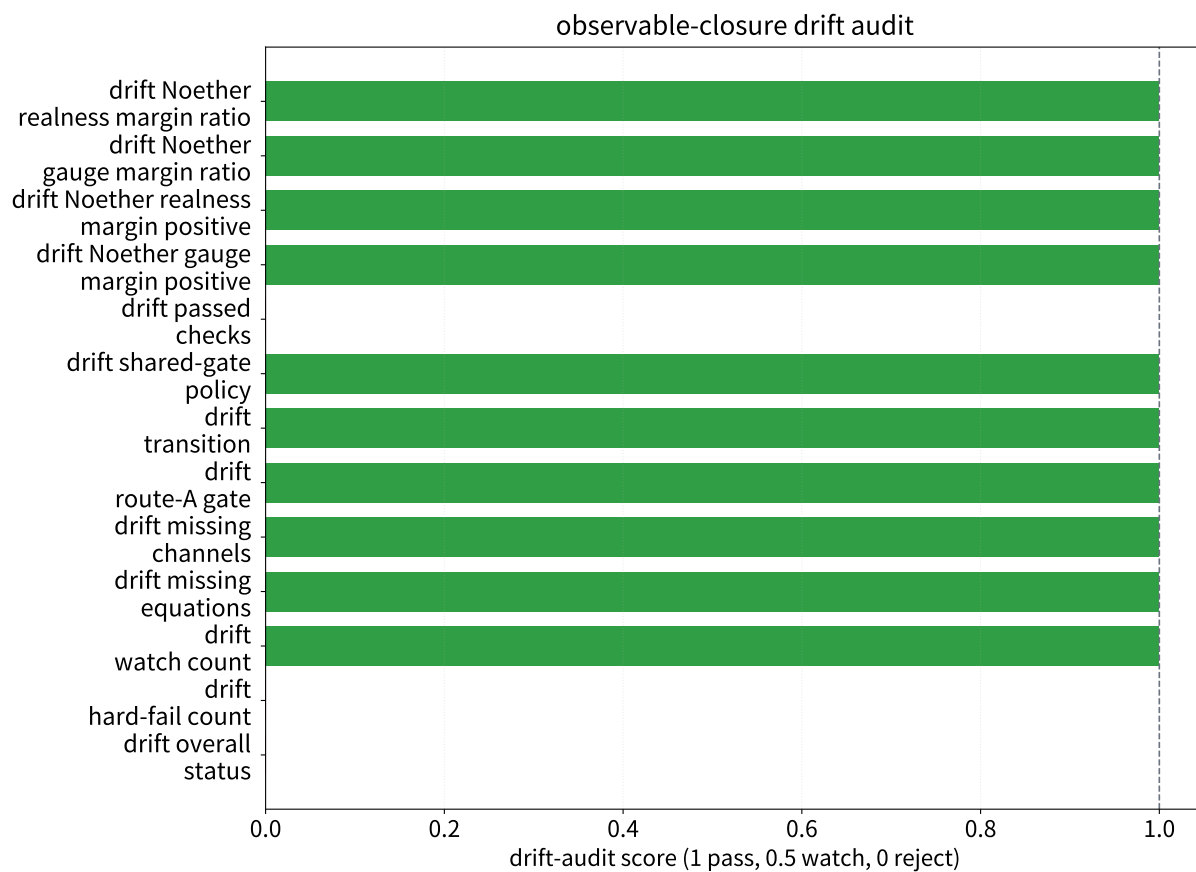


Figure 44: Lagrangian-Noether observable-closure drift audit

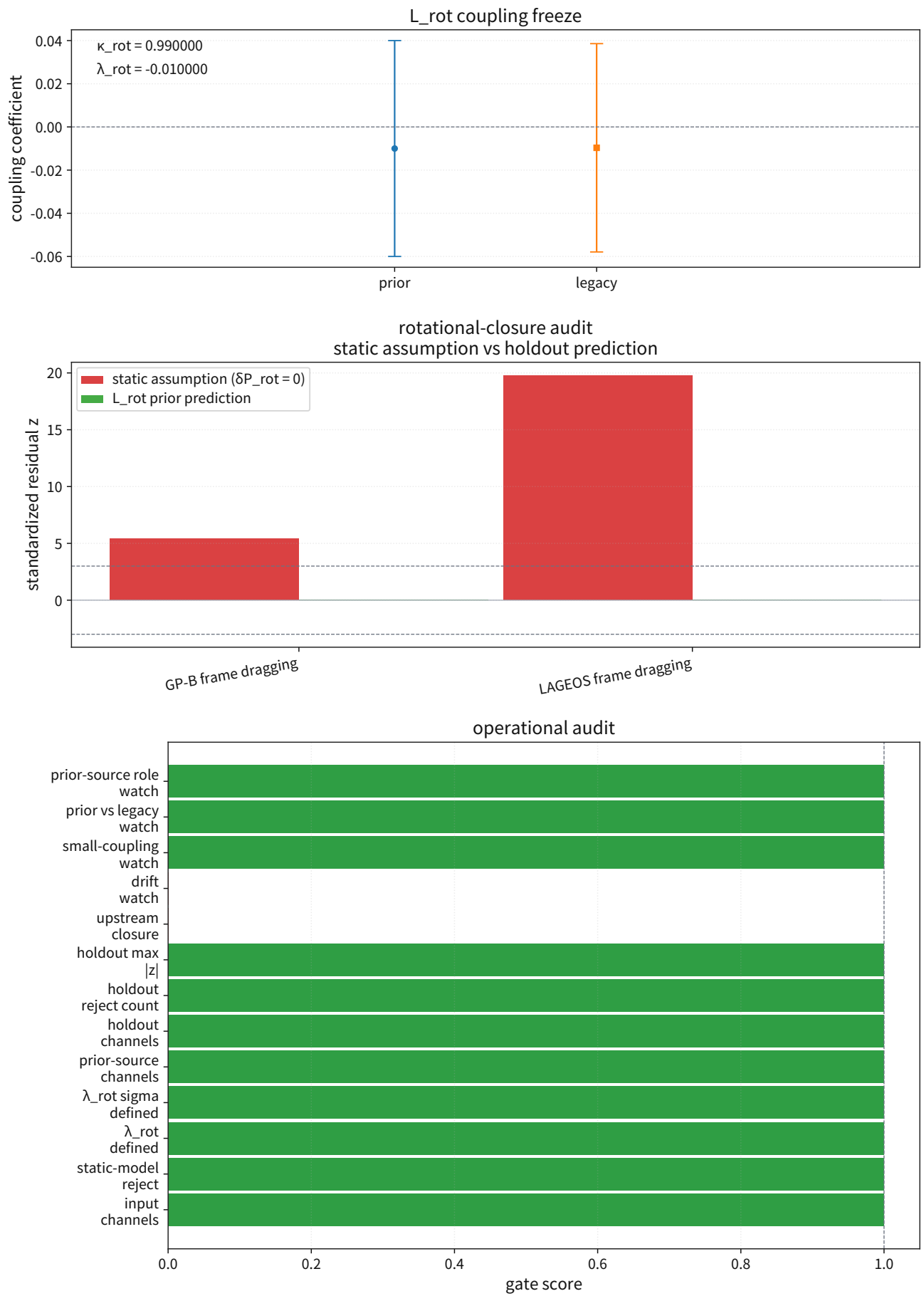


Figure 45: Rotational-coupling closure audit

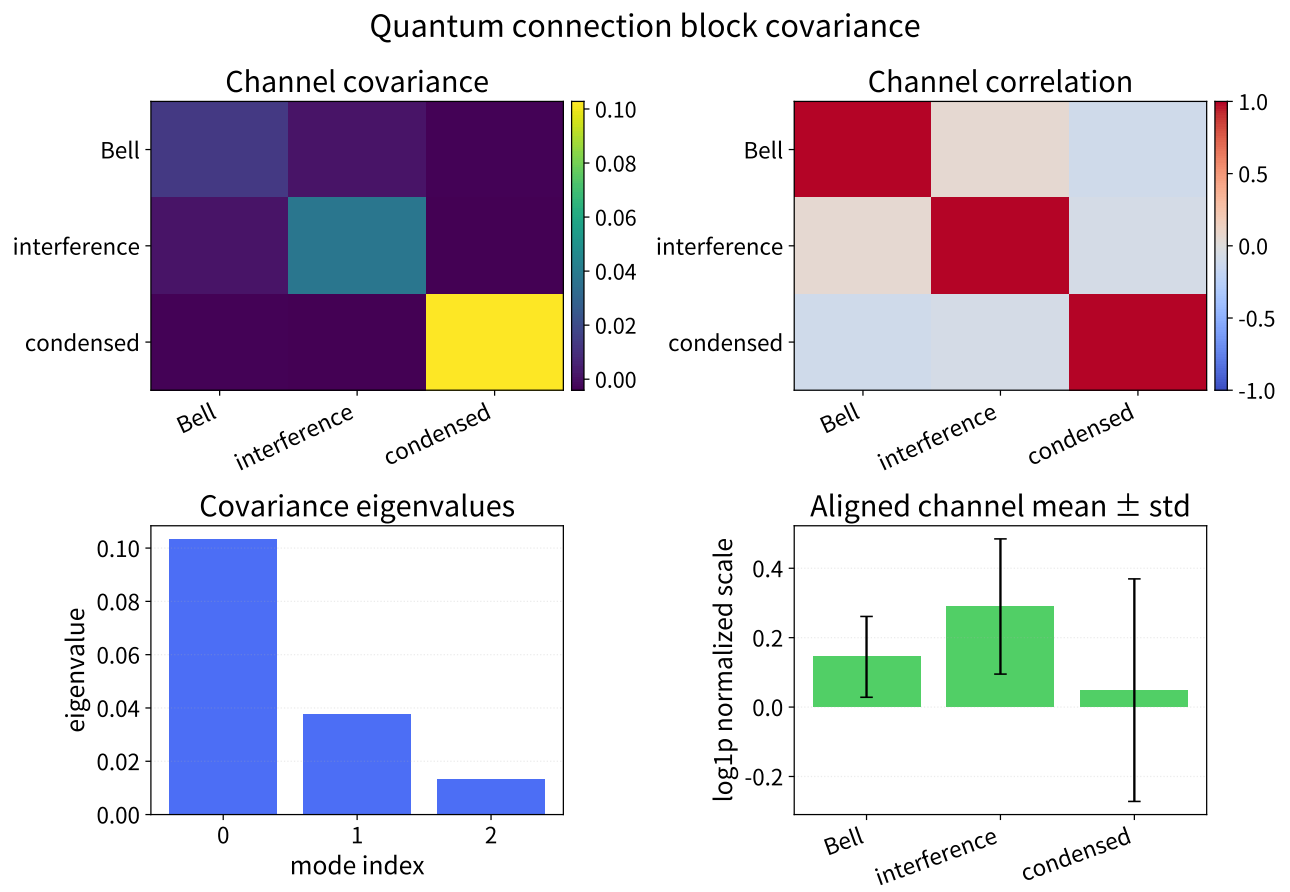


Figure 46: Quantum-connection block-covariance audit

dynamic-collapse simulation

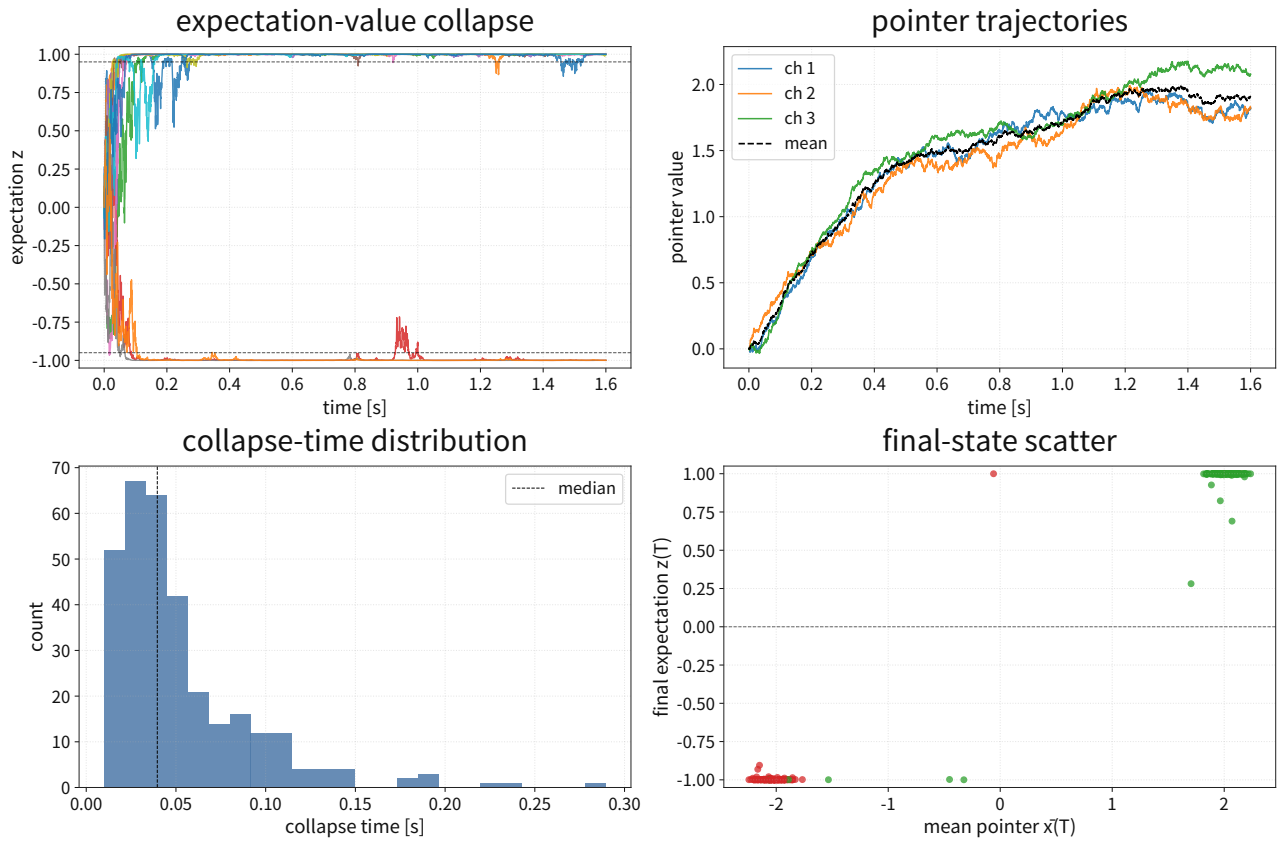


Figure 47: Dynamic-collapse simulation of the measurement problem

Support surface of the fine-structure selection theorem

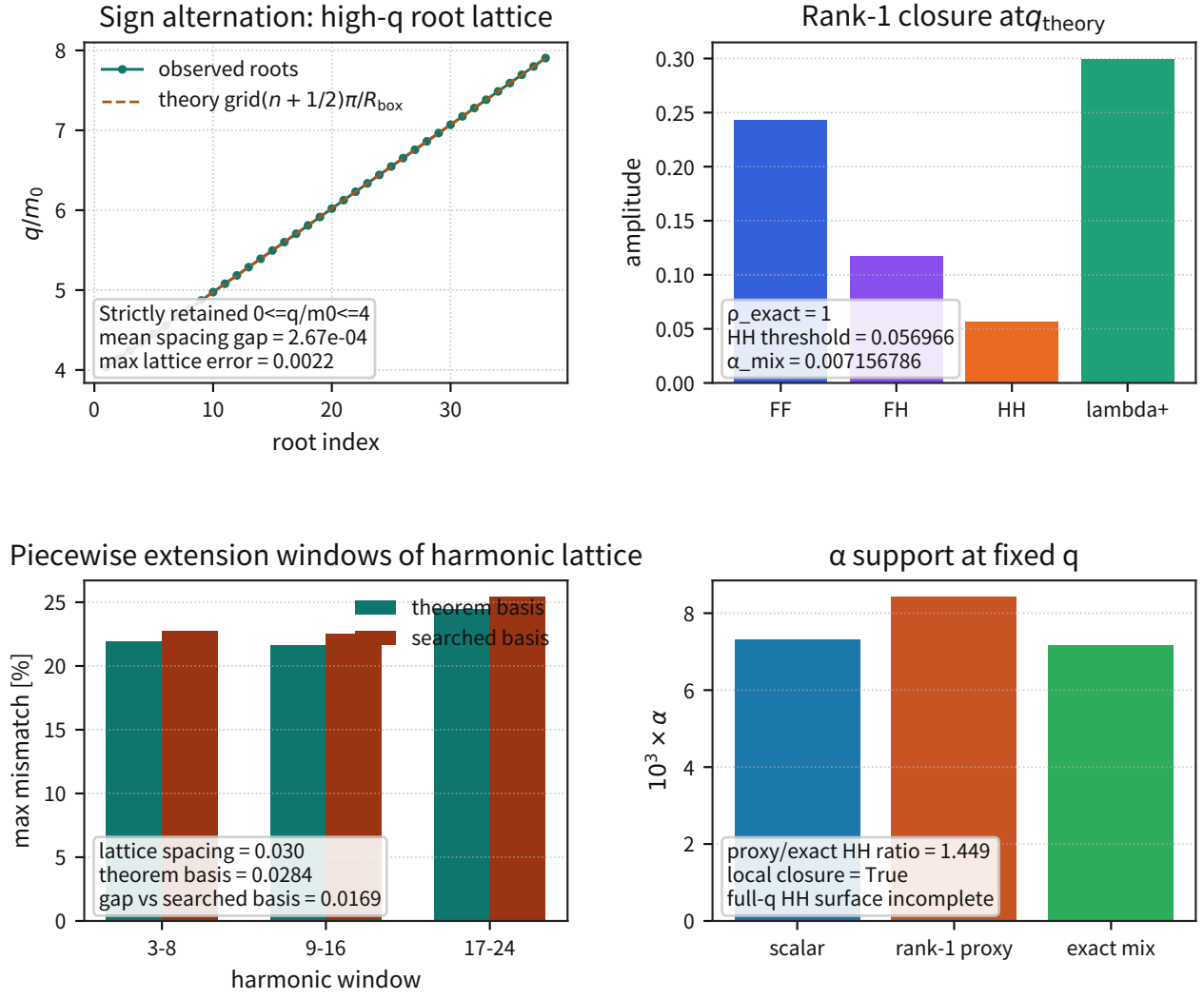


Figure 48: Support surface for the fine-structure-constant selection theorem

weak-checkpoint summary from coupled localization
at the W/Z anchor points

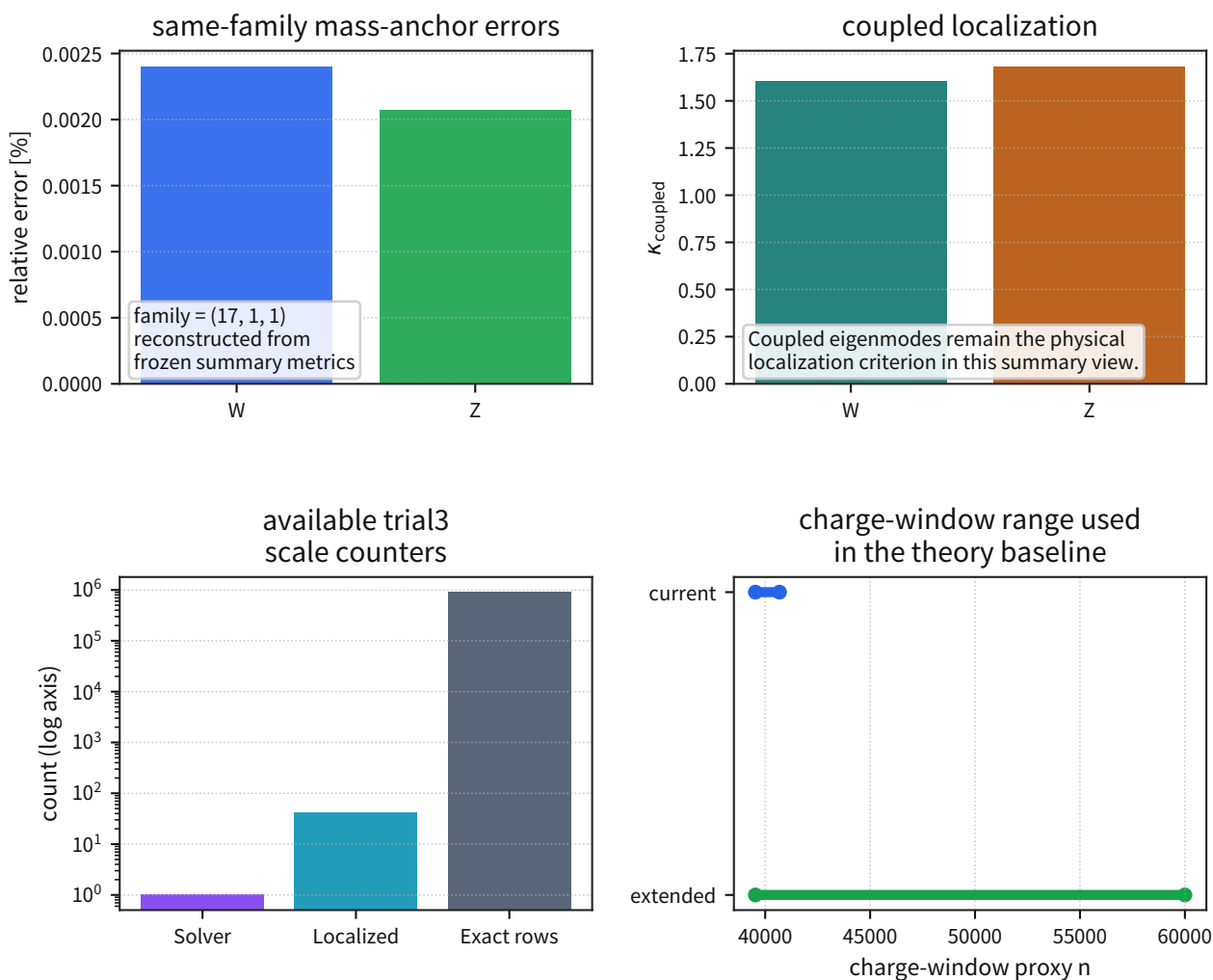


Figure 49: Theoretical standard under the coupled-localization condition at the W/Z mass checkpoints

gravity-induced decoherence: observed quantity and noise budget

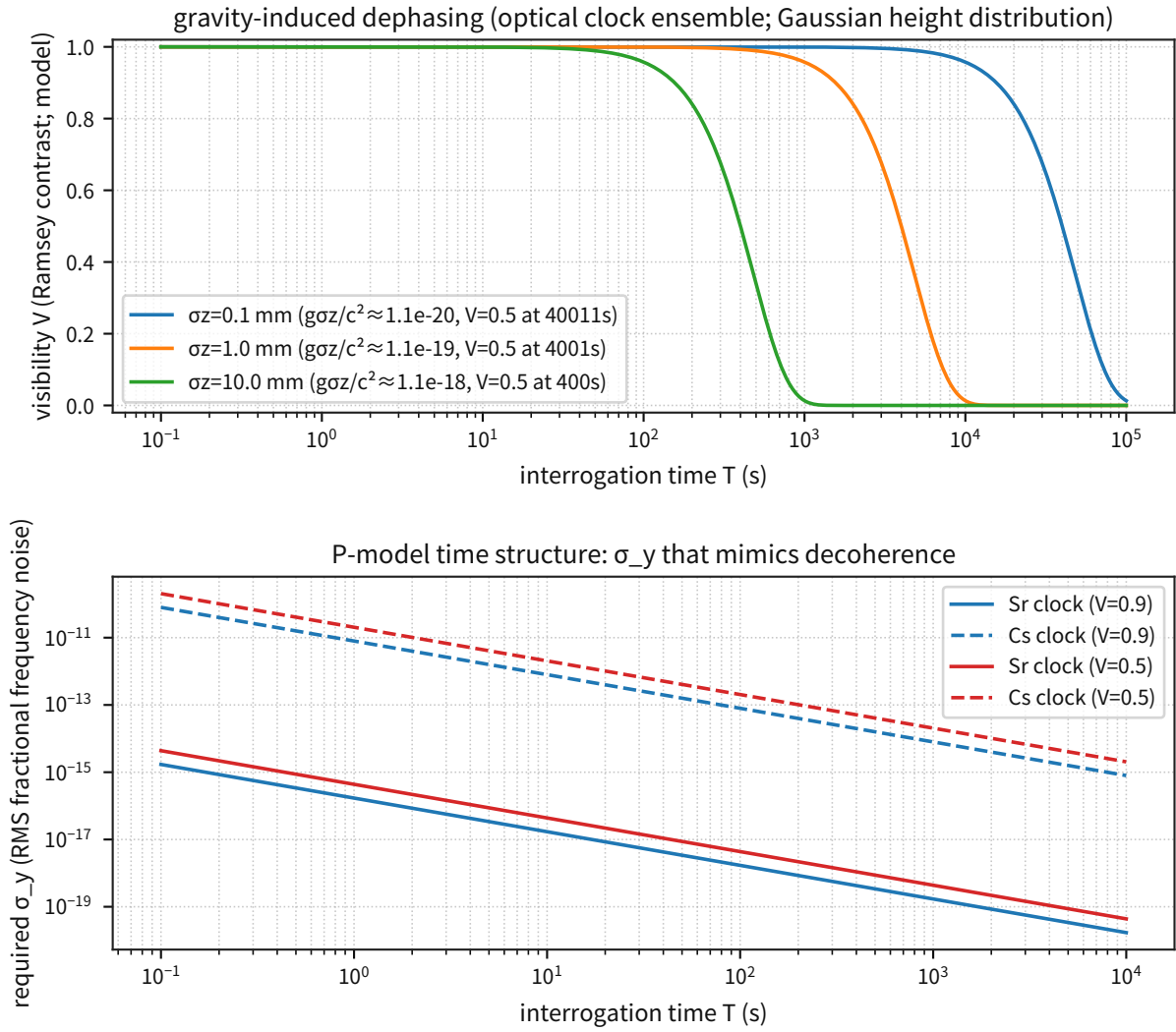


Figure 50: Gravity-induced decoherence audit

Item	Contents
Derived parameter falsification conditions	route_a_gate=A_continue, transition=A_stay. All main gates pass, watch is limited to insufficient interference accuracy and physical properties/thermal holdout rejects

Continued on next page

Item	Contents
Derivation chain lock audit	hard consistency gate all passes. After watch-policy route <code>_a_gate=A_continue</code> , transition= <code>A_stay</code> , overall_status= <code>watch</code> , shared_gate_policy= <code>watch_if_bell_pairing_only</code> , demoted_to_watch= <code>true</code> . watchlist is <code>hom_noise_psd_shape</code> , <code>kwiat_delay_signature</code> , <i>shared :: overall_status</i>

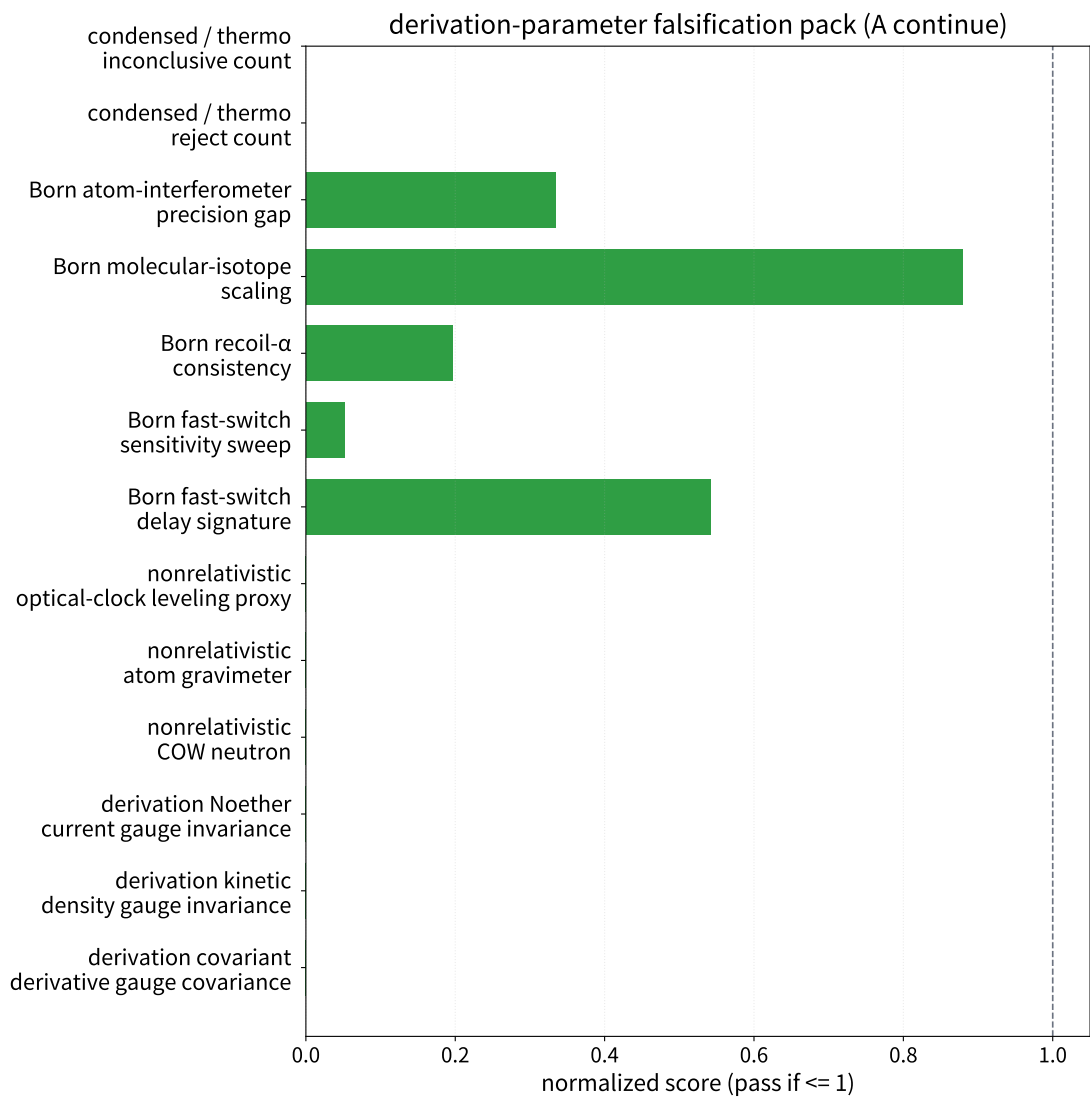


Figure 51: Comparison result for derivation parameter falsification pack.

Falsification conditions:

- If `decision.route_a_gate` of `derivation_parameter_falsification_pack.json` is `A_reject`, reject derivation challenge A.
- If `decision.hard_unknown_ids` of the same pack is non-empty, reject the A continuation judgment and record it as subject to B transition review.
- If `decision.part4_quantum_sync_status` of `quantum_part4_closeout_sync_pack.json` is not pass, `decision.closeout_overall_status` is not `completed_scope_fixed`, or `decision.closeout_remaining_blocking_items_n` is > 0 , reject the closeout boundary synchronization in Part III.

Item	Contents
Reference skeleton of action + closure of derived extensions of Part III-A	<code>overall_status=pass</code> , <code>hard_fail_ids_n=0</code> , <code>watch_ids_n=0</code> , <code>missing_equations_n=0</code> , <code>missing_nonrel_channels_n=0</code> , <code>route_a_gate=A_continue</code> , <code>transition=A_stay</code>
Closure drift audit	<code>overall_status=pass</code> , <code>recalc_required=false</code> , <code>hard_fail_row_ids_n=0</code> , <code>watch_row_ids_n=0</code> , <code>drift_reject_row_ids_n=0</code>
Closure operation cycle	<code>overall_status=watch</code> , <code>decision=operational_cycle_stable__</code> <code>with_ckm_watch</code> , <code>ckm_pmns_closure=watch</code> , <code>closure_drift=pass</code> , <code>recalc_required=false</code>
Rotational combination closure (independent prior + holdout prediction)	<code>overall_status=pass</code> , <code>decision=l_rot_closure_pass</code> , $\lambda_{\text{rot}} = -0.0100 \pm 0.0500$, <code>priorsource=1</code> , $\text{holdout} = 1$, $\text{holdoutmax} z =0.0267$, <code>staticbranchreject=2/2</code> , $\text{scalarreductiongate}(J^i = 0 \Rightarrow P_i = 0) =$ <i>pass</i>

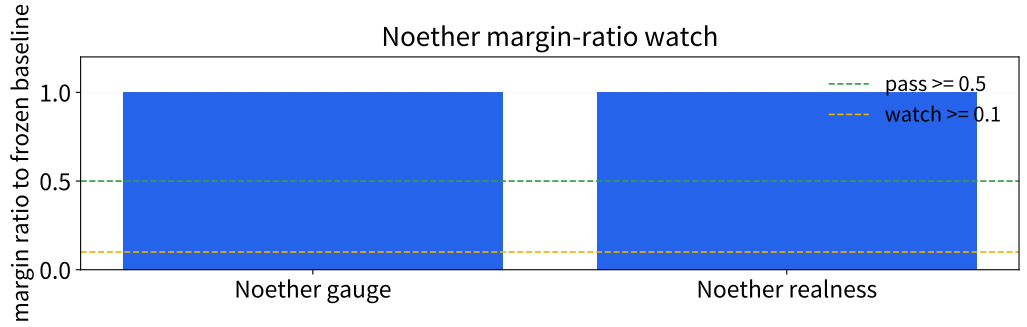
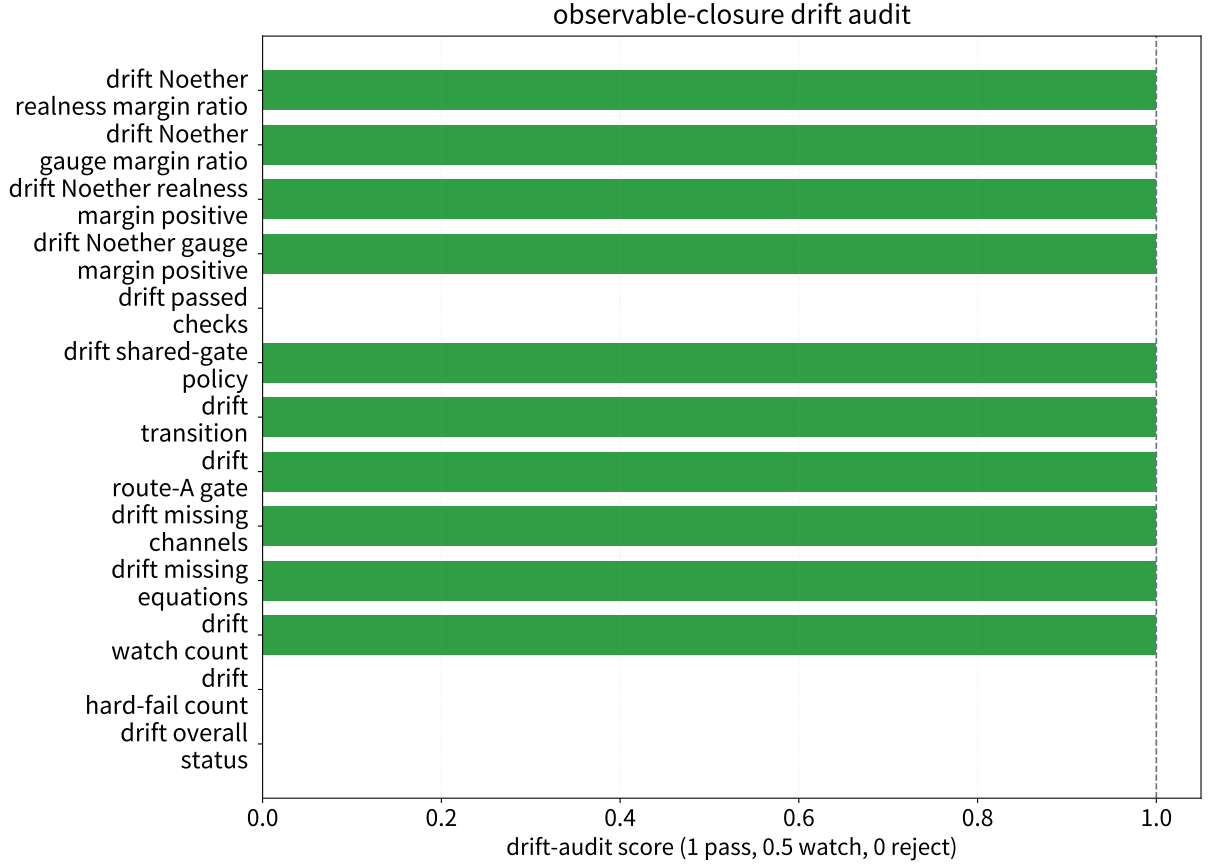


Figure 52: Comparison result for lagrangian noether observable closure drift audit.

Falsification conditions:

- If `decision.overall_status` of `lagrangian_noether_observable_closure_audit.json` is `reject`, reject the closure claim of $L_{\text{total}} - > EL - > \text{observables}$ for the derived extension of action reference skeleton + Part III-A.
- If `decision.recalc_required=true` or `decision.hard_fail_row_ids` of `lagrangian_noether_observable_closure_drift_audit.json` is non-empty, reject the closure operation and record it as

a recalculation required state. The same procedure is used during explicit execution.

- If, in the same drift audit, `decision.overall_status=reject` is established or `decision.drift_reject_row_ids` is non-empty, or if `lagrangian_noether_rotational_closure_audit.json` has `decision.overall_status=reject` (or `prediction_gate.holdout_reject_n` is > 0 / `holdout_max_abs_z` is > 3), reject the frame-dragging connection by the current A-route continuation and the rotational coupling L_{rot} in Part III 2.6.2.

Item	Contents
Dynamic collapse of measurement (env-coupled)	collapse fraction=1.00, τ_{50} =39.6 ms, pointer consensus=0.966, branch reversal=0.075, post-sign consistency=1.000 (2 states + 3 channels instruction degrees of freedom + environment dissipation, n=320)
Dynamic collapse of measurements (retest stability)	overall_status=pass, decision=env_seed_stability_confirmed, n_runs=18, τ_{50} range = 38.8-43.0 ms($cv = 0.0380$), pointer_consensusmin=0.9531, branch_reversalmax=0.0677 (seed $\times$ env_scale, each run n=192)

dynamic-collapse stability audit



Figure 53: dynamic-collapse stability audit for quantum measurement

Falsification conditions:

- If `summary.collapse_fraction` of `quantum_measurement_dynamic_collapse_simulation_metrics.json` decreases significantly (operational guideline: < 0.95), reject the minimum dynamic collapse model.
- If `summary.overall_status` of `quantum_measurement_dynamic_collapse_stability_audit.json` is reject (or `summary.status_counts.reject` is > 0), reject the claim of test stability.
- In the same stability summary, if `pointer_consensus_stats.min` is < 0.90 , `branch_reversal_stats.max` is > 0.20 , or `tau50_stats_s.cv` is > 0.35 , the description of branch stability after collapse is rejected.

Item	Contents
quantum closeout boundary audit	<code>overall_closeout_status=completed_ scope_fixed</code> , blocking item number = 0, <code>Born=p_specific_derivation_closed_ common_probability_residual, measurement=effective_derivation_ closed_statmech_residual, entanglement=effective_audit_closed_ selection_watch_retained</code>
Part IV Synchronization Policy	<code>part4_quantum_sync_status=pass, legacy route=A_continue/A_stay, shared_gate_policy=watch_if_bell_ pairing_only</code> , <code>demoted_to_watch=true</code> , <code>selection_watch_policy=nonblocking_ watch_retained</code> , residual that does not hinder <code>progress=single_event_probability_ status=residual_not_p_specific + irreversibility_status=residual_not_p_ specific</code> , quantum information=minimal_ <code>connection_entry_fixed</code> , not covered by protocol mathematics=standard_hilbert_ <code>math_scope_out</code>

output/public/quantum/quantum_part4_closeout_sync_pack.json

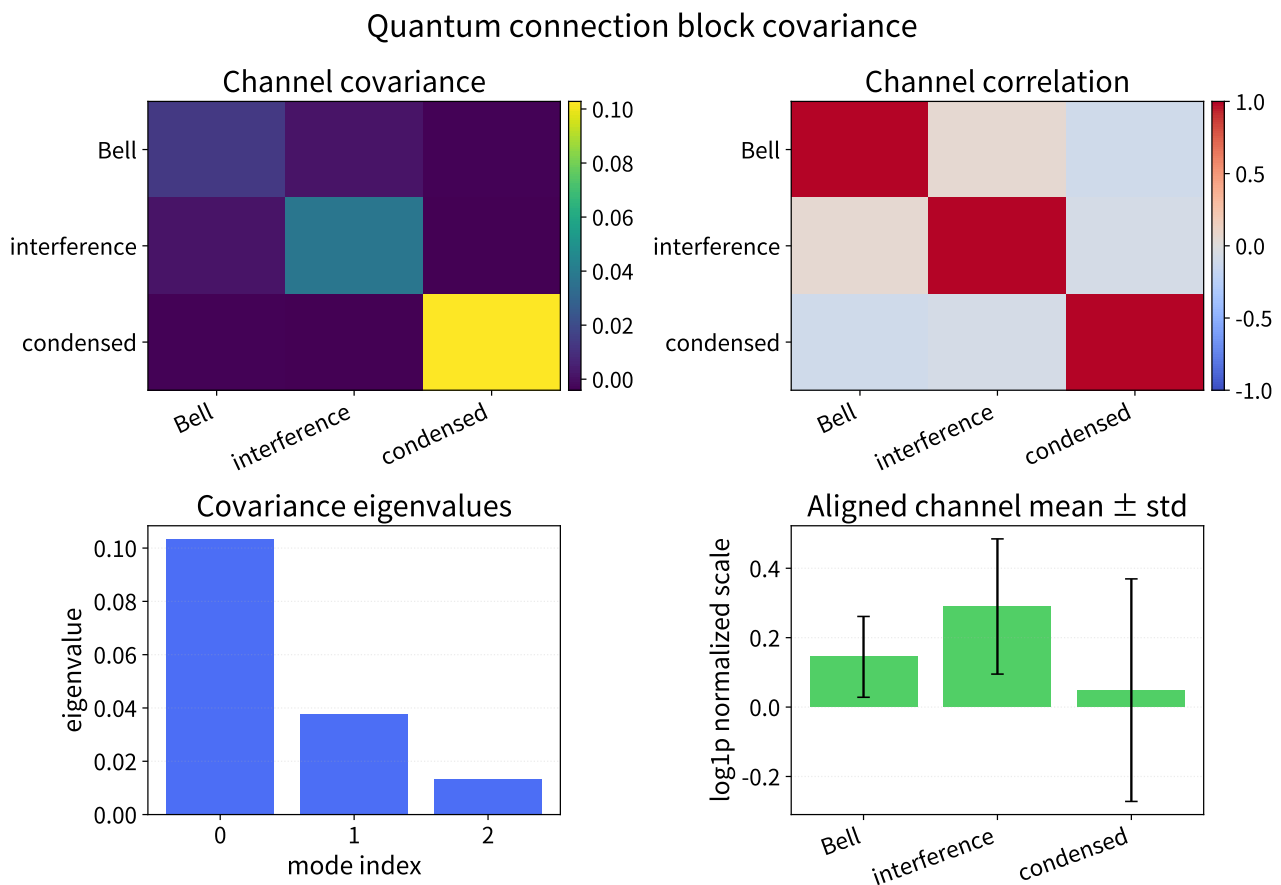


Figure 54: Comparison result for quantum connection block covariance.

The observable-side addition of the P-model-specific fine-structure constant, together with the origin, charge quantization, and charge normalization of local U(1), remains a theoretical residual discussed in Part III-A Section 3.2 and is not included in this Part IV synchronization pack.

Falsification conditions:

- If `decision.part4_quantum_sync_status` of `quantum_part4_closeout_sync_pack.json` is not pass, reject the quantum closeout synchronization in Part IV.
- If `decision.selection_watch_policy` of the same pack is not `nonblocking_watch_retained` or `decision.nonblocking_residual` is not `residual_not_p_specific`, the delineation of Bell selection watch / measurement residual is rejected.
- If `decision.v11_plus_scope_boundary_fixed` of `quantum_v11_plus_scope_closeout_audit_metrics.json` is not true or `decision.remaining_blocking_items` is non-empty, reject the quantum closeout boundary claim.

Reproduction entry points (scripts/GitHub):

- [scripts/quantum/bell_primary_products.py](#)
- `scripts/quantum/born_route_a_proxy_constraints.py`
- `scripts/quantum/action_principle_el_derivation_audit.py`

- `scripts/quantum/nonrelativistic_reduction_schrodinger_mapping_audit.py`
- `scripts/quantum/derivation_parameter_falsification_pack.py`
- `scripts/quantum/derivation_observable_chain_lock_audit.py`
- `scripts/quantum/lagrangian_noether_observable_closure_audit.py`
- `scripts/quantum/lagrangian_noether_observable_closure_drift_audit.py`
- `scripts/quantum/lagrangian_noether_closure_operational_cycle.py`
- `scripts/quantum/lagrangian_noether_rotational_closure_audit.py`
- `scripts/quantum/quantum_v11_plus_scope_closeout_audit.py`
- `scripts/quantum/quantum_part4_closeout_sync_pack.py`
- `scripts/quantum/v2_derivation_gap_figures.py`
- [scripts/quantum/quantum_connection_shared_kpi.py](#)
- [scripts/quantum/quantum_connection_bridge_table.py](#)
- [scripts/quantum/quantum_connection_born_ab_gate.py](#)
- [scripts/quantum/quantum_connection_block_covariance.py](#)
- `scripts/quantum/quantum_measurement_dynamic_collapse_simulation.py`
- `scripts/quantum/gravity_induced_decoherence.py`

11 Quantum (Nuclear): Z-N Residual Map and Differential Predictions (Constrained by Independent Observables)

Purpose: We investigated how the residual of binding energy (B.E.) changes systematically in the vicinity of the magic number and in the deformed core region. Visualize it as a “residual surface” on the Z–N plane, and further bind it with independent quantities such as separation energy.

Public artifacts (GitHub):

- Directory: [output/public/quantum/](#)
- AME2020 Z–N residual map (binding energy) for all nuclides: [nuclear_binding_energy_frequency_mapping_ame2020_all_nuclei_zn_residual_map.pdf](#)
- Differential prediction table (multiple independent cross-checks): [nuclear_binding_energy_frequency_mapping_differential_predictions.csv](#)
- Final nuclear summary (falsification conditions, freeze, main indicators): [nuclear_binding_energy_frequency_mapping_falsification_pack.json](#)
- Weak interaction (β decay) A/B quantitative audit (DoF same number + holdout): Reference directory: [output/public/quantum/](#) File: [weak_interaction_beta_decay_route_ab_audit.json](#) / [weak_interaction_beta_decay_route_ab_audit_summary.csv](#) / [weak_interaction_beta_decay_route_ab_audit.png](#)
- Weak interaction (β decay) A/B equalization summary: File: [weak_interaction_beta_decay_route_ab_equalized_summary.csv](#) / [weak_interaction_beta_decay_route_ab_equalized_holdout.png](#) (Currently: A/B both $k = 2$, holdout split `hash mod 5`, A is hard pass, B is hard reject + overfit guard fail)
- BBN main gate: File: [bbn_multi_isotope_gate_metrics.json](#) / [bbn_multi_isotope_gate_summary.csv](#) / [bbn_multi_isotope_gate.png](#)
- BBN watch convergence audit: File: [bbn_he4_watch_convergence_audit_metrics.json](#) / [bbn_he4_watch_convergence_audit_summary.csv](#) / [bbn_he4_watch_convergence_audit.png](#)
- Weak interaction (CKM primary input) update monitoring pack: File: [weak_interaction_beta_decay_route_ab_audit.json](#) (diagnostics.ckm_primary_update_watchpack) (Currently: `update_event_type=no_change`, `event_counter=0`, `next_action=wait_for_ckm_primary_input_hash_change_then_rerun_step_8_7_22`) `next_action` here is recorded as a re-execution candidate, and instead of being automatically executed when a change in the input hash is detected, it is explicitly re-executed when necessary.
- Weak interaction (β decay) Route-A watch outlier audit: File: [weak_interaction_beta_decay_route_a_watch_outliers.json](#) / [weak_interaction_beta_decay_route_a_watch_outliers.csv](#) / [weak_interaction_beta_decay_route_a_watch_outliers.png](#) (Currently: 4 cases, concentrated on A=249 chain. `review_action=relabel_route_a_mode_consistent_keep_watch` is 4 cases in transition definition re-audit, `cause_tag` is `high_precision_small_offset_watch=2` and `mode_or_definition_watch=2`)
- Weak interaction (CKM primary line) audit: File: [weak_interaction_ckm_first_row_audit.json](#) / [weak_interaction_ckm_first_row_audit_summary.csv](#) / [weak_interaction_ckm_first_row_audit.png](#) (Currently: $\Delta_{\text{CKM}} = -1.6 \times 10^{-3}$, $|z| = 2.29$, hard pass / watch)

- Weak interaction (PMNS primary line) audit: File: weak_interaction_pmns_first_row_audit.json / weak_interaction_pmns_first_row_audit_summary.csv / weak_interaction_pmns_first_row_audit.png (Currently: $\Delta_{\text{PMNS}} = -7.07 \times 10^{-4}$, $|z| = 0.043$, hard pass / watch pass)

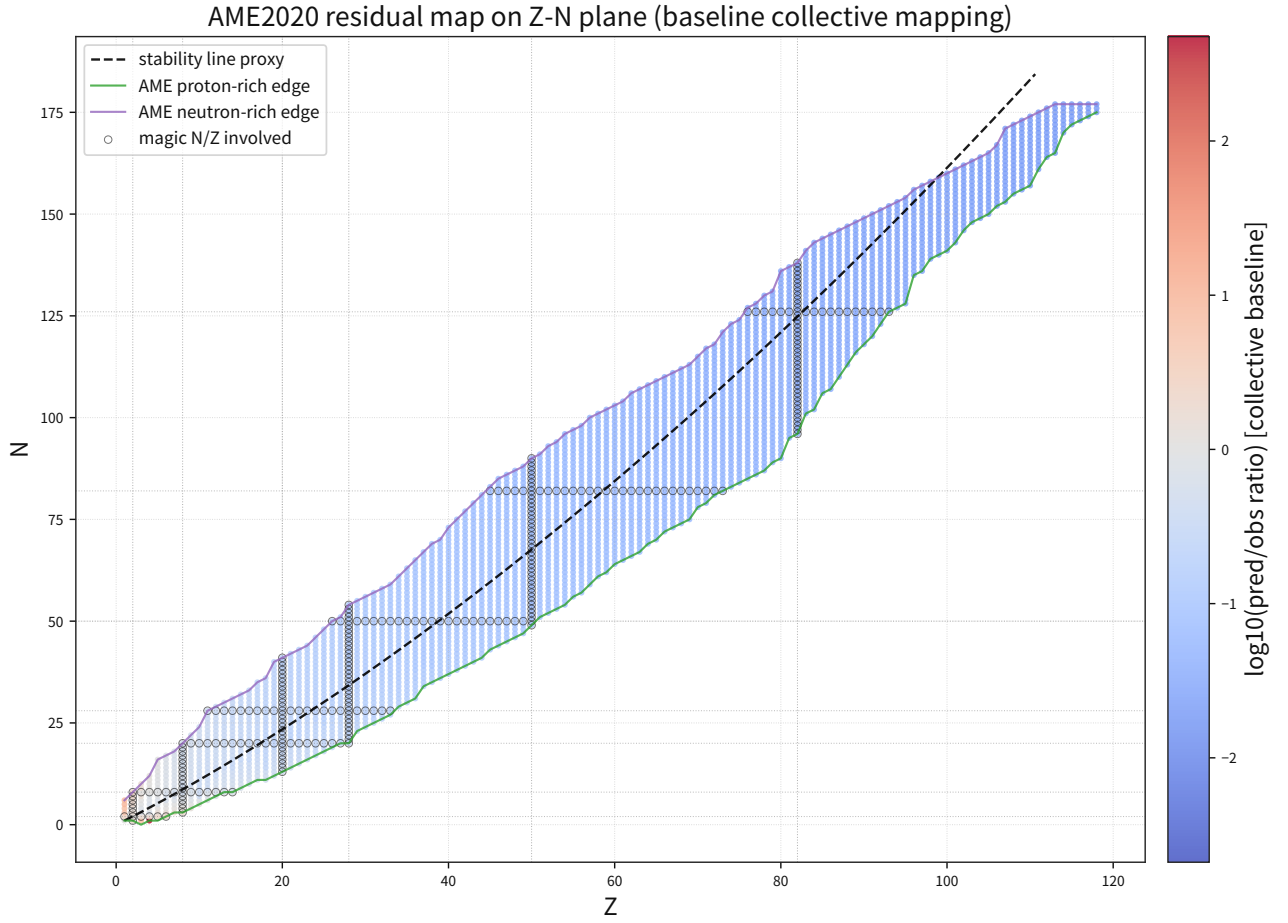


Figure 55: AME2020 Z-N residual map of all nuclei

differential predictions (distance proxy audit)

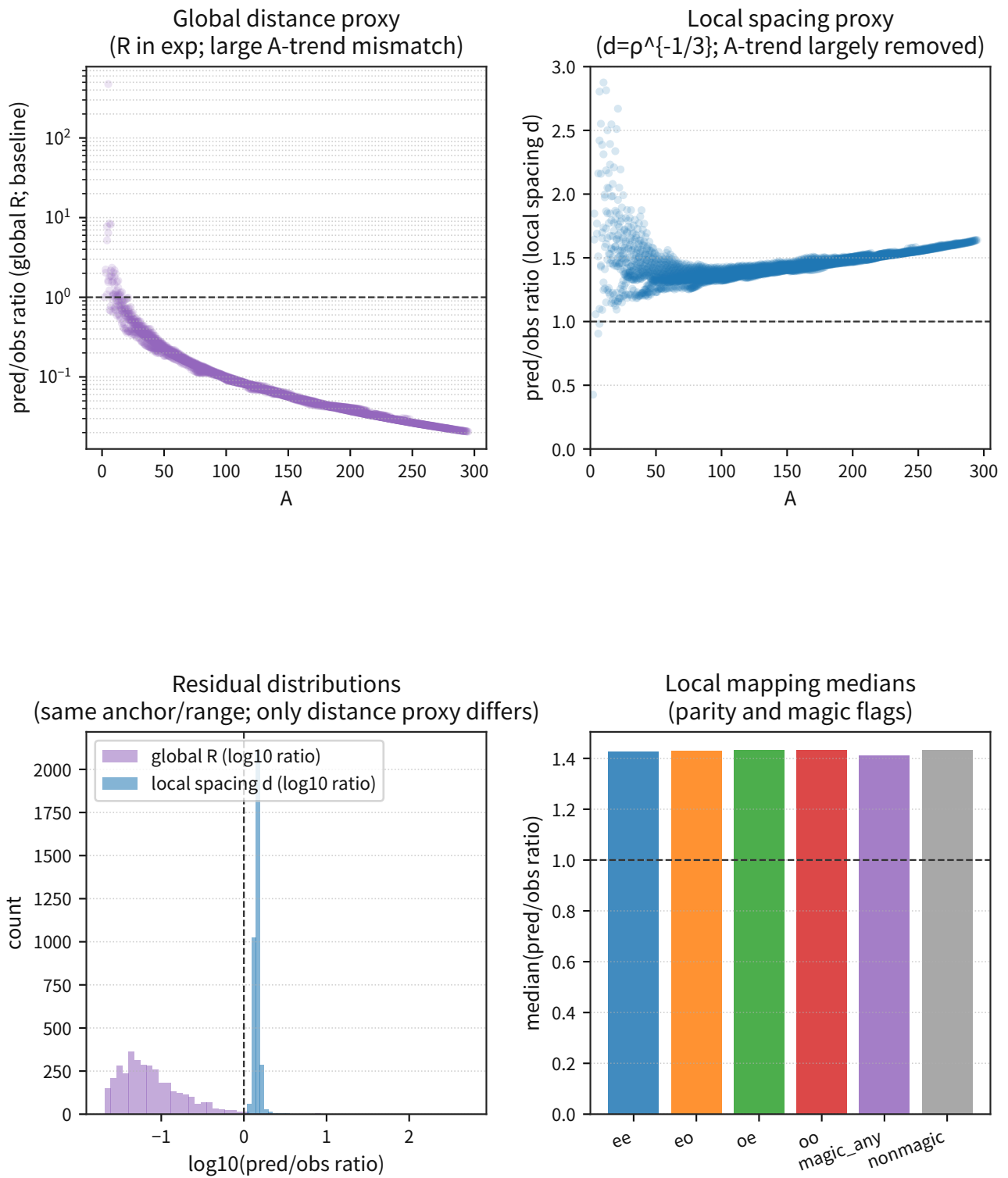


Figure 56: Nuclear differential prediction audit

falsification pack and independent cross-check

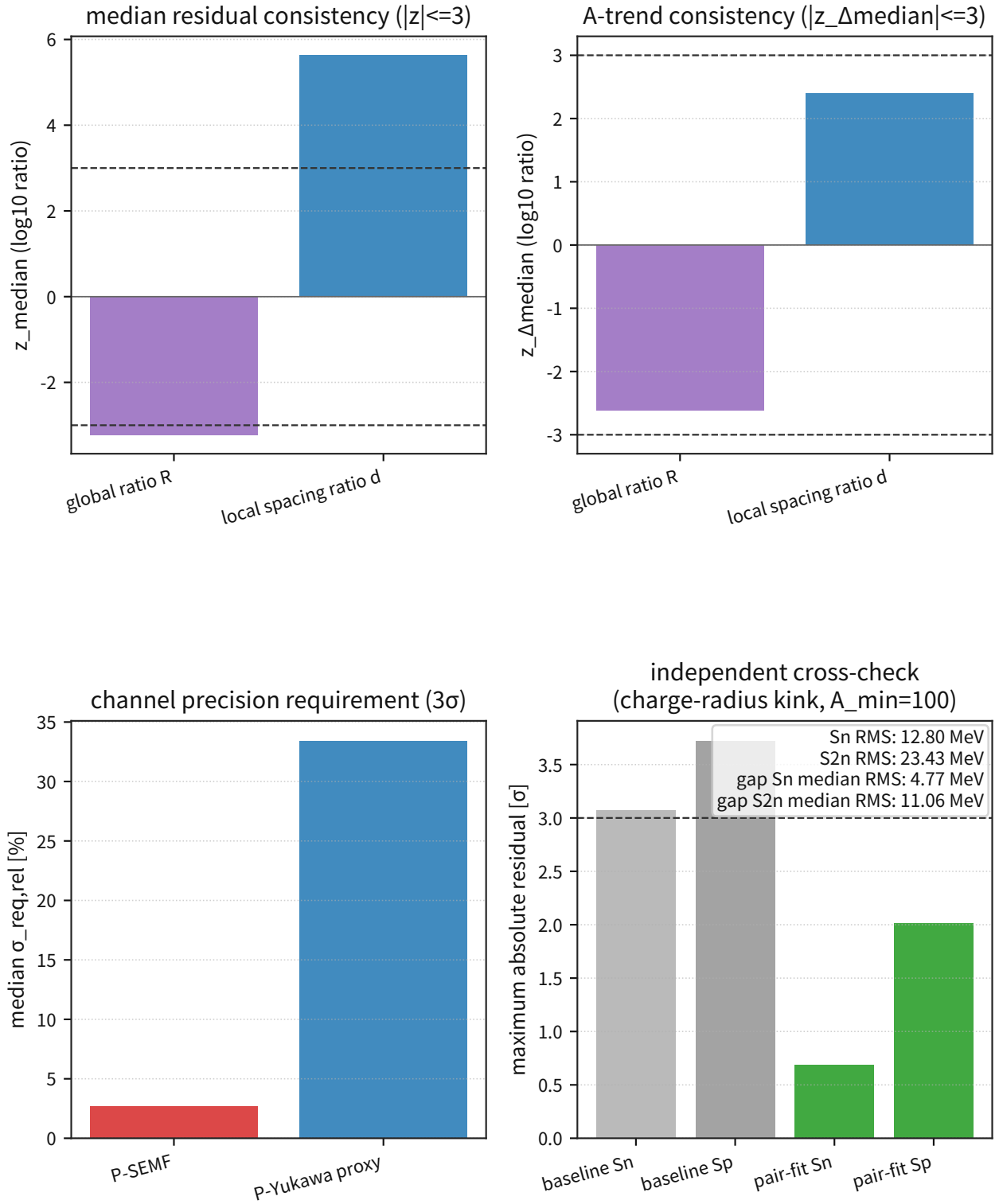


Figure 57: Nuclear falsification condition pack summary

Reproduction entry points (scripts/GitHub):

- [scripts/quantum/nuclear_binding_energy_frequency_mapping_ame2020_all_nuclei.py](#)
- [scripts/quantum/nuclear_binding_energy_frequency_mapping_differential_predictions.py](#)
- [scripts/quantum/weak_interaction_beta_decay_route_ab_audit.py](#)
- [scripts/quantum/weak_interaction_ckm_first_row_audit.py](#)
- [scripts/quantum/weak_interaction_pmns_first_row_audit.py](#)
- [scripts/quantum/bbn_multi_isotope_gate.py](#)
- [scripts/quantum/bbn_he4_watch_convergence_audit.py](#)

On the theoretical side, Part III-A Section 3.2 derives the two-component coupled Q-ball and retains the W/Z mass reference points on the same branch as the theoretical standard. On the observational side, this section takes `weak_interaction_beta_decay_route_ab_audit.json` as canonical; at this stage, the observational verdict is fixed by that audit as retaining Route A and rejecting Route B.

Falsification conditions (weak interaction A/B):

- As long as `decision.route_b_hard_pass=false` of `weak_interaction_beta_decay_route_ab_audit.json` continues, Part III B route (P-specific replacement) will remain rejected.
- If `decision.route_a_hard_pass` is reversed to false in the same JSON, stop continuation of A route and reexamine the weak interaction clause.
- If the same JSON is reversed to `decision.dof_equalization.equal_k=false` or `decision.overfit_gate.route_a_pass=false`, the assumption of A/B comparison (same degrees of freedom + generalization) is not satisfied, so the determination is invalidated and re-audited.
- If `decision.ckm_gate.hard_pass=false` in the same JSON, it will be rejected in the CKM primary line (weak section fail).
- If `decision.pmns_gate.hard_pass=false` in the same JSON, it will be rejected in the PMNS primary line (weak section fail).
- If `decision.ckm_pmns_closure.status=watch` (current status) in the same JSON, the observation side's adoption or rejection of the weak interaction sector will be treated as continuous monitoring.
- If `summary.by_review_action` to `keep_mode_definition_watch` or `summary.by_cause_tag` to `residual_watch` increases in Route-A outlier audit JSON, reclassify it as a model residual rather than a simple relabel, and reevaluate the watch judgment for route A.
- If `overall_status=reject` of `bbn_multi_isotope_gate_metrics.json` or main gate failure occurs, fix the BBN main gate as rejected.
- If the watch convergence condition breaks in `bbn_he4_watch_convergence_audit_metrics.json`, the BBN operation judgment will be returned to Inconclusive and re-audited.

12 Condensed Matter: Baselines and Falsification Conditions (Hold-out Audit)

Purpose: The baseline (reference model) of the condensed system is fixed, and the rejection conditions are clarified by data division (holdout). The detailed ansatz trial log will be saved in the appendix (Part III Appendix), and the main text will be placed in the minimum set (baseline + rejection conditions).

Public artifacts (GitHub):

- Directory: [output/public/quantum/](#) (condensed_ prefix)
- Final summary of physical properties (falsification conditions, freezing, main indicators): [condensed_falsification_pack.json](#)
- holdout audit (summary of splits and rejections): [condensed_holdout_audit_summary.json](#) / Figure (Summary): [condensed_holdout_audit.pdf](#)
- Fixed KPI judgment rule: the $Si\rho(T)$ coefficient is evaluated in holdout with $\sigma_{\text{proxy}} = \max(\sigma_{\text{train}}, \sigma_{\text{floor}}^{\text{global}})$. The Debye+Einstein basis for $Si\alpha(T)$ is excluded from KPI aggregation because it is retained only for diagnosis (stress test); the KPI-side judgment is instead based on the $\text{DOS}+\gamma(\omega)$ basis.

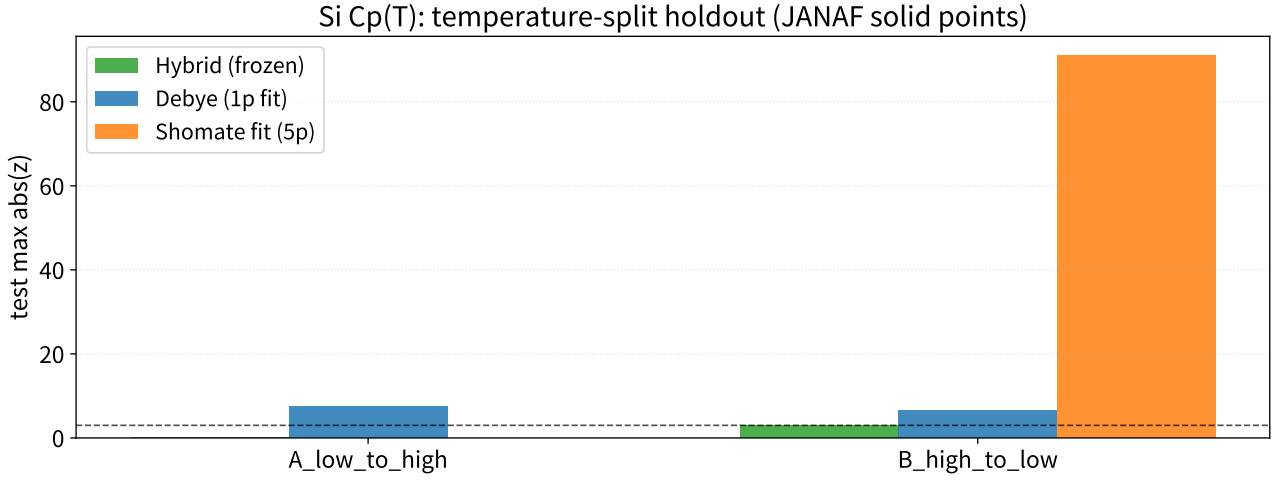


Figure 58: Si specific heat holdout split audit

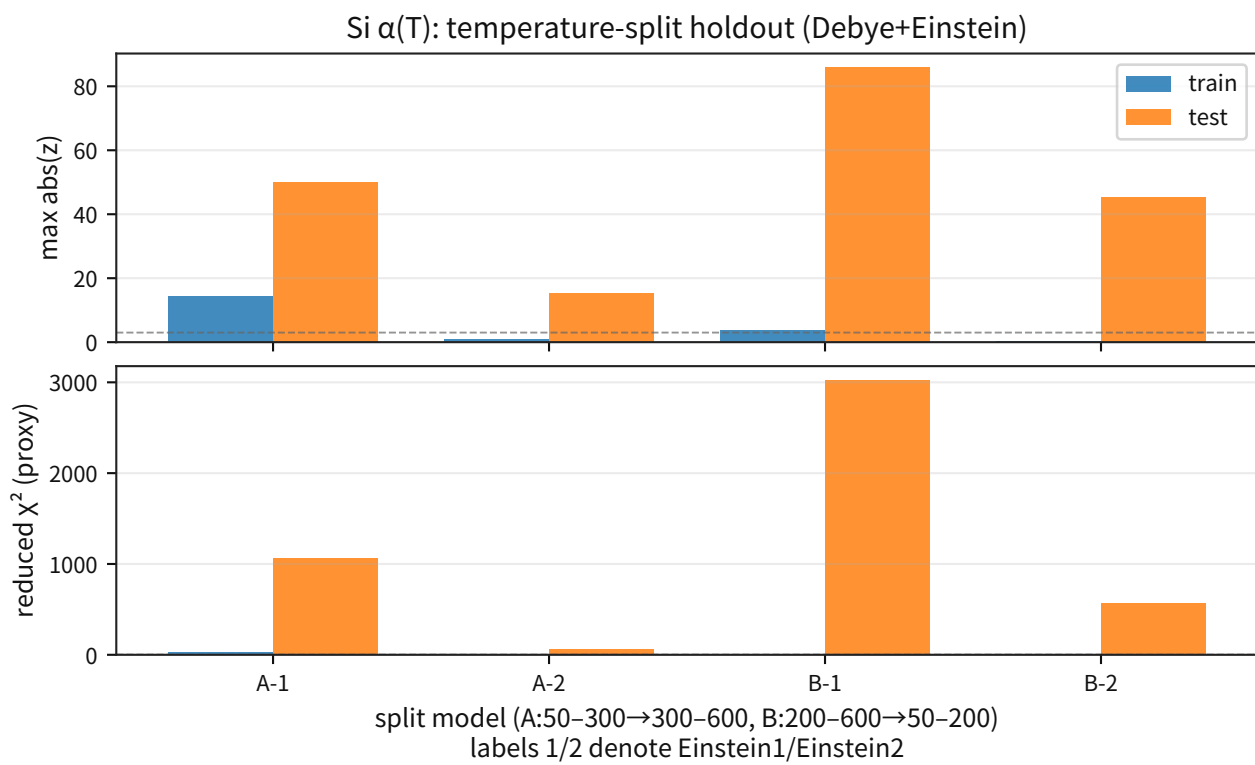


Figure 59: Si thermal expansion holdout split audit

Reproduction entry points (scripts/GitHub):

- [scripts/quantum/condensed_falsification_pack.py](#)
- [scripts/quantum/condensed_holdout_audit.py](#)

13 Thermal (Blackbody): Baselines and Minimal Audit

Purpose: Fix the blackbody standard equation (baseline) and check whether the mapping on the P-model side is consistent with the observational number at a minimum.

Public artifacts (GitHub):

- Directory: [output/public/quantum/](#) (`thermo_` prefix)
- Radiation baseline: [thermo_blackbody_radiation_baseline.csv](#) / Figure: [thermo_blackbody_radiation_baseline.pdf](#)
- Entropy baseline: [thermo_blackbody_entropy_baseline.csv](#) / Figure: [thermo_blackbody_entropy_baseline.pdf](#)
- Holdout audit (4 representative figures in text): [output/public/quantum/thermo_blackbody_momentum_density_holdout_splits_metrics.json](#) / Figure: [output/public/quantum/locales/en/thermo_blackbody_momentum_density_holdout_splits.pdf](#) [output/public/quantum/thermo_blackbody_helmholtz_free_energy_density_holdout_splits_metrics.json](#) / Figure: [output/public/quantum/locales/en/thermo_blackbody_helmholtz_free_energy_density_holdout_splits.pdf](#) [output/public/quantum/thermo_blackbody_peak_frequency_wavelength_product_holdout_splits_metrics.json](#) / Figure: [output/public/quantum/locales/en/thermo_blackbody_peak_frequency_wavelength_product_holdout_splits.pdf](#) [output/public/quantum/thermo_blackbody_photon_flux_holdout_splits_metrics.json](#) / Figure: [output/public/quantum/locales/en/thermo_blackbody_photon_flux_holdout_splits.pdf](#)

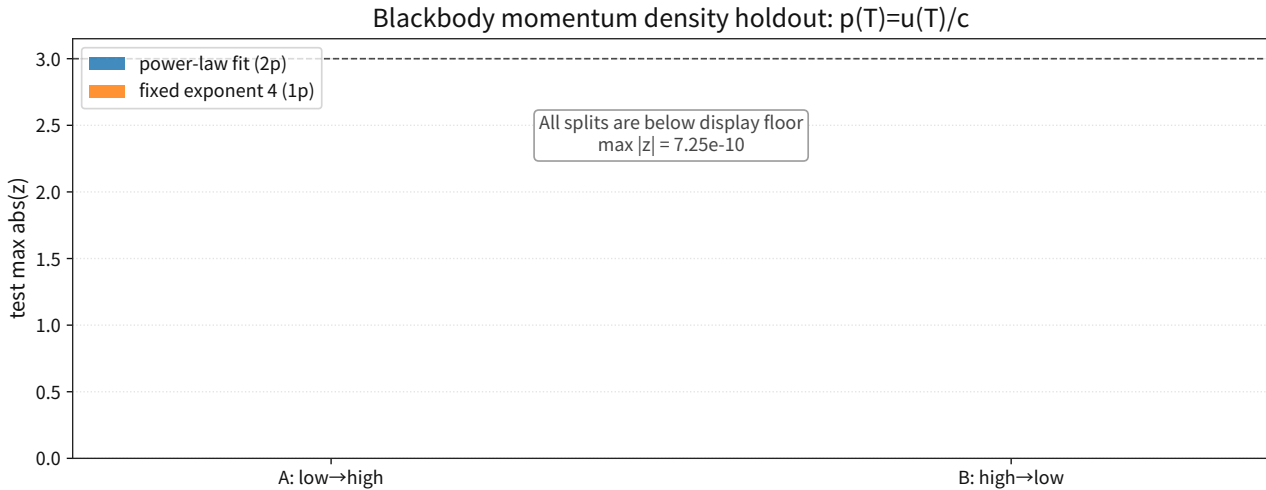


Figure 60: Blackbody momentum-density holdout audit

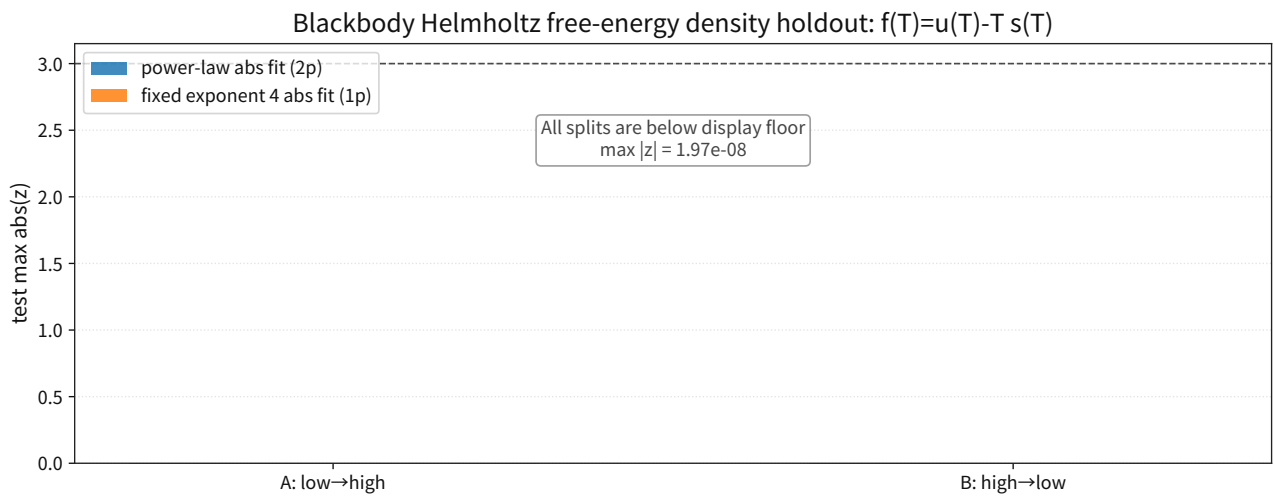


Figure 61: Blackbody Helmholtz free-energy-density holdout audit

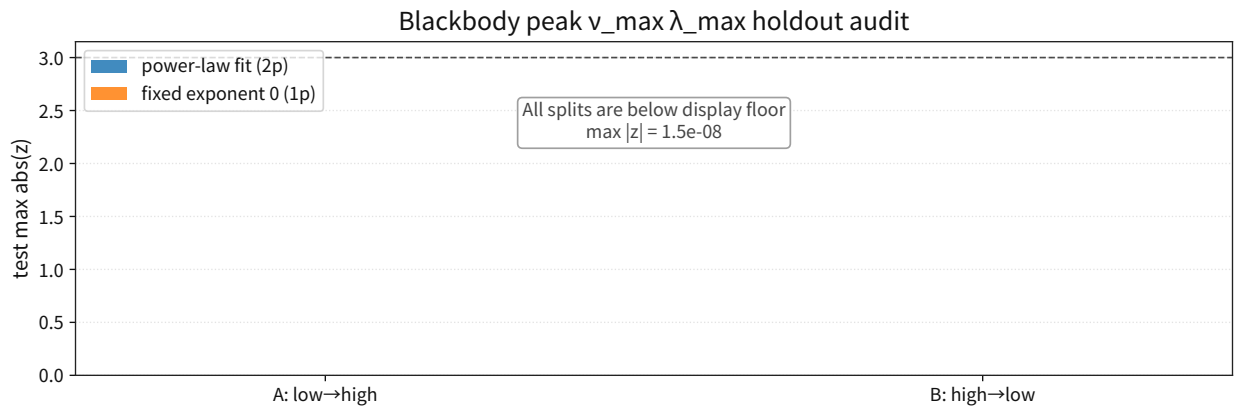


Figure 62: Blackbody peak-frequency-wavelength-product holdout audit

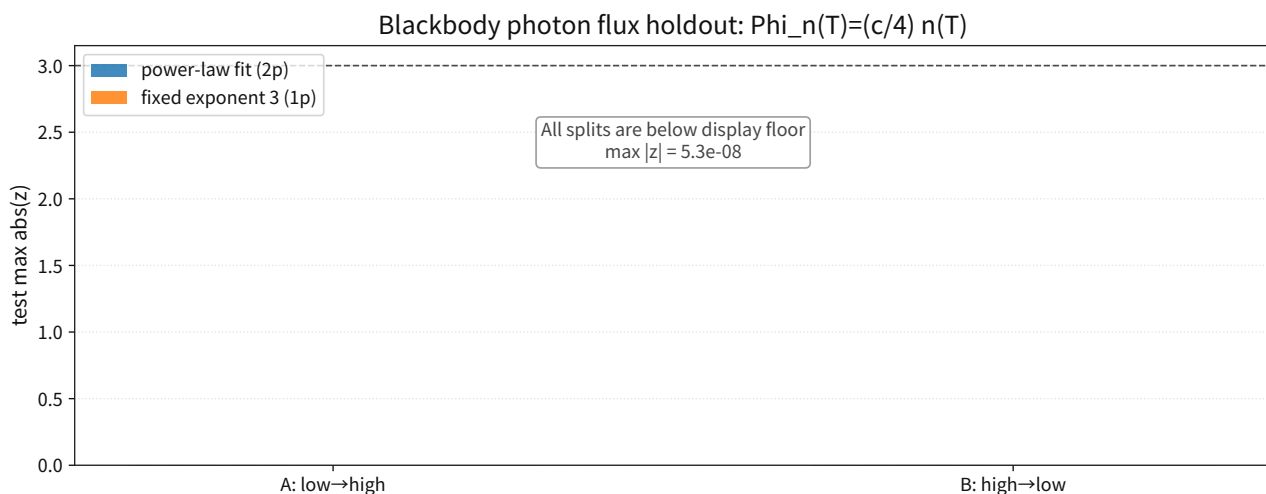


Figure 63: Blackbody photon-flux holdout audit

Primary gate: A holdout audit was conducted for 78 quantities in total, including the four representative figures above. The integrated result is summarized in [condensed_holdout_audit_summary.json](#), and the main holdout gate passes in the current snapshot. The main text retains only four representative figures covering the fundamental quantity, thermodynamic potential, observed quantity, and radiation quantity; the remaining individual figures and indicator lists are collected in Appendix A, "Complete List of Blackbody Holdout Audits." The condensed-system audit ledger also includes the blackbody baseline and blackbody holdout audits, so the canonical integrated ledgers remain [condensed_holdout_audit_summary.json](#) and [condensed_falsification_pack.json](#).

Reproduction entry points (scripts/GitHub):

- [scripts/quantum/thermo_blackbody_radiation_baseline.py](#)
- [scripts/quantum/thermo_blackbody_entropy_baseline.py](#)
- For a complete list of holdout extension scripts, see Appendix A, "Complete List of Blackbody Holdout Audits."

14 Toward Part V and Update Operations

Role: Part V is the canonical roadmap for future observations, including the ordering of priorities and the timing of possible conclusion updates. Part IV only maintains the monitoring packs, artifacts, and rerun entry points needed to reproduce those future updates.

Delegated to:

- Future differential predictions and observation priorities for strong fields, gravitational waves, and cosmology are based on Part V.
- The future gate of quantum, nuclear, and gravitational quantum also uses the prediction list and timeline in Part V as the original.
- Part IV maintains the JSON/CSV/PDF and re-execution entry corresponding to those predictions as a ledger.

Fixed operations:

- For future observation re-evaluation, the input hash update of **watchpack** is recorded in a machine-readable manner along with `update_event_type` / `update_event_detected` / `event_counter` as re-execution candidates, and is explicitly re-executed using the same I/F only when necessary.
- This section does not reproduce future predictions themselves. See Part V for settlement conditions and priorities for current readers.
- The judgment in this section does not overwrite the current main text result, and connects to the same freezing value and same gate name as the prediction section in Part V.

15 Appendix A: Complete List of Blackbody Holdout Audits

This appendix compiles a complete list of holdout audits conducted during the blackbody baseline audit in Chapter 12. In the main text, we will leave only the four representative figures, and here we will summarize the remaining individual figures and the holdout extension script. The canonical integrated-judgment sources are `condensed_holdout_audit_summary.json` and `condensed_falsification_pack.json`, which are collectively managed including blackbody items.

Individual deliverables (excluding 4 main figures):

- `output/public/quantum/thermo_blackbody_momentum_energy_ratio_holdout_splits_metrics.json` /

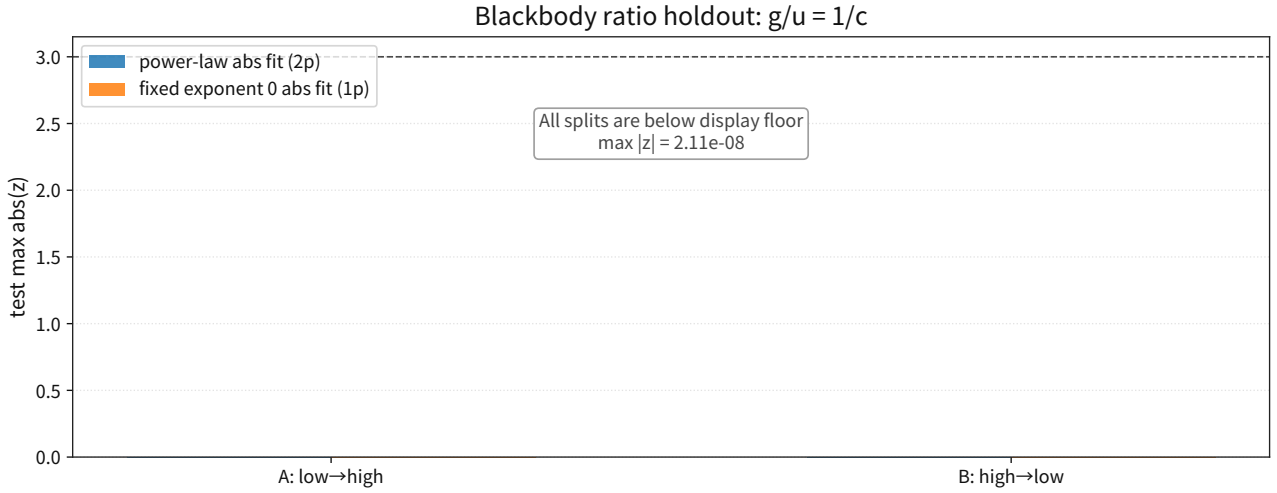


Figure 64: Comparison result for thermodynamics blackbody momentum energy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_momentum_pressure_ratio_holdout_splits_metrics.json` /

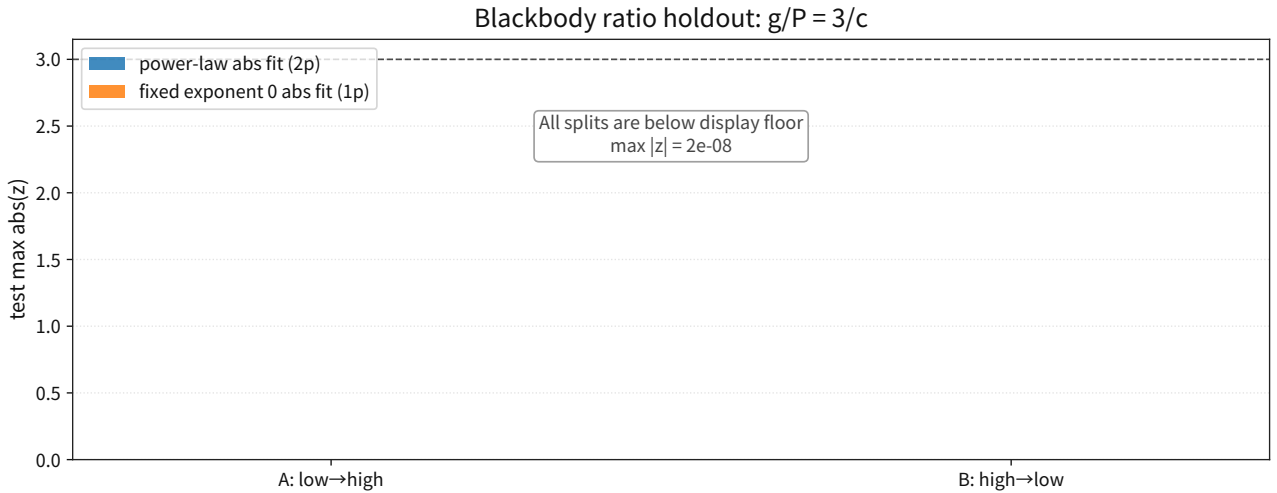


Figure 65: Comparison result for thermodynamics blackbody momentum pressure ratio holdout splits.

- output/public/quantum/thermo_blackbody_pressure_momentum_ratio_holdout_splits_metrics.json /

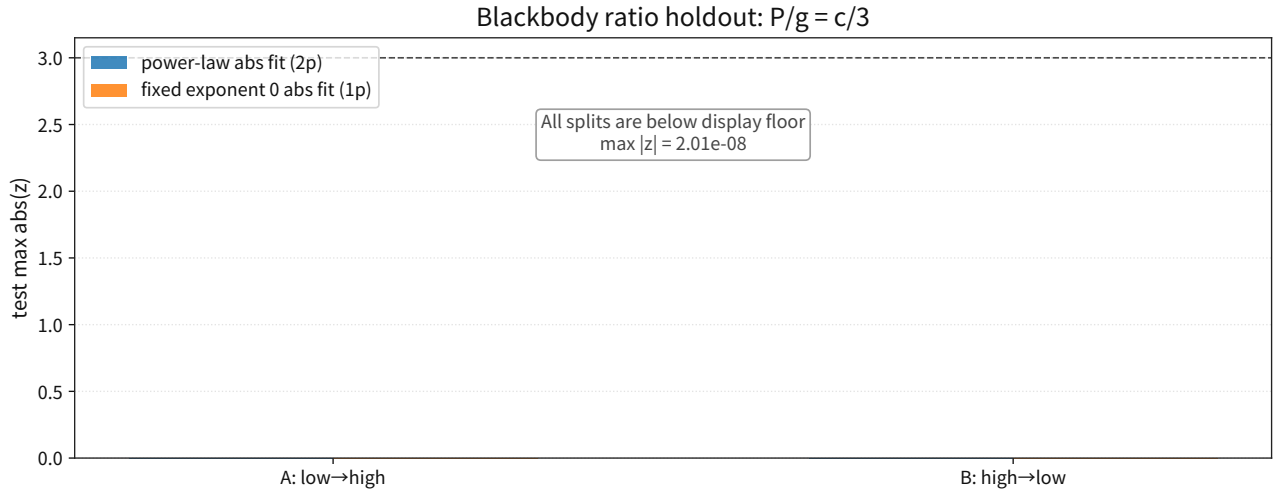


Figure 66: Comparison result for thermodynamics blackbody pressure momentum ratio holdout splits.

- output/public/quantum/thermo_blackbody_momentum_entropy_ratio_holdout_splits_metrics.json /

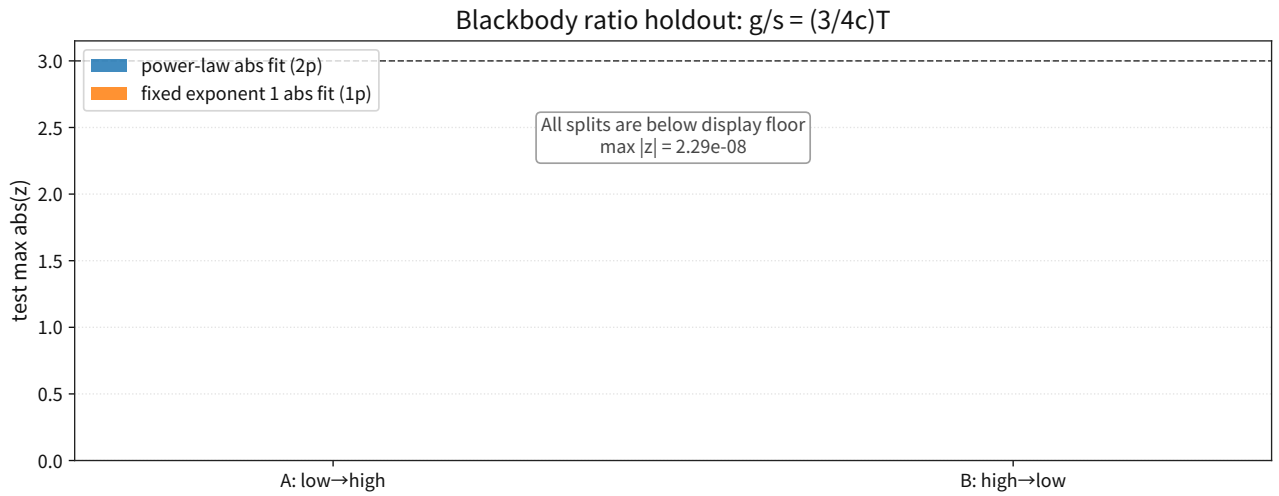


Figure 67: Comparison result for thermodynamics blackbody momentum entropy ratio holdout splits.

- output/public/quantum/thermo_blackbody_momentum_per_photon_holdout_splits_metrics.json /

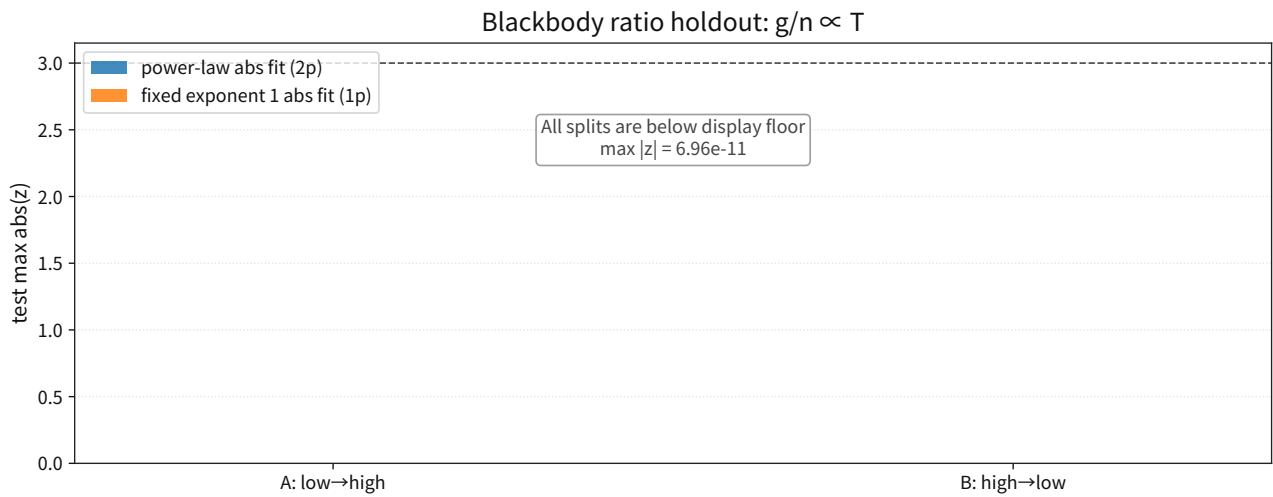


Figure 68: Comparison result for thermodynamics blackbody momentum per photon holdout splits.

- `output/public/quantum/thermo_blackbody_momentum_per_photon_flux_holdout_splits_metrics.json` /

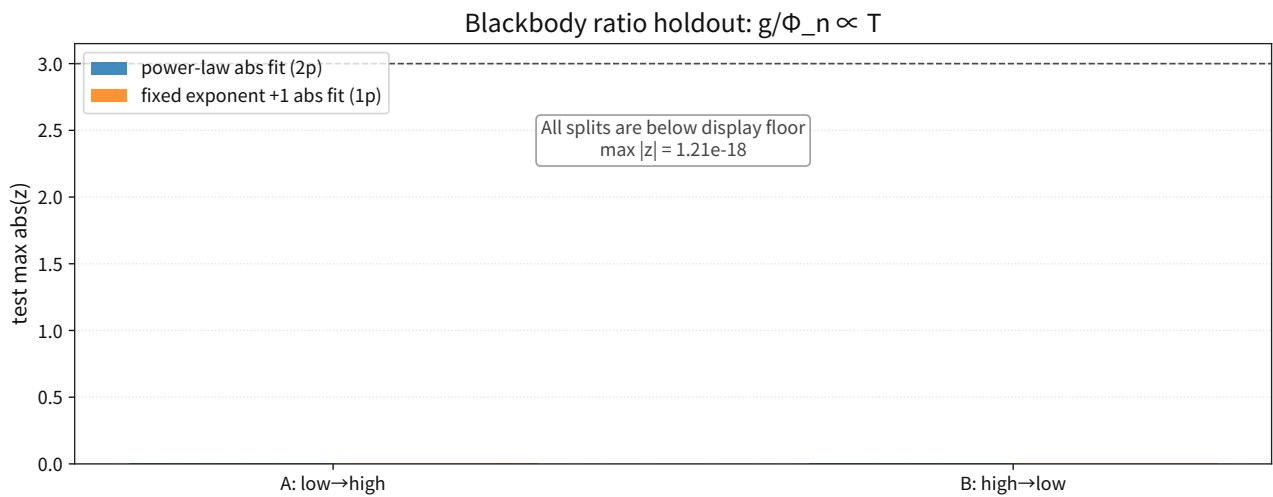


Figure 69: Comparison result for thermodynamics blackbody momentum per photon flux holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_per_photon_holdout_splits_metrics.json` /

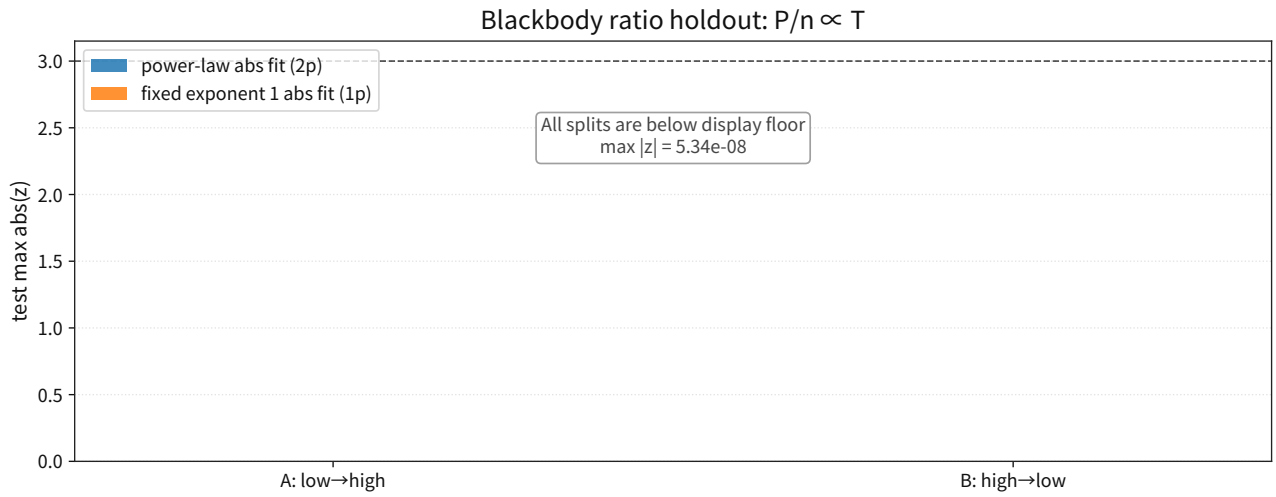


Figure 70: Comparison result for thermodynamics blackbody pressure per photon holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_pressure_holdout_splits_metrics.json` /

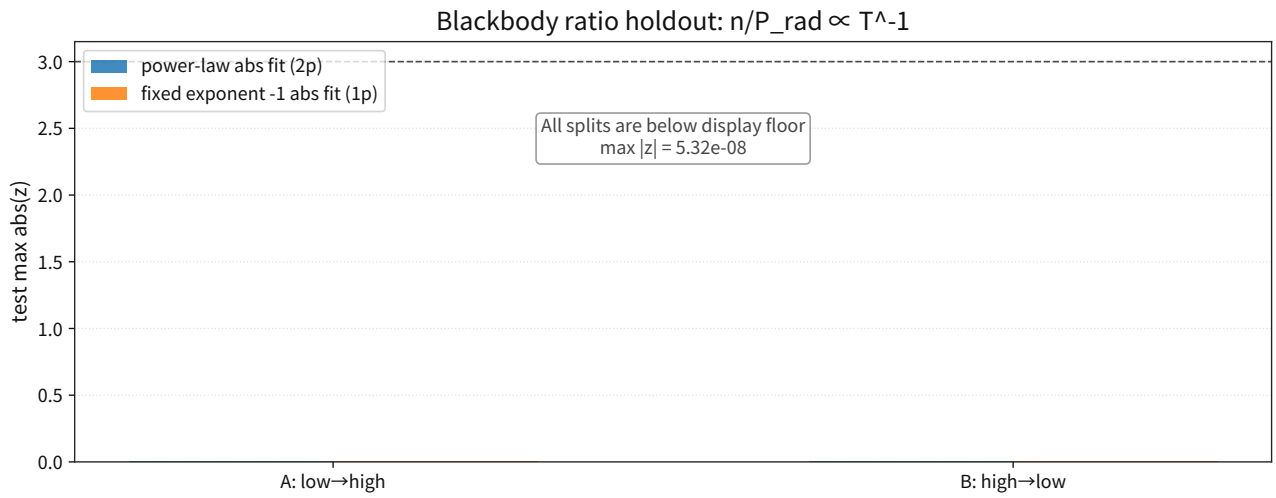


Figure 71: Comparison result for thermodynamics blackbody photon density per pressure holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_energy_density_holdout_splits_metrics.json` /

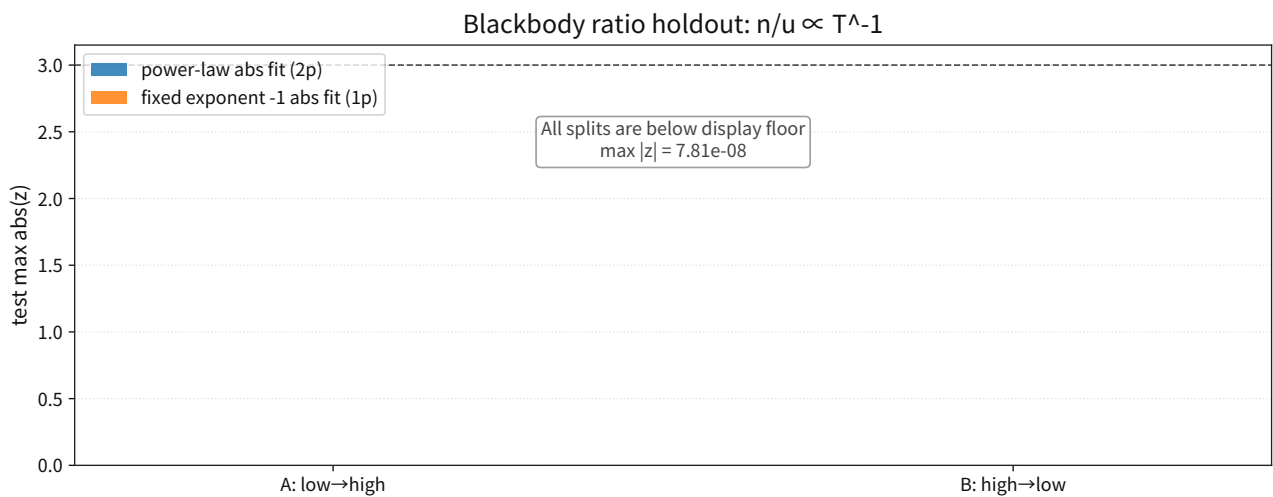


Figure 72: Comparison result for thermodynamics blackbody photon density per energy density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_enthalpy_density_holdout_splits_metrics.json /`

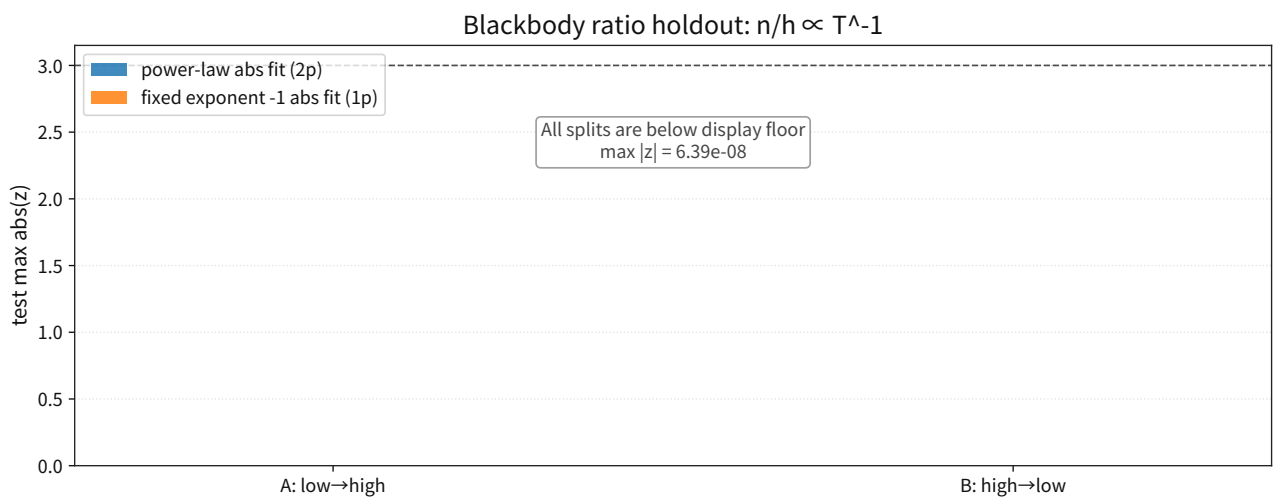


Figure 73: Comparison result for thermodynamics blackbody photon density per enthalpy density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_momentum_density_holdout_splits_metrics.json /`

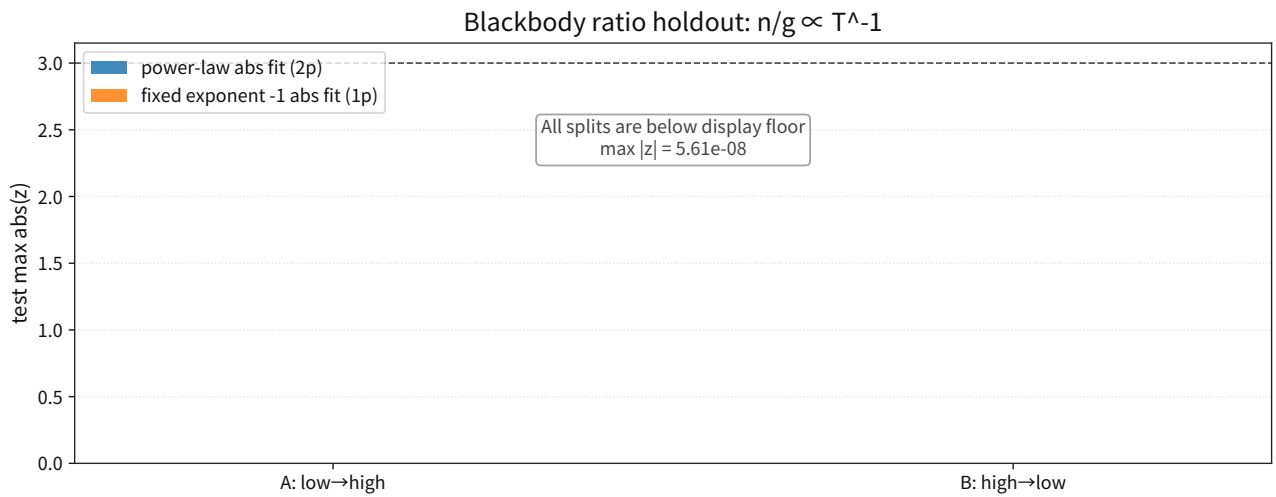


Figure 74: Comparison result for thermodynamics blackbody photon density per momentum density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_energy_density_holdout_splits_metrics.json /`

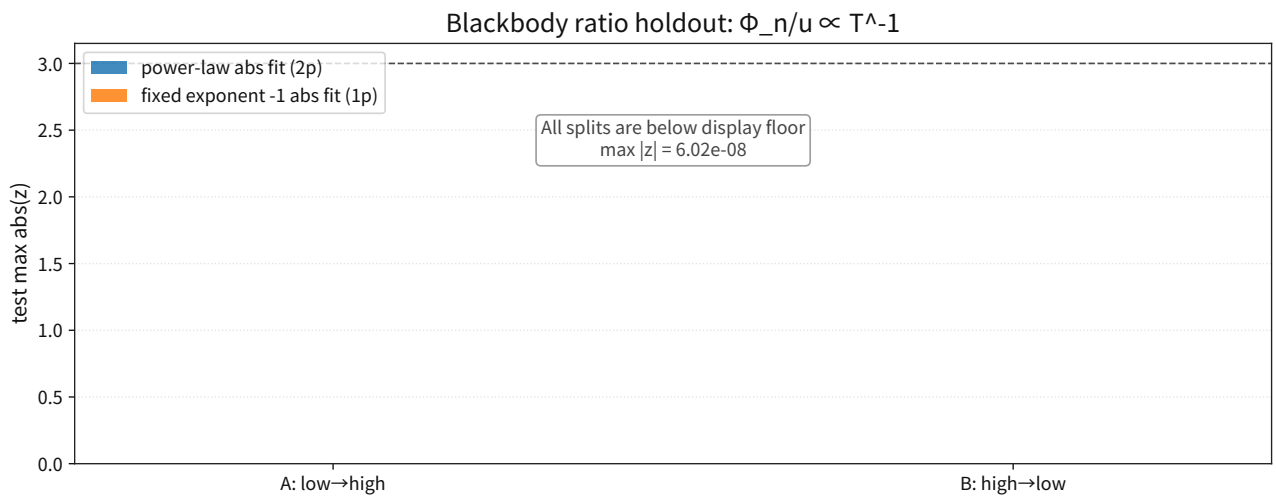


Figure 75: Comparison result for thermodynamics blackbody photon flux per energy density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_enthalpy_density_holdout_splits_metrics.json /`

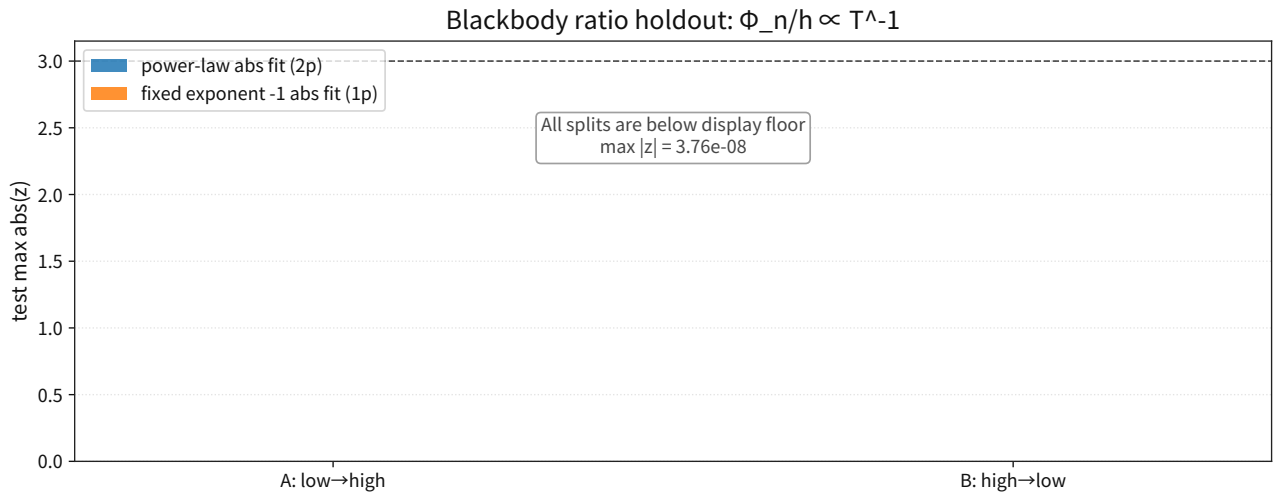


Figure 76: Comparison result for thermodynamics blackbody photon flux per enthalpy density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_pressure_holdout_splits_metrics.json /`

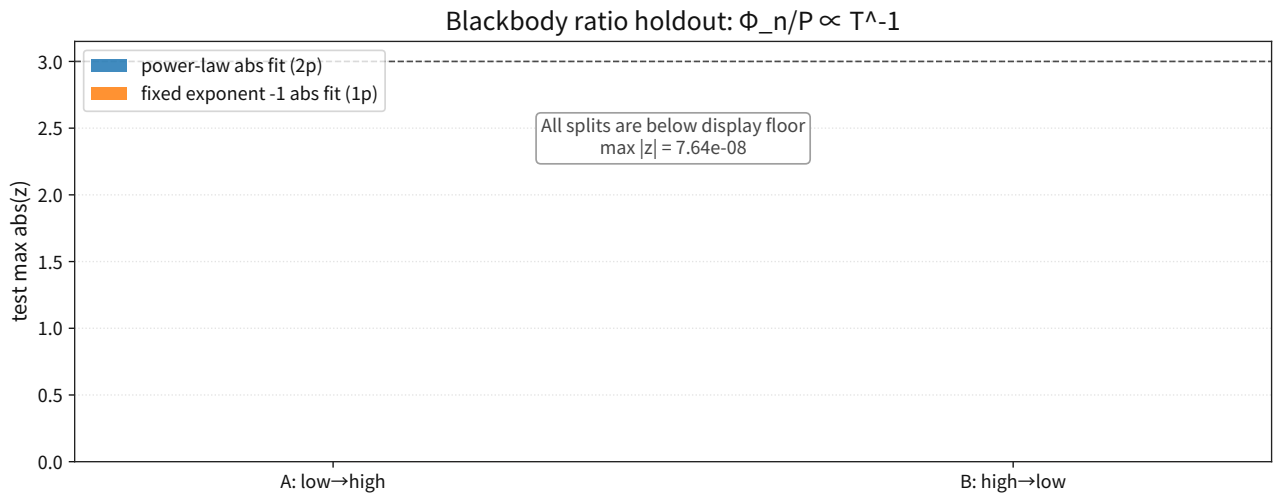


Figure 77: Comparison result for thermodynamics blackbody photon flux per pressure holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_heat_capacity_density_holdout_splits_metrics.json /`

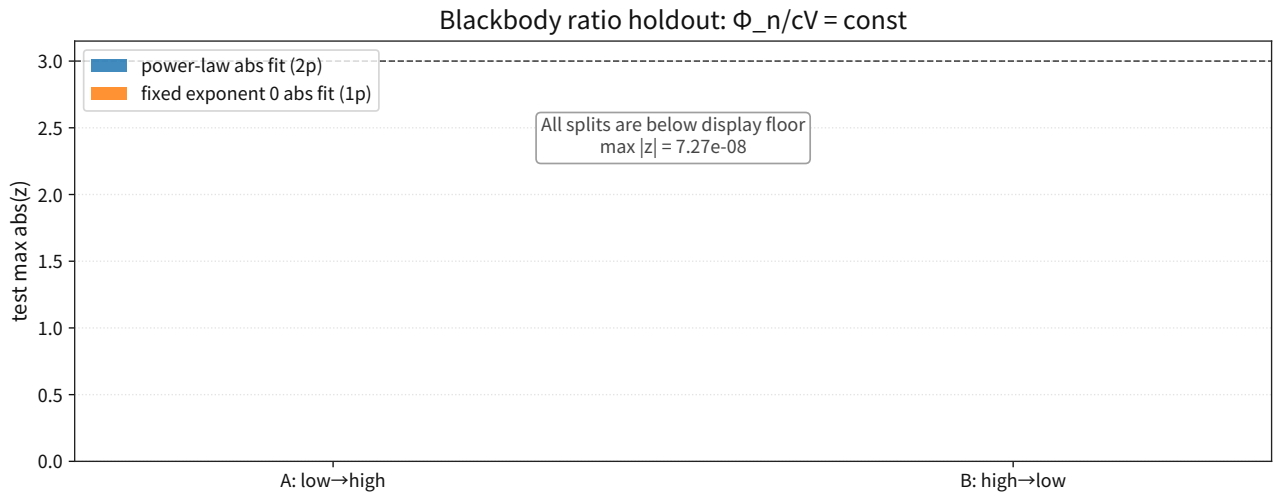


Figure 78: Comparison result for thermodynamics blackbody photon flux per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_heat_capacity_flux_holdout_splits_metrics.json` /

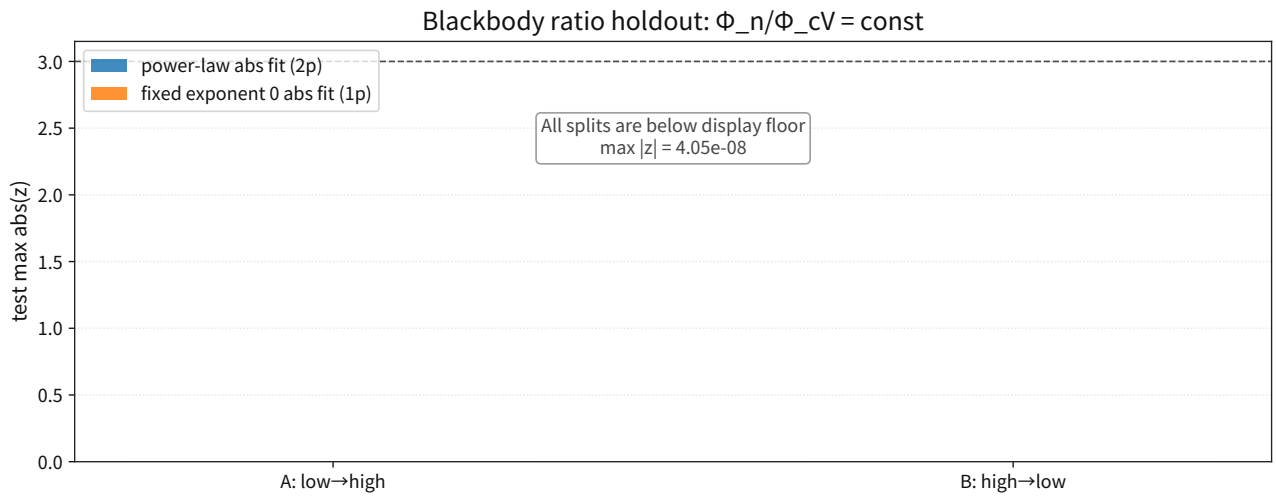


Figure 79: Comparison result for thermodynamics blackbody photon flux per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_enthalpy_flux_holdout_splits_metrics.json` /

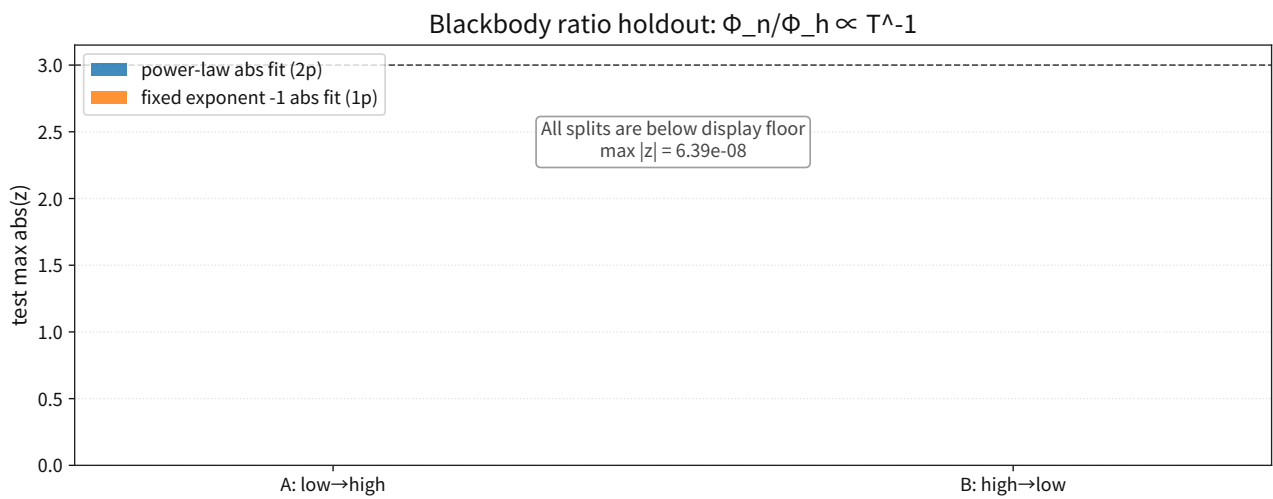


Figure 80: Comparison result for thermodynamics blackbody photon flux per enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_entropy_density_holdout_splits_metrics.json /`

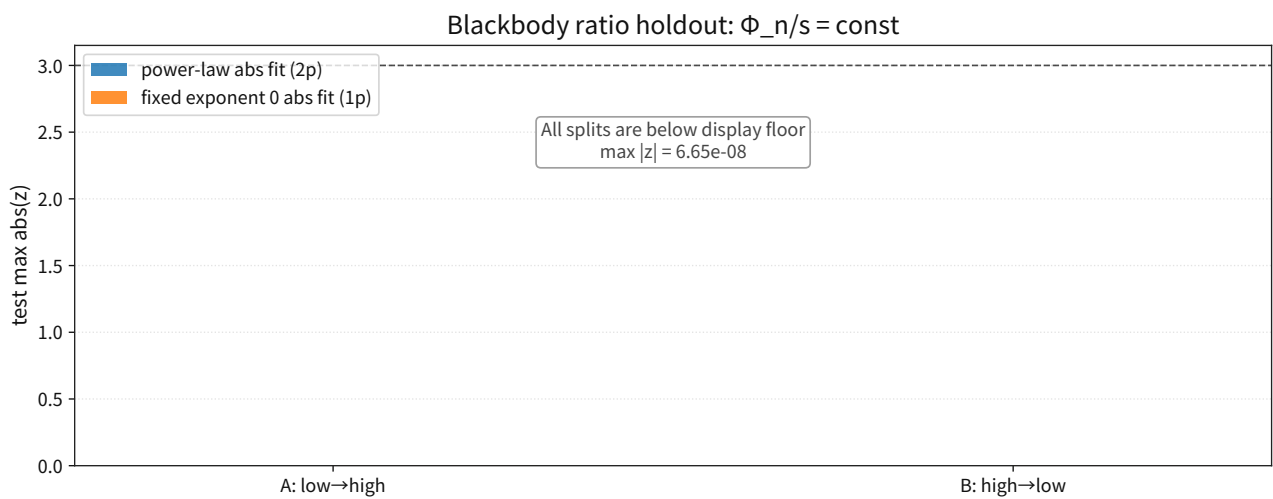


Figure 81: Comparison result for thermodynamics blackbody photon flux per entropy density holdout splits.

- `output/public/quantum/thermo_blackbody_photon_flux_per_momentum_density_holdout_splits_metrics.json /`

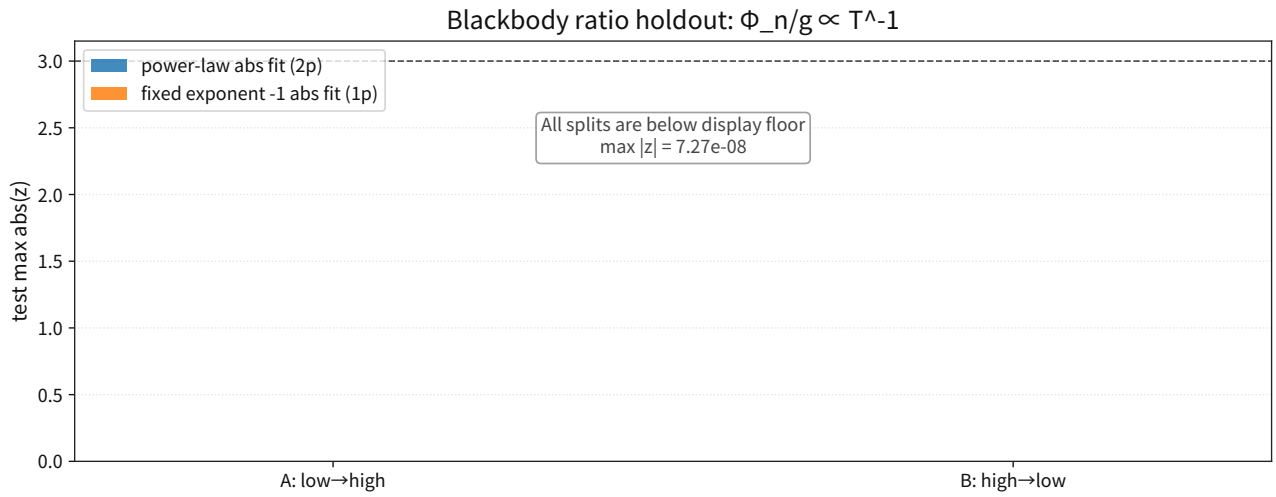


Figure 82: Comparison result for thermodynamics blackbody photon flux per momentum density holdout splits.

- output/public/quantum/thermo_blackbody_pressure_per_photon_flux_holdout_splits_metrics.json /

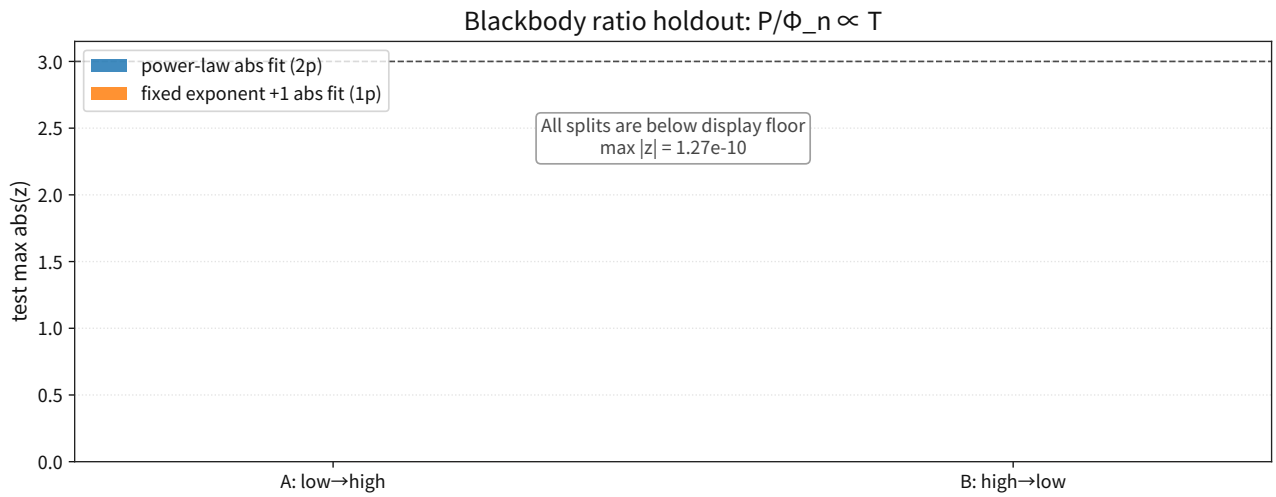


Figure 83: Comparison result for thermodynamics blackbody pressure per photon flux holdout splits.

- output/public/quantum/thermo_blackbody_pressure_per_entropy_flux_holdout_splits_metrics.json /

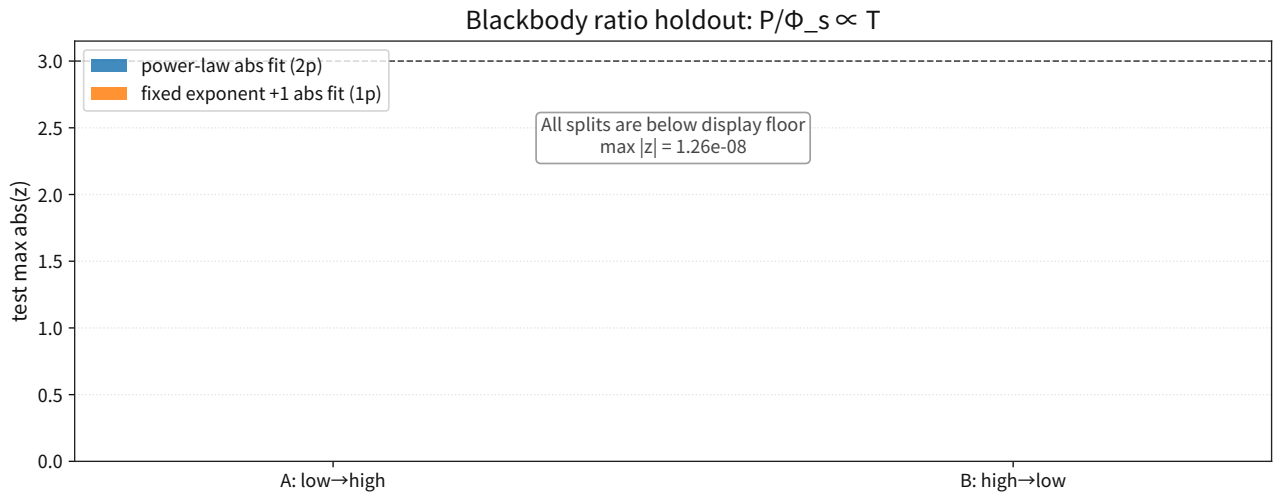


Figure 84: Comparison result for thermodynamics blackbody pressure per entropy flux holdout splits.

- `output/public/quantum/thermo_blackbody_energy_per_entropy_flux_holdout_splits_metrics.json` /

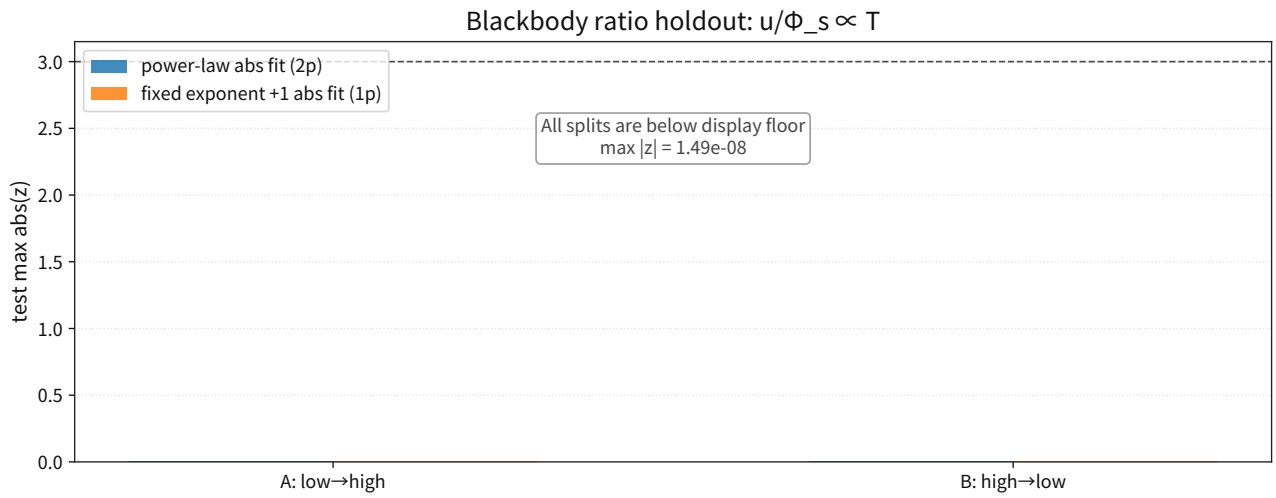


Figure 85: Comparison result for thermodynamics blackbody energy per entropy flux holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_per_entropy_flux_holdout_splits_metrics.json` /

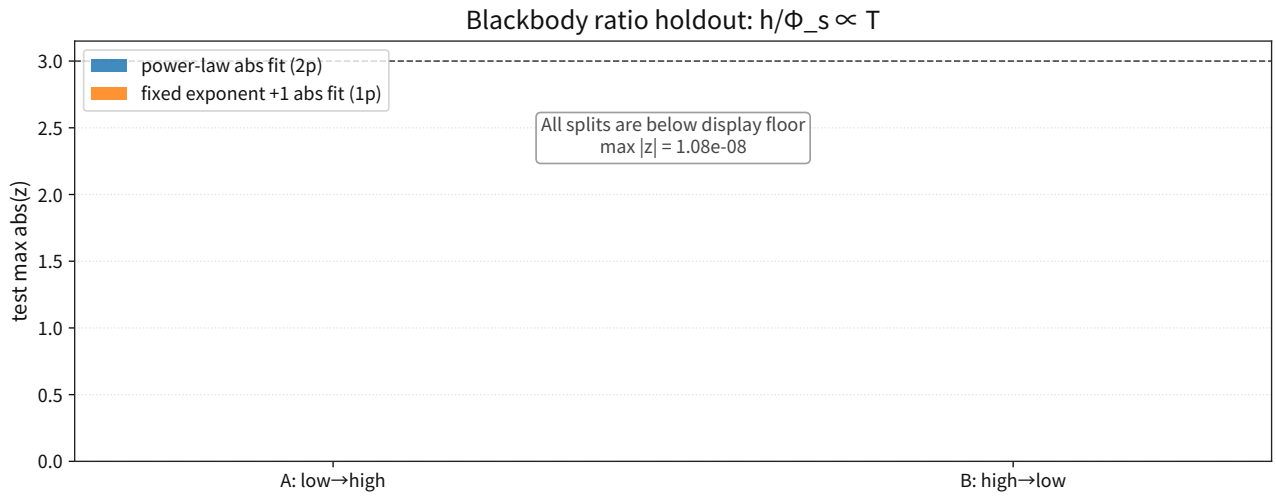


Figure 86: Comparison result for thermodynamics blackbody enthalpy per entropy flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_entropy_flux_holdout_splits_metrics.json` /

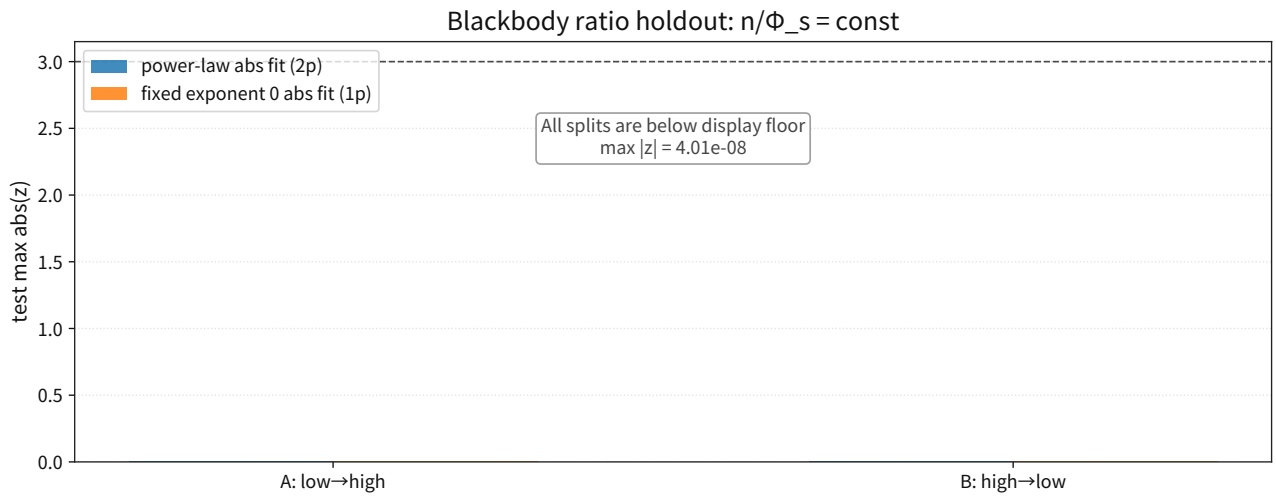


Figure 87: Comparison result for thermodynamics blackbody photon density per entropy flux holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_per_enthalpy_flux_holdout_splits_metrics.json` /

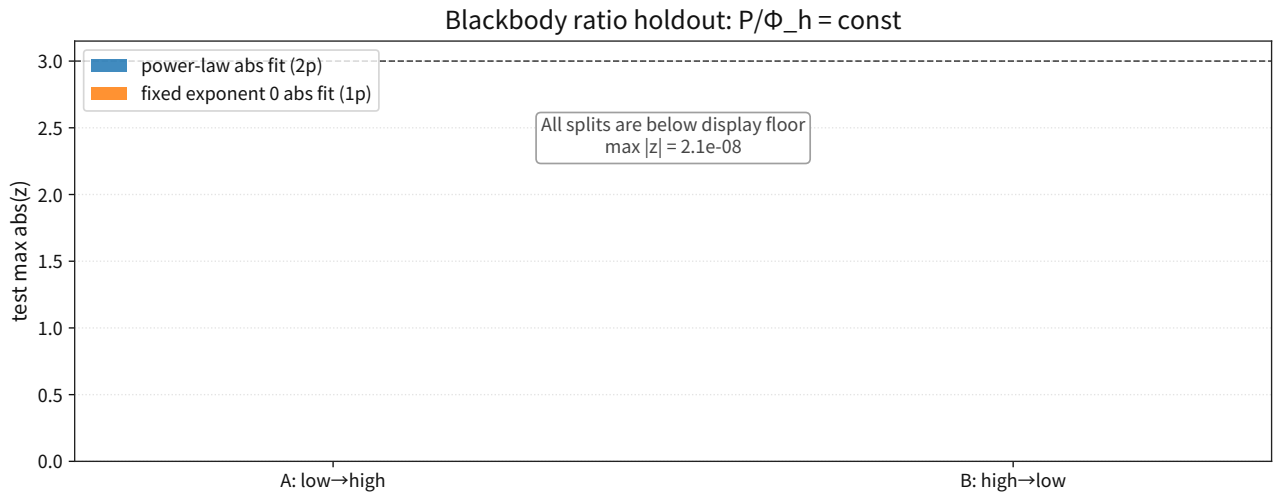


Figure 88: Comparison result for thermodynamics blackbody pressure per enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_entropy_per_enthalpy_flux_holdout_splits_metrics.json` /

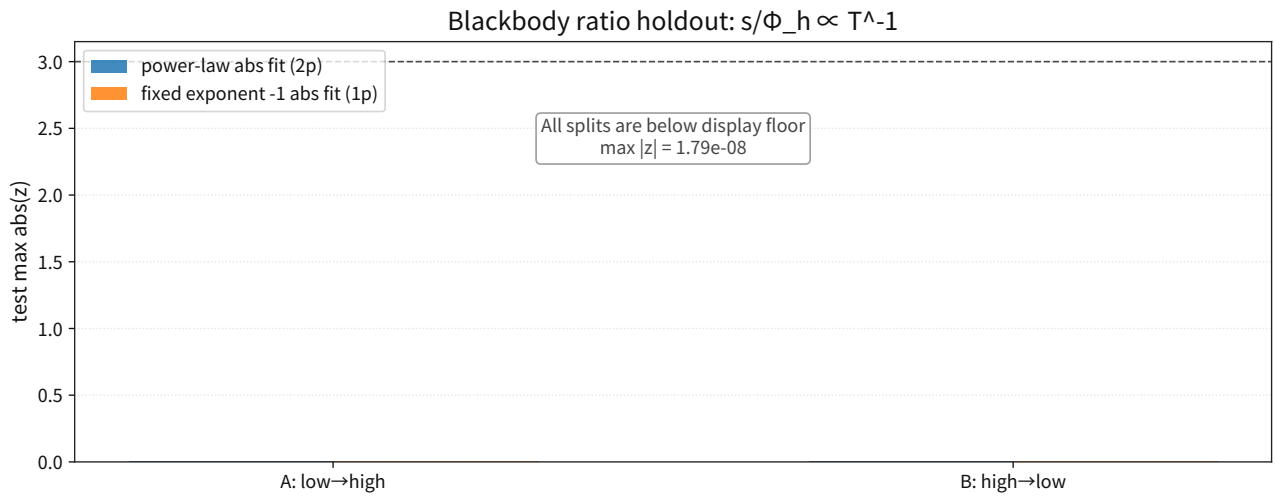


Figure 89: Comparison result for thermodynamics blackbody entropy per enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_entropy_per_enthalpy_density_holdout_splits_metrics.json` /

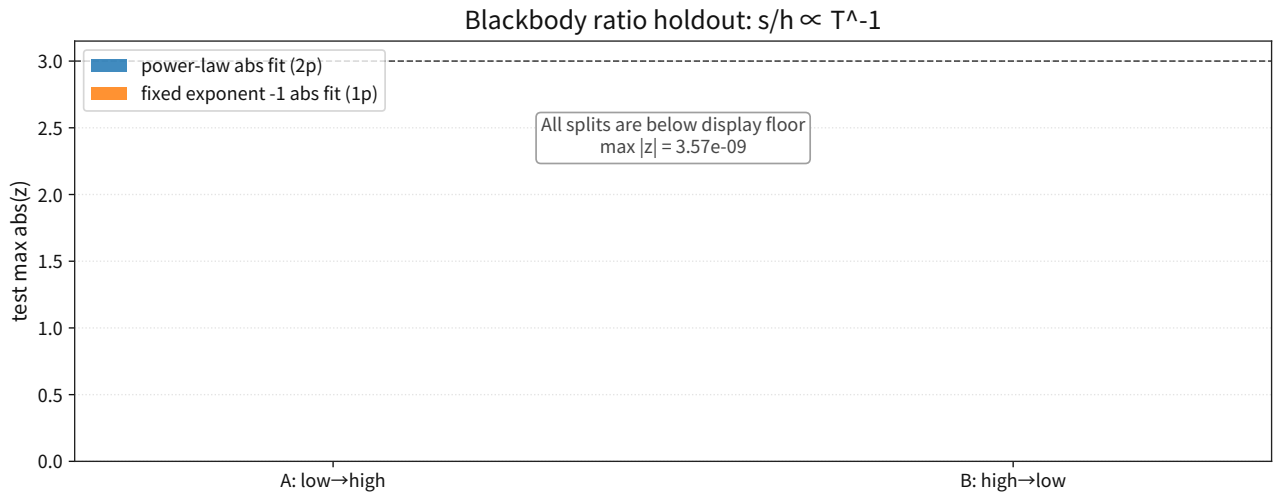


Figure 90: Comparison result for thermodynamics blackbody entropy per enthalpy density holdout splits.

- output/public/quantum/thermo_blackbody_energy_per_enthalpy_flux_holdout_splits_metrics.json /

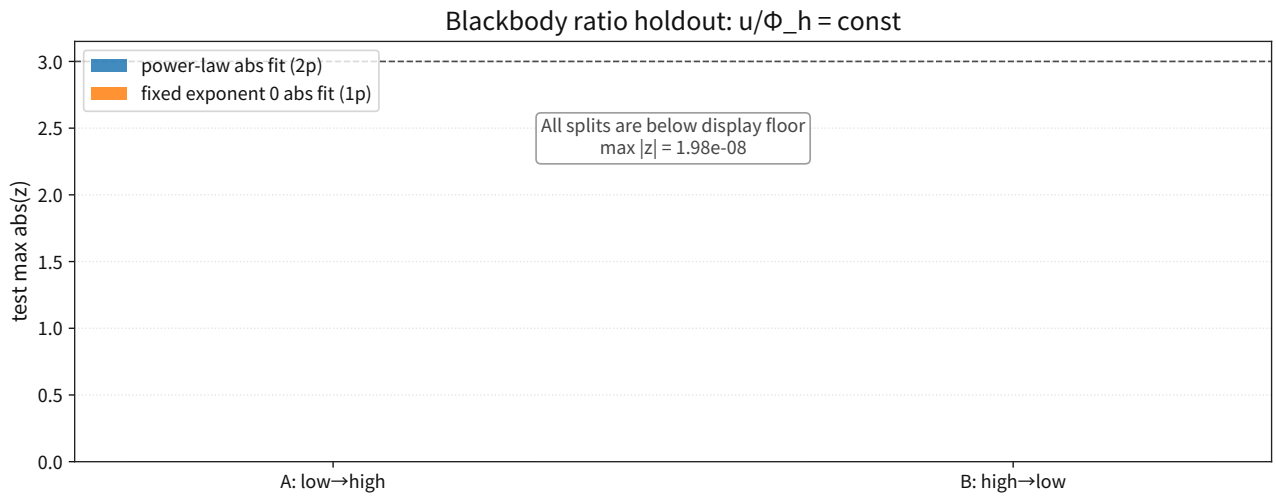


Figure 91: Comparison result for thermodynamics blackbody energy per enthalpy flux holdout splits.

- output/public/quantum/thermo_blackbody_heat_capacity_per_enthalpy_flux_holdout_splits_metrics.json /

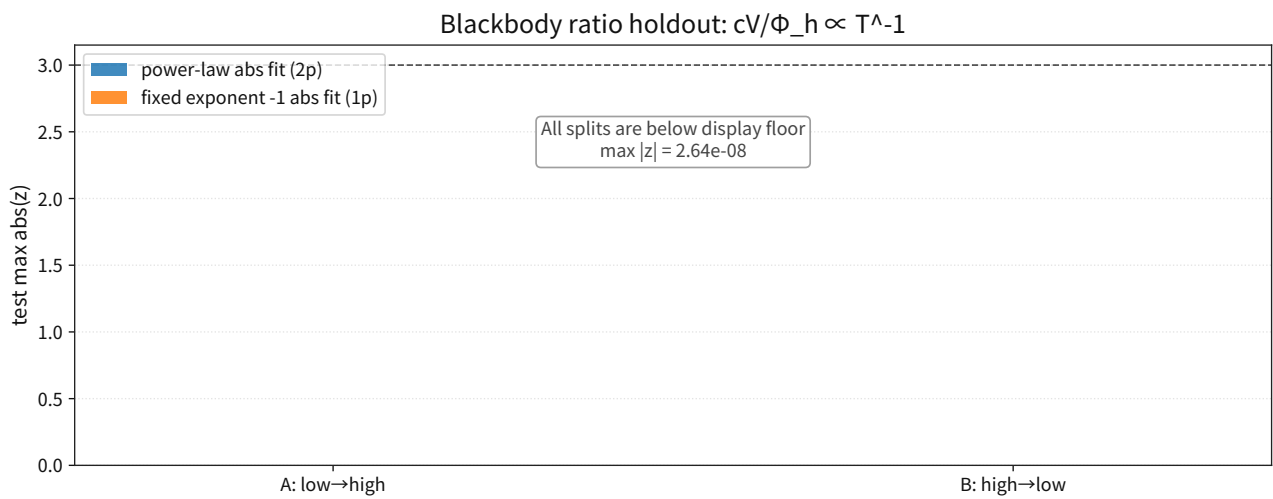


Figure 92: Comparison result for thermodynamics blackbody heat capacity per enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_enthalpy_flux_holdout_splits_metrics.json` /

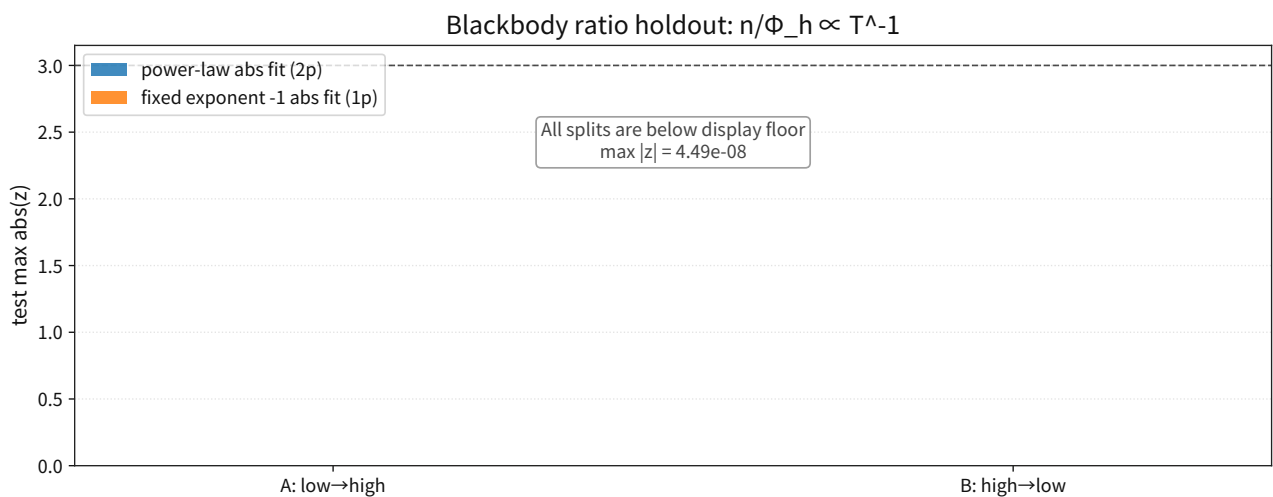


Figure 93: Comparison result for thermodynamics blackbody photon density per enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_per_heat_capacity_flux_holdout_splits_metrics.json` /

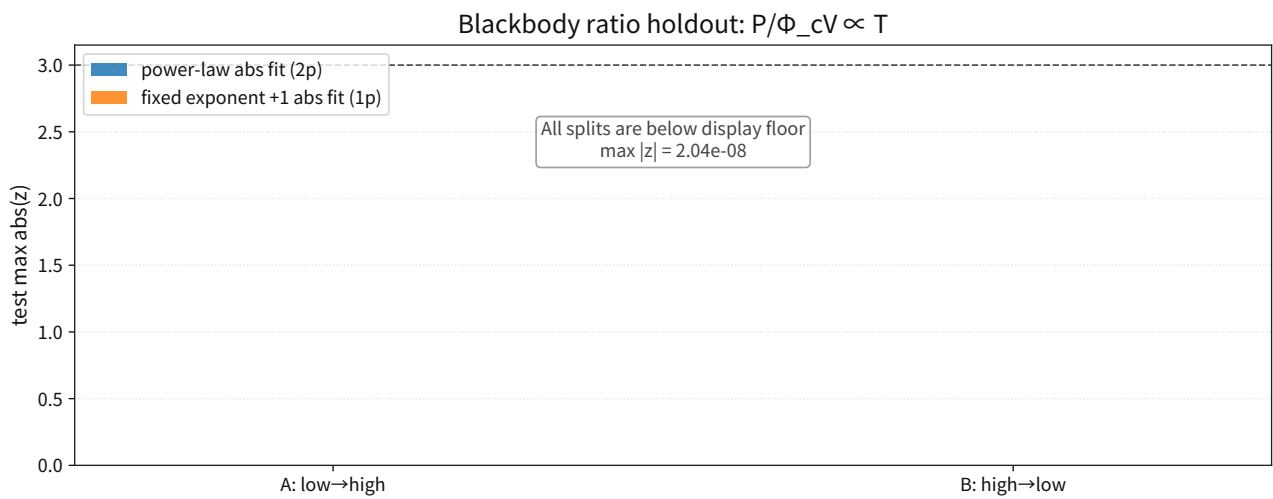


Figure 94: Comparison result for thermodynamics blackbody pressure per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_per_heat_capacity_density_holdout_splits_metrics.json /`

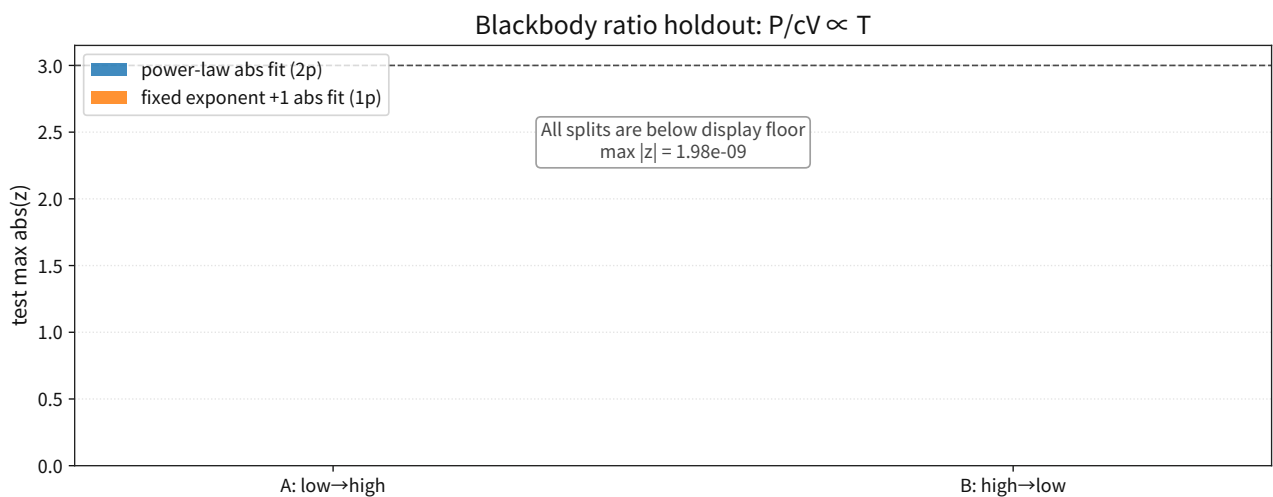


Figure 95: Comparison result for thermodynamics blackbody pressure per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_energy_per_heat_capacity_density_holdout_splits_metrics.json /`

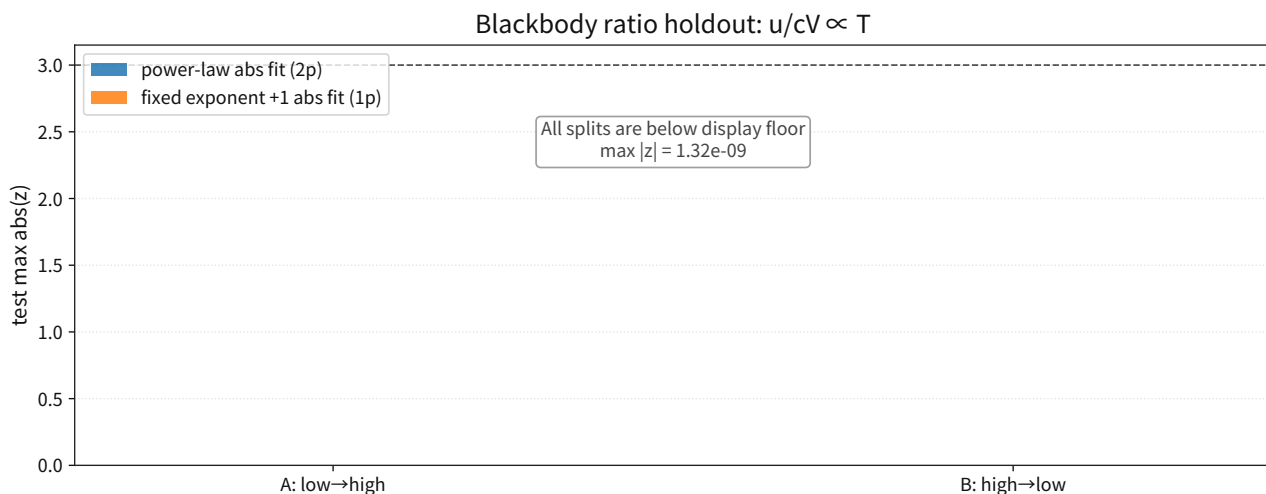


Figure 96: Comparison result for thermodynamics blackbody energy per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_per_heat_capacity_density_holdout_splits_metrics.json` /

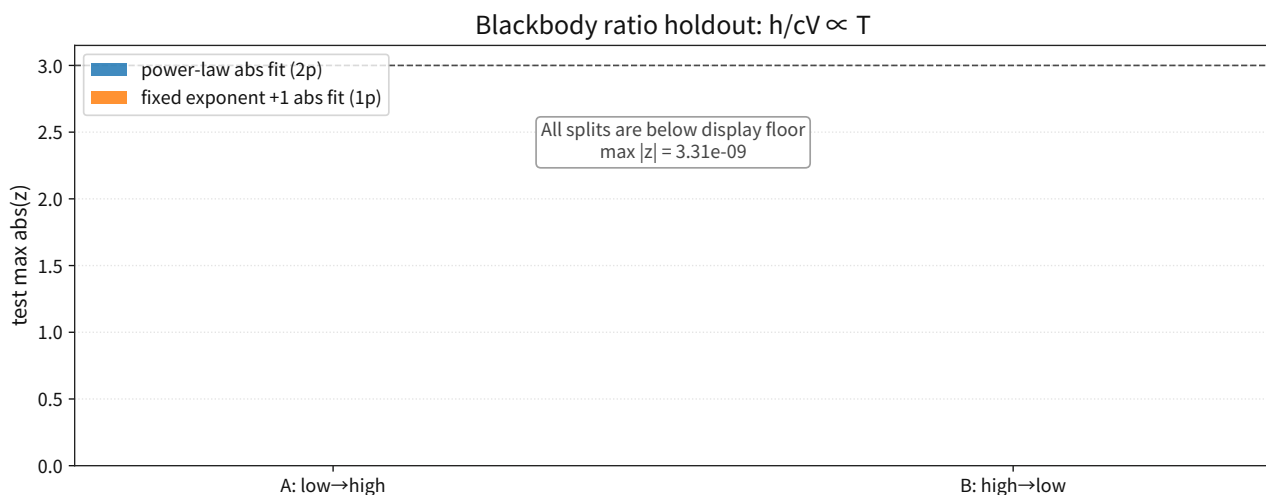


Figure 97: Comparison result for thermodynamics blackbody enthalpy per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_entropy_per_heat_capacity_density_holdout_splits_metrics.json` /

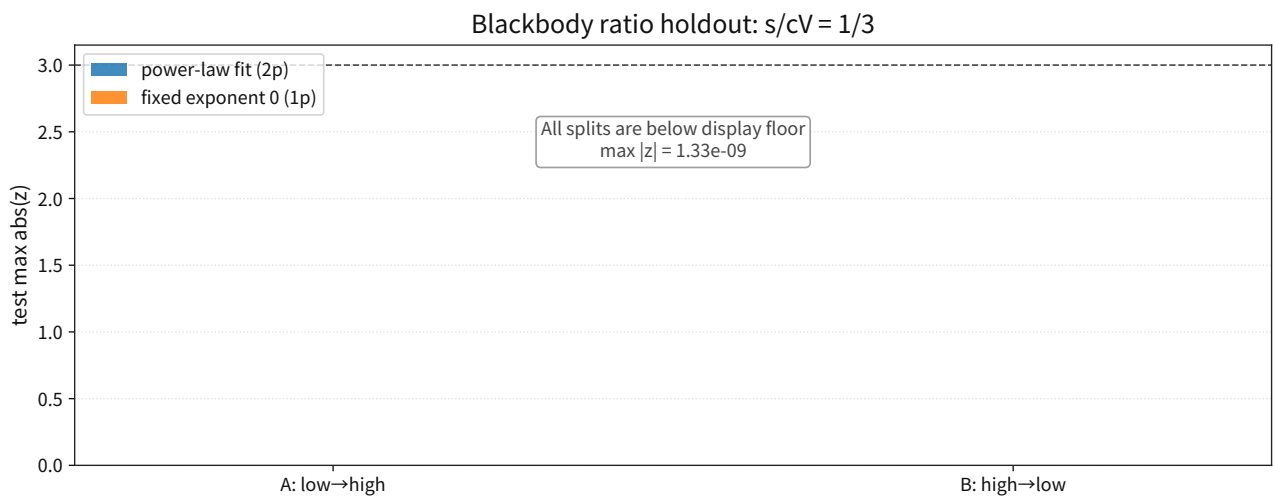


Figure 98: Comparison result for thermodynamics blackbody entropy per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_energy_per_heat_capacity_flux_holdout_splits_metrics.json` /

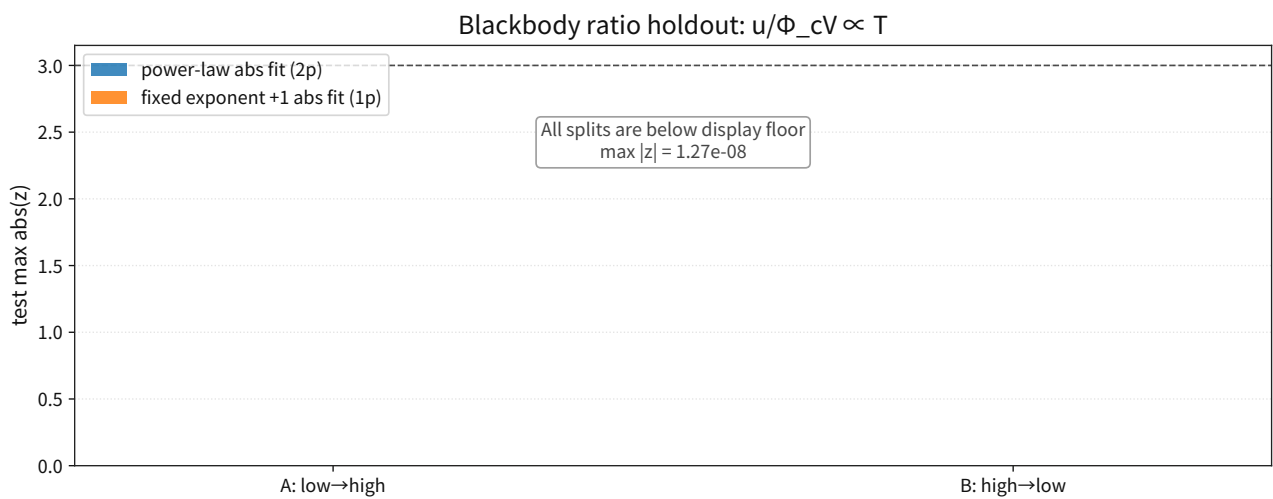


Figure 99: Comparison result for thermodynamics blackbody energy per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_per_heat_capacity_flux_holdout_splits_metrics.json` /

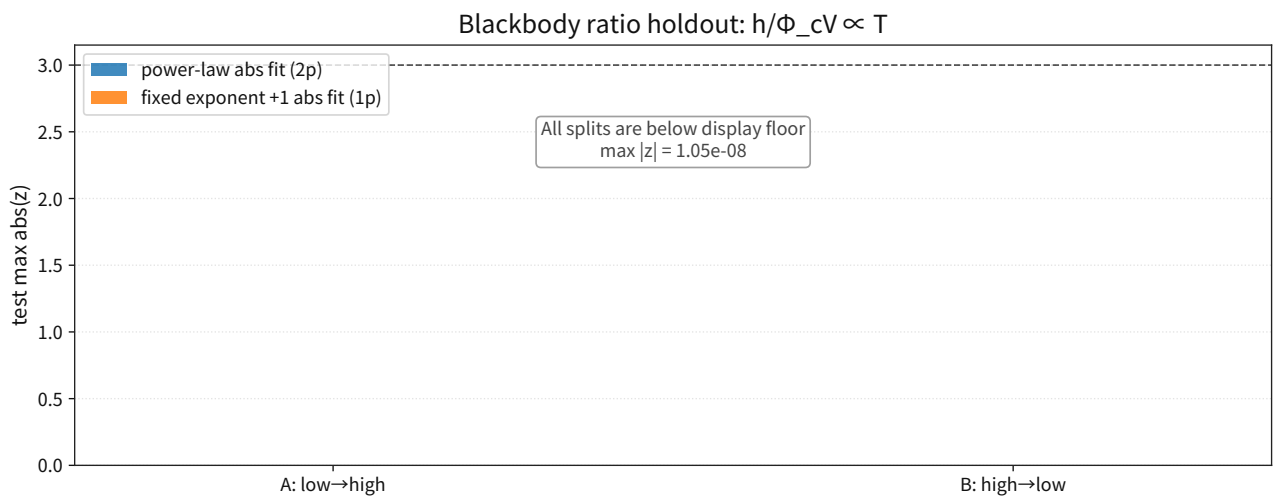


Figure 100: Comparison result for thermodynamics blackbody enthalpy per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_entropy_per_heat_capacity_flux_holdout_splits_metrics.json` /

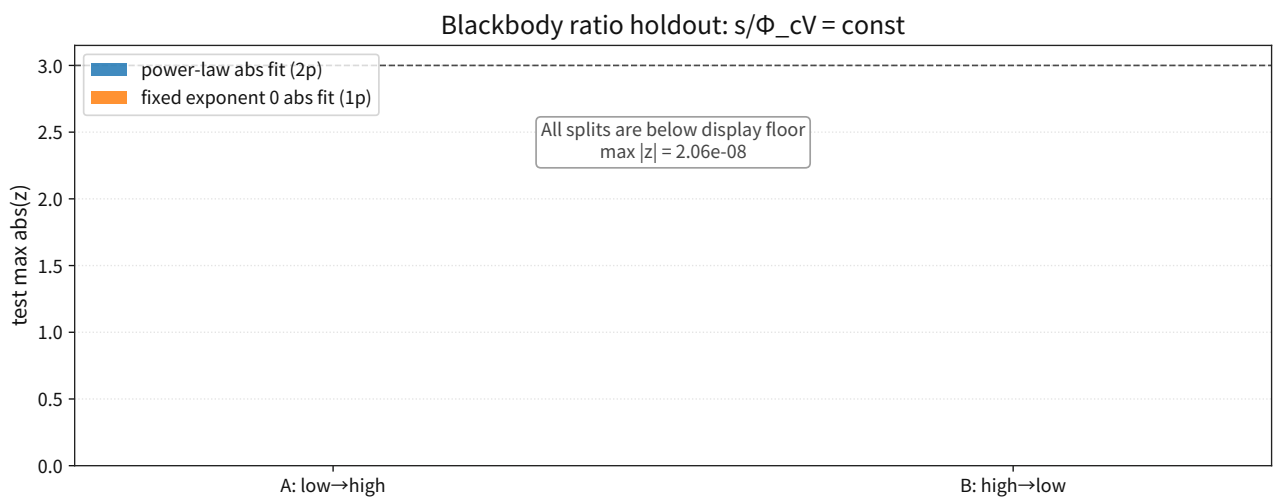


Figure 101: Comparison result for thermodynamics blackbody entropy per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_heat_capacity_flux_holdout_splits_metrics.json` /

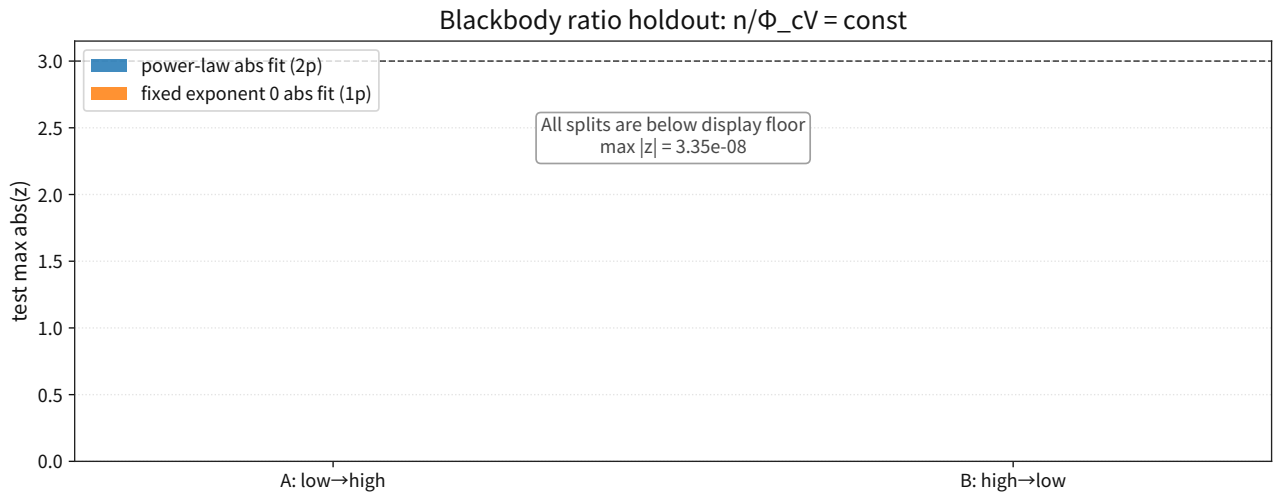


Figure 102: Comparison result for thermodynamics blackbody photon density per heat capacity flux holdout splits.

- `output/public/quantum/thermo_blackbody_photon_density_per_heat_capacity_density_holdout_splits_metrics.json` /

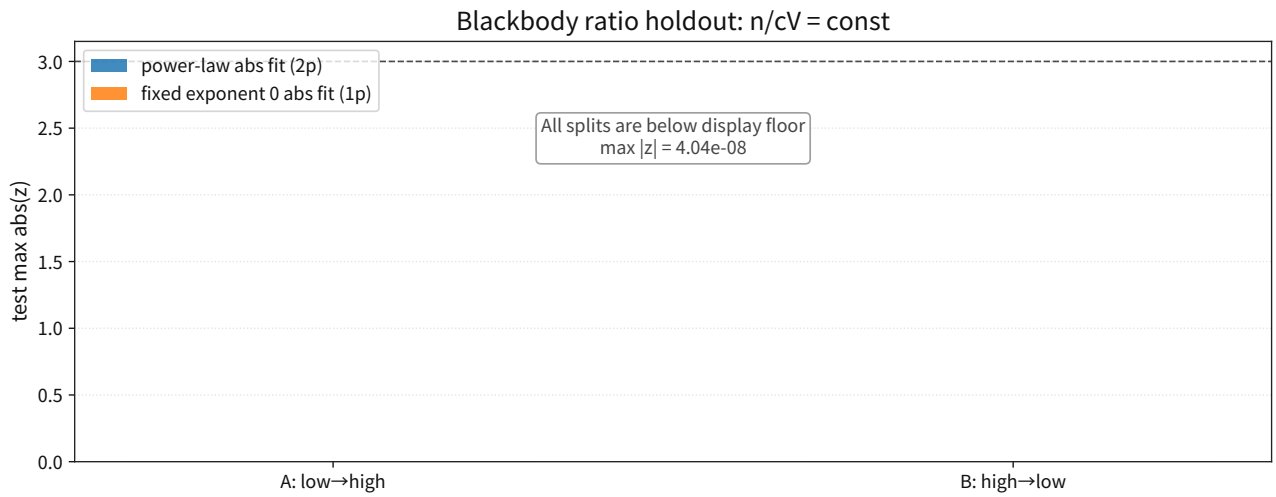


Figure 103: Comparison result for thermodynamics blackbody photon density per heat capacity density holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_per_photon_holdout_splits_metrics.json` /

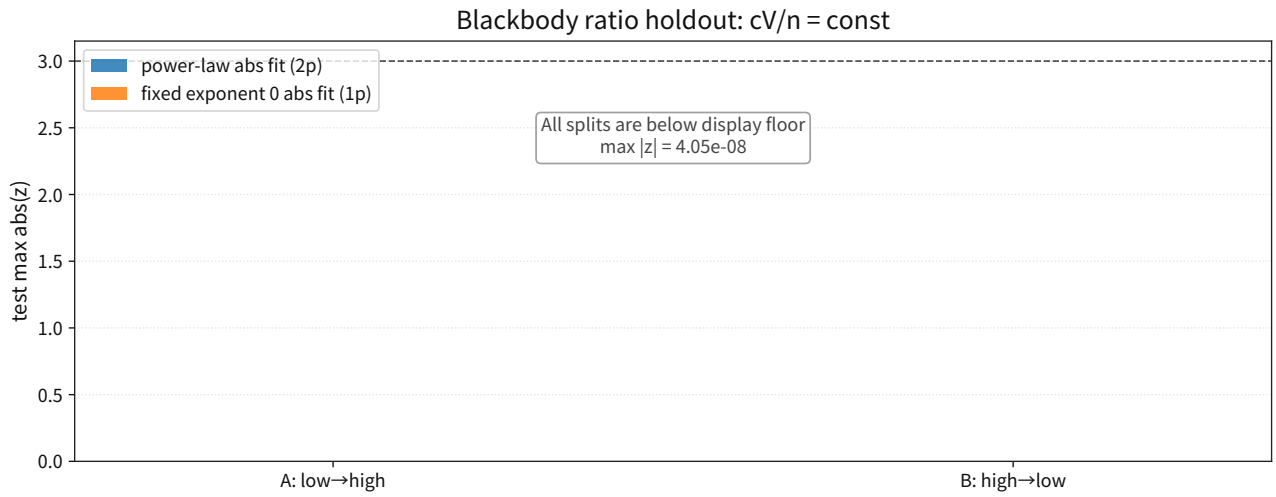


Figure 104: Comparison result for thermodynamics blackbody heat capacity per photon holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_per_photon_flux_holdout_splits_metrics.json` /

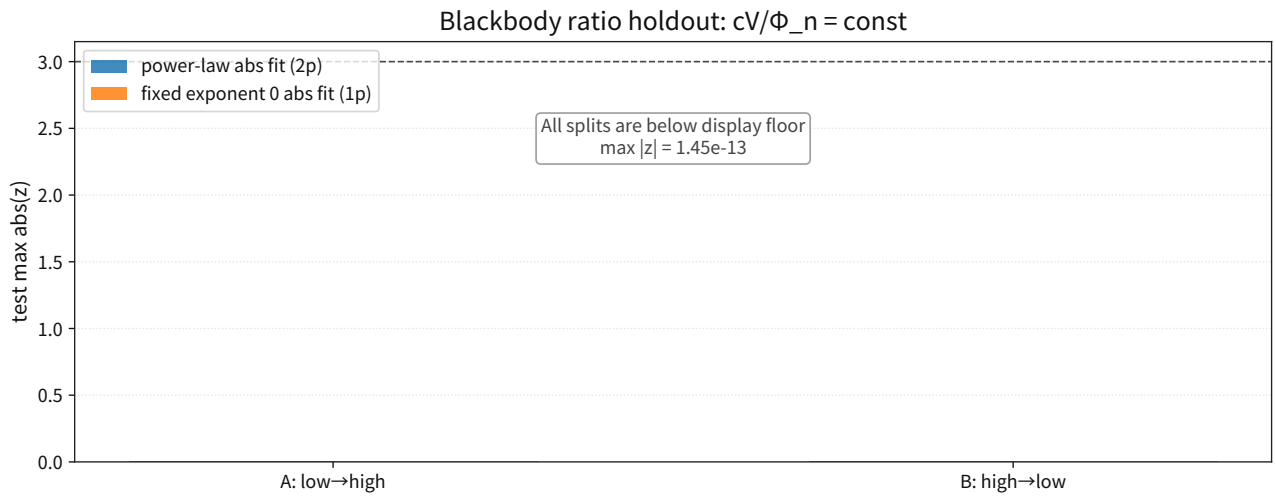


Figure 105: Comparison result for thermodynamics blackbody heat capacity per photon flux holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_per_entropy_flux_holdout_splits_metrics.json` /

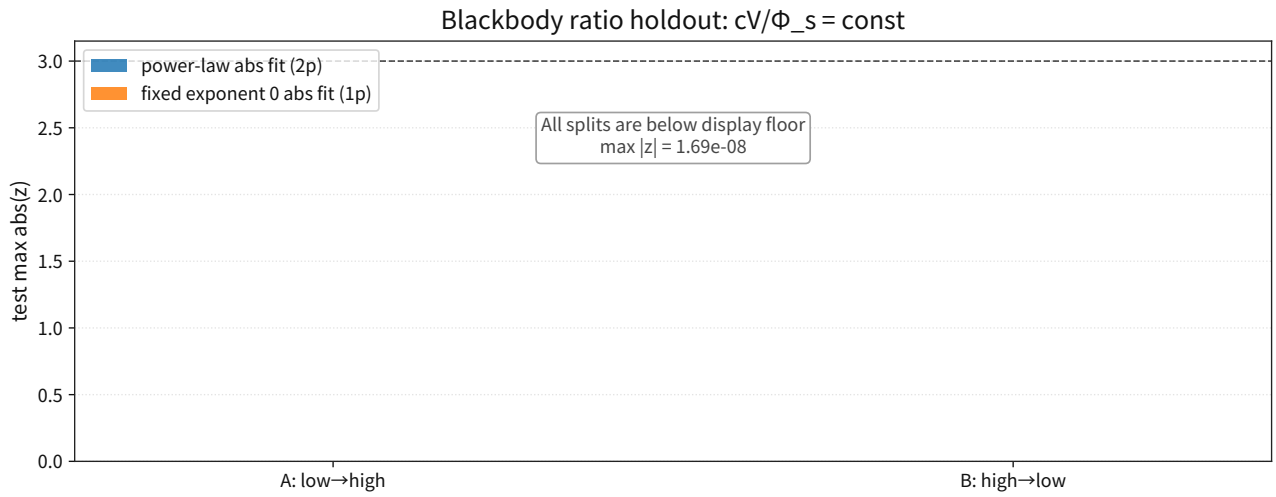


Figure 106: Comparison result for thermodynamics blackbody heat capacity per entropy flux holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_entropy_ratio_holdout_splits_metrics.json` /

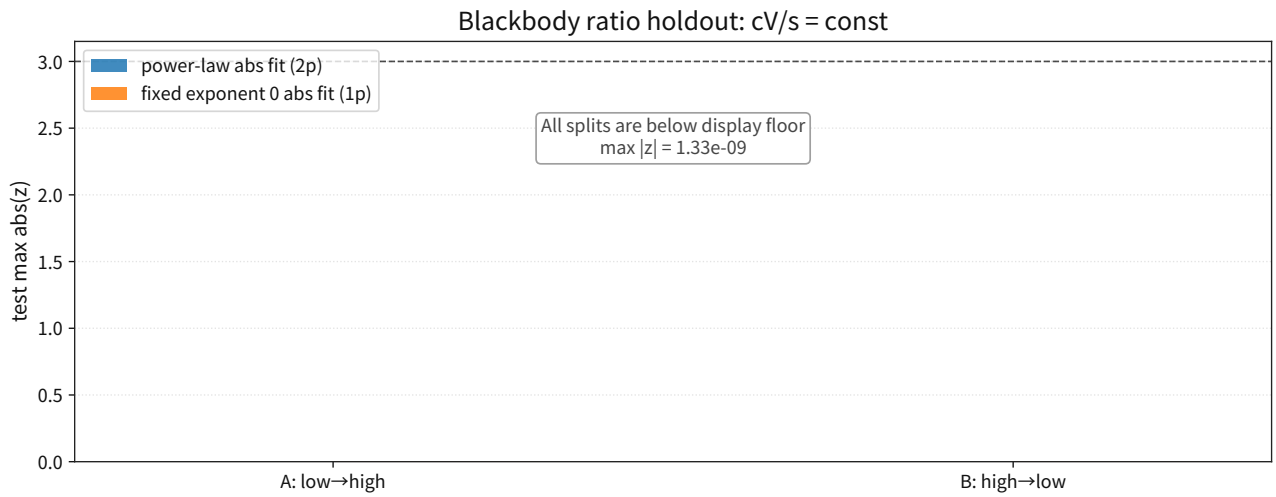


Figure 107: Comparison result for thermodynamics blackbody heat capacity entropy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_flux_ratio_holdout_splits_metrics.json` /

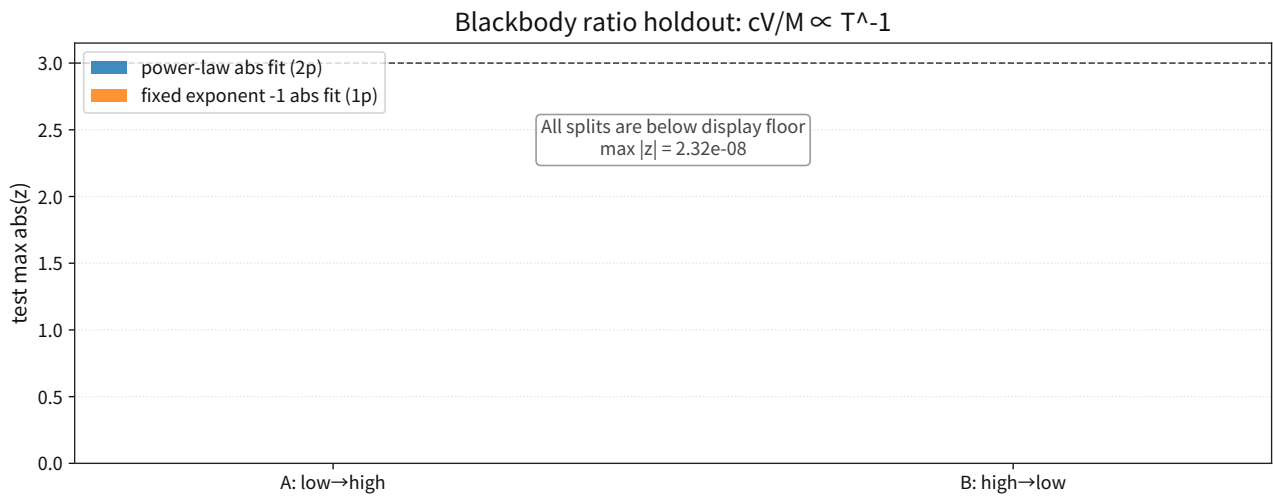


Figure 108: Comparison result for thermodynamics blackbody heat capacity flux ratio holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_momentum_ratio_holdout_splits_metrics.json` /

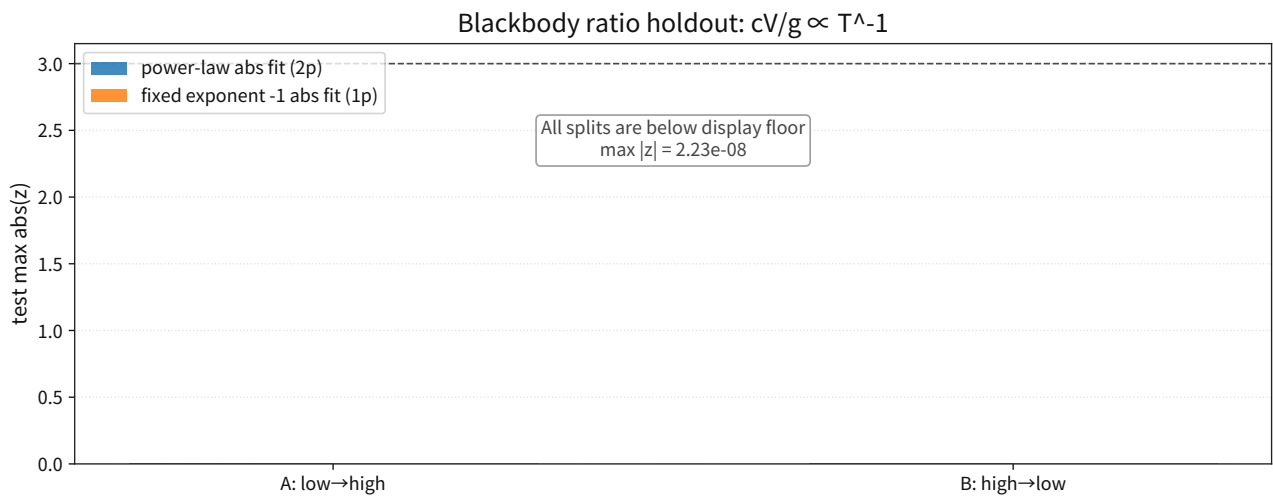


Figure 109: Comparison result for thermodynamics blackbody heat capacity momentum ratio holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_energy_ratio_holdout_splits_metrics.json` /

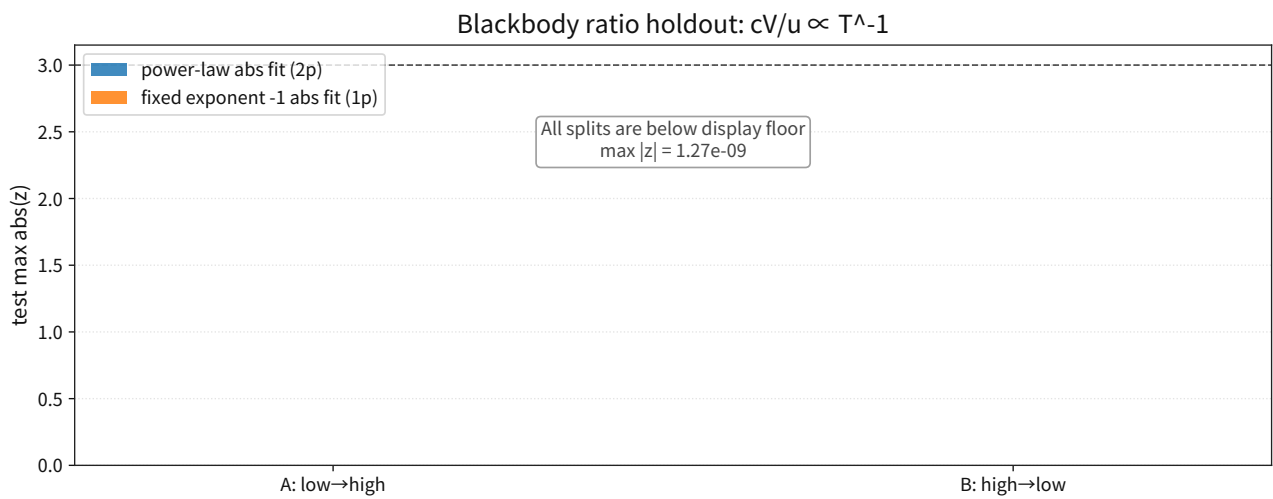


Figure 110: Comparison result for thermodynamics blackbody heat capacity energy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_pressure_ratio_holdout_splits_metrics.json` /

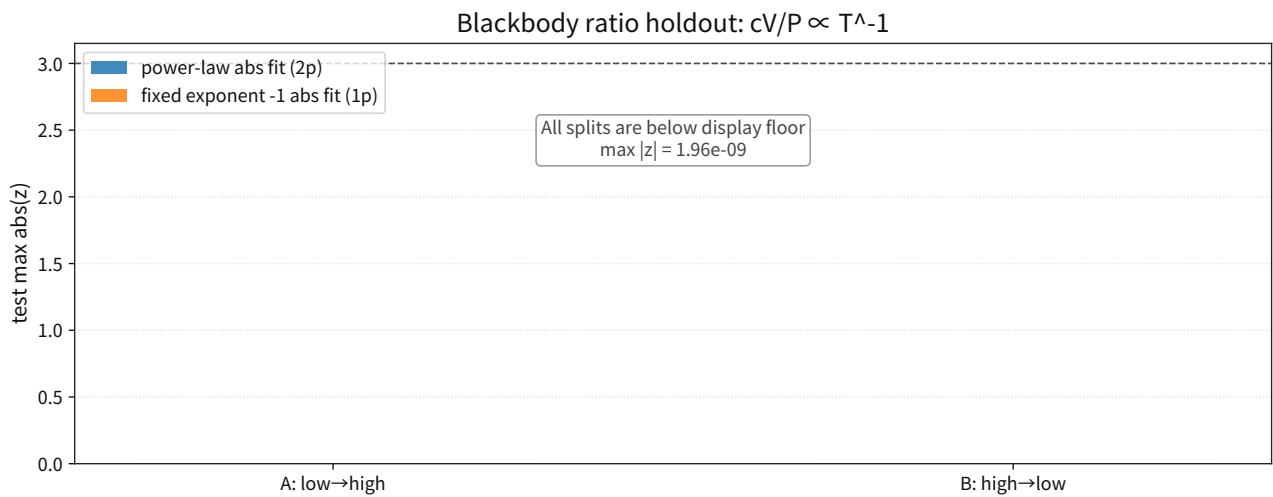


Figure 111: Comparison result for thermodynamics blackbody heat capacity pressure ratio holdout splits.

- `output/public/quantum/thermo_blackbody_heat_capacity_enthalpy_ratio_holdout_splits_metrics.json` /

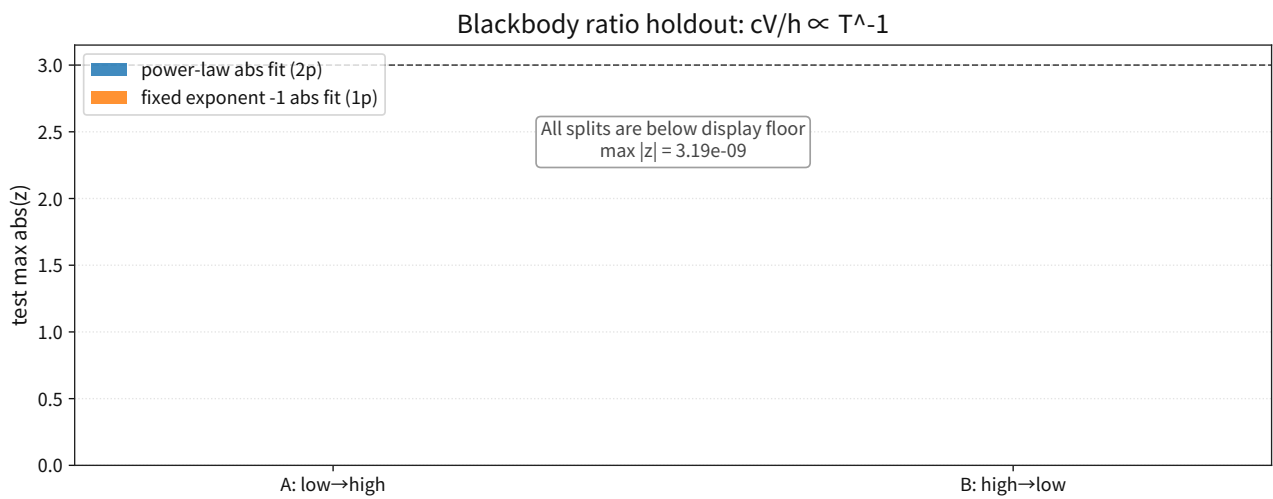


Figure 112: Comparison result for thermodynamics blackbody heat capacity enthalpy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_per_photon_holdout_splits_metrics.json /`

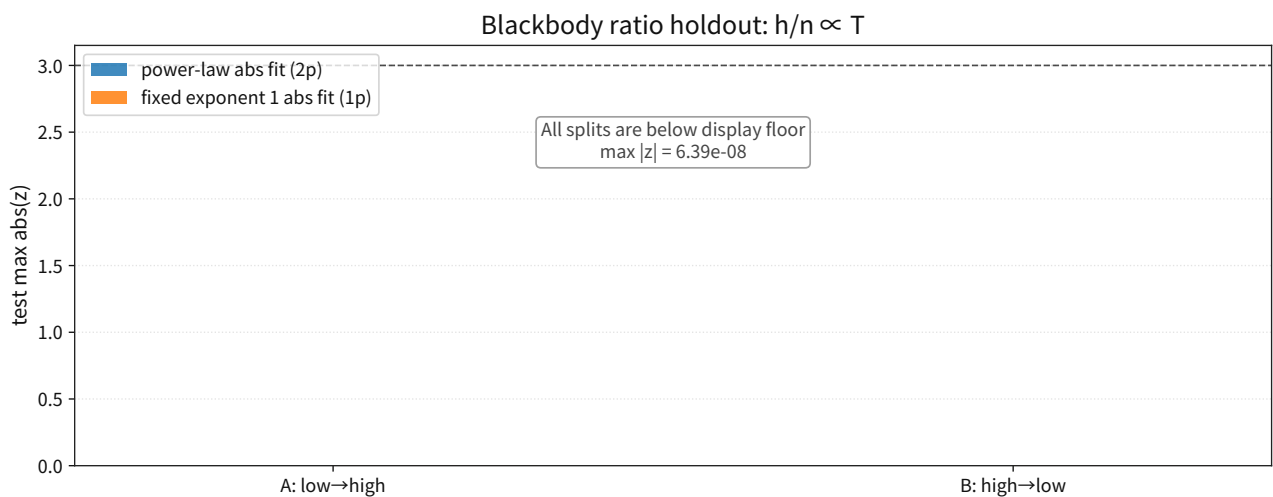


Figure 113: Comparison result for thermodynamics blackbody enthalpy per photon holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_per_photon_flux_holdout_splits_metrics.json /`

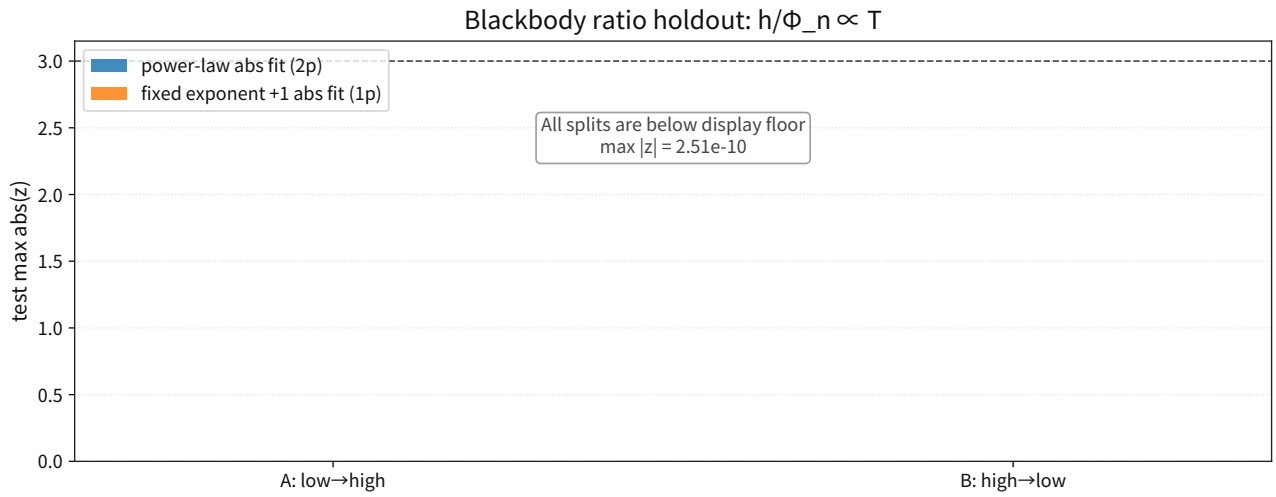


Figure 114: Comparison result for thermodynamics blackbody enthalpy per photon flux holdout splits.

- `output/public/quantum/thermo_blackbody_energy_per_photon_flux_holdout_splits_metrics.json` /

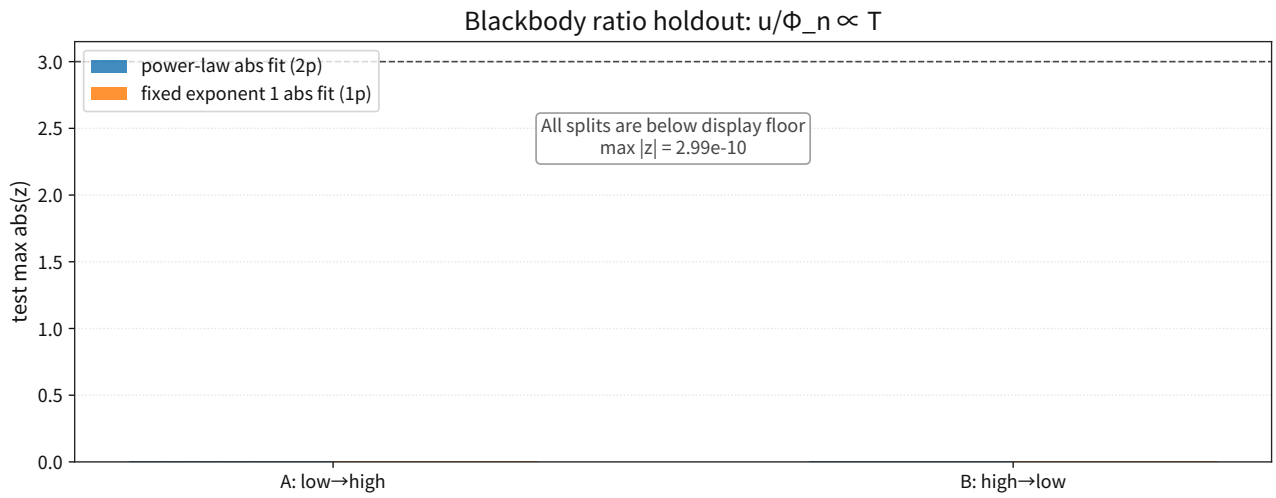


Figure 115: Comparison result for thermodynamics blackbody energy per photon flux holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_per_photon_holdout_splits_metrics.json` /

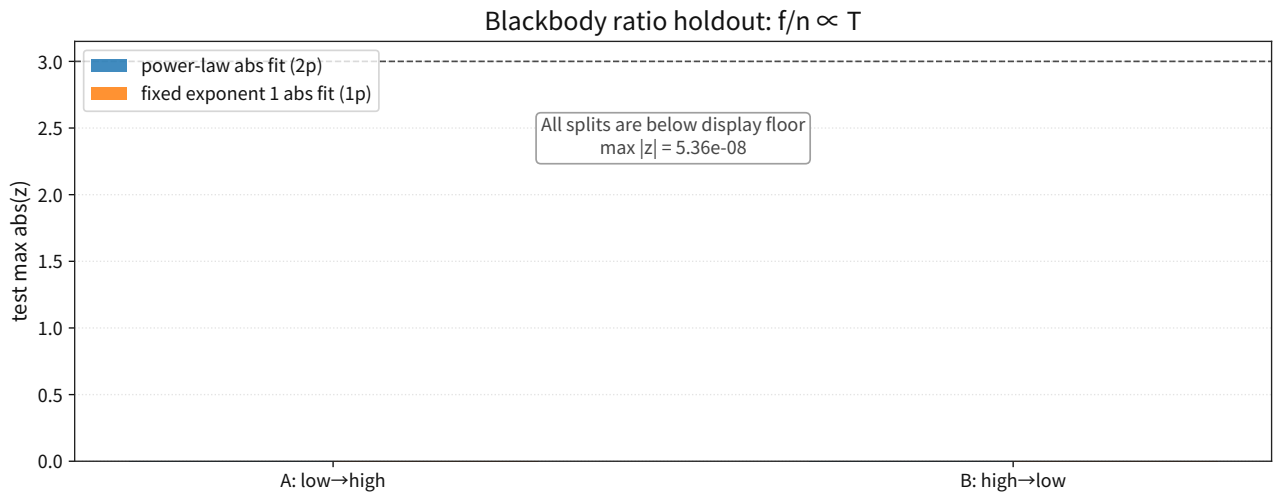


Figure 116: Comparison result for thermodynamics blackbody Helmholtz per photon holdout splits.

- `output/public/quantum/thermo_blackbody_entropy_per_photon_flux_holdout_splits_metrics.json` /

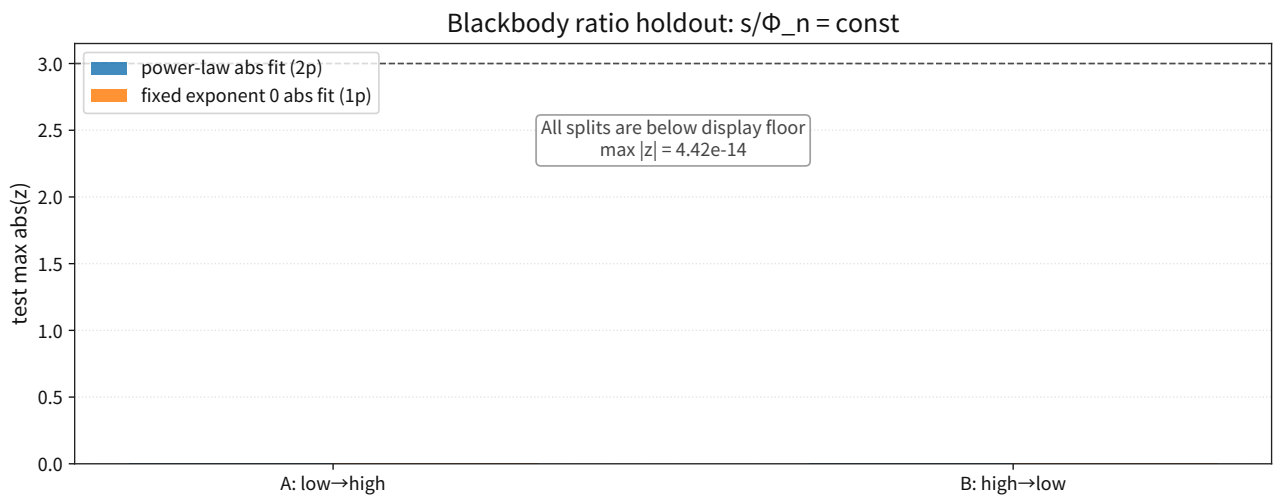


Figure 117: Comparison result for thermodynamics blackbody entropy per photon flux holdout splits.

- `output/public/quantum/thermo_blackbody_flux_per_photon_holdout_splits_metrics.json` /

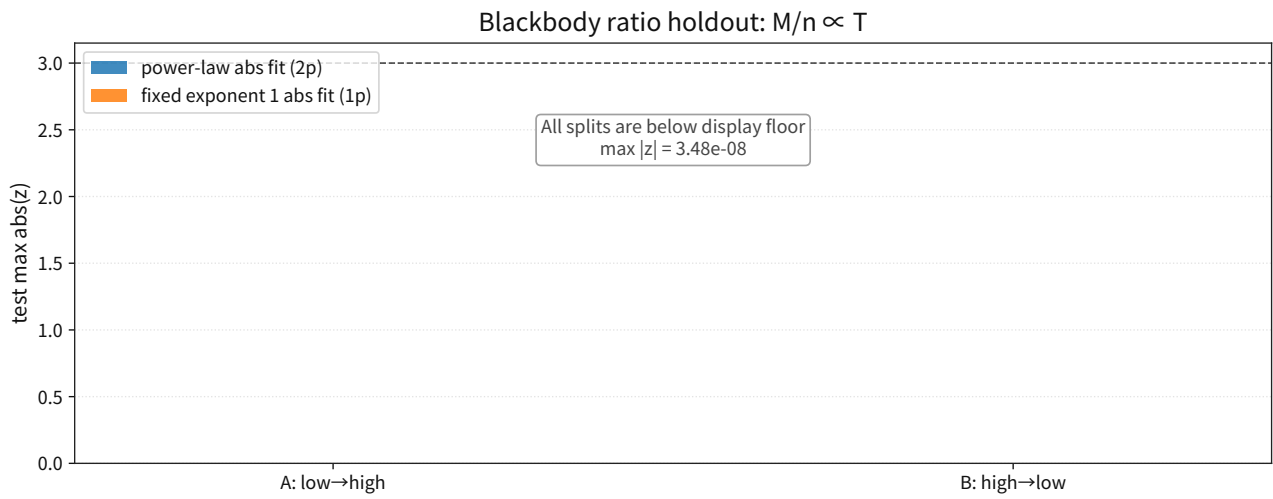


Figure 118: Comparison result for thermodynamics blackbody flux per photon holdout splits.

- [output/public/quantum/thermo_blackbody_entropy_flux_holdout_splits_metrics.json](#) /

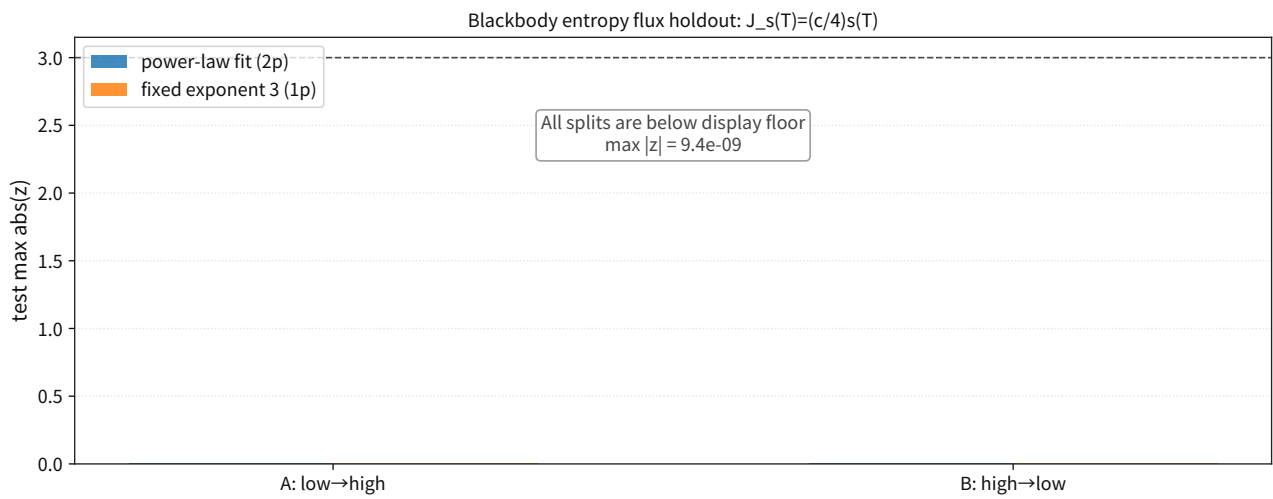


Figure 119: Comparison result for thermodynamics blackbody entropy flux holdout splits.

- [output/public/quantum/thermo_blackbody_enthalpy_flux_holdout_splits_metrics.json](#) /

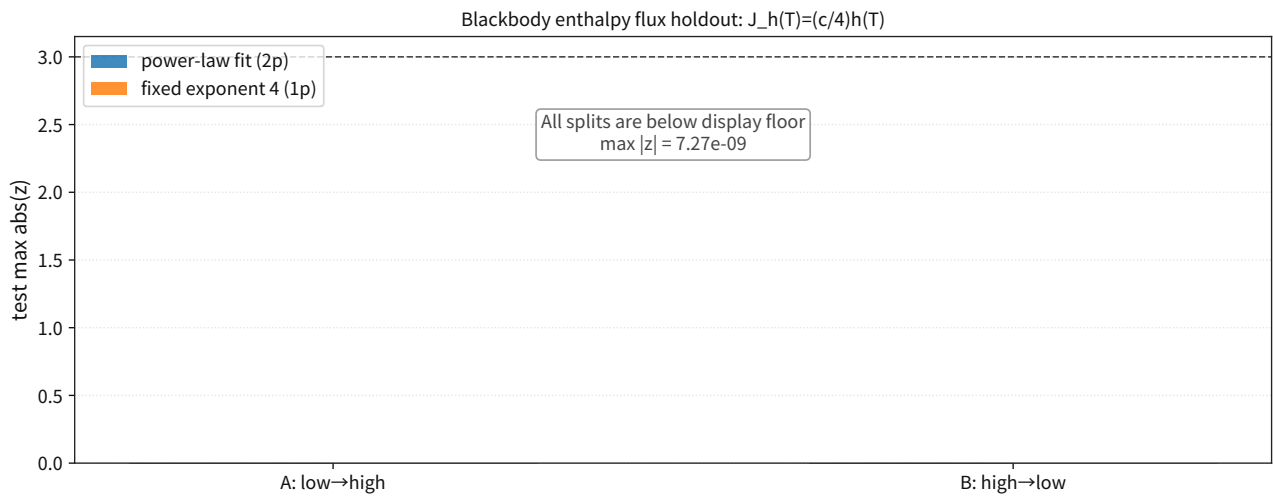


Figure 120: Comparison result for thermodynamics blackbody enthalpy flux holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_flux_ratio_holdout_splits_metrics.json` /

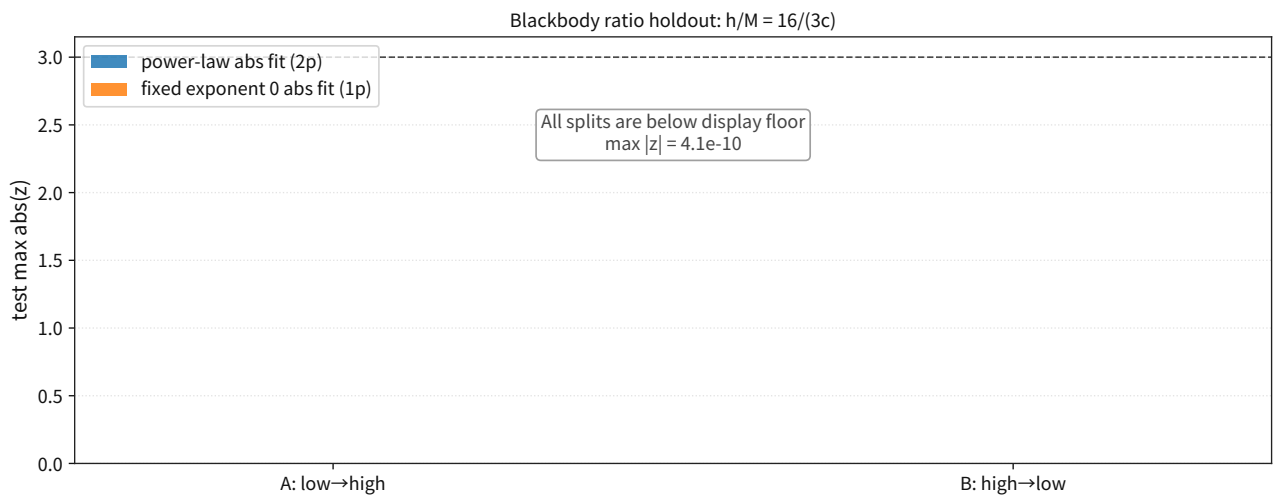


Figure 121: Comparison result for thermodynamics blackbody enthalpy flux ratio holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_momentum_ratio_holdout_splits_metrics.json` /

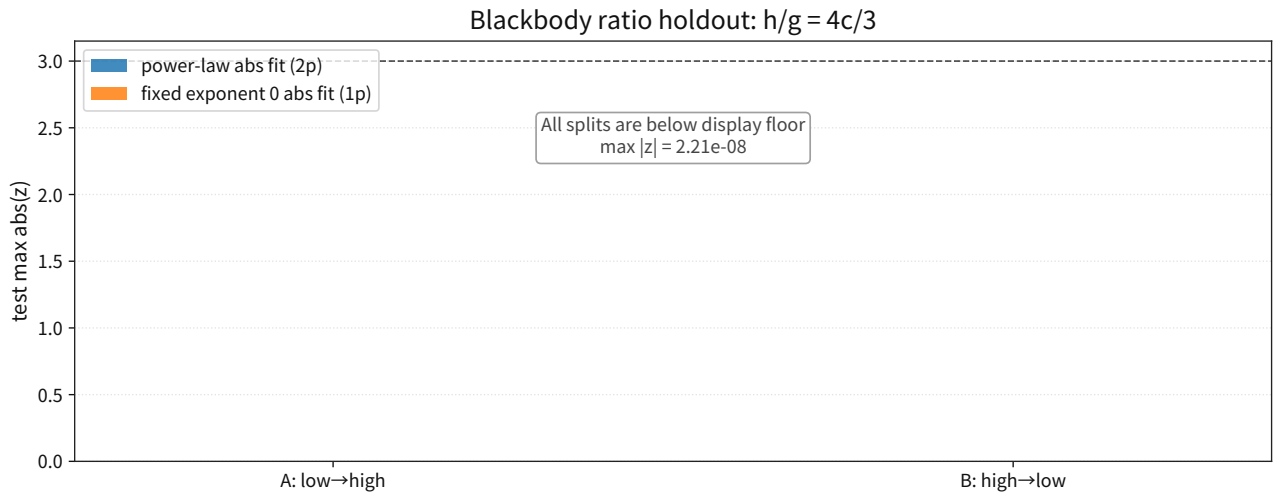


Figure 122: Comparison result for thermodynamics blackbody enthalpy momentum ratio holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_energy_ratio_holdout_splits_metrics.json` /

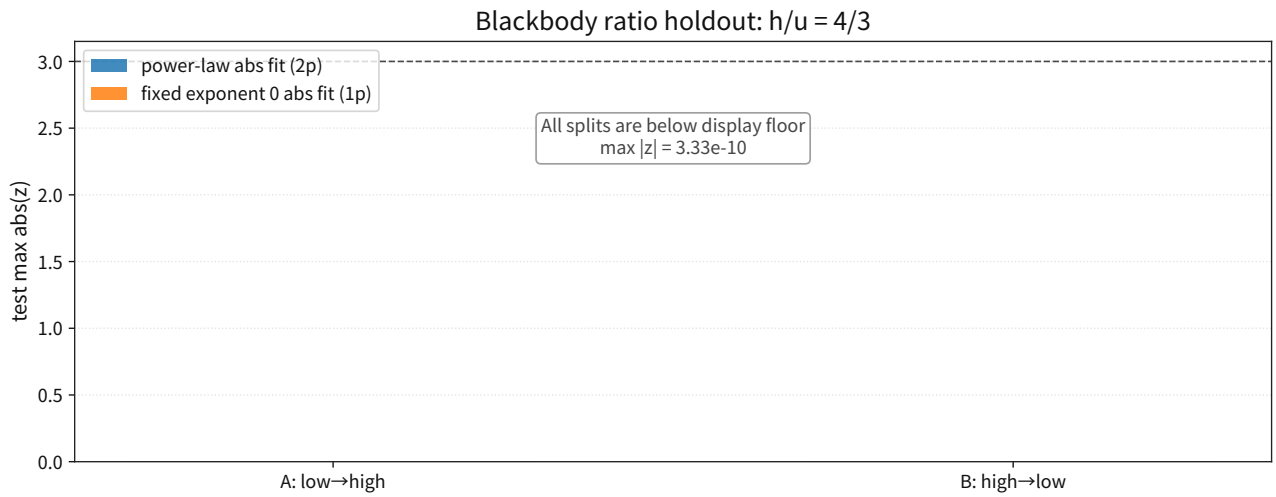


Figure 123: Comparison result for thermodynamics blackbody enthalpy energy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_pressure_ratio_holdout_splits_metrics.json` /

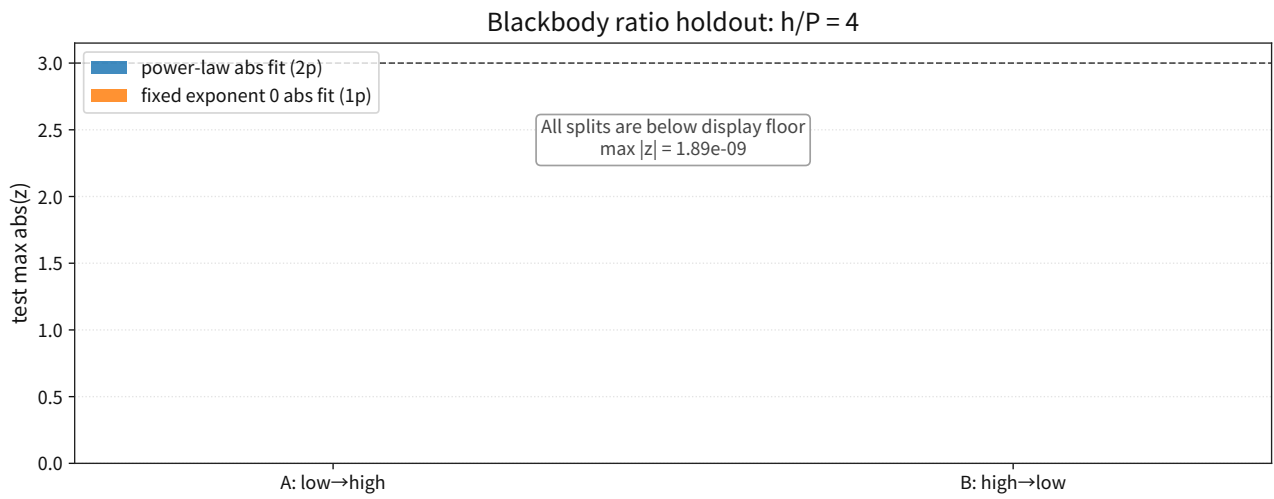


Figure 124: Comparison result for thermodynamics blackbody enthalpy pressure ratio holdout splits.

- `output/public/quantum/thermo_blackbody_enthalpy_entropy_ratio_holdout_splits_metrics.json` /

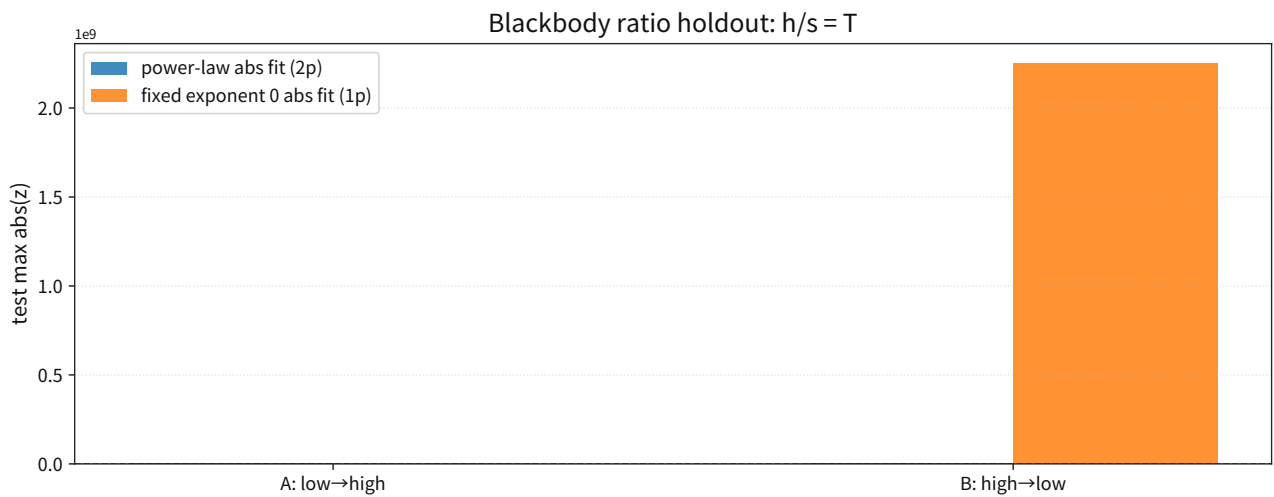


Figure 125: Comparison result for thermodynamics blackbody enthalpy entropy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_entropy_ratio_holdout_splits_metrics.json` /

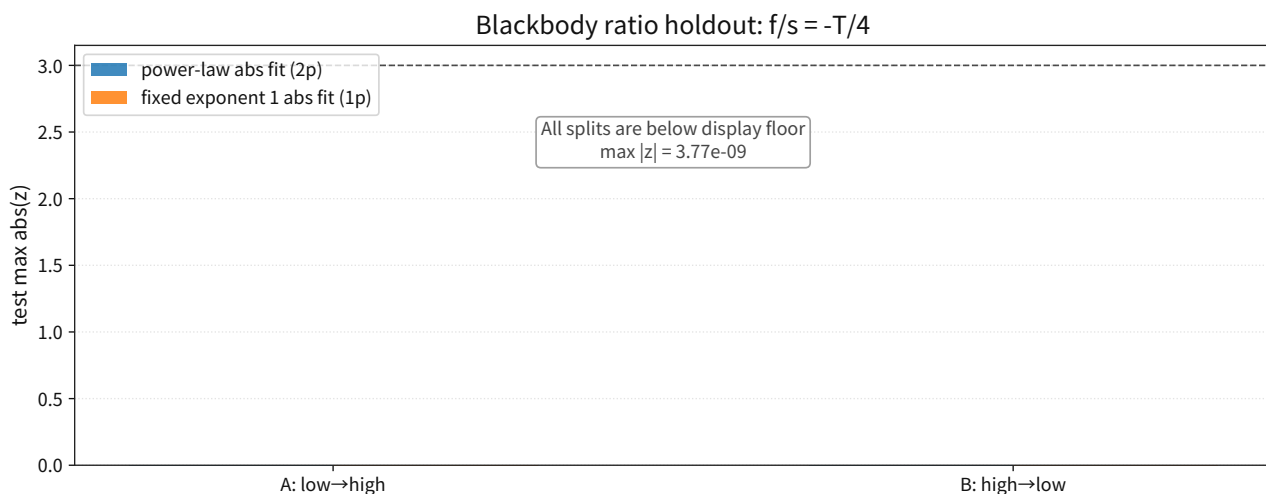


Figure 126: Comparison result for thermodynamics blackbody Helmholtz entropy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_energy_ratio_holdout_splits_metrics.json` /

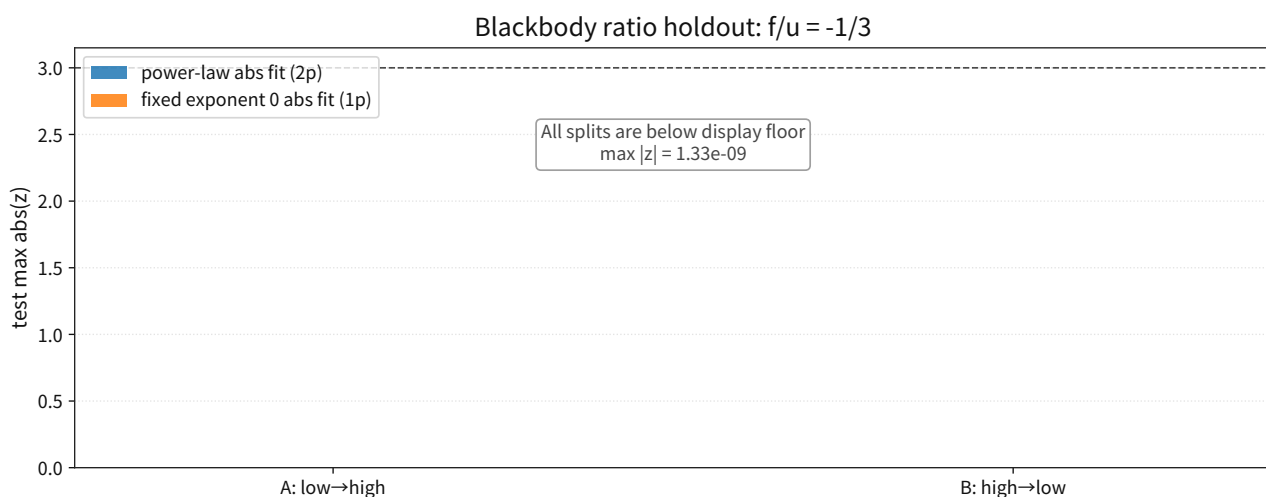


Figure 127: Comparison result for thermodynamics blackbody Helmholtz energy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_enthalpy_ratio_holdout_splits_metrics.json` /

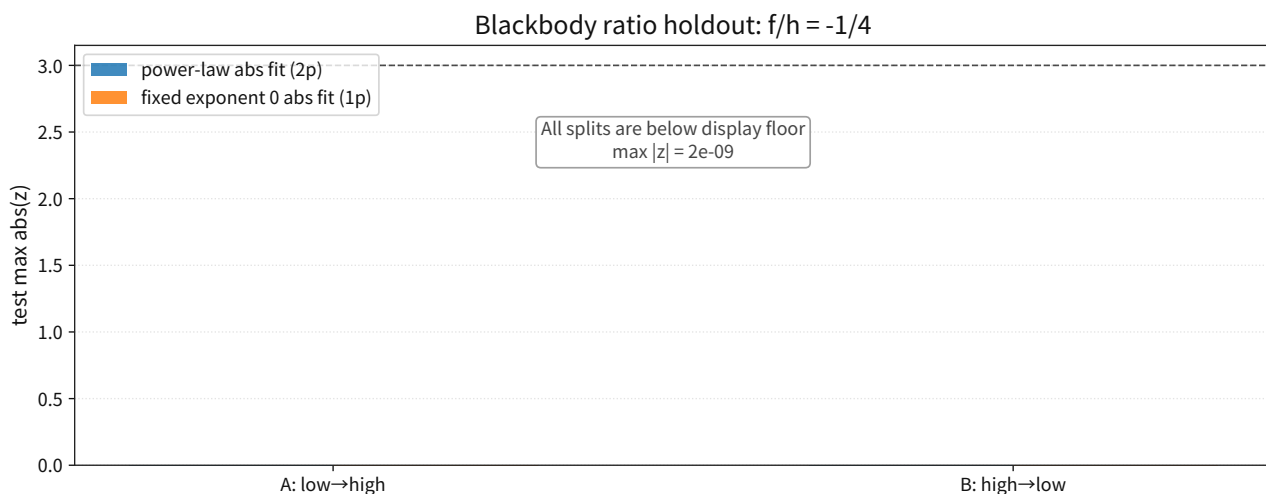


Figure 128: Comparison result for thermodynamics blackbody Helmholtz enthalpy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_momentum_ratio_holdout_splits_metrics.json` /

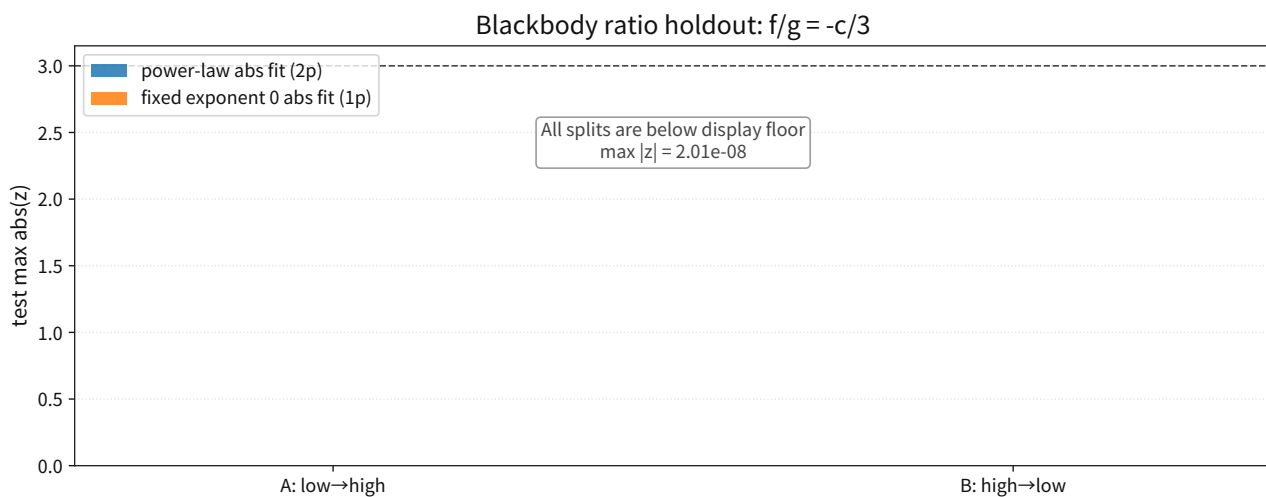


Figure 129: Comparison result for thermodynamics blackbody Helmholtz momentum ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_pressure_ratio_holdout_splits_metrics.json` /

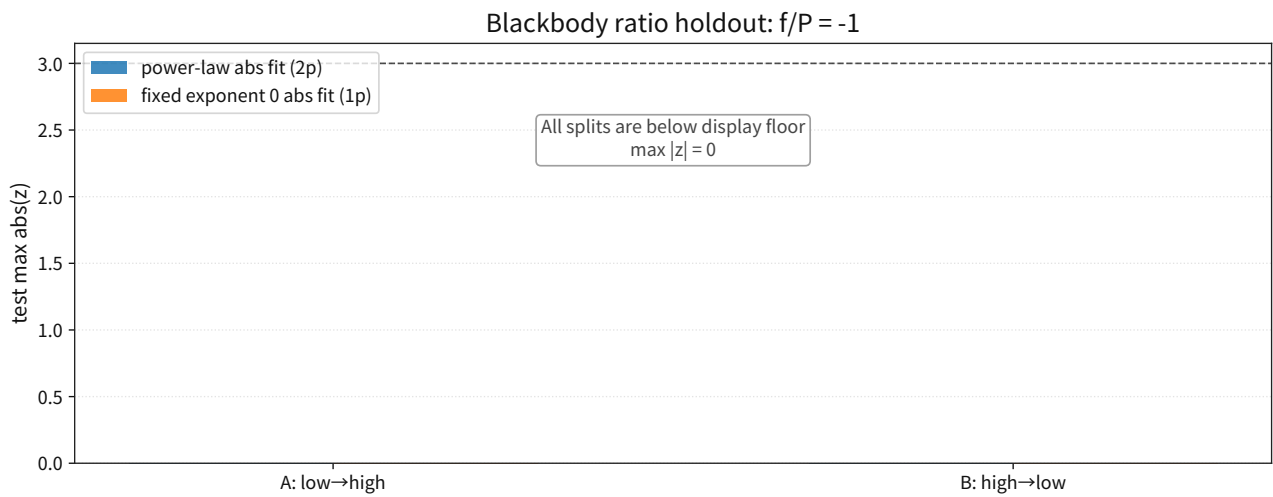


Figure 130: Comparison result for thermodynamics blackbody Helmholtz pressure ratio holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_entropy_ratio_holdout_splits_metrics.json` /

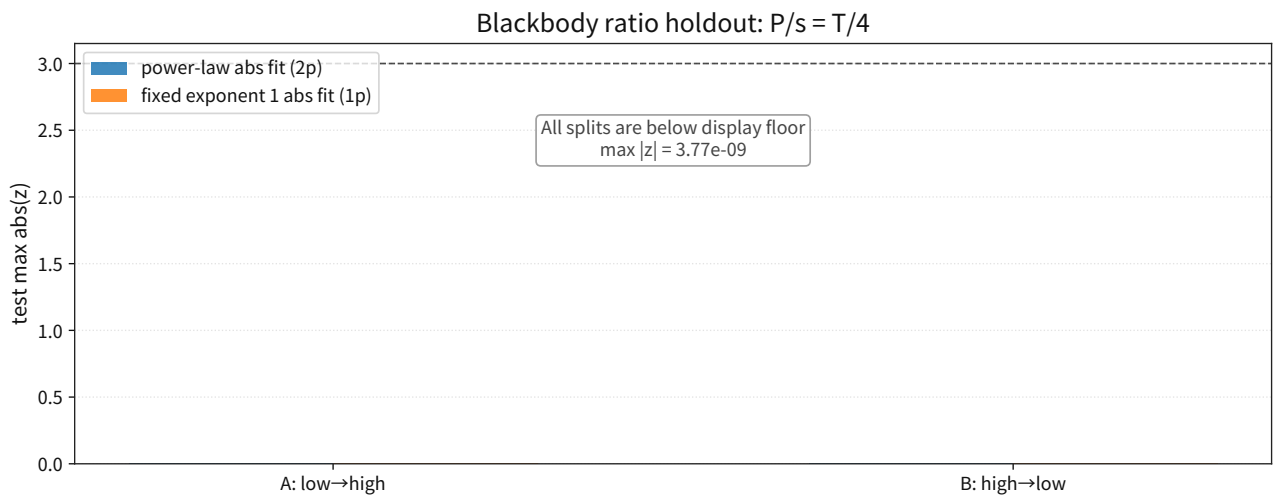


Figure 131: Comparison result for thermodynamics blackbody pressure entropy ratio holdout splits.

- `output/public/quantum/thermo_blackbody_pressure_flux_ratio_holdout_splits_metrics.json` /

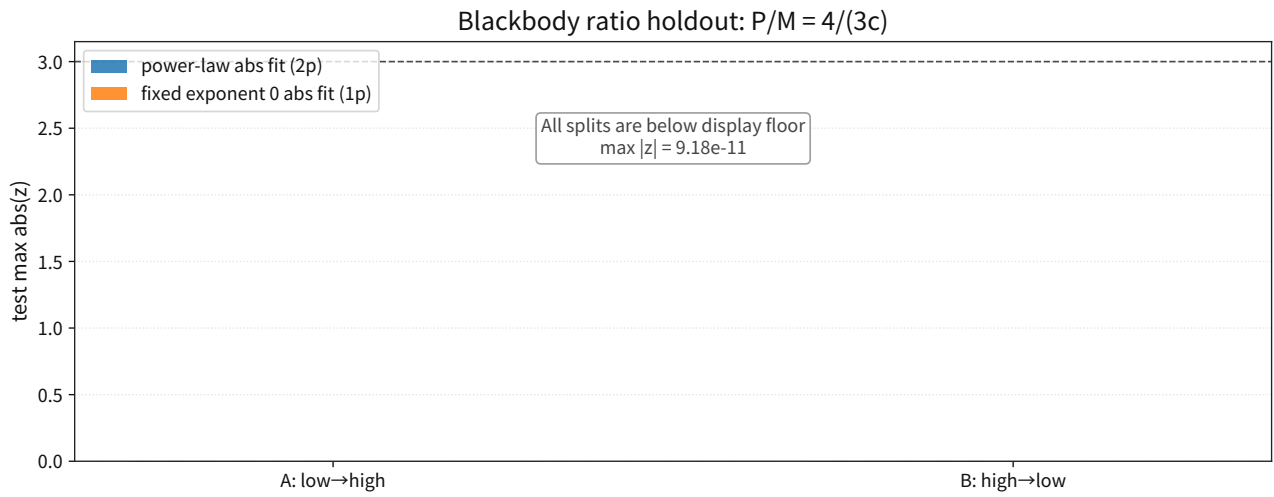


Figure 132: Comparison result for thermodynamics blackbody pressure flux ratio holdout splits.

- `output/public/quantum/thermo_blackbody_momentum_flux_ratio_holdout_splits_metrics.json` /

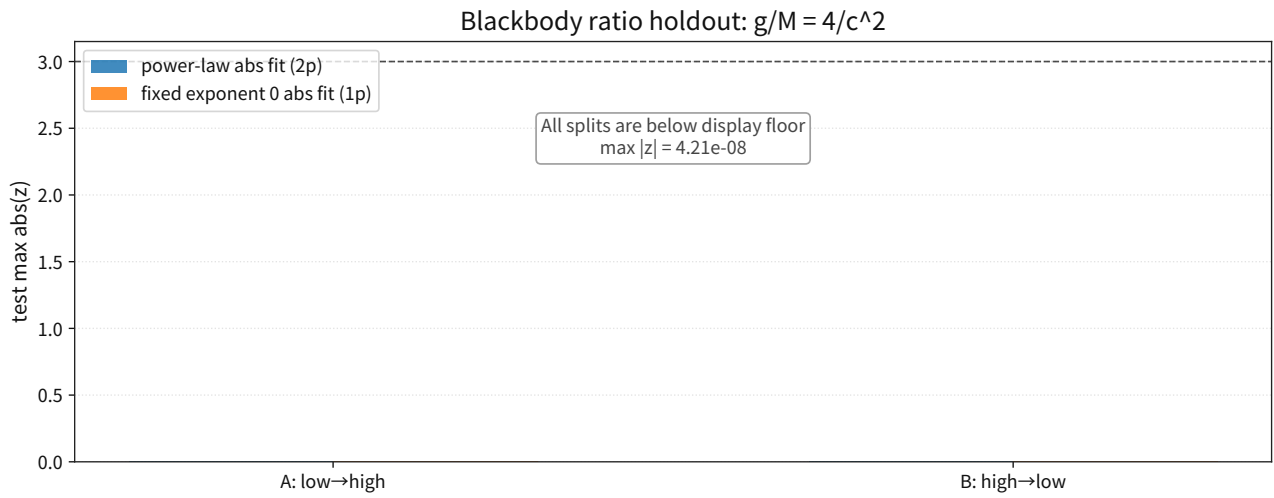


Figure 133: Comparison result for thermodynamics blackbody momentum flux ratio holdout splits.

- `output/public/quantum/thermo_blackbody_helmholtz_free_energy_flux_holdout_splits_metrics.json` /

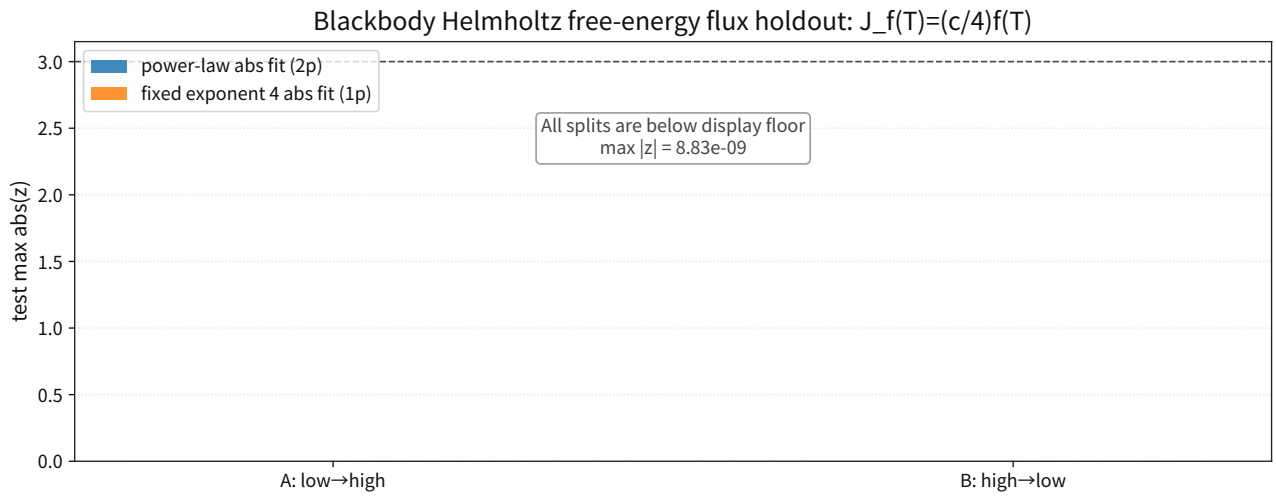


Figure 134: Comparison result for thermodynamics blackbody Helmholtz free energy flux holdout splits.

- output/public/quantum/thermo_blackbody_helmholtz_flux_ratio_holdout_splits_metrics.json /

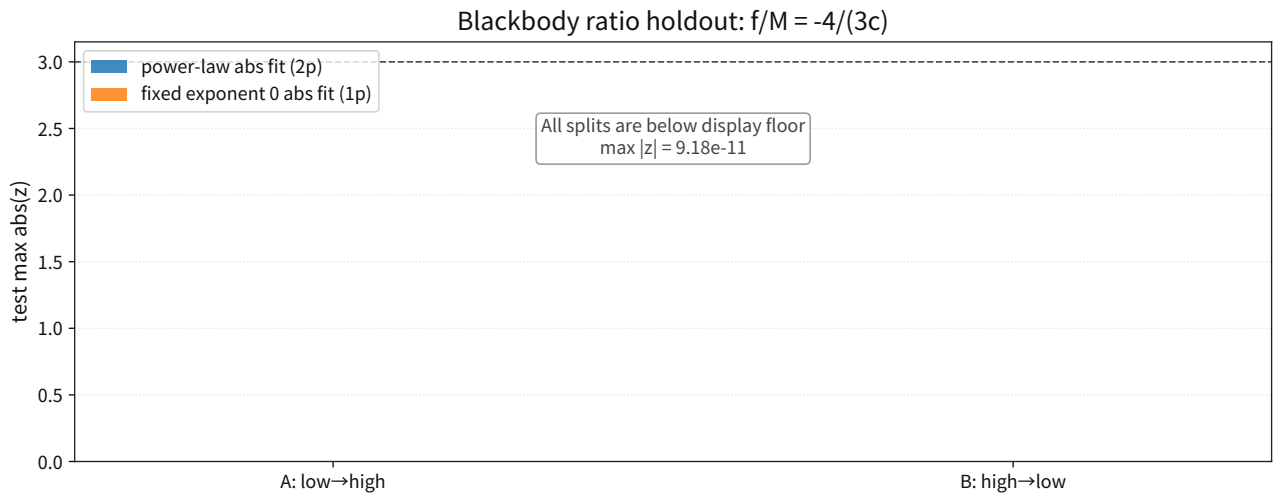


Figure 135: Comparison result for thermodynamics blackbody Helmholtz flux ratio holdout splits.

- output/public/quantum/thermo_blackbody_entropy_per_photon_holdout_splits_metrics.json /

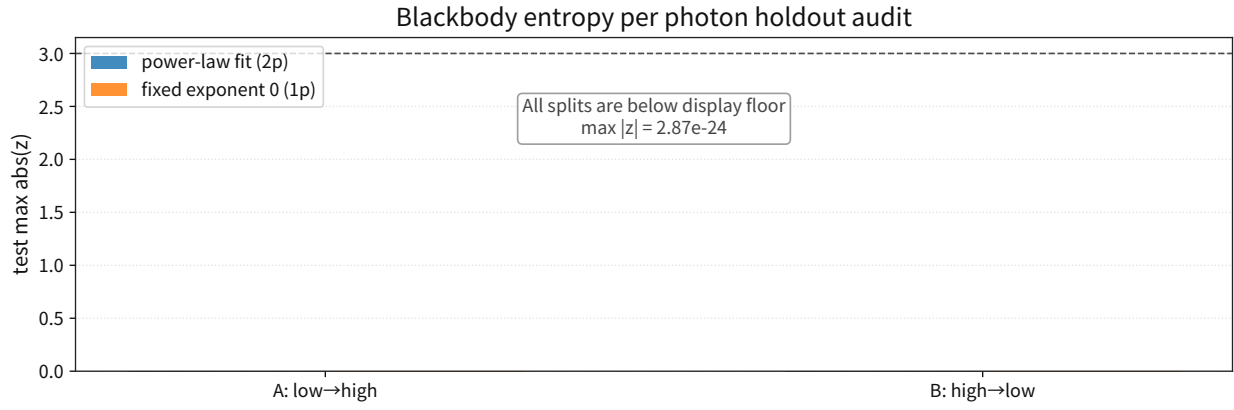


Figure 136: Comparison result for thermodynamics blackbody entropy per photon holdout splits.

- `output/public/quantum/thermo_blackbody_peak_frequency_per_wavelength_holdout_splits_metrics.json` /

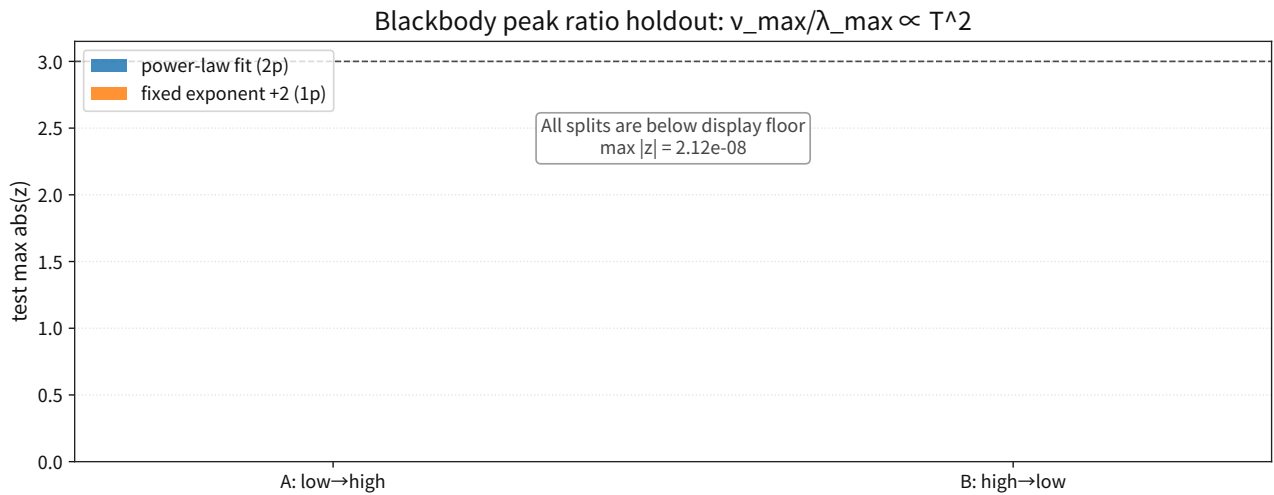


Figure 137: Comparison result for thermodynamics blackbody peak frequency per wavelength holdout splits.

List of holdout extension scripts:

- `scripts/quantum/thermo_blackbody_momentum_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_momentum_energy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_momentum_pressure_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_momentum_ratio_holdout_splits.py` (holdout extension)

- `scripts/quantum/thermo_blackbody_momentum_entropy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_momentum_per_photon_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_momentum_per_photon_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_photon_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_pressure_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_energy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_enthalpy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_momentum_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_energy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_enthalpy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_pressure_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_entropy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_flux_per_momentum_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_photon_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_entropy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_energy_per_entropy_flux_holdout_splits.py` (holdout extension)

- `scripts/quantum/thermo_blackbody_enthalpy_per_entropy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_entropy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_entropy_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_entropy_per_enthalpy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_energy_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_heat_capacity_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_enthalpy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_energy_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_enthalpy_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_entropy_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_energy_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_enthalpy_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_entropy_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_heat_capacity_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_photon_density_per_heat_capacity_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_heat_capacity_per_photon_holdout_splits.py` (holdout extension)

- scripts/quantum/thermo_blackbody_heat_capacity_per_photon_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_per_entropy_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_entropy_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_flux_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_momentum_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_energy_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_pressure_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_heat_capacity_enthalpy_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_per_photon_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_per_photon_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_energy_per_photon_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_helmholtz_per_photon_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_entropy_per_photon_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_flux_per_photon_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_photon_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_entropy_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_flux_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_flux_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_momentum_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_energy_ratio_holdout_splits.py (holdout extension)
- scripts/quantum/thermo_blackbody_enthalpy_pressure_ratio_holdout_splits.py (holdout extension)

- `scripts/quantum/thermo_blackbody_enthalpy_entropy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_free_energy_density_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_entropy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_energy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_enthalpy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_momentum_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_pressure_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_entropy_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_pressure_flux_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_momentum_flux_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_free_energy_flux_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_helmholtz_flux_ratio_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_entropy_per_photon_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_peak_frequency_wavelength_product_holdout_splits.py` (holdout extension)
- `scripts/quantum/thermo_blackbody_peak_frequency_per_wavelength_holdout_splits.py` (holdout extension)

16 References (Minimum)

Part IV is supporting material for verification, and the main context of the theoretical definition and primary observations is left to Parts I-III. In addition, the public verification materials and representative comparison reference documents on which this material is based are Let the following be true. [1][2][3][4][5][6]

References

- [1] P-model Verification Materials (supplementary material for reproduction commands, fixed artifact lists, audit gates, hashes, and manifests). Web: [EnterOgawa/p-model](#) (Releases: [GitHub Releases](#)). Accessed: 2026-02-15.
- [2] Will, C. M. “The Confrontation between General Relativity and Experiment.” *Living Reviews in Relativity* **17**, 4 (2014). DOI: 10.12942/lrr-2014-4. Accessed: 2026-01-03.
- [3] Ashby, N. “Relativity in the Global Positioning System.” *Living Reviews in Relativity* **6**, 1 (2003). DOI: 10.12942/lrr-2003-1. Accessed: 2026-01-03.
- [4] Event Horizon Telescope Collaboration. “First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole.” *The Astrophysical Journal Letters* **875**, L1 (2019). DOI: 10.3847/2041-8213/ab0ec7. Accessed: 2026-01-22.
- [5] Abbott, B. P., et al. “Observation of Gravitational Waves from a Binary Black Hole Merger.” *Physical Review Letters* **116**, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102. Accessed: 2026-01-06.
- [6] Martinelli, M., et al. “Euclid: Forecast constraints on the cosmic distance duality relation with complementary external probes.” *Astronomy & Astrophysics* (2021). DOI: 10.1051/0004-6361/202039637. arXiv:2007.16153. Accessed: 2026-01-08.